

Turing universality, computability, and incompleteness in hypergraph Turán theory

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Abstract

Given a finite family $\mathcal{F}$ of forbidden r -graphs, the Turán problem asks for the largest asymptotic edge density of $\mathcal{F}$ -free r -graphs and for the structure of those whose densities are close to this maximum. We show that both parts of this problem can encode arbitrary computation. Fix a universal Turing machine U . For every sufficiently large fixed r , there is a rational threshold $\tau_r \in (0, 1)$ such that from each binary word β , one can construct a finite family $\mathcal{F}_{r,\beta}$ of r -graphs whose Turán density equals τ_r when U does not halt on β and is strictly larger than τ_r when U halts.

The same halting distinction also governs the hypergraphs attaining, or nearly attaining, the extremal density. For every sufficiently large fixed r , we construct a finite family $\mathcal{G}_{r,\beta}$, together with a continuous statistic that depends only on r , such that, when U does not halt, the problem has a unique extremal limit and satisfies Erdős–Simonovits stability; when U halts, the extremal limit space instead splits into two nonempty compact phases on which the fixed statistic has opposite signs. Thus, uniqueness and connectedness of the extremal space, symmetry breaking, the existence of two separated phases, and Erdős–Simonovits stability are all undecidable. Both reductions are obtained by encoding finite local rules of computation into a weighted hypergraph problem and then removing the auxiliary roots, types, and labels without changing the exact extremal value.

The reductions are explicit enough to be verified within ZFC through finite certificates based on balanced blowups. Consequently, for every consistent computably axiomatized theory extending ZFC and every sufficiently large fixed r , one can construct a finite family $\mathcal{F}$ of r -graphs for which the true equality $\pi(\mathcal{F}) = \tau_r$ is neither provable nor refutable in that theory. Analogous independence statements hold for all five structural properties just listed.

We also include effective approximation for finite-family Turán densities and determine the complexity of exact comparison for finite and effectively presented forbidden families. Quantitatively, the smallest improvement witnesses have Busy-Beaver growth, and the density gains admit no uniform computable positive lower bound.

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1 Introduction

For an integer $r \geq 2$, an r -graph is a finite family of r -subsets of a finite vertex set. For an r -graph H , write $V(H)$ for its vertex set and $v(H) := |V(H)|$ for its order. We identify H with its edge set, and hence write $|H|$ for its number of edges. Given a finite family $\mathcal{F}$ of r -graphs, let $\text{ex}(n, \mathcal{F})$ be the largest number of edges in an n -vertex $\mathcal{F}$ -free r -graph and let $\pi(\mathcal{F}) := \lim_{n \rightarrow \infty} \text{ex}(n, \mathcal{F}) / \binom{n}{r}$ be its *Turán density*. By the averaging argument of Katona, Nemetz, and Simonovits [34], this limit exists. We call $\mathcal{F}$ *nondegenerate* if $\pi(\mathcal{F}) > 0$. Throughout this work, every forbidden r -graph is assumed to contain at least one edge. This excludes the special convention needed when a forbidden family contains an edgeless hypergraph.

For ordinary graphs, the theorem of Erdős and Stone [18], in the form refined by Erdős and Simonovits [17], reduces the density problem for a finite forbidden family to the smallest chromatic number among its members. No comparable finite invariant is known in higher uniformity, despite decades of work on hypergraph Turán problems. Even the density of the tetrahedron $K_4^{(3)}$, posed by Turán in 1941 [70], remains unknown. Kostochka [37] constructed exponentially many pairwise nonisomorphic examples of the conjectured extremal size on $3k$ vertices as k grows. Frohmader [22] later extended this phenomenon to the other two residue classes modulo 3 and obtained still larger families of examples.

A general method for studying such problems is Razborov’s flag-algebra method [58], which reduces the search for many extremal-density bounds to finite semidefinite programs. Razborov’s hypergraph applications [59], together with later work of Baber and Talbot [3] and Falgas-Ravry and Vaughan [19], determine or tightly bound the Turán densities of many small forbidden hypergraphs. The surveys of Füredi [23], Sidorenko [64], and Keevash [35] describe the development of hypergraph Turán theory and its principal open problems.

Beyond individual forbidden families, another line of work studies the set of all possible Turán densities. Brown and Simonovits [10] showed that the density of an arbitrary forbidden family is the infimum of the densities of its finite subfamilies. A number $\alpha \in [0, 1)$ is a *jump* for r if there exists $c > 0$ such that no number in $(\alpha, \alpha + c)$ is the Turán density of a family of r -graphs. Disproving Erdős’s conjecture [16],

Frankl and Rödl [21] famously proved the existence of non-jumps for every $r \geq 3$. Frankl, Peng, Rödl, and Talbot [20] found further smaller non-jumps, whereas Baber and Talbot [2] found the first jumps above the elementary initial interval for 3-graphs. More recently, Liu and Mubayi [41] proved that $4/9$ is a non-jump for 3-graphs; it is currently the smallest known non-jump for 3-graphs.

The study of the full set of Turán densities also includes questions about its gaps, topology, and arithmetic structure. Pikhurko [55] proved that the initial gap $(0, r!/r^r)$ is strictly longer than every other gap in the set of r -graph Turán densities. Chung and Graham [12, p. 95] asked whether a finite family of uniform hypergraphs could have irrational Turán density and conjectured that the answer was no. Baber and Talbot [3] gave the first counterexamples by constructing finite families of 3-graphs with irrational Turán density; Pikhurko [54] independently disproved the same conjecture by a different method. Pikhurko [54] and Grosu [26] also studied the topology and algebra of the set of densities, while Liu and Pikhurko [46] proved that it contains the interval $[1 - \delta_r, 1]$ for some $\delta_r > 0$ and every $r \geq 3$, thereby answering questions of Frankl, Peng, Rödl, and Talbot [20] and Grosu [26]. On the arithmetic side, Liu and Pikhurko [44] constructed finite-family densities of arbitrarily large algebraic degree, and Li, Liu, and Liu [38, Theorem 1.2] proved that, for every sufficiently large r , some finite family of r -graphs has transcendental Turán density, thereby answering Fox’s question for finite forbidden families [53, Section 3, Question 2].

Besides determining the Turán density, another central problem is to describe the structure of extremal and near-extremal hypergraphs and to establish stability. For nondegenerate graph Turán problems, the classical stability theorem, proved independently by Erdős and Simonovits [15, 65], gives a canonical asymptotic structure determined by the smallest chromatic number among the forbidden graphs. Hypergraph Turán problems can behave differently. A sequence of works by Liu and Mubayi [40], Liu, Mubayi, and Reiher [43], and Hou, Li, Liu, Mubayi, and Zhang [30] shows that describing the extremal behavior of a finite forbidden family may require two, any prescribed finite number, or infinitely many constructions that are mutually separated in edit distance. The mixing-pattern theorem of Liu and Pikhurko [45] gives further examples whose extremal hypergraphs arise by recursively mixing a prescribed finite collection of minimal patterns.

From an algorithmic perspective, the main difficulty in determining exact Turán densities and classifying extremal structures is the passage from finite optimization to asymptotic information. Note that for each fixed n , one can compute $\text{ex}(n, \mathcal{F})$ by checking every r -graph on $[n]$ and taking the largest edge count among those that are $\mathcal{F}$ -free. By contrast, determining $\pi(\mathcal{F})$ exactly requires controlling these finite optimization problems for arbitrarily large n , while classifying the extremal structure additionally requires describing all nearly extremal hypergraphs. For nondegenerate graph Turán problems, the Erdős–Stone theorem [18] and the classical stability theorem of Erdős and Simonovits [15, 65] show that the smallest chromatic number among the forbidden graphs determines both the limiting density and the asymptotic structure of nearly extremal graphs. Thus, for finite forbidden families of graphs, both the limiting density and the near-extremal structure can be determined algorithmically. The phenomena described above show that the corresponding numerical and structural questions for hypergraphs are considerably more intricate. This leads to the following questions.

- (i) Can the number $\pi(\mathcal{F})$ be approximated effectively from a finite forbidden family?
- (ii) Can one decide exact relations such as $\pi(\mathcal{F}) = q$ or $\pi(\mathcal{F}) = \pi(\mathcal{G})$?
- (iii) Can one decide whether all extremal constructions have one asymptotic structure, or whether several macroscopically separated phases occur?

These questions need not have the same algorithmic behavior. Finite computations may approximate a Turán density to any prescribed accuracy without ever certifying whether it is exactly equal to a given number. At every fixed precision, an arbitrarily small positive gap is indistinguishable from equality. Even when the extremal value is known, one may still be unable to decide whether all nearly extremal hypergraphs resemble one canonical construction or split into several incompatible phases. Our results show that numerical approximation, exact comparison, and structural classification have different logical complexity, although they are proved here by related reductions.

We recall the basic facts from computability theory needed to state the main results. A *Turing machine* is a finite algorithmic procedure that, on a binary input word, either halts after finitely many steps or continues forever. A *universal Turing machine* can simulate any such procedure once the machine and its input have been encoded as one binary word. The halting problem provides a common form for every

computational search whose success admits a finite certificate. Indeed, one can construct a machine that examines all possible certificates and halts precisely when it finds a valid one. After this search procedure is encoded as an input word for a universal machine, the original property holds precisely when the universal machine halts on that word. Thus the halting problem is universal among problems whose positive instances can be recognized by finite computation.

By interleaving simulations on all binary inputs, one can enumerate every input on which a fixed universal machine eventually halts. The absence of an input from the list gives no certificate of nonhalting, since its computation may halt much later. In fact, no algorithm can correctly decide between eventual halting and nonhalting for every input. Because a fixed universal machine can simulate every effective search, all computational instances can be encoded in its input words. Since its halting problem remains undecidable, it is enough for our purposes to encode the halting behavior of this one machine.

Following Turing [71], we fix a universal Turing machine U . We write $\{0, 1\}^*$ for the set of all finite binary words. We choose U so that its word-halting domain

$$K_U := \{\beta \in \{0, 1\}^* : U \text{ halts on input } \beta\}$$

is Σ_1^0 -complete in the modern recursion-theoretic terminology; see Rogers [62] for the standard theory of universal machines and effective recodings. Here Σ_1^0 means that membership can be confirmed by a finite computation, although the search for such a confirmation may run forever on a nonmember. Dually, a set is Π_1^0 when its complement is Σ_1^0 , so nonmembership has a finite computational certificate. Completeness means that K_U is universal among such sets, in the sense that membership in any one of them can be translated effectively into the question whether U halts. Here a program index is simply a natural-number code for a program. Concretely, we require a total computable machine-input translator that maps standard indices and inputs to binary words while preserving halting. By such a recoding, we also arrange that a fixed explicit word β_∞ does not halt. Write $U(\beta) \downarrow$ when it halts on β and $U(\beta) \uparrow$ otherwise. Our construction assigns a finite forbidden family to each word β , so an algorithm deciding the resulting exact Turán relation would decide whether U halts on β .

The two main construction theorems are stated below. The first expresses halting as an exact Turán-density dichotomy. Its main point is that neither the uniformity nor the comparison value encodes the computation. For every sufficiently large fixed uniformity, one rational threshold is chosen in advance, while the finite forbidden family varies with the input word. The second construction uses this numerical dichotomy to obtain a one-phase versus two-phase classification of the extremal space, including edit stability of nearly extremal finite r -graphs. Further properties concerning effective presentations and finite recursive value schemes are recorded after these numerical and structural constructions.

Here and below, *effectively computable* means that the proof gives a terminating algorithm that produces the stated integer from the fixed finite data. Thus the lower bound r_{num} below can actually be found, rather than being asserted to exist without a procedure for determining it. A *total algorithm* halts and produces the prescribed output on every admissible input. In particular, the algorithm below must construct the family $\mathcal{F}_{r,\beta}$ in finite time whether or not U halts on β .

Theorem 1.1. *There exist an effectively computable integer r_{num} and a total algorithm with the following property. On input an integer $r \geq r_{\text{num}}$ and a binary word β , the algorithm outputs a rational number $\tau_r \in \mathbb{Q} \cap (0, 1)$, which depends only on r , and a finite family $\mathcal{F}_{r,\beta}$ of r -graphs such that the following hold.*

- (i) U does not halt on β if and only if $\pi(\mathcal{F}_{r,\beta}) = \tau_r$.
- (ii) U halts on β if and only if $\pi(\mathcal{F}_{r,\beta}) > \tau_r$.

Pikhurko [54, Question 29] asked whether $\pi(\mathcal{F}) \leq \alpha$ is decidable when the input is a finite forbidden family $\mathcal{F}$ and a rational number α . In a related analytic setting, Hatami and Norine [28, Theorem 2.12] proved undecidability for universal linear inequalities in graph homomorphism densities. For ordinary Turán densities, Li, Liu, and Liu [38, Theorem 1.1] proved that, for every sufficiently large fixed uniformity, rational-threshold comparison and comparison of two finite forbidden families are undecidable. Their proof combines the effective Diophantine representation of Davis, Putnam, and Robinson [13] and Matiyasevich [49] with a finite-sampling construction.

Theorem 1.1 does not claim the undecidability of density comparison as a new result. Its additional feature is that the comparison value is independent of the encoded input. Indeed, since the explicit word β_∞ does not halt, $\pi(\mathcal{F}_{r,\beta_\infty}) = \tau_r$. Thus, for every fixed $r \geq r_{\text{num}}$, it is undecidable, given a finite family $\mathcal{F}$ of r -graphs, whether $\pi(\mathcal{F}) = \pi(\mathcal{F}_{r,\beta_\infty})$. For each fixed $r \geq r_{\text{num}}$, the same rational threshold τ_r is used

for every word β . The compiler encodes the computation directly in finite local hypergraph constraints rather than first representing it by a Diophantine equation. Several later results in this work also sharpen this fixed-threshold conclusion. [Theorem 1.5](#) and [Proposition 19.2](#) determine the exact Σ_1^0/Π_1^0 complexity of two-family comparison and comparison with τ_r , as well as the corresponding fixed-template sharpness problem, and [Lemma 17.1](#) gives a fixed baseline construction that is extremal exactly when U does not halt. In the halting case, finite witnesses with larger density exist, [Theorems 20.1](#) and [21.1](#) show explicit bounds on their order and on the positive density gap.

[Theorem 1.1](#) turns halting into the distinction between equality and strict inequality of Turán densities. We next show that the same numerical distinction can be detected from the structure of the extremal hypergraphs.

For this purpose, we use Austin’s exchangeability approach [\[1\]](#) and the measure-theoretic hypergraph-limit theory of Elek and Szegedy [\[14\]](#). The descriptions below serve only as working definitions for the Introduction; precise formulations are given in [Section 8](#).

Let $\mathfrak{W}_r$ denote the compact space of dense r -graph limits, equipped with the *homomorphism-density topology*, that is, the topology of coordinatewise convergence of all homomorphism densities [\[75, Section 1.4\]](#). Let $\mathcal{E}(\mathcal{F})$ be the set of $\mathcal{F}$ -free limits of edge density $\pi(\mathcal{F})$. A *quantum r -graph statistic* is a finite real linear combination of homomorphism densities. In this work, a *finite-step limit* means a limit generated by finitely many vertex types, fixed limiting type proportions, and a fixed list of permitted multisets of r types; repeated occurrences of one type in an edge profile are allowed. In the statements below, such a limit W is accompanied by one fixed finite-profile construction that generates it. We say that a sequence (G_n) of r -graphs is $o(v(G_n)^r)$ -close in edit distance to W if, on the vertex set of each G_n , there is a standard integer realization of this construction differing from G_n in $o(v(G_n)^r)$ edges.

Theorem 1.2. *There exist an effectively computable integer r_{str} and a total algorithm with the following property. On input an integer $r \geq 2$ and a binary word β , the algorithm outputs a finite family $\mathcal{G}_{r,\beta}$ of r -graphs. For every $r \geq r_{\text{str}}$ there are a finite-step limit $W_{0,r} \in \mathfrak{W}_r$ and a continuous quantum r -graph statistic $\text{Sep}_r: \mathfrak{W}_r \rightarrow \mathbb{R}$, depending only on r , with $\text{Sep}_r(W_{0,r}) = 0$, such that the following hold.*

- (i) *If U does not halt on β , then $\mathcal{E}(\mathcal{G}_{r,\beta}) = \{W_{0,r}\}$. Moreover, every asymptotically extremal $\mathcal{G}_{r,\beta}$ -free sequence (G_n) is $o(v(G_n)^r)$ -close in edit distance to $W_{0,r}$.*
- (ii) *If U halts on β , then there are nonempty compact sets $\mathcal{E}_{r,+}(\beta), \mathcal{E}_{r,-}(\beta) \subseteq \mathfrak{W}_r$ and $c_{r,\beta} > 0$ such that $\mathcal{E}(\mathcal{G}_{r,\beta}) = \mathcal{E}_{r,+}(\beta) \dot{\cup} \mathcal{E}_{r,-}(\beta)$, while $\text{Sep}_r(W) = c_{r,\beta}$ for every $W \in \mathcal{E}_{r,+}(\beta)$ and $\text{Sep}_r(W) = -c_{r,\beta}$ for every $W \in \mathcal{E}_{r,-}(\beta)$. Moreover, every asymptotically extremal $\mathcal{G}_{r,\beta}$ -free sequence (G_n) is $o(v(G_n)^r)$ -close in edit distance to finite canonical completed models representing one of these two phases, in the precise sense of [Proposition 15.3](#).*

The two alternatives in [Theorem 1.2](#) lead directly to structural undecidability. If $\mathsf{U}(\beta)$ does not halt, the extremal limit space is a singleton and all asymptotically extremal sequences follow one fixed finite profile, which gives the usual Erdős–Simonovits stability conclusion. If $\mathsf{U}(\beta)$ halts, the space splits into two nonempty compact phases separated by the fixed statistic Sep_r , and no single profile describes both. Thus an algorithm deciding any property that separates these cases would decide the halting problem.

For the next corollary, fix $r \geq r_{\text{str}}$ and let Sep_r be the statistic supplied by [Theorem 1.2](#) (note that it is fixed before the forbidden family is given). For a finite family $\mathcal{F}$ of r -graphs, we say that its extremal problem has a *unique extremal limit* when $|\mathcal{E}(\mathcal{F})| = 1$, and a *connected extremal limit space* when $\mathcal{E}(\mathcal{F})$ is connected in the homomorphism-density topology. It exhibits *Sep_r -symmetry breaking* when Sep_r takes both positive and negative values on $\mathcal{E}(\mathcal{F})$. It has a *Sep_r -separated two-phase decomposition* when there is $\eta > 0$ such that every $W \in \mathcal{E}(\mathcal{F})$ satisfies either $\text{Sep}_r(W) \geq \eta$ or $\text{Sep}_r(W) \leq -\eta$, and both alternatives occur. Finally, it has *Erdős–Simonovits stability* when there is one fixed finite-type construction, with fixed limiting type proportions and a fixed list of permitted multisets of r types, whose limit is extremal and such that every asymptotically extremal $\mathcal{F}$ -free sequence (G_n) has edit distance $o(v(G_n)^r)$ from the class of its standard integer realizations on $v(G_n)$ vertices.

Corollary 1.3. *For every fixed $r \geq r_{\text{str}}$, each of the following decision problems is undecidable. In each case, the input is a finite family $\mathcal{F}$ of r -graphs, and the question is whether the corresponding extremal problem has*

- (i) *a unique extremal limit;*
- (ii) *a connected extremal limit space;*

- (iii) *Sep_r-symmetry breaking*;
- (iv) *a Sep_r-separated two-phase decomposition*;
- (v) *Erdős–Simonovits stability*.

The numerical construction also supplies, for each $r \geq r_{\text{num}}$, a fixed baseline template $K_{*,r}$ whose uniform weighting is optimal and whose balanced blowups have limiting density τ_r . This template is asymptotically optimal for $\mathcal{F}_{r,\beta}$ exactly when $\mathcal{U}$ does not halt on β ; in that case, its balanced blowups are asymptotically extremal for $\mathcal{F}_{r,\beta}$. This gives a fixed-template normal form for every Π_1^0 predicate.

This fixed-template normal form has a purely finite certificate at every admissible uniformity. For $r \geq r_{\text{num}}$, write the corresponding threshold in lowest terms as $\tau_r = p/q$. For a finite r -graph H such that every blowup of H is $\mathcal{F}$ -free, the integer inequality $qr!|H| > pv(H)^r$ says that the balanced blowups of H have limiting density larger than τ_r . We prove that such an H exists if and only if $\pi(\mathcal{F}) > \tau_r$. For the compiled families, this correspondence preserves certificates in both directions: a finite halting trace produces such an H , and any such H forces the encoded computation to halt within $v(H)$ steps. All parameters in this certificate are computable from r before the input word or formal theory is supplied. This makes the proof-theoretic content of the compiler explicit.

This finite certificate allows the undecidability construction to be combined with the Gödel–Rosser argument [24, 63]. Applied to Zermelo–Fraenkel set theory with the axiom of choice (ZFC), Rosser’s theorem effectively produces a sentence ρ_{ZFC} that ZFC can neither prove nor refute if ZFC proves no contradiction. We apply the numerical compiler to a program that searches for a finite counterexample to ρ_{ZFC} . The certificate-preserving form of the compiler can be verified in ZFC and shows there that ρ_{ZFC} is equivalent to the equality $\pi(\mathcal{F}) = \tau_r$. Rosser independence therefore transfers to this Turán-density equality, while the fixed baseline template gives the ZFC-provable lower bound $\pi(\mathcal{F}) \geq \tau_r$.

Theorem 1.4. *Let r_{num} and the thresholds τ_r be as in Theorem 1.1. There is a total algorithm which, on input an integer $r \geq r_{\text{num}}$, outputs a finite r -graph $K_{*,r}$ and a finite family $\mathcal{F}_{r,\text{ZFC}}$ of r -graphs. The uniform weighting of $K_{*,r}$ is optimal, and its balanced blowups have limiting density τ_r . If ZFC is consistent, then the following statements hold.*

- (i) *ZFC proves that every blowup of $K_{*,r}$ is $\mathcal{F}_{r,\text{ZFC}}$ -free, that the uniform weighting is optimal with normalized value τ_r , and hence that $\pi(\mathcal{F}_{r,\text{ZFC}}) \geq \tau_r$.*
- (ii) *The equality $\pi(\mathcal{F}_{r,\text{ZFC}}) = \tau_r$ is true, so $K_{*,r}$ is asymptotically optimal for $\mathcal{F}_{r,\text{ZFC}}$.*
- (iii) *ZFC proves neither $\pi(\mathcal{F}_{r,\text{ZFC}}) = \tau_r$ nor $\pi(\mathcal{F}_{r,\text{ZFC}}) > \tau_r$. Consequently, the asymptotic optimality of $K_{*,r}$ is independent of ZFC.*

Theorem 1.4 is the most concrete specialization of two results that are uniform in the choice of theory. A theory is *computably axiomatized* if an algorithm can enumerate its axioms. Given an integer $r \geq r_{\text{num}}$ and an index a for such an enumerator, one total algorithm produces a finite family $\mathcal{F}_{r,a}$. Whenever the enumerated theory T_a is a consistent extension of ZFC, parts (i)–(iii) hold with T_a and $\mathcal{F}_{r,a}$ in place of ZFC and $\mathcal{F}_{r,\text{ZFC}}$. The threshold τ_r and the baseline template $K_{*,r}$ depend only on r , and the algorithm does not need to determine whether T_a satisfies these hypotheses. Applying the same Rosser input to the structural compiler makes each of the five properties in Corollary 1.3 independent of T_a , with uniqueness, connectedness, and Erdős–Simonovits stability holding and the two phase-separation properties failing. The full numerical and structural statements are proved in Theorems 18.2 and 18.6.

We next state two further consequences concerning exact comparison and the size of finite witnesses. Using the effective coding of finite forbidden families fixed in Section 2, we regard the following comparisons as decision problems on natural-number codes. Completeness is with respect to many-one reductions.

Finite-family Turán densities can be approximated uniformly to arbitrary precision (see Corollary 19.1), but approximation alone need not determine whether two limiting values are exactly equal. Operationally, a Σ_1^0 property is one whose positive instances eventually receive a finite computational certificate, while a Π_1^0 property has such certificates for its negative instances. The theorem below shows that strict separation is finitely detectable, equality and the corresponding non-strict comparisons lie on the complementary side, and none of these classifications can be improved.

Theorem 1.5. *For every fixed $r \geq r_{\text{num}}$, the following statements hold for finite families $\mathcal{F}$ and $\mathcal{G}$ of r -graphs.*

- (i) *The relations $\pi(\mathcal{F}) > \pi(\mathcal{G})$, $\pi(\mathcal{F}) < \pi(\mathcal{G})$, and $\pi(\mathcal{F}) \neq \pi(\mathcal{G})$ are Σ_1^0 -complete.*

(ii) The relations $\pi(\mathcal{F}) \leq \pi(\mathcal{G})$, $\pi(\mathcal{F}) \geq \pi(\mathcal{G})$, and $\pi(\mathcal{F}) = \pi(\mathcal{G})$ are Π_1^0 -complete.

The hardness remains even when one side of the comparison is fixed. For $>$, $\neq$, $=$, and $\leq$, one may fix $\mathcal{G} = \mathcal{F}_{r,\beta_\infty}$; for $<$ and $\geq$, one may instead fix $\mathcal{F} = \mathcal{F}_{r,\beta_\infty}$. Comparison with the fixed rational threshold τ_r is likewise Σ_1^0 -complete for $>$ and $\neq$, and Π_1^0 -complete for $=$ and $\leq$.

Theorem 1.5 classifies whether a strict density improvement exists. We next ask the corresponding quantitative question. When an improvement exists, how large must the smallest finite template witnessing it be? To state the result, let $\lambda_r(G)$ denote the largest limiting edge density among the blowups of a finite r -graph G . We call G $\mathcal{F}$ -hom-free if no member of $\mathcal{F}$ admits a hypergraph homomorphism to G ; equivalently, every blowup of G is $\mathcal{F}$ -free. For a word β , let $T(\beta)$ be the running time of U on β , with value ∞ when the computation does not halt. For fixed $r \geq r_{\text{num}}$, define

$$\omega_r(\beta) := \min \{v(G) : G \text{ is } \mathcal{F}_{r,\beta}\text{-hom-free and } \lambda_r(G) > \lambda_r(K_{*,r}) = \tau_r\},$$

with value ∞ when the set is empty. Thus $\omega_r(\beta)$ is the order of the smallest finite template whose blowups avoid $\mathcal{F}_{r,\beta}$ while attaining a density strictly above the fixed baseline τ_r . Equivalently, it measures the size of the smallest finite certificate of a strict density improvement. To capture the worst-case growth of this minimum certificate size, we maximize it over all inputs of length at most m for which an improvement exists. In the spirit of Radó's Busy Beaver functions [57], put

$$\begin{aligned} \text{BB}_U(m) &:= \max \{T(\beta) : |\beta| \leq m \text{ and } T(\beta) < \infty\}, \\ \text{BB}_r^{\text{Tur}}(m) &:= \max \{\omega_r(\beta) : |\beta| \leq m \text{ and } \omega_r(\beta) < \infty\}, \end{aligned}$$

where each maximum is interpreted as 0 when its defining set is empty. We write $\langle \mathcal{F} \rangle$ for the canonical binary code of a finite family fixed in Section 2 and $|\langle \mathcal{F} \rangle|$ for its length.

Thus $\text{BB}_r^{\text{Tur}}(m)$ records the largest minimum witness order among inputs of length at most m for which an improving template exists. The construction shows that such a template exists exactly when U halts. Any improving template must be large enough to recover the encoded computation, whereas a halting computation yields a periodic tableau and hence a template whose order is at most quadratic in the running time and input length. The next theorem makes this comparison precise.

Theorem 1.6. *For every fixed $r \geq r_{\text{num}}$, there is a constant $C_r > 0$ such that, for every $m \in \mathbb{N}$, we have*

$$\text{BB}_U(m) \leq \text{BB}_r^{\text{Tur}}(m) \leq C_r (\text{BB}_U(m) + m + 1)^2 + C_r.$$

Two noncomputability consequences are worth recording. Like Radó's classical Busy Beaver function [57], the function BB_r^{Tur} is not bounded above by any total computable function. Moreover, there is no total computable function $g : \mathbb{N} \rightarrow \mathbb{N}$ such that $\omega_r(\beta) \leq g(|\langle \mathcal{F}_{r,\beta} \rangle|)$ for every input β on which U halts. Informally, no computable bound in the input length controls how large the smallest improving template may have to be. More intrinsically, the same remains true when the size of the instance is measured by the length of the compiled forbidden-family code.

Taken together, the numerical, structural, and incompleteness results show that a fully effective Erdős–Stone-type classification of ordinary nondegenerate hypergraph Turán problems cannot exist in general. No algorithm can recover from every finite forbidden family both its exact Turán density and the asymptotic structure of its extremal configurations. The obstruction appears at three distinct levels. Turán densities can be approximated uniformly, but exact comparison is undecidable; deciding whether the extremal limit space has one canonical phase or two separated phases is likewise undecidable; and concrete true density equalities can be independent of a prescribed consistent computably axiomatized theory. Thus the results separate computable approximation from exact comparison, numerical value from extremal structure, and mathematical truth from formal provability.

We now outline the two constructions underlying the main theorems.

Finite local tiling rules have long provided a way to encode computation. Wang introduced the colored-edge tile formalism and posed the associated domino problem [72]. Berger proved this problem undecidable by encoding Turing-machine behavior into locally matching tiles [7], and Robinson later gave a streamlined construction in which the horizontal and vertical directions represent tape position and time, respectively [60]. The construction below follows this space–time paradigm, but its initialization, reset, and error-handling rules are tailored to make halting equivalent to the existence of a finite periodic tableau.

The proof of [Theorem 1.1](#) has four stages. First, we augment the space–time simulation of the fixed universal machine U with initialization, reset, and error-handling phases. The reset returns a halting computation to its marked initial row, so halting is equivalent to the existence of a finite periodic space–time tableau. A finite five-rail system of partial matchings then enforces the local tableau rules, and every closed relation component containing a coordinate of the marked five-tuple reconstructs a genuine periodic tableau; see [Lemmas 3.2](#) and [3.5](#). Second, we associate with the five rails a homogeneous polynomial of degree five in the vertex weights. A simple unmarked five-rail realization attains a fixed input-independent baseline, and the five-rail energy inequality shows that the defects caused by missing prescribed relation edges absorb any possible gain from an incomplete marked record. Consequently, the optimum equals the baseline in the nonhalting case and is strictly larger in the halting case; see [Theorem 4.1](#).

Third, tagged five-set records represent this weighted polynomial without loss, and fixed root-support matchings encode their types and tags as edges of simple uncolored rooted r -graphs. Balancing and calibrating the root weights convert the weighted baseline into the class Lagrangian $\tau_r/r!$; see [Lemma 5.1](#) and [Theorem 6.3](#). Finally, a bounded persistent-witness argument eliminates the auxiliary roots and produces an effective finite obstruction family. Pair-covering extensions then convert the class Lagrangian into the Turán density of the finite family $\mathcal{F}_{r,\beta}$, supplying the factor $r!$ and hence the threshold τ_r ; see [Lemmas 2.9](#) and [7.2](#). A useful feature of the construction is that it encodes the computation directly by finite local tableau constraints rather than first translating the halting problem into an arithmetic surrogate such as a Diophantine equation. The four-stage compiler is summarized in [Figure 1](#) at the beginning of [Part I](#).

For the structural construction, set $k := r_{\text{num}}$, $\tau_{\text{base}} := \tau_k$, and $\mathcal{F}_\beta := \mathcal{F}_{k,\beta}$. The construction uses only the finite description of $\mathcal{F}_\beta$. In the analysis, $z_\beta := \pi(\mathcal{F}_\beta)$ equals τ_{base} when U does not halt and exceeds it when U halts, by [Theorem 1.1](#).

After this setup, the proof of [Theorem 1.2](#) proceeds in five stages. First, a symmetric phase objective converts the numerical excess $z_\beta - \tau_{\text{base}}$ into a choice of phases. The phase-roof lemma gives one neutral outer maximizer when the excess is zero and exactly two pure-sign outer maximizers when it is positive; see [Lemma 9.1](#). Allowing the inner optimizer to vary turns the two pure-sign choices into two compact phase sets, rather than necessarily two individual extremal limits.

Second, a finite asymmetric root frame and regular root-support designs realize this phase optimization at the level of completed weighted models. Data splitting and root balancing prevent additional outer maximizers. The resulting weighted compiler also supplies a fixed quantum r -graph statistic Sep_r , independent of β , which vanishes on the neutral optimizer and has opposite nonzero values on the two active phases; see [Theorem 11.1](#).

Third, pair-covering extensions of the rooted obstruction family, together with finite star blockers, produce the ordinary finite forbidden family $\mathcal{G}_{r,\beta}$. Pair covering preserves the exact extremal value and forces symmetrized $\mathcal{G}_{r,\beta}$ -free r -graphs into the canonical class, while the star blockers rule out vertices that cannot be assigned a canonical role; see [Propositions 13.1](#), [14.2](#) and [14.3](#).

Fourth, the criterion of Liu, Mubayi, and Reiher [42], recorded in [Theorem 15.1](#), upgrades symmetrized stability and vertex extendability to degree stability. This is then converted into ordinary edit stability, and the quantitative phase estimate places every asymptotically extremal sequence near the neutral finite-profile construction in the zero-gap case or near the union of the two pure-sign phase classes in the positive-gap case; see [Theorem 15.2](#) and [Proposition 15.3](#).

Finally, [Theorem 16.1](#) identifies the extremal limit space of $\mathcal{G}_{r,\beta}$ with the weighted optimizer space. This transfers the singleton-versus-two-phase dichotomy and the fixed separating statistic to the ordinary Turán problem and completes the proof of [Theorem 1.2](#).

The five-stage proof chain is summarized in [Figure 11](#) at the beginning of [Part II](#); the analytic phase transition is illustrated later in [Figure 12](#).

We conclude the introduction with a brief guide to the rest of the paper.

[Section 2](#) sets up the common framework of homomorphisms, blowups, Lagrangians, finite coding, and rooted hypergraphs, including the finite-template characterization of Turán density. It also proves the finite-root elimination and pair-covering transfer principles used by both main constructions. The main body is then divided into four parts. [Part I](#) gives the numerical construction. It encodes a reset computation by finite local records, proves the five-rail weighted estimate, removes all type labels and

roots, and transfers the resulting optimum to the ordinary Turán problem in [Theorem 1.1](#). [Part II](#) begins by introducing limit spaces and edit stability. It then presents the combinatorial phase gadget, isolates the analytic one-phase/two-phase mechanism, and realizes the resulting phase function by finite rooted hypergraphs. Star blockers and pair-covering extensions yield degree and edit stability, after which the extremal limit space is identified exactly with the weighted optimizer space, proving [Theorem 1.2](#).

The final two parts address incompleteness and further properties of Turán densities. [Part III](#) develops balanced finite certificates and a primitive recursive normal form, then uses the explicit finite approximation from [Section 2](#) to express exact density equality. It gives the numerical and structural coding needed to formalize the construction in ZFC and proves the corresponding Gödel–Rosser incompleteness theorems, including [Theorem 1.4](#). [Part IV](#) records uniform approximation and exact-comparison results, quantitative properties of improvement witnesses and density gaps, the spectra of recursive and c.e. forbidden families, and the limitations of two finite semialgebraic value schemes.

2 Basic definitions and preliminary tools

This section fixes the homomorphism, blowup, cloning, Lagrangian, and computability conventions used throughout this work. It also records the standard finite-template characterization of Turán density, an explicit finite-order approximation, and two transfer principles needed by both constructions. The first eliminates auxiliary roots, and the second converts the Lagrangian of a monotone class defined by finitely many forbidden r -graphs into the Turán density of a finite ordinary forbidden family.

Fix an integer $r \geq 2$ and retain the r -graph, vertex-set, order, edge-count, extremal-number, and Turán-density conventions from the Introduction. For a set V and an integer $s \geq 0$, write $\binom{V}{s} := \{S \subseteq V : |S| = s\}$; when V is finite, this family has cardinality $\binom{|V|}{s}$. For an integer $t \geq 0$, write $[t] := \{1, \dots, t\}$, with $[0] = \emptyset$. A class of r -graphs is *monotone* if it is closed under taking subgraphs. For a family $\mathcal{F}$ of r -graphs, an r -graph H is $\mathcal{F}$ -free if it contains no member of $\mathcal{F}$ as a subgraph. Let $\text{Forb}(\mathcal{F})$ denote the class of all finite $\mathcal{F}$ -free r -graphs. When $\mathcal{F}$ is finite, it is convenient to regard the finite edge lists as the input. When $\mathcal{F}$ is infinite, no effectiveness is assumed unless explicitly stated.

For a binary relation $R \subseteq X \times Y$, write

$$\text{dom } R := \{x \in X : (x, y) \in R \text{ for some } y \in Y\} \quad \text{and} \quad \text{im } R := \{y \in Y : (x, y) \in R \text{ for some } x \in X\}.$$

For a partial map $f: X \rightarrow Y$, we apply the same notation to the associated binary relation $\{(x, f(x)) : x \in \text{dom } f\} \subseteq X \times Y$.

Following Brown and Simonovits [[10](#), Definition 3, p. 150], a *blowup* of an r -graph G is specified by pairwise disjoint finite vertex classes $(V_u)_{u \in V(G)}$, which may be empty. Its vertex set is $\bigcup_{u \in V(G)} V_u$, and an r -set is an edge exactly when it consists of one vertex from each of $V_{u_1}, \dots, V_{u_r}$ for some $\{u_1, \dots, u_r\} \in G$. A class of r -graphs is *blowup closed* if it contains every blowup of each of its members.

For $x \in V(G)$, *cloning x into a nonempty set C* means taking the special blowup in which C replaces x and every other vertex class is a singleton. We call C the *clone class* replacing x and its members *clones* of x . Equivalently, cloning retains every edge not containing x and replaces each edge $e \ni x$ by the edges $(e \setminus \{x\}) \cup \{y\}$ for $y \in C$; in particular, no edge contains two vertices of C . If $|C| = 1$, cloning changes G only up to relabeling. An *iterated cloning* consists of finitely many such operations. Every blowup with nonempty vertex classes is isomorphic to a hypergraph obtained by iterated cloning; in general, one first deletes the vertices whose blowup classes are empty.

We next recall the homomorphism language that records containment in blowups. For a map $\varphi: V(F) \rightarrow V(G)$ and a set $e \subseteq V(F)$, put $\varphi(e) := \{\varphi(v) : v \in e\}$. For finite r -graphs F and G , a map $\varphi: V(F) \rightarrow V(G)$ is a *hypergraph homomorphism* from F to G if $\varphi(e) \in G$ for every $e \in F$. Write $\text{Hom}(F, G)$ for the set of all such maps and put $\text{hom}(F, G) := |\text{Hom}(F, G)|$. We write $F \rightarrow G$ when there is a hypergraph homomorphism from F to G . Equivalently, $F \rightarrow G$ if and only if $\text{Hom}(F, G) \neq \emptyset$, or if and only if $\text{hom}(F, G) > 0$. Every hypergraph homomorphism is injective on each edge. For $\varphi \in \text{Hom}(F, G)$, its *image r -graph* $\varphi(F)$ has vertex set $\varphi(V(F))$ and edge set $\{\varphi(e) : e \in F\}$; repeated image edges are merged. An r -graph G is $\mathcal{F}$ -*hom-free* if $\text{Hom}(F, G) = \emptyset$ for every $F \in \mathcal{F}$.

For a finite r -graph F , let $\mathcal{Q}(F)$ be the family, taken up to isomorphism, of all image r -graphs $\varphi(F)$ of homomorphisms from F into finite r -graphs. Every such image has at most $v(F)$ vertices, so $\mathcal{Q}(F)$ is finite. It can be constructed effectively by enumerating the partitions of $V(F)$ whose blocks meet every

edge in at most one vertex and forming the corresponding image r -graphs. For a family $\mathcal{F}$ of r -graphs, put $\mathcal{Q}(\mathcal{F}) := \bigcup_{F \in \mathcal{F}} \mathcal{Q}(F)$. Then an r -graph is $\mathcal{Q}(\mathcal{F})$ -free exactly when it is $\mathcal{F}$ -hom-free.

We shall also discard isolated vertices before taking this closure. For a finite r -graph F , let F° be obtained by deleting its isolated vertices, and for a finite family $\mathcal{F}$ of r -graphs put $\mathcal{F}^\circ := \{F^\circ : F \in \mathcal{F}\}$. Thus the members of $\mathcal{Q}(\mathcal{F}^\circ)$ are the edgewise-injective homomorphic images of F° for $F \in \mathcal{F}$, with repeated image edges merged. Here *edgewise injective* means injective on every edge; vertices belonging to different edges may still be identified. The family $\mathcal{Q}(\mathcal{F}^\circ)$ is finite and can be enumerated effectively from $\mathcal{F}$.

The following immediate consequence of the blowup definition will be used repeatedly.

Lemma 2.1. *For finite r -graphs F and G , we have $F \rightarrow G$ if and only if F embeds into some blowup of G . Consequently, every blowup of an $\mathcal{F}$ -hom-free r -graph is $\mathcal{F}$ -free.*

Blowups also connect finite r -graphs to continuous optimization through Lagrangians. The graph Lagrangian was introduced by Motzkin and Straus [51], and Frankl and Rödl [21] introduced the corresponding framework for uniform hypergraphs. See Talbot [68] for a survey.

For any finite uniform hypergraph G and a nonnegative vector $\mathbf{x} = (x_v)_{v \in V(G)}$, define its Lagrangian polynomial by

$$\lambda(G; \mathbf{x}) := \sum_{e \in G} \prod_{v \in e} x_v.$$

A *probability weighting*, or simply a *weighting*, on a finite nonempty set V is a vector $\mathbf{x} = (x_v)_{v \in V} \in \mathbb{R}_{\geq 0}^V$ satisfying $\sum_{v \in V} x_v = 1$. A weighting of a finite hypergraph or finite relation structure means a weighting on its vertex set. For $V(G) \neq \emptyset$, put

$$\lambda(G) := \max \{ \lambda(G; \mathbf{x}) : \mathbf{x} \text{ is a weighting of } G \},$$

and set $\lambda(G) := 0$ when $V(G) = \emptyset$. When G is r -uniform, put $\lambda_r(G) := r! \lambda(G)$. The value $\lambda_r(G)$ is the largest limiting edge density of blowups of G . For a class $\mathcal{C}$ of r -graphs containing an r -graph with nonempty vertex set, put $\Lambda(\mathcal{C}) := \sup \{ \lambda(G) : G \in \mathcal{C} \text{ and } |V(G)| > 0 \}$ and $\Lambda_r(\mathcal{C}) := r! \Lambda(\mathcal{C})$. Thus Λ is the unnormalized class Lagrangian, while λ_r and Λ_r use the ordinary edge-density normalization for r -graphs and their classes, respectively.

For a finite indexed vector $\mathbf{x} = (x_i)_{i \in I}$ and an integer $j \geq 0$, write $e_j(\mathbf{x}) := \sum_{A \in \binom{I}{j}} \prod_{i \in A} x_i$ for its j th elementary symmetric polynomial. We take the empty product to be 1, so $e_0(\mathbf{x}) = 1$, while $e_j(\mathbf{x}) = 0$ when $j > |I|$. When $\mathbf{x}$ is fixed and $U \subseteq I$, abbreviate $e_j((x_i)_{i \in U})$ to $e_j(U)$.

We also use a polynomial for labeled ordered relations. Let $\ell \geq 1$, let V be finite, and let $\mathcal{R} = (R_\alpha)_{\alpha \in A}$ be a finite labeled family of relations $R_\alpha \subseteq V^\ell$. For a nonnegative vector $\mathbf{x} = (x_v)_{v \in V}$, define the *relation polynomial* by

$$\lambda_{\text{rel}}(\mathcal{R}; \mathbf{x}) := \sum_{\alpha \in A} \sum_{(v_1, \dots, v_\ell) \in R_\alpha} \prod_{h \in [\ell]} x_{v_h}.$$

A single relation is regarded as a one-label family, and the empty labeled family has value 0. Thus the same ordered tuple occurring under different labels is counted once for each label. Coordinate order is retained, and no factorial normalization is used; the labels may also encode types and coordinate roles.

The next standard observation expresses the monotonicity of the Lagrangian under hypergraph homomorphisms; see the Lagrangian framework of Frankl and Rödl [21].

Lemma 2.2. *For finite r -graphs F and G , if $F \rightarrow G$, then $\lambda(F) \leq \lambda(G)$ and hence $\lambda_r(F) \leq \lambda_r(G)$.*

We next record two consequences of the blowup and continuity theorems of Brown and Simonovits. Pikhurko [54, p. 417] also records the finite-subfamily approximation.

Theorem 2.3 ([10, Theorems 1 and 3]). *For every family $\mathcal{F}$ of r -graphs, the following statements hold.*

- (i) *Replacing $\mathcal{F}$ by all homomorphic images of its members does not change its Turán density; in symbols, $\pi(\mathcal{Q}(\mathcal{F})) = \pi(\mathcal{F})$.*
- (ii) *The density satisfies $\pi(\mathcal{F}) = \inf \{ \pi(\mathcal{F}') : \mathcal{F}' \subseteq \mathcal{F} \text{ finite} \}$.*

The following standard consequence of the approximation theorem of Brown and Simonovits [10, Theorem 6] packages the finite homomorphic closure used throughout the structural construction; see also Pikhurko [54, Lemma 23].

Lemma 2.4. *For every finite family $\mathcal{F}$ of r -graphs, deleting isolated vertices and then taking all homomorphic images preserves the Turán density of $\mathcal{F}$, and the resulting free class is blowup closed. More precisely, we have*

$$\pi(\mathcal{Q}(\mathcal{F}^\circ)) = \pi(\mathcal{F}), \quad \text{and} \quad \Lambda_r(\text{Forb}(\mathcal{Q}(\mathcal{F}^\circ))) = \pi(\mathcal{F}).$$

Proof. Deleting isolated vertices does not change Turán density, and $\mathcal{Q}(\mathcal{F}^\circ)$ is the family of homomorphic images in Theorem 2.3, up to isomorphism and the merging of repeated image edges. Hence $\pi(\mathcal{Q}(\mathcal{F}^\circ)) = \pi(\mathcal{F})$. By Lemma 2.1, an r -graph G belongs to $\text{Forb}(\mathcal{Q}(\mathcal{F}^\circ))$ exactly when every blowup of G is $\mathcal{F}$ -free. Thus this class is blowup closed, and the approximation theorem of Brown and Simonovits gives $\Lambda_r(\text{Forb}(\mathcal{Q}(\mathcal{F}^\circ))) = \pi(\mathcal{F})$. ■

The following standard Lagrangian characterization is a consequence of the approximation theorem of Brown and Simonovits [10, Theorem 6]; see also Keevash [35, Section 3, p. 87].

Corollary 2.5. *For every finite family $\mathcal{F}$ of r -graphs, we have*

$$\pi(\mathcal{F}) = \sup \{ \lambda_r(G) : G \text{ is a finite } \mathcal{F}\text{-hom-free } r\text{-graph} \}.$$

Consequently, if $a \in \mathbb{R}$ and $a < \pi(\mathcal{F})$, then there is a finite $\mathcal{F}$ -hom-free r -graph G such that $\lambda_r(G) > a$.

The same finite homomorphic closure gives an explicit finite-order approximation. Let $\mathcal{F}$ be a finite family of r -graphs. For $m \geq r$, put

$$\pi^{\text{hom}}(m, \mathcal{F}) := \frac{\text{ex}(m, \mathcal{Q}(\mathcal{F}))}{\binom{m}{r}}, \quad \text{and} \quad \theta_{r,m} := \frac{r! \binom{m}{r}}{m^r} = \prod_{i=0}^{r-1} \left(1 - \frac{i}{m} \right). \quad (2.1)$$

By Theorem 2.3, we have $\pi(\mathcal{Q}(\mathcal{F})) = \pi(\mathcal{F})$. The rational number $\pi^{\text{hom}}(m, \mathcal{F})$ is effectively computable by enumerating the r -graphs on $[m]$, testing whether they contain a member of $\mathcal{Q}(\mathcal{F})$, and retaining the largest edge count.

Proposition 2.6. *For every integer $r \geq 2$, every finite family $\mathcal{F}$ of r -graphs, and every $m \geq r$, we have*

$$\theta_{r,m} \pi^{\text{hom}}(m, \mathcal{F}) \leq \pi(\mathcal{F}) \leq \pi^{\text{hom}}(m, \mathcal{F}).$$

Consequently, $0 \leq \pi^{\text{hom}}(m, \mathcal{F}) - \pi(\mathcal{F}) \leq 1 - \theta_{r,m} \leq r(r-1)/(2m)$.

Proof. By Theorem 2.3, we have $\pi(\mathcal{F}) = \pi(\mathcal{Q}(\mathcal{F}))$. The averaging argument of Katona, Nemetz, and Simonovits [34] shows that the Turán density sequence for the fixed family $\mathcal{Q}(\mathcal{F})$ is nonincreasing. Therefore, $\pi(\mathcal{F}) = \pi(\mathcal{Q}(\mathcal{F})) \leq \text{ex}(m, \mathcal{Q}(\mathcal{F})) / \binom{m}{r} = \pi^{\text{hom}}(m, \mathcal{F})$. This proves the upper bound.

For the lower bound, choose an m -vertex $\mathcal{Q}(\mathcal{F})$ -free r -graph G with $|G| = \pi^{\text{hom}}(m, \mathcal{F}) \binom{m}{r}$. The r -graph G is $\mathcal{F}$ -hom-free. By Lemma 2.1, every blowup of G is $\mathcal{F}$ -free. Take balanced blowups with every vertex class of size t . Their edge density tends, as $t \rightarrow \infty$, to $r!|G|/m^r = \pi^{\text{hom}}(m, \mathcal{F}) r! \binom{m}{r} / m^r = \theta_{r,m} \pi^{\text{hom}}(m, \mathcal{F})$. Hence the displayed lower bound holds.

Since $0 \leq \pi^{\text{hom}}(m, \mathcal{F}) \leq 1$, the length of the displayed interval is at most $1 - \theta_{r,m}$. Finally, $1 - \prod_{i=0}^{r-1} (1 - i/m) \leq \sum_{i=0}^{r-1} i/m = r(r-1)/(2m)$, using $1 - \prod_i (1 - a_i) \leq \sum_i a_i$ for $a_i \in [0, 1]$. ■

The factorials in the Lagrangian definitions convert the edge polynomial, in which each unordered edge occurs once, to the ordinary edge-density normalization. The conventions used throughout this work are collected in Table 1.

We now fix the computability and coding conventions shared by the later parts of the paper.

The standard definitions in this paragraph may be found in Rogers [62]. For $k \geq 1$, a partial function $\psi: \mathbb{N}^k \rightarrow \mathbb{N}$ is *partial computable* if some Turing-machine program halts exactly on the inputs in $\text{dom } \psi$ and, on each such input $\mathbf{n}$, outputs $\psi(\mathbf{n})$. A partial computable function ψ is *computable* if $\text{dom } \psi = \mathbb{N}^k$. Equivalently, there is a Turing-machine program that halts on every input $\mathbf{n} \in \mathbb{N}^k$ and outputs $\psi(\mathbf{n})$. We also call such a function *total computable*. The *primitive recursive functions* form the smallest class of total functions of finite arity that contains the zero, successor, and projection functions and is closed under composition and primitive recursion. For every $k \geq 0$, primitive recursion constructs $\zeta: \mathbb{N}^{k+1} \rightarrow \mathbb{N}$ from functions $\gamma: \mathbb{N}^k \rightarrow \mathbb{N}$ and $\eta: \mathbb{N}^{k+2} \rightarrow \mathbb{N}$ according to $\zeta(\mathbf{n}, 0) = \gamma(\mathbf{n})$ and $\zeta(\mathbf{n}, t+1) = \eta(\mathbf{n}, t, \zeta(\mathbf{n}, t))$. A predicate $R \subseteq \mathbb{N}^k$ is *primitive recursive* if its characteristic function, equal to 1 on R and 0 off R , is

Quantity	Definition	Relation used in this work
$\lambda(G; \mathbf{x})$	$\sum_{e \in G} \prod_{v \in e} x_v$	Lagrangian polynomial evaluated at a nonnegative vector $\mathbf{x}$.
$\lambda(G)$	$\max_{\mathbf{x} \text{ a weighting of } G} \lambda(G; \mathbf{x})$	Unnormalized hypergraph Lagrangian; every unordered edge contributes one monomial.
$\lambda_r(G)$	$r! \lambda(G)$	Largest limiting ordinary edge density among blowups of G .
$\Lambda(\mathcal{C})$	$\sup_{G \in \mathcal{C}, V(G) > 0} \lambda(G)$	Unnormalized class Lagrangian.
$\Lambda_r(\mathcal{C})$	$r! \Lambda(\mathcal{C})$	Normalized class Lagrangian; the direct pair-covering transfer gives $\pi(\mathcal{F}) = \Lambda_r(\mathcal{C})$.
$\pi(\mathcal{F})$	$\lim_{n \rightarrow \infty} \text{ex}(n, \mathcal{F}) / \binom{n}{r}$	Ordinary normalized Turán density.

Table 1: Normalization conventions.

primitive recursive. In particular, every primitive recursive function is total and computable. We use fixed primitive recursive encodings of finite tuples and finite words by natural numbers, and apply the same terminology to functions on these codes.

Following Weihrauch [73, Chapter 4], a real number α is *computable* if an algorithm, on input k , produces a rational q_k satisfying $|q_k - \alpha| < 2^{-k}$.

We fix a computable canonical encoding of every finite r -graph that records its number of vertices, including isolated vertices, and its edge list. Write $\langle G \rangle$ and $\langle \mathcal{F} \rangle$ for the canonical binary codes of an r -graph and a finite family, respectively, and write $|\langle \mathcal{F} \rangle|$ for the length of the finite-family code. Let dec_r be the total computable decoder that returns the encoded r -graph when the code is canonical and the encoded r -graph has at least one edge, and returns $\perp$ on syntactically invalid codes or codes of edgeless r -graphs. We also fix a total decoder for finite sequences. When a natural number n is used as a finite-family code, each entry of the decoded sequence is passed through dec_r and every entry sent to $\perp$ is discarded; a syntactically invalid outer sequence code denotes the empty family. Write $\text{DecFam}_r(n)$ for the resulting finite family. Thus every natural number denotes a finite forbidden family, rather than only well-formed inputs doing so. We write $\text{CanFam}_r(n)$ when n is the canonical finite-sequence code of a finite family of canonical codes for r -graphs having at least one edge; this is a primitive recursive predicate. These two conventions will also be used in the formalized arguments later in the paper.

We follow the modern recursion-theoretic conventions of Soare [66]. A set $A \subseteq \mathbb{N}$ is *computably enumerable*, or *c.e.*, if it is the domain of a partial computable function; equivalently, some algorithm enumerates exactly the elements of A . It is *co-c.e.* if its complement is c.e. The arithmetical hierarchy was introduced independently by Kleene [36] and Mostowski [50]. At its first level, a set is Σ_1^0 exactly when it is c.e., and it is Π_1^0 exactly when it is co-c.e. A *many-one reduction* from $A \subseteq \mathbb{N}$ to $B \subseteq \mathbb{N}$ is a total computable map $f: \mathbb{N} \rightarrow \mathbb{N}$ such that $n \in A$ if and only if $f(n) \in B$ for every $n \in \mathbb{N}$. A set is *hard* for a class if every set in that class many-one reduces to it, and it is *complete* if it is also a member of that class. All completeness statements in this work use many-one reductions.

We now turn to the finite elimination of roots, which turns persistent local obstructions for partial rootings into a finite family of ordinary forbidden r -graphs. Both the numerical construction for Theorem 1.1 and the structural construction for Theorem 1.2 first use finitely many distinguished roots to fix auxiliary positions and express local constraints. This is analogous to the labeled vertices in Razborov’s flag-algebra formalism [58, Section 2.1]. Here we allow only a subset of a fixed set of root labels to be present because the root-removal argument adds roots one at a time. The main results, however, concern ordinary Turán problems defined by finite forbidden families with no distinguished vertices. We therefore isolate the general root-removal principle used in both constructions and give an effective bound on the order of the resulting forbidden r -graphs.

The two transfers below form one pipeline:

$$\begin{aligned}
&\text{rooted local rules} \longrightarrow \text{a finite family of ordinary forbidden } r\text{-graphs} \\
&\quad \longrightarrow \text{pair-covering extensions} \longrightarrow \text{an exact density identity.}
\end{aligned}$$

The first step removes the roots, while the second converts the resulting class Lagrangian into an ordinary Turán density.

Let $\mathcal{M}$ be a finite set of root labels. A *partial $\mathcal{M}$ -rooting* of an r -graph G is an injective partial map $\mu: \mathcal{M} \rightarrow V(G)$. Write $\nu \supseteq \mu$ when ν extends μ . Rooted isomorphisms preserve every root label. The notation $\text{Adm}(G, \mu)$ denotes an abstract condition saying that the rooted r -graph (G, μ) satisfies the local rules under consideration; its concrete meaning will be specified separately in the two constructions. We assume that this condition is invariant under rooted isomorphisms and let $\mathcal{D} := \{G: \text{Adm}(G, \mu) \text{ for some partial } \mathcal{M}\text{-rooting } \mu\}$.

An r -graph can admit many partial rootings, so checking them one by one gives no a priori finite obstruction bound. Starting from the empty rooting, however, each failed rooting will supply a small obstruction that persists until a new root is placed inside it. There are finitely many root labels, and branching over those possible new placements gives a bounded search tree.

Definition 2.7. For an integer $w \geq 1$, the predicate Adm has the *w -bounded-witness property* if every inadmissible (G, μ) has a subgraph $F_\mu \subseteq G$ with $|V(F_\mu)| \leq w$ such that, whenever $F_\mu \subseteq H \subseteq G$, ν is a partial $\mathcal{M}$ -rooting of H extending μ , and $\text{Adm}(H, \nu)$, some newly used root label is mapped to a vertex of F_μ that was unrooted under μ ; equivalently,

$$\nu(\text{dom } \nu \setminus \text{dom } \mu) \cap (V(F_\mu) \setminus \text{im } \mu) \neq \emptyset.$$

Note that in this definition it is not enough that failure be visible on at most w vertices. The same witness must persist in every intermediate subgraph unless a new root is placed inside it.

The resulting bounded search tree is formalized by the following finite-root elimination principle of Li, Liu, and Liu [38, Proposition 4.7]. For completeness, we include its bounded search-tree proof, including the effective assertion.

Proposition 2.8 ([38, Proposition 4.7]). *Suppose that $\mathcal{D}$ is monotone and Adm has the w -bounded-witness property for an integer $w \geq 1$. Then $\mathcal{D} = \text{Forb}(\mathcal{A})$ for a finite family $\mathcal{A}$ whose members have at most $w \sum_{j=0}^{|\mathcal{M}|} (|\mathcal{M}|w)^j$ vertices. If $r, \mathcal{M}, w$ are given effectively and $\text{Adm}(G, \mu)$ is decidable uniformly on finite inputs, then $\mathcal{A}$ is effectively computable.*

Proof. Fix an r -graph $G \notin \mathcal{D}$. We construct a finite rooted search tree whose nodes are partial $\mathcal{M}$ -rootings of G . The root is the empty rooting. At a node μ , the pair (G, μ) is inadmissible because $G \notin \mathcal{D}$, so choose a witness F_μ supplied by the w -bounded-witness property. For every $a \in \mathcal{M} \setminus \text{dom } \mu$ and every $v \in V(F_\mu) \setminus \text{im } \mu$, add the child $\mu \cup \{(a, v)\}$. Every branch has length at most $|\mathcal{M}|$, and every node has at most $|\mathcal{M}|w$ children. Hence the tree has at most $\sum_{j=0}^{|\mathcal{M}|} (|\mathcal{M}|w)^j$ nodes. Let F be the union, as a subgraph of G , of the witnesses F_μ over all nodes μ of the tree. Then $v(F) \leq w \sum_{j=0}^{|\mathcal{M}|} (|\mathcal{M}|w)^j$.

We claim that $F \notin \mathcal{D}$. Suppose instead that ν is a partial $\mathcal{M}$ -rooting with $\text{Adm}(F, \nu)$. Starting with the empty rooting μ_0 , assume that a node μ_j with $\nu \supseteq \mu_j$ has been reached. Since $F_{\mu_j} \subseteq F \subseteq G$, the bounded-witness property, applied with $H = F$, gives a label $a \in \text{dom } \nu \setminus \text{dom } \mu_j$ such that $\nu(a) \in V(F_{\mu_j}) \setminus \text{im } \mu_j$. Thus $\mu_{j+1} = \mu_j \cup \{(a, \nu(a))\}$ is a child of μ_j in the search tree and satisfies $\nu \supseteq \mu_{j+1}$. Repeating this step eventually gives $\mu_j = \nu$, at which point the bounded-witness property would require a label in $\text{dom } \nu \setminus \text{dom } \nu$, a contradiction. This proves the claim.

Let $\mathcal{A}$ contain one representative of every isomorphism class of r -graphs satisfying the displayed order bound and not belonging to $\mathcal{D}$. The family $\mathcal{A}$ is finite. By monotonicity, no member of $\mathcal{D}$ contains a member of $\mathcal{A}$. Conversely, the construction above shows that every $G \notin \mathcal{D}$ contains such an $F \notin \mathcal{D}$, and hence contains a member of $\mathcal{A}$. Therefore $\mathcal{D} = \text{Forb}(\mathcal{A})$.

For the effective assertion, enumerate the finitely many r -graphs satisfying the displayed order bound. For each such graph, enumerate its finitely many partial $\mathcal{M}$ -rootings and use the decision procedure for Adm to determine membership in $\mathcal{D}$. Retaining one canonical representative of each isomorphism class outside $\mathcal{D}$ computes $\mathcal{A}$. ■

The other general transfer used by both main constructions is pair covering, an extension technique used by Mubayi [52, Proof 3 of Theorem 1]; a closely related formulation also appears in Keevash's treatment of hypergraph Lagrangians [35, Section 3]. An r -graph *covers pairs* if every pair of distinct vertices lies

in an edge. For an r -graph J , put $B_{\text{pc}}(J) := v(J) + (r-2)\binom{v(J)}{2}$. For an r -graph Q and an injection $\iota: V(J) \rightarrow V(Q)$, consider the following conditions.

- (i) The map ι maps every edge of J to an edge of Q .
- (ii) Every pair of vertices in $\iota(V(J))$ lies in an edge of Q .
- (iii) The r -graph Q has no isolated vertices.
- (iv) We have $v(Q) \leq B_{\text{pc}}(J)$.

The first two conditions say that (Q, ι) is a *J -pair-covering extension*. Informally, Q contains the distinguished copy $\iota(J)$, and additional edges, possibly using auxiliary vertices, cover every pair of vertices in the core $\iota(V(J))$. The definition neither prescribes one new edge for each previously uncovered pair nor requires pairs involving auxiliary vertices to be covered. Let $\text{Ext}(J)$ contain one representative Q of every isomorphism class of underlying r -graphs for which some injection ι satisfies all four conditions. For every $Q \in \text{Ext}(J)$, fix one witnessing injection $\iota_{J,Q}$ and call $\iota_{J,Q}(V(J))$ its *distinguished J -core*. Note that these finite choices can be made effectively. Indeed, the order bound $B_{\text{pc}}(J)$ reduces the construction of $\text{Ext}(J)$ to a finite enumeration.

If J has an edge, then $\text{Ext}(J)$ is nonempty. For each pair of vertices of J not already covered by an edge, add $r-2$ fresh vertices and the resulting edge. This produces a no-isolated extension within the stated order bound. The definition of $\text{Ext}(J)$ includes all extensions up to this bound, not just this particular one.

The following result is an effective finite-family form of this extension method. Its minimum-support Lagrangian argument goes back to Frankl and Rödl [21]. For completeness, we include the short proof.

Lemma 2.9 ([38, Lemma 3.3]). *Let $\mathcal{A}$ be a finite family of r -graphs, each with an edge, and suppose that $\text{Forb}(\mathcal{A})$ contains an r -graph with an edge. Put $\mathcal{F} := \bigcup_{J \in \mathcal{A}} \text{Ext}(J)$. Then $\pi(\mathcal{F}) = \Lambda_r(\text{Forb}(\mathcal{A}))$. Moreover, if H is an $\mathcal{F}$ -free r -graph with $\lambda(H) > 0$ and $\mathbf{x}$ is a maximizing weighting of H with minimum support, then $H[\text{supp}(\mathbf{x})]$ belongs to $\text{Forb}(\mathcal{A})$, covers pairs, and has the same Lagrangian as H . The family $\mathcal{F}$ is effectively computable from $\mathcal{A}$.*

Proof. Put $\mathcal{C} := \text{Forb}(\mathcal{A})$. We first prove the lower bound. Let $G \in \mathcal{C}$ have nonempty vertex set, and let B be any blowup of G with blowup projection $\rho: V(B) \rightarrow V(G)$. We claim that B is $\mathcal{F}$ -free. Otherwise some $Q \in \text{Ext}(J)$ with $J \in \mathcal{A}$ embeds in B . Every pair of vertices in the distinguished J -core of Q lies in an edge of Q . Hence the composition of the embedding with ρ is injective on that core, since an edge of a blowup contains at most one vertex from each blowup class. Its restriction to the distinguished core therefore embeds J in G , a contradiction. Rational blowups of G approximating an arbitrary probability weighting $\mathbf{x}$ are consequently $\mathcal{F}$ -free, and their ordinary edge densities tend to $r!\lambda(G; \mathbf{x})$. Taking the supremum over $G \in \mathcal{C}$ and $\mathbf{x}$ gives

$$\pi(\mathcal{F}) \geq \Lambda_r(\mathcal{C}).$$

For the reverse bound, let H be an n -vertex $\mathcal{F}$ -free r -graph with $n \geq r$, choose a Lagrangian-maximizing weighting $\mathbf{x}$ of minimum support, and put $D := H[\text{supp}(\mathbf{x})]$. The graph D covers pairs. Indeed, if two support vertices u and v lie in no common edge, then, with all other weights and the sum $x_u + x_v$ fixed, the edge polynomial is affine in x_u . One of the two endpoints of this interval therefore has value at least its value at $\mathbf{x}$. Since $\mathbf{x}$ is maximizing, moving all of $x_u + x_v$ to the corresponding vertex preserves the maximum and decreases the support, a contradiction. The restricted weighting attains $\lambda(H)$ on D , while every weighting of D extends by zero to a weighting of H ; hence $\lambda(D) = \lambda(H)$.

Suppose that D contains a copy of some $J \in \mathcal{A}$. For every pair of vertices in this copy, choose an edge of D containing that pair, and retain the copy together with all the chosen edges and precisely the vertices occurring in them. Every auxiliary vertex lies in a chosen edge, and every core vertex lies in one because $v(J) \geq r \geq 2$ and all pairs of core vertices were covered. The resulting subgraph has no isolated vertices, covers every pair in its distinguished J -core, and has at most

$$v(J) + (r-2)\binom{v(J)}{2} = B_{\text{pc}}(J)$$

vertices. It is therefore isomorphic to a member of $\text{Ext}(J)$, contradicting that H is $\mathcal{F}$ -free. Thus $D \in \mathcal{C}$. This proves the additional assertion of the lemma and gives $\lambda(H) \leq \Lambda(\mathcal{C})$.

Applying the last inequality to the uniform weighting of an n -vertex $\mathcal{F}$ -free graph H gives

$$\frac{|H|}{\binom{n}{r}} \leq \frac{\Lambda_r(\mathcal{C})}{\theta_{r,n}}.$$

Since $\theta_{r,n} \rightarrow 1$, letting $n \rightarrow \infty$ yields $\pi(\mathcal{F}) \leq \Lambda_r(\mathcal{C})$, and hence equality. Finally, for each $J \in \mathcal{A}$, one can enumerate all r -graphs through order $B_{\text{pc}}(J)$, test the finitely many injections of $V(J)$ against the four defining conditions, and retain canonical isomorphism representatives. This computes $\text{Ext}(J)$ and therefore $\mathcal{F}$ effectively. $\blacksquare$

Part I

Direct computational universality

This part proves [Theorem 1.1](#) by constructing the direct numerical compiler. The first half encodes the computation of $\mathbf{U}$ by finite local matching records and establishes the resulting weighted dichotomy. The second half converts this weighted system into a finite family of ordinary r -graphs while preserving its exact extremal value. The diagram below records how the sections carry out these transformations from the input word β to the forbidden family $\mathcal{F}_{r,\beta}$.

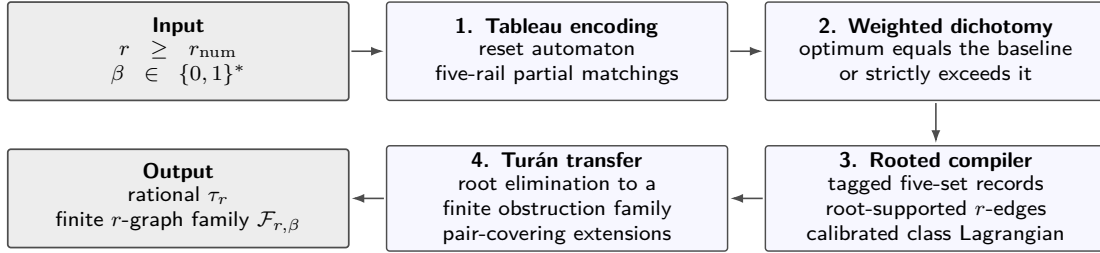


Figure 1: The four stages of the direct numerical compiler. From (r, β) it produces the threshold τ_r and the family $\mathcal{F}_{r,\beta}$ without deciding whether $\mathbf{U}(\beta)$ halts, with the threshold depending only on r .

3 A direct finite-tableau simulation

We use the following elementary terminology throughout this part. For disjoint finite sets X and Y , a relation $D \subseteq X \times Y$ is a *partial matching* if both coordinate projections are injective. Equivalently, D defines a bijection $\text{dom } D \rightarrow \text{im } D$, viewed as a partial map from X to Y .

This section encodes the computation of $\mathbf{U}$ on an input word $\beta \in \{0,1\}^*$ as a finite typed matching system governed by local obstructions. The starting point is its space–time table. A row records one machine configuration, a column follows one tape cell through time, and a bounded neighborhood determines whether two consecutive rows obey the transition rule. We convert this dynamic description into a finite static structure in five steps.

- (i) We add initialization, reset, and error-handling phases to the fixed universal Turing machine $\mathbf{U}$ and implement the resulting process by a radius-one cellular automaton. [Lemma 3.1](#) verifies that its local rule is well defined and effectively computable. The reset returns the automaton to a recognizable initial row after a halt.
- (ii) We prove that the reset automaton returns to its marked initial row during a finite wall-bounded computation on input β exactly when the fixed machine $\mathbf{U}$ halts on β . Thus halting is equivalent to the existence of a finite periodic space–time table containing the marked initial row, as stated in [Lemma 3.2](#).
- (iii) We encode horizontal adjacency, the passage from one row to the next, the symbol in each cell, and the marked cell by short paths of edge-indexed binary relations. Each rail has eight logical types and their eight hatted balancing copies. Its logical edges, corresponding edges on the balancing

layer, and bridges form a regular type multidigraph. We then take five disjoint rails and add a five-ary symbol whose interpretation joins their marked cells. The local obstructions force the binary interpretations and this five-ary relation to be partial matchings, giving the fixed typed matching signature in [Lemma 3.5](#).

- (iv) We express each invalid local configuration by a finite typed local obstruction. Enumerating all allowed identifications among its displayed positions ensures that the test still detects a violation when nominal positions happen to coincide. These form the local-test part of the finite list $\mathcal{P}_\beta^{\text{phys}}$ in [Lemma 3.5](#).
- (v) In a legal five-rail realization, a relation component in which every vertex participates in every binary relation prescribed at its type is closed and reconstructs a genuine periodic table. If a prescribed relation is absent at some vertex, the relation component is grounded. Consequently, in the nonhalting case, every coordinate of every marked five-tuple belongs to a grounded relation component, as stated in part (iii) of [Lemma 3.5](#).

3.1 A radius-one reset automaton

We realize the fixed universal machine U by a one-dimensional cellular automaton. Each row records one complete configuration of U , and the bounded locality of its transition rule makes the next row a function of bounded neighborhoods. The additional phases below initialize this simulation, reset it after a halt, and detect an attempted crossing of the finite right boundary.

We use the standard radius-one cellular-automaton formalism. Such an automaton has a finite alphabet Σ_{CA} and a local update rule $\varphi: \Sigma_{CA}^3 \rightarrow \Sigma_{CA}$; at each time step, every cell is updated from its own state and those of its two neighbors. The shift-dynamical framework goes back to Hedlund [29], while Kari [32, 33] discusses its algorithmic theory and the standard correspondence between Turing machines and cellular automata. The resulting automaton evolves through the following four stages.

- (i) It initializes the work tape.
- (ii) It simulates U .
- (iii) It resets the tape after U halts.
- (iv) It enters a permanent crash state if the simulated head crosses an artificial right boundary.

The reset makes a finite periodic orbit equivalent to halting.

We now specify the normalization of U . Using the conventions of [Section 2](#), fix a standard effective programming system $(\psi_j)_{j \in \mathbb{N}}$ enumerating the unary partial computable functions, chosen so that parameters can be compiled into program indices by a primitive recursive procedure. Also fix a primitive recursive encoding $\text{enc}: \mathbb{N}^2 \rightarrow \{0, 1\}^*$ of program–input pairs as binary words. Choose an underlying universal Turing machine U_0 with a finite transition table satisfying $U_0(\text{enc}(j, x)) \downarrow$ exactly when $\psi_j(x)$ is defined. The chosen transition table is fixed throughout the construction and built into every algorithm below. Any of the standard finite-table universal-machine constructions gives such a choice, and replacing it by another one changes only the fixed constants. We make the following effective normalization and denote the resulting machine by U . On the empty input and on every input beginning with 0, the machine U enters a permanent loop; on input 1γ , it simulates U_0 on γ . The permanent loop is implemented by a new state q_∞ whose explicit transition rule is given below.

With juxtaposition denoting word concatenation, define the total primitive recursive translator $\text{tr}_U: \mathbb{N}^2 \rightarrow \{0, 1\}^*$ by $\text{tr}_U(j, x) := 1\text{enc}(j, x)$. Then $\psi_j(x)$ is defined exactly when $U(\text{tr}_U(j, x))$ halts. We choose the programming system and its universal simulator in the following standard trace-faithful form. Its evaluator is the fixed deterministic finite-table machine U_0 , so a source trace for $\psi_j(x)$ is, by definition, a trace of U_0 on $\text{enc}(j, x)$. Implement the 1-branch of U by a deterministic initialization followed by a step-for-step simulation of U_0 , with a distinguished state marking the first simulated configuration. The forward trace map prefixes the initialization and relabels the source configurations, while the reverse map discards that prefix and reads the simulated U_0 -configurations. Both procedures are bounded scans of the supplied finite trace and hence are primitive recursive. Only these transformations of finite traces, and no bound on a nonhalting computation, will be used below. Hence the word-halting domain K_U is c.e. and Σ_1^0 -hard, and is therefore Σ_1^0 -complete. Moreover, $\beta_\infty := 0$ is a fixed, explicitly known nonhalting word. The normalized transition table is fixed in advance and is not part of the input.

The normalized machine U has a one-sided tape with cells $1, 2, \dots$, a read-only track with alphabet $\mathcal{I} := \{0, 1, \$, \text{pad}\}$, and a work track with finite alphabet Γ , whose distinguished blank symbol is $\square$.

The symbol $\$$ marks the end of the input, and pad fills the rest of the read-only track. Both symbols remain unchanged throughout the computation and are distinct from the writable blank symbol $\square$.

The transition function of U reads the read-only and work symbols in the current cell, writes only on the work track, and moves one cell left or right. A requested left move from cell 1 leaves the head there. The machine has initial state q_0 and a unique halting state q_h ; its transition map δ is defined on every (q, i, a) with $q \neq q_h$, $i \in \mathcal{I}$, and $a \in \Gamma$, and has no entry with state q_h . To implement the permanent loop, U enters q_∞ with its head at cell 1 and sets $\delta(q_\infty, i, a) = (q_\infty, a, -1)$ for every $i \in \mathcal{I}$ and $a \in \Gamma$. The head then remains at cell 1 and the complete configuration repeats forever. In particular, ZFC proves by induction on the running time that $U(0)^\uparrow$. These conventions can be incorporated into the finite state set without affecting undecidability.

Regard every head symbol (q, a) as a new formal symbol disjoint from Γ , and choose all the control symbols displayed below to be pairwise distinct and disjoint from both Γ and the head symbols. Let $\mathcal{D}$ consist of the symbols of Γ , the head symbols (q, a) with q a state of U and $a \in \Gamma$, and

Start, Init, BootR, BootL, SweepL, SweepR, SweepB, Crash.

These control symbols have three roles.

- *Initialization.* The symbol **Start** marks the first cell of the distinguished initial row, and **Init** fills the remaining cells. The signal **BootR** moves from left to right, replacing **Init** by blank work symbols; **BootL** then returns to the left and creates the initial head in state q_0 .
- *Reset.* When U reaches q_h , **SweepL** starts at the halting head and moves to the left boundary. The signal **SweepR** then moves to the right and clears the work track, and **SweepB** moves back to the left, restoring **Start** in the first cell and **Init** in all other cells.
- *Error handling.* The symbol **Crash** is permanent and is used when the simulated head attempts to cross the artificial right boundary or when a local neighborhood does not match any valid phase.

The cellular-automaton alphabet is $\Sigma_{CA} = \{L, R\} \sqcup (\mathcal{I} \times \mathcal{D})$. We will repeatedly group its interior states by one of their two coordinates. For $s \in \mathcal{D}$ and $\iota \in \mathcal{I}$, put

$$\Sigma_{\text{dyn}}(s) := \{(\iota', s) : \iota' \in \mathcal{I}\}, \quad \text{and} \quad \Sigma_{\text{inp}}(\iota) := \{(\iota, s') : s' \in \mathcal{D}\}.$$

With this alphabet, the simulation phase realizes the correspondence between machine configurations and cellular-automaton rows. The read-only coordinate stores the input track, the dynamic coordinate stores the work symbol, and the unique cell scanned by the head carries a symbol (q, a) recording both the machine state q and the work symbol a . The local rule described below updates these symbols according to one transition of U , so successive rows form the space-time table of U . The additional control symbols above place initialization, reset, and error-handling phases around the simulation.

The first coordinate of an interior state is read-only and remains unchanged. The two wall states are fixed. When no confusion is possible, the name of a dynamic symbol $s \in \mathcal{D}$ denotes an interior state whose dynamic coordinate is s ; every update below retains the read-only coordinate of the middle cell. We now define a radius-one rule $\varphi: \Sigma_{CA}^3 \rightarrow \Sigma_{CA}$ by an explicit finite template procedure.

For $W \geq 2$ and $\mathbf{i} = (i_1, \dots, i_W) \in \mathcal{I}^W$, a *phase row* has walls at positions $0, W+1$, fixed read-only word $\mathbf{i}$, and one of the following dynamic words. Here $\mathbf{w} = (w_1, \dots, w_W) \in \Gamma^W$, $h, j \in [W]$, and q is a state of U .

$$\begin{aligned} S_W &:= \text{Start Init}^{W-1}, \\ B_{j,W}^R &:= \square^{j-1} \text{BootR Init}^{W-j}, \\ B_{j,W}^L &:= \square^{j-1} \text{BootL} \square^{W-j}, \\ C_{h,q}(\mathbf{w}) &:= w_1 \cdots w_{h-1} (q, w_h) w_{h+1} \cdots w_W, \\ L_j(\mathbf{w}) &:= w_1 \cdots w_{j-1} \text{SweepL} w_{j+1} \cdots w_W, \\ R_j(\mathbf{w}) &:= \square^{j-1} \text{SweepR} w_{j+1} \cdots w_W, \\ B_{j,W} &:= \square^{j-1} \text{SweepB Init}^{W-j}. \end{aligned}$$

We next specify the intended successor of every phase row. On the boot and backward-sweep rows it is

$$\begin{aligned}
S_W &\mapsto B_{1,W}^R, \\
B_{j,W}^R &\mapsto B_{j+1,W}^R \quad \text{for } j < W, \\
B_{W,W}^R &\mapsto B_{W,W}^L, \\
B_{j,W}^L &\mapsto B_{j-1,W}^L \quad \text{for } j > 1, \\
B_{1,W}^L &\mapsto C_{1,q_0}(\square, \dots, \square), \\
B_{j,W} &\mapsto B_{j-1,W} \quad \text{for } j > 1, \\
B_{1,W} &\mapsto S_W.
\end{aligned}$$

The remaining successor rules fall into three cases.

- *Simulation.* For a nonhalting state $q \neq q_h$, write $\delta(q, i_h, w_h) = (q', b, \varepsilon)$, where $\varepsilon \in \{-1, +1\}$. If the requested move remains within $[W]$, the old head cell is replaced by b , and cell $h + \varepsilon$ receives the head in state q' carrying its previous work symbol. If $\varepsilon = -1$ and $h = 1$, cell 1 instead becomes (q', b) . If $\varepsilon = +1$ and $h = W$, the old head cell becomes **Crash** and all other cells retain their work symbols.
- *Beginning the reset.* When the machine is in its halting state, $C_{h,q_h}(w) \mapsto L_h(w)$.
- *Completing the reset.* If $j > 1$, the successor of $L_j(w)$ has **SweepL** in cell $j - 1$, $\square$ in cell j , and agrees with w elsewhere; the successor of $L_1(w)$ is $R_1(w)$. For $j < W$, we have $R_j(w) \mapsto R_{j+1}(w)$, whereas $R_W(w) \mapsto B_{W,W}$.

All three cases retain the fixed read-only word i .

The phase transitions are summarized in Figure 2.

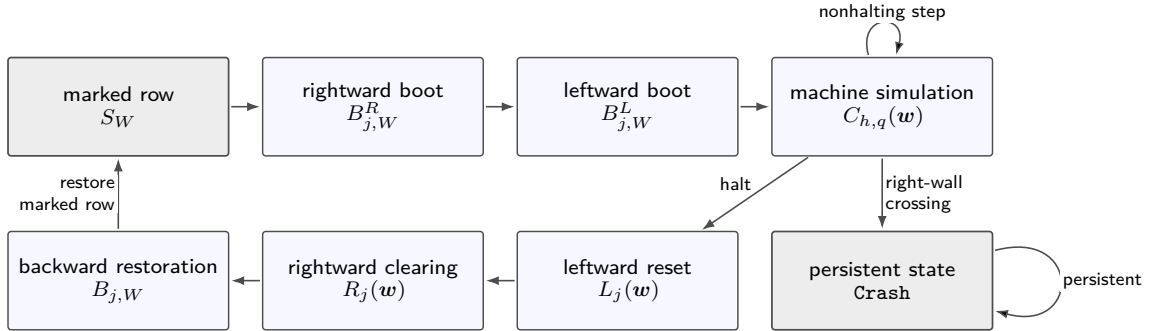


Figure 2: The phase flow of the wall-bounded reset automaton. Along the canonical orbit, a halting simulation enters the three reset sweeps and returns to the marked row S_W , whereas an attempted crossing of the right boundary enters the persistent crash state.

For a radius-one neighborhood $\mathbf{a} = (a_-, a_0, a_+) \in \Sigma_{CA}^3$ with interior middle state, define its set of *phase requests* $\text{Req}(\mathbf{a})$ as follows. An element b belongs to $\text{Req}(\mathbf{a})$ if, for some $W \geq 2$, some fixed read-only word, some phase row above, and some interior position j , the old neighborhood at j is $\mathbf{a}$ and the state at j in its prescribed successor row is b .

The following local determinacy statement supplies the finite check needed in the definition of φ .

Lemma 3.1. *For every radius-one neighborhood $\mathbf{a}$ with interior middle state, we have $|\text{Req}(\mathbf{a})| \leq 1$. Moreover, the request sets, and hence the lookup table of φ , are effectively computable from the finite transition table of $\mathcal{U}$.*

Proof. Every phase row has exactly one *distinguished symbol*, namely **Start**, one of the five boot or sweep symbols, or a head symbol (q, a) . These symbols are pairwise disjoint and none can occur as a background symbol. Consequently, if a window from a phase row contains the distinguished symbol, the window itself determines both its phase family and whether the distinguished symbol is in the left, middle, or right position.

We first list all signal cases. The named entries in the table give the new dynamic coordinate; the read-only coordinate of the middle cell is always retained. The word “unchanged” means that the complete middle state is retained. These are all the possible outputs at the middle cell.

distinguished symbol	in the left position	in the middle position	in the right position
Start	unchanged	BootR	impossible
BootR	BootR	$\square$, or BootL next to R	unchanged
BootL	unchanged	$\square$, or $(q_0, \square)$ next to L	BootL
SweepL	unchanged	$\square$, or SweepR next to L	SweepL
SweepR	SweepR	$\square$, or SweepB next to R	unchanged
SweepB	unchanged	Init , or Start next to L	SweepB

For example, a **BootR** in the left position moves into the middle cell, whereas a **BootR** in the right position moves away from it. If the signal is in the middle position, the presence and side of an end wall are part of the same radius-one window, so the exceptional boundary output in the table is also determined by $\mathbf{a}$. The row for **Start** uses the fact that **Start** occurs only in cell 1. The formally unused entries of $\mathbf{w}$ hidden below **SweepL**, or below and to the left of **SweepR**, do not affect any entry of the prescribed successor. Thus the nonunique parameterizations of $L_j(\mathbf{w})$ and $R_j(\mathbf{w})$ cannot produce different requests.

It remains to check head symbols. Suppose first that the distinguished symbol is (q, a) with $q \neq q_h$. Its full interior state displays the read-only symbol i in the head cell and therefore determines $\delta(q, i, a) = (q', b, \varepsilon)$. If the head is in the middle position, the new dynamic symbol there is b , except that a left move next to L gives (q', b) and a right move next to R gives **Crash**. If the head is in the left position, the middle cell receives (q', u) exactly when $\varepsilon = +1$, where the old middle work symbol u is displayed by $\mathbf{a}$; otherwise it is unchanged. The analogous statement holds with left and right interchanged when the head is in the right position and $\varepsilon = -1$. The complete Σ_{CA}^3 -window also fixes the read-only coordinate of this target cell, which is retained in its output. If $q = q_h$, only the head cell changes, from (q_h, a) to **SweepL**, and both neighboring cells are unchanged. Thus a window containing a head also has one uniquely determined middle output.

Finally, suppose that a phase-row window contains no distinguished symbol. The middle cell is then a work symbol or **Init**. The distinguished symbol lies at distance at least two from it, and every displayed successor changes only the distinguished cell and, at most, one adjacent cell. Hence the middle state is unchanged. A neighboring wall does not create another case. Its side is visible in $\mathbf{a}$, and a wall-centered window is excluded from the definition of Req.

This classification also handles possible overlaps. Two occurrences of the same window either contain the same distinguished symbol in the same position, in which case the preceding list gives the same output, or contain no distinguished symbol, in which case both retain the middle state. A window with two distinguished symbols, a **Crash** symbol, or a distinguished symbol beside background data excluded by its phase form occurs in no phase row and has empty request set. The same is true of a marker-free mixture of work and **Init** symbols. In the only phase families using both kinds of background, the unique signal separates them. Therefore every request set is empty or a singleton.

We next remove the apparently unbounded quantifier over W from the effectivity assertion. After suppressing the read-only coordinates, the dynamic words of the phase rows belong to the finite union

$$\begin{aligned} & \text{StartInit}^+, \quad \square^* \text{BootRInit}^*, \quad \square^* \text{BootL} \square^*, \quad \Gamma^* \mathcal{D}_{\text{head}} \Gamma^*, \\ & \Gamma^* \text{SweepL} \Gamma^*, \quad \square^* \text{SweepR} \Gamma^*, \quad \square^* \text{SweepBInit}^*, \end{aligned}$$

where $\mathcal{D}_{\text{head}} = \{(q, a) : q \text{ is a state of } \mathbf{U} \text{ and } a \in \Gamma\}$; in the last six languages we retain only words of length at least two. Adjoin L and R at the two ends. These are regular languages over a finite alphabet, so their length-three factors with an interior middle letter can be enumerated effectively. The read-only coordinates range over the finite set $\mathcal{I}$, so their assignments can also be enumerated; the middle coordinate is retained in every request, and the read-only coordinate at a head selects the relevant entry of the finite table δ . Together with the preceding cases, this enumerates exactly every pair $(\mathbf{a}, b)$ with $b \in \text{Req}(\mathbf{a})$ and hence computes every request set. $\blacksquare$

We can now give the complete lookup table with its priority made explicit.

(R1) If $a_0 \in \{\mathbf{L}, \mathbf{R}\}$, then $\varphi(\mathbf{a}) = a_0$.

- (R2) Otherwise, if one of a_-, a_0, a_+ is an interior state with dynamic coordinate **Crash**, then $\varphi(\mathbf{a})$ is the interior state with the read-only coordinate of a_0 and dynamic coordinate **Crash**.
- (R3) Otherwise compute the finite set $\text{Req}(\mathbf{a})$.
- (R4) If $\text{Req}(\mathbf{a}) = \{b\}$, set $\varphi(\mathbf{a}) = b$.
- (R5) If the request set is empty, set $\varphi(\mathbf{a})$ equal to the interior **Crash** state with the read-only coordinate of a_0 .

Lemma 3.1 shows, in particular, that every phase row is updated exactly to its displayed successor. A malformed neighborhood with two heads, two moving signals, or symbols from incompatible phases occurs in no phase template. Unless its middle cell is a wall or the neighborhood already contains **Crash**, in which cases (R1) or (R2) applies, it is therefore covered by (R5). Thus (R1)–(R5) define every entry of the finite table of φ without any block recoding.

A wall-centered update ignores both neighbors. Consequently, wall-bounded compartments can be concatenated on a horizontal cycle without interacting.

Finally, every all-work window is a stationary window of a sufficiently long phase row; hence

$$\varphi((i_-, u_-), (i_0, u_0), (i_+, u_+)) = (i_0, u_0) \quad \text{for all } i_-, i_0, i_+ \in \mathcal{I} \text{ and } u_-, u_0, u_+ \in \Gamma.$$

In particular, the definition fixes the constant background state by

$$\varphi((\text{pad}, \square), (\text{pad}, \square), (\text{pad}, \square)) = (\text{pad}, \square). \quad (3.1)$$

3.2 Periodic reset tables

For $W \geq 2$, a configuration is a map $c: \{0, 1, \dots, W+1\} \rightarrow \Sigma_{\text{CA}}$ with $c_0 = \text{L}$, $c_{W+1} = \text{R}$, and $c_j \in \mathcal{I} \times \mathcal{D}$ for $j \in [W]$. Define its successor by $(\Phi_W(c))_0 = \text{L}$, $(\Phi_W(c))_{W+1} = \text{R}$, and $(\Phi_W(c))_j = \varphi(c_{j-1}, c_j, c_{j+1})$ for $j \in [W]$. The definition of φ shows that $\Phi_W(c)$ is again a configuration of this form. Thus the walls are held fixed and no state outside this interval is consulted. Write $\beta = \beta_1 \cdots \beta_{\ell_\beta}$, where $\ell_\beta = |\beta|$. If $W \geq \ell_\beta + 2$, define $c_{\beta, W}$ by

$$\begin{aligned} \text{input track: } & \beta_1 \cdots \beta_{\ell_\beta} \$ \underbrace{\text{pad} \cdots \text{pad}}_{W - \ell_\beta - 1 \text{ copies}}, \\ \text{dynamic track: } & \text{Start} \underbrace{\text{Init} \cdots \text{Init}}_{W-1 \text{ copies}}. \end{aligned}$$

Here the marked cell is the first interior cell, which carries **Start**; the left wall at position 0 serves only as a fixed boundary and is not itself marked.

The two tracks of the initial configuration and the wall-bounded evolution are shown schematically in **Figure 3**.

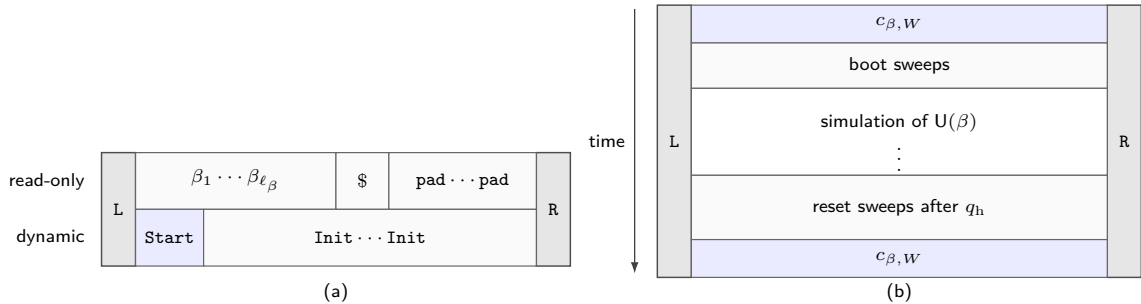


Figure 3: The canonical configuration and its periodic reset tableau. (a) The fixed read-only track contains $\beta \$ \text{pad} \cdots$, while the dynamic track contains **Start** **Init** $\cdots$, with the first interior cell marked by **Start**. (b) Time runs downward between the fixed walls. After initialization and simulation, a halt triggers the reset sweeps and restores exactly $c_{\beta, W}$, giving $\Phi_W^p(c_{\beta, W}) = c_{\beta, W}$.

To expand panel (b), consider a width-five example with $|\beta| \leq 3$. Write $i_1 \cdots i_5 = \beta \$ \text{pad}^{4-|\beta|}$ for the fixed read-only word, and suppose that $U(\beta)$ halts after T simulated steps with its head in cell 3. During

the simulation, let $d_{t,j} \in \mathcal{D}$ be the dynamic coordinate in cell j at time t . Thus exactly one $d_{t,j}$ is a head symbol. The complete phase sequence is displayed in Table 2; the row indexed by $1 \leq t < T$ ranges over all intermediate simulation times.

phase	left wall	cell 1	cell 2	cell 3	cell 4	cell 5	right wall
S_5	L	(i_1, Start)	(i_2, Init)	(i_3, Init)	(i_4, Init)	(i_5, Init)	R
$B_{1,5}^R$	L	(i_1, BootR)	(i_2, Init)	(i_3, Init)	(i_4, Init)	(i_5, Init)	R
$B_{2,5}^R$	L	$(i_1, \square)$	(i_2, BootR)	(i_3, Init)	(i_4, Init)	(i_5, Init)	R
$B_{3,5}^R$	L	$(i_1, \square)$	$(i_2, \square)$	(i_3, BootR)	(i_4, Init)	(i_5, Init)	R
$B_{4,5}^R$	L	$(i_1, \square)$	$(i_2, \square)$	$(i_3, \square)$	(i_4, BootR)	(i_5, Init)	R
$B_{5,5}^R$	L	$(i_1, \square)$	$(i_2, \square)$	$(i_3, \square)$	$(i_4, \square)$	(i_5, BootR)	R
$B_{1,5}^L$	L	$(i_1, \square)$	$(i_2, \square)$	$(i_3, \square)$	$(i_4, \square)$	(i_5, BootL)	R
$B_{2,5}^L$	L	$(i_1, \square)$	$(i_2, \square)$	$(i_3, \square)$	(i_4, BootL)	$(i_5, \square)$	R
$B_{3,5}^L$	L	$(i_1, \square)$	$(i_2, \square)$	(i_3, BootL)	$(i_4, \square)$	$(i_5, \square)$	R
$B_{4,5}^L$	L	$(i_1, \square)$	(i_2, BootL)	$(i_3, \square)$	$(i_4, \square)$	$(i_5, \square)$	R
$B_{5,5}^L$	L	(i_1, BootL)	$(i_2, \square)$	$(i_3, \square)$	$(i_4, \square)$	$(i_5, \square)$	R
$C_{1,q_0}(\square^5)$	L	$(i_1, (q_0, \square))$	$(i_2, \square)$	$(i_3, \square)$	$(i_4, \square)$	$(i_5, \square)$	R
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$C_{h_t, q_t}(\mathbf{w}^{(t)})$ $1 \leq t < T$	L	$(i_1, d_{t,1})$	$(i_2, d_{t,2})$	$(i_3, d_{t,3})$	$(i_4, d_{t,4})$	$(i_5, d_{t,5})$	R
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$C_{3,q_h}(\mathbf{w})$	L	(i_1, w_1)	(i_2, w_2)	$(i_3, (q_h, w_3))$	(i_4, w_4)	(i_5, w_5)	R
$L_3(\mathbf{w})$	L	(i_1, w_1)	(i_2, w_2)	(i_3, SweepL)	(i_4, w_4)	(i_5, w_5)	R
L_2	L	(i_1, w_1)	(i_2, SweepL)	$(i_3, \square)$	(i_4, w_4)	(i_5, w_5)	R
L_1	L	(i_1, SweepL)	$(i_2, \square)$	$(i_3, \square)$	(i_4, w_4)	(i_5, w_5)	R
R_1	L	(i_1, SweepR)	$(i_2, \square)$	$(i_3, \square)$	(i_4, w_4)	(i_5, w_5)	R
R_2	L	$(i_1, \square)$	(i_2, SweepR)	$(i_3, \square)$	(i_4, w_4)	(i_5, w_5)	R
R_3	L	$(i_1, \square)$	$(i_2, \square)$	(i_3, SweepR)	(i_4, w_4)	(i_5, w_5)	R
R_4	L	$(i_1, \square)$	$(i_2, \square)$	$(i_3, \square)$	(i_4, SweepR)	(i_5, w_5)	R
R_5	L	$(i_1, \square)$	$(i_2, \square)$	$(i_3, \square)$	$(i_4, \square)$	(i_5, SweepR)	R
$B_{5,5}$	L	$(i_1, \square)$	$(i_2, \square)$	$(i_3, \square)$	$(i_4, \square)$	(i_5, SweepB)	R
$B_{4,5}$	L	$(i_1, \square)$	$(i_2, \square)$	$(i_3, \square)$	(i_4, SweepB)	(i_5, Init)	R
$B_{3,5}$	L	$(i_1, \square)$	$(i_2, \square)$	(i_3, SweepB)	(i_4, Init)	(i_5, Init)	R
$B_{2,5}$	L	$(i_1, \square)$	(i_2, SweepB)	(i_3, Init)	(i_4, Init)	(i_5, Init)	R
$B_{1,5}$	L	(i_1, SweepB)	(i_2, Init)	(i_3, Init)	(i_4, Init)	(i_5, Init)	R
S_5	L	(i_1, Start)	(i_2, Init)	(i_3, Init)	(i_4, Init)	(i_5, Init)	R

Table 2: A complete period of the wall-bounded tableau in Figure 3 for $W = 5$. Every interior entry is the full cellular-automaton state (i_j, d) , combining the fixed read-only coordinate i_j with the displayed dynamic coordinate d . The row indexed by $1 \leq t < T$ represents every intermediate machine configuration in chronological order. The first and last rows are the same marked configuration $c_{\beta,5} = S_5$.

The reset mechanism is designed so that a sufficiently wide wall-bounded configuration returns to its distinguished initial configuration exactly when the simulated computation halts. The following lemma makes this equivalence precise.

Lemma 3.2. *For every input β , $\mathcal{U}$ halts on β if and only if there are $W \geq |\beta| + 2$ and $p > 0$ such that $\Phi_W^p(c_{\beta,W}) = c_{\beta,W}$.*

Proof. Fix $W \geq \ell_\beta + 2$. The following phase invariant follows inductively from the displayed successor table. Starting at $c_{\beta,W}$, the dynamic row passes through

$$S_W, \quad B_{1,W}^R, \dots, B_{W,W}^R, \quad B_{W,W}^L, \dots, B_{1,W}^L, \quad (3.2)$$

and then becomes $C_{1,q_0}(\square, \dots, \square)$. Thus at every boot time there is exactly one boot signal; to the left of **BootR** all cells are blank and to its right all cells carry **Init**, while all cells other than **BootL** are blank during the leftward boot phase. The fixed read-only word remains $\beta \text{pad}^{W-\ell_\beta-1}$. Consequently the last row in (3.2) is followed by the genuine initial configuration of $\mathcal{U}(\beta)$ on the first W cells.

During the simulation phase the invariant is $C_{h,q}(\mathbf{w})$, so there is exactly one head and every other dynamic symbol belongs to Γ . A nonhalting transition which stays inside the interval gives the next phase row of the same form. A requested right move from cell W creates **Crash**. If the unique head has

state q_h , the successive phase forms are

$$L_h(\mathbf{w}), L_{h-1}(\mathbf{w}^{(h-1)}), \dots, L_1(\mathbf{w}^{(1)}), \quad R_1(\mathbf{v}), \dots, R_W(\mathbf{v}), \quad B_{W,W}, \dots, B_{1,W}, S_W. \quad (3.3)$$

Here the superscripts merely record the work word left after each **SweepL** move, and $\mathbf{v} \in \Gamma^W$ is any completion of the work symbols in cells $2, \dots, W$ when the rightward sweep begins (the value of v_1 is immaterial). More explicitly, there is exactly one **SweepL** and each cell it has crossed is blank; in the **SweepR** phase every cell to the left of the signal is blank; and in the **SweepB** phase every cell to its right carries **Init** while every cell to its left is blank. Therefore the final move in (3.3) places **Start** in cell 1 and leaves **Init** in cells $2, \dots, W$.

Each row just listed is one of the phase templates defining φ , and Lemma 3.1 shows that all its radius-one windows receive the prescribed successor. This proves the invariant without an implicit lookup-table convention and shows that the default clause (R5) is not used on the intended orbit before a right-wall crossing.

If $U(\beta)$ halts and its computation visits at most the first s cells, choose $W > \max\{s, \ell_\beta + 1\}$. No head reaches the right wall. After the halt, the three reset sweeps blank the work tape and then write **Init** in cells $2, \dots, W$ and **Start** in cell 1. The read-only track and walls have not changed, so the automaton has returned exactly to $c_{\beta,W}$.

Conversely, start from $c_{\beta,W}$. After the boot sweeps, the orbit is the genuine computation of $U(\beta)$ until it either halts or attempts to cross the artificial right wall. An attempted crossing creates **Crash**, which is persistent and therefore precludes a return. Along this orbit, ordinary simulation and boot phases never create **Start**; the first later occurrence of **Start** can only be produced by the last $B_{1,W} \rightarrow S_W$ reset move, and that reset phase is entered only after a head in state q_h has occurred. Since $c_{\beta,W}$ contains **Start**, a return to it is therefore possible only after a genuine halt. This assertion concerns the canonical orbit from $c_{\beta,W}$; no claim is made about malformed configurations. This proves the reverse implication. ■

The symbols **Init** in cells $2, \dots, W$ allow the entire marked initial row to be certified by finitely many local tests. The anchored test checks only the fixed input prefix, while the marked-row adjacency rules force the **Init** suffix to continue to the right wall, independently of W . Thus no bounded test has to inspect an arbitrarily long suffix of ordinary blank cells, which also occur during the simulation.

3.3 The d -regular five-rail multidigraph

We first define the complete finite directed multidigraph that underlies the construction. Fix once and for all an integer $d \geq 7$ divisible by 625. For each $i \in [5]$, let

$$\mathcal{T}_i := \{C_i, A_i, E_i^-, E_i^+, N_i^-, N_i^+, B_i^-, B_i^+, \hat{C}_i, \hat{A}_i, \hat{E}_i^-, \hat{E}_i^+, \hat{N}_i^-, \hat{N}_i^+, \hat{B}_i^-, \hat{B}_i^+\}.$$

Take the five sets $\mathcal{T}_i$ pairwise disjoint, and put $\mathcal{T} := \bigsqcup_{i \in [5]} \mathcal{T}_i$. At this point the members of $\mathcal{T}$ are only vertex names.

For each $i \in [5]$, Figure 4 shows the directed multigraph $\mathcal{H}_i$ on $\mathcal{T}_i$. The vertices C_i and $\hat{C}_i$ are placed at the centers of the upper and lower layers, respectively. Each gray bundle represents a family of parallel bridge edges joining corresponding vertices. The three arrows in a bundle indicate parallelism only; the actual multiplicities are specified in the edge list below.

We regard the parallel edges of $\mathcal{H}_i$ as distinct. When no confusion can arise, we also identify $\mathcal{H}_i$ with this indexed edge multiset; thus $e \in \mathcal{H}_i$ means that e is one of its directed edges. Each directed edge will index one binary relation in a candidate realization of the five-rail signature defined below. We write a nonparallel edge as (s, t) and the ℓ th edge in a parallel family as $(s, t; \ell)$. Thus the endpoint types and, when necessary, the parallel-edge index distinguish all edges.

(i) The ten directed edges on the unhatted vertices are

$$(C_i, E_i^-), (E_i^-, E_i^+), (E_i^+, C_i), (C_i, N_i^-), (N_i^-, N_i^+), (N_i^+, C_i), (C_i, B_i^-), (B_i^-, B_i^+), (B_i^+, C_i), (A_i, C_i).$$

(ii) The ten directed edges on the hatted vertices are

$$(\hat{C}_i, \hat{E}_i^-), (\hat{E}_i^-, \hat{E}_i^+), (\hat{E}_i^+, \hat{C}_i), (\hat{C}_i, \hat{N}_i^-), (\hat{N}_i^-, \hat{N}_i^+), (\hat{N}_i^+, \hat{C}_i), (\hat{C}_i, \hat{B}_i^-), (\hat{B}_i^-, \hat{B}_i^+), (\hat{B}_i^+, \hat{C}_i), (\hat{A}_i, \hat{C}_i).$$

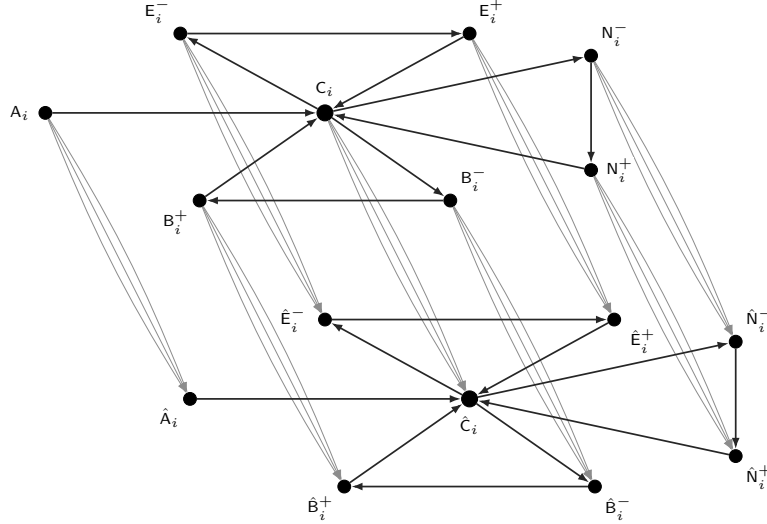


Figure 4: The directed multigraph $\mathcal{H}_i$, representing the i th copy in the five-rail construction. Black arrows join vertices within the unhatted and hatted vertex sets. Each group of three gray arrows schematically represents one family of parallel edges from s_i to $\hat{s}_i$, with the indexed parallel edges specified below.

(iii) The remaining edges form the following eight parallel families:

$$\begin{aligned} & \{(C_i, \hat{C}_i; \ell) : \ell \in [d-7]\}, \quad \{(A_i, \hat{A}_i; \ell) : \ell \in [d-1]\}, \quad \{(E_i^-, \hat{E}_i^-; \ell) : \ell \in [d-2]\}, \quad \{(E_i^+, \hat{E}_i^+; \ell) : \ell \in [d-2]\}, \\ & \{(N_i^-, \hat{N}_i^-; \ell) : \ell \in [d-2]\}, \quad \{(N_i^+, \hat{N}_i^+; \ell) : \ell \in [d-2]\}, \quad \{(B_i^-, \hat{B}_i^-; \ell) : \ell \in [d-2]\}, \quad \{(B_i^+, \hat{B}_i^+; \ell) : \ell \in [d-2]\}. \end{aligned}$$

Fact 3.3. *The following assertions hold.*

- (i) *For every $i \in [5]$, the underlying undirected multigraph of $\mathcal{H}_i$ is loopless and d -regular.*
- (ii) *For every $i \in [5]$, the multigraph $\mathcal{H}_i$ has sixteen vertices and $8d$ directed edges, counted with multiplicity.*
- (iii) *The multigraph $\mathcal{H} := \bigsqcup_{i \in [5]} \mathcal{H}_i$ is loopless and d -regular on 80 vertices and has $40d$ directed edges, counted with multiplicity.*

We call the submultigraph induced by the eight unhatted vertices of $\mathcal{H}_i$ the *logical layer*, the submultigraph induced by the eight hatted vertices the *balancing layer*, and the eight parallel edge families joining s_i to $\hat{s}_i$ the *bridge families*. The five subgraphs $\mathcal{H}_i$ are the *rails*, and the vertices of $\mathcal{H}$ are the *types*. The eight unhatted types have the following roles.

- A cell of the space-time tableau has type c .
- An anchor has type A . After it is linked to a vertex of type c , that cell is required to lie in the marked initial row.
- Horizontal succession between two cells in the same row is encoded by a path through the intermediate types E^- and E^+ .
- Time succession between corresponding cells in consecutive rows is encoded by a path through the intermediate types N^- and N^+ .
- A symbol from Σ_{CA} is assigned to a cell along a path through the intermediate types B^- and B^+ .

The superscripts $-$ and $+$ distinguish the first and second intermediate types along an oriented path; they do not denote signs.

3.4 Realizations and local obstructions

We view $\mathcal{H}$ as a blueprint for a finite typed relational structure. To couple the five rails, we supplement it with one five-ary relation symbol M whose i th coordinate has type A_i for every $i \in [5]$. The symbol M is not indexed by an edge of $\mathcal{H}$. We call the five edges (B_i^-, B_i^+) , $i \in [5]$, the *label edges*; all other edges are *ordinary*.

A finite *candidate realization* $\mathfrak{R}$ of $(\mathcal{H}, \mathbf{M})$ on a finite vertex set V consists of the following data.

- (i) A map $V \rightarrow \mathcal{T}$ assigning a type to every vertex. For each type $U \in \mathcal{T}$, let $V_U(\mathfrak{R})$ be the inverse image of U under this map. These vertex classes partition V ; when $\mathfrak{R}$ is fixed, we write simply V_U .
- (ii) For every ordinary nonparallel directed edge $e = (s, t)$ of $\mathcal{H}$, a binary relation $D_e = D_{s,t} \subseteq V_s \times V_t$.
For every ordinary parallel directed edge $e = (s, t; \ell)$ of $\mathcal{H}$, a binary relation $D_e = D_{s,t}^{(\ell)} \subseteq V_s \times V_t$.
- (iii) For every rail $i \in [5]$ and cellular-automaton color $a \in \Sigma_{\text{CA}}$, a binary relation $D_{\mathbb{B}_i^-, \mathbb{B}_i^+}^{[a]} \subseteq V_{\mathbb{B}_i^-} \times V_{\mathbb{B}_i^+}$.

The underlying relation of the label edge is $D_{\mathbb{B}_i^-, \mathbb{B}_i^+} := \bigcup_{a \in \Sigma_{\text{CA}}} D_{\mathbb{B}_i^-, \mathbb{B}_i^+}^{[a]}$.

- (iv) A five-ary relation $\mathbf{M} \subseteq V_{A_1} \times \dots \times V_{A_5}$.

For a directed edge e from s to t , we also regard $D_e \subseteq V_s \times V_t$ as a directed bipartite graph whose edges are oriented from V_s to V_t . We call a directed edge (x, y) of D_e a *relation instance* and also write it as $D_e(x, y)$. For a label edge, we write $D_{\mathbb{B}_i^-, \mathbb{B}_i^+}^{[a]}(u, v)$ for membership in its color- a relation and call it a *color- a label pair*. A member of $\mathbf{M}$ is an $\mathbf{M}$ -*tuple*. The superscript $^{[a]}$ records a cellular-automaton color, whereas $^{(\ell)}$ distinguishes parallel bridge edges. Although the label relation is subdivided by color, $\mathcal{H}_i$ contains only one label edge $(\mathbb{B}_i^-, \mathbb{B}_i^+)$, counted once in its degree sequence, and its hatted counterpart $(\hat{\mathbb{B}}_i^-, \hat{\mathbb{B}}_i^+)$ indexes one ordinary relation. Apart from the displayed typing and union requirements, all these relations are arbitrary at this stage.

The three directed paths in the logical layer encode horizontal successor, time successor, and cell label, while $A_i \rightarrow c_i$ indexes the anchor-to-cell relation. Only these unhatted relations occur in (Q1)–(Q5). The hatted relations and bridges are used for balancing, and no hatted relation is required to agree with its unhatted counterpart.

We first impose three matching requirements on all relations.

- (PM1) For every ordinary edge e , the edges of the directed bipartite graph D_e form a matching.
- (PM2) For each label edge on rail i , every source vertex u admits at most one pair (a, v) for which $D_{\mathbb{B}_i^-, \mathbb{B}_i^+}^{[a]}(u, v)$ holds, and every target vertex v admits at most one pair (u, a) for which $D_{\mathbb{B}_i^-, \mathbb{B}_i^+}^{[a]}(u, v)$ holds. Equivalently, the edges of the directed bipartite graph $D_{\mathbb{B}_i^-, \mathbb{B}_i^+}$ form a matching and the relations $D_{\mathbb{B}_i^-, \mathbb{B}_i^+}^{[a]}$, $a \in \Sigma_{\text{CA}}$, are pairwise disjoint.
- (PM3) The relation $\mathbf{M}$ is a partial five-matching, meaning that two distinct $\mathbf{M}$ -tuples cannot agree in any coordinate.

The remaining requirements concern the tableau carried by the unhatted relations. Fix a rail $i \in [5]$ and cell vertices $x, y \in V_{c_i}$. We use the following notation.

- (i) The predicate $D_{\text{hor}}(x, y)$ holds if there are $u \in V_{\mathbb{E}_i^-}$ and $v \in V_{\mathbb{E}_i^+}$ such that $D_{c_i, \mathbb{E}_i^-}(x, u)$, $D_{\mathbb{E}_i^-, \mathbb{E}_i^+}(u, v)$, and $D_{\mathbb{E}_i^+, c_i}(v, y)$ all hold. That is, $(x, u) \in D_{c_i, \mathbb{E}_i^-}$, $(u, v) \in D_{\mathbb{E}_i^-, \mathbb{E}_i^+}$, and $(v, y) \in D_{\mathbb{E}_i^+, c_i}$.
- (ii) The predicate $D_{\text{time}}(x, y)$ holds if there are $u \in V_{N_i^-}$ and $v \in V_{N_i^+}$ such that $D_{c_i, N_i^-}(x, u)$, $D_{N_i^-, N_i^+}(u, v)$, and $D_{N_i^+, c_i}(v, y)$ all hold. That is, $(x, u) \in D_{c_i, N_i^-}$, $(u, v) \in D_{N_i^-, N_i^+}$, and $(v, y) \in D_{N_i^+, c_i}$.
- (iii) For $a \in \Sigma_{\text{CA}}$, the predicate $L_{i,a}(x, y)$ holds if there are $u \in V_{\mathbb{B}_i^-}$ and $v \in V_{\mathbb{B}_i^+}$ such that $D_{c_i, \mathbb{B}_i^-}(x, u)$, $D_{\mathbb{B}_i^-, \mathbb{B}_i^+}^{[a]}(u, v)$, and $D_{\mathbb{B}_i^+, c_i}(v, y)$ all hold. Thus $L_{i,a}(x, y)$ records a fully present color- a label path from x to y . The notation $\text{label}(x) = a$ means that this path is closed, that is, that $L_{i,a}(x, x)$ holds.

When $a \notin \{\mathbf{L}, \mathbf{R}\}$, it has the form $a = (\iota, d) \in \mathcal{I} \times \mathcal{D}$. Thus $\text{label}(x) = a$ records both the fixed read-only symbol ι and the dynamic symbol d of the cell. The two coordinates are not represented by separate cell vertices.

The three paths are shown schematically in Figure 5.

Each circle represents the indicated vertex class, and the labels x, y, u, v name vertices chosen from these classes. An arrow from a vertex of type s to a vertex of type t represents an instance of $D_{s,t}$. The middle arrow in the right-hand diagram, marked $[a]$, represents an instance of $D_{\mathbb{B}_i^-, \mathbb{B}_i^+}^{[a]}$. The diagram identifies the initial and final c_i -vertices and hence depicts $L_{i,a}(x, x)$; the general predicate $L_{i,a}(x, y)$ allows distinct

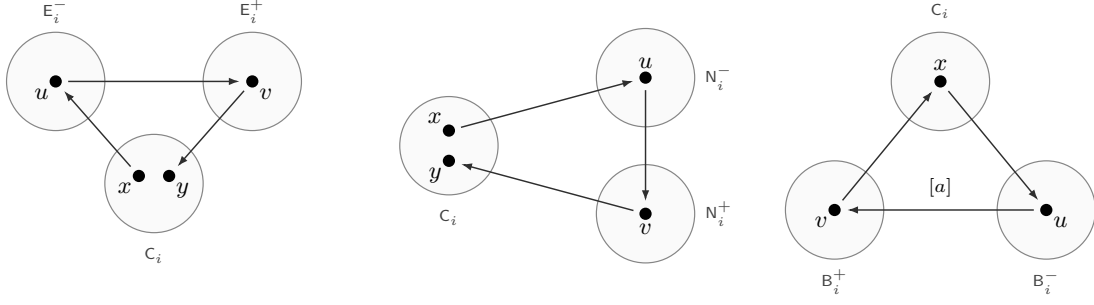


Figure 5: The three relation paths on rail i . From left to right, the diagrams show the horizontal-successor predicate $D_{\text{hor}}(x, y)$, the time-successor predicate $D_{\text{time}}(x, y)$, and the closed special case $L_{i,a}(x, x)$ of the color- a label-path predicate.

endpoints. Thus each notation describes a complete three-edge path and does not introduce an additional edge-indexed relation. Whenever one rail is fixed or its index is immaterial, we suppress the common rail index from the endpoint subscripts of D .

Conditions (Q1)–(Q5) below are guarded. A condition applies only when every relation path and label mentioned in its hypothesis is present. Each condition is imposed separately on every rail, with (Q5) additionally guarded by an M -tuple.

(Q1) *Label closure*. For every $a \in \Sigma_{CA}$ and $x, y \in V_{C_i}$, one has $L_{i,a}(x, y) \implies y = x$.

(Q2) *Commutation*. For cell vertices $x, x_E, x_N, y, z \in V_{C_i}$, whenever the four paths

$$D_{\text{hor}}(x, x_E), \quad D_{\text{time}}(x, x_N), \quad D_{\text{time}}(x_E, y), \quad \text{and} \quad D_{\text{hor}}(x_N, z)$$

are present, one has $y = z$. Figure 6 displays the four paths in the hypothesis before this equality is imposed, including the two intermediate vertices in each path. In panel (a), each circle represents one

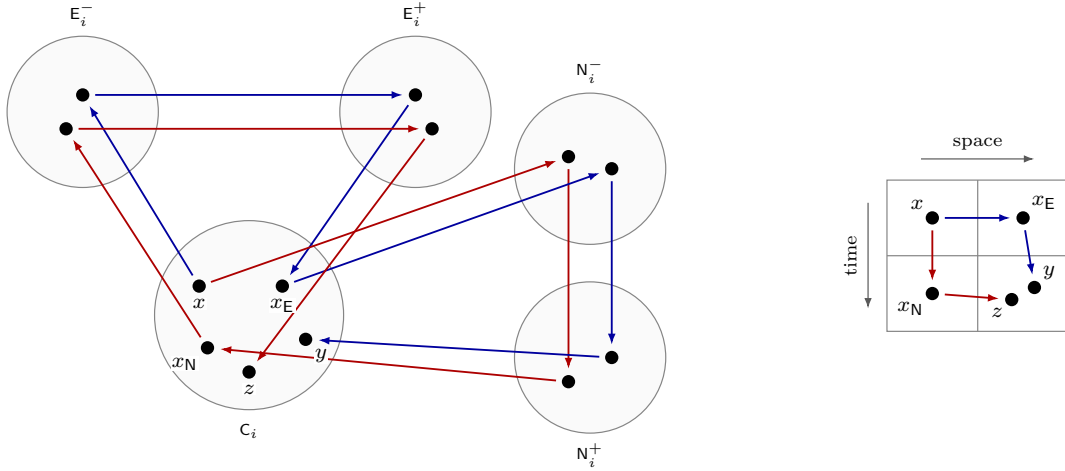


Figure 6: (a) Typed relation instances for condition (Q2). The blue edges form the paths $D_{\text{hor}}(x, x_E)$ and $D_{\text{time}}(x_E, y)$, while the red edges form $D_{\text{time}}(x, x_N)$ and $D_{\text{hor}}(x_N, z)$. (b) Their intended positions in the space-time tableau.

vertex class. The five labeled points in the C_i -circle are the cell vertices, and each of the other four circles contains the two intermediate vertices of the indicated type used by the two corresponding paths. Each arrow represents the binary relation determined by the types at its endpoints. Panel (b) shows the intended relative positions of the five cell vertices in the space-time tableau. The blue route is $x \rightarrow x_E \rightarrow y$, and the red route is $x \rightarrow x_N \rightarrow z$. The points y and z are drawn separately in the lower-right cell because the conclusion has not yet been imposed; condition (Q2) requires them to coincide.

(Q3) *Evolution*. Whenever the paths and labels

$$D_{\text{hor}}(x_-, x), \quad D_{\text{hor}}(x, x_+), \quad D_{\text{time}}(x, x'), \quad \text{label}(x_-) = a_-, \\ \text{label}(x) = a_0, \quad \text{label}(x_+) = a_+, \quad \text{and} \quad \text{label}(x') = b$$

are present, one has $b = \varphi(a_-, a_0, a_+)$. Figure 7 expands every relation path in this condition.

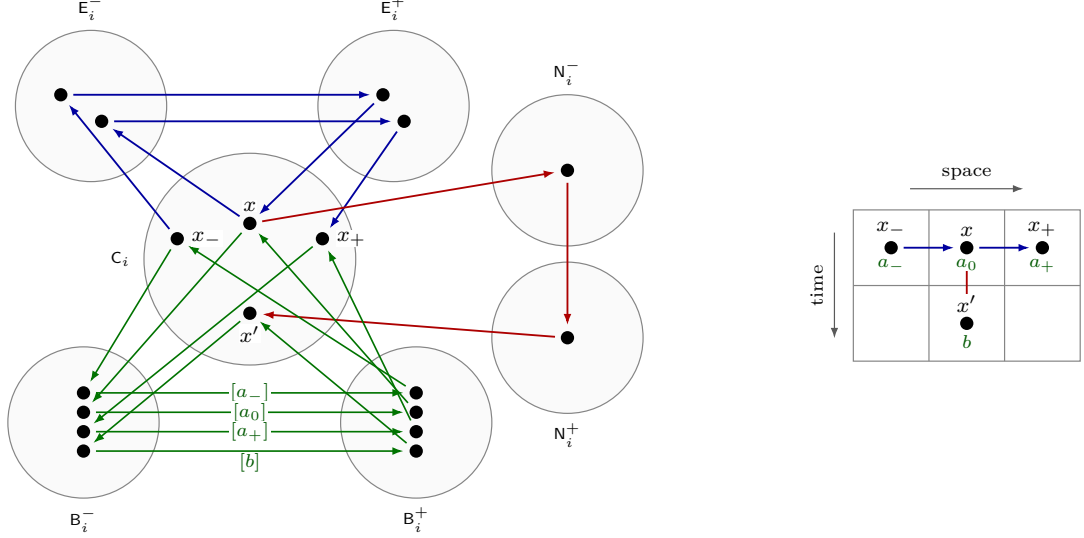


Figure 7: (a) Typed relation instances for condition (Q3). Blue edges form the two horizontal-successor paths, red edges form the time-successor path, and green edges form the four closed label paths, with the middle green edges carrying the cellular-automaton colors a_- , a_0 , a_+ , b . (b) The corresponding local configuration in the space-time tableau. Condition (Q3) requires $b = \varphi(a_-, a_0, a_+)$.

(Q4) *Marked-row grammar*. Using Σ_{dyn} and Σ_{inp} , whenever $D_{\text{hor}}(x, y)$ is present and both x and y are labeled, the following implications must hold.

$$\begin{aligned} \text{label}(x) = \mathbf{R} &\implies \text{label}(y) = \mathbf{L}; \\ \text{label}(x) \in \Sigma_{\text{dyn}}(\mathbf{Start}) &\implies \text{label}(y) \in \Sigma_{\text{dyn}}(\mathbf{Init}); \\ \text{label}(x) \in \Sigma_{\text{dyn}}(\mathbf{Init}) &\implies \text{label}(y) \in \Sigma_{\text{dyn}}(\mathbf{Init}) \cup \{\mathbf{R}\}; \\ \text{label}(x) \in \Sigma_{\text{inp}}(\$) &\implies \text{label}(y) \in \Sigma_{\text{inp}}(\mathbf{pad}); \\ \text{label}(x) \in \Sigma_{\text{inp}}(\mathbf{pad}) &\implies \text{label}(y) \in \Sigma_{\text{inp}}(\mathbf{pad}) \cup \{\mathbf{R}\}. \end{aligned}$$

There is no global restriction on the successor of $\mathbf{L}$. In particular $\mathbf{L} \rightarrow \mathbf{Start}$ is permitted on a reset row, while $\mathbf{L} \rightarrow$ an ordinary simulation state is permitted on every other row. The marked anchor, rather than a global rule, identifies which occurrence of $\mathbf{L} \rightarrow \mathbf{Start}$ is canonical. The sets $\Sigma_{\text{inp}}(\iota)$ and $\Sigma_{\text{dyn}}(s)$ contain only interior colors, so requiring $\text{label}(z)$ to lie in either set automatically excludes the two wall colors. A failed membership condition is therefore one of finitely many forbidden color alternatives.

(Q5) *The input-dependent anchor*. Recall that $\beta = \beta_1 \cdots \beta_{\ell_\beta}$. For each $i \in [5]$, guard the following conditions by an $\mathbf{M}$ -tuple $\mathbf{M}(a_1, \dots, a_5)$ and a present instance $D_{A_i, C_i}(a_i, x)$ of the anchor relation on rail i , where $x \in V_{C_i}$.

- (a) if $D_{\text{hor}}(y, x)$ and the two labels are present, then $\text{label}(y) = \mathbf{L}$ and $\text{label}(x) \in \Sigma_{\text{dyn}}(\mathbf{Start})$;
- (b) for $1 \leq j \leq \ell_\beta$, if the relation path from x to $D_{\text{hor}}^{j-1}x$ and its label are present, then $\text{label}(D_{\text{hor}}^{j-1}x) \in \Sigma_{\text{inp}}(\beta_j)$;
- (c) if the relation path from x to $D_{\text{hor}}^{\ell_\beta}x$ and its label are present, then $\text{label}(D_{\text{hor}}^{\ell_\beta}x) \in \Sigma_{\text{inp}}(\$)$;
- (d) for $1 \leq \ell \leq \ell_\beta + 3$, no fully present relation path $D_{\text{hor}}^\ell(x, x)$ occurs.

Here $D_{\text{hor}}^0x = x$, and every displayed D_{hor}^j is expanded into j concatenated copies of the three-step horizontal path. For $\ell_\beta = 0$, part (c) is imposed at x itself. Notice that each condition uses no binary relation instance on a rail other than i .

We call a candidate realization *legal* for the input β if it satisfies (PM1)–(PM3) and (Q1)–(Q5). We next encode every violation by a finite typed local obstruction.

A *typed local obstruction* is a finite candidate realization regarded as a forbidden pattern. We say that a candidate realization $\mathfrak{R}$ *contains a copy* of an obstruction Q if there is a type-preserving injection from $V(Q)$ to $V(\mathfrak{R})$ that preserves every binary-relation instance and every M -tuple of Q .

Lemma 3.4. *For every input word β , one can effectively construct a finite list $\mathcal{P}_\beta^{\text{phys}}$ of typed local obstructions such that a candidate realization is legal for β if and only if it contains no copy of a member of $\mathcal{P}_\beta^{\text{phys}}$.*

We remark that the matching-conflict obstructions and the obstructions arising from (Q1)–(Q4) are independent of β . Every β -dependent obstruction arises from (Q5) and has size $O(|\beta| + 1)$. In particular, legality is hereditary under deleting binary-relation instances, M -tuples, or vertices.

Proof. We first record failures of (PM1)–(PM3) by matching-conflict obstructions. In the formulas below, each variable is a formal vertex of the type determined by its position in the relation, and an inequality declares two formal vertices of the same type to be distinct.

- (i) For every ordinary edge $e = (u, v)$ or $(u, v; \ell)$ in $\mathcal{H}$, include $\{D_e(x, y), D_e(x, y')\}$ whenever $y \neq y'$, and include $\{D_e(x, y), D_e(x', y)\}$ whenever $x \neq x'$.
- (ii) For every $i \in [5]$ and $a, b \in \Sigma_{CA}$, include the obstruction consisting of $D_{B_i^-, B_i^+}^{[a]}(x, y)$ and $D_{B_i^-, B_i^+}^{[b]}(x, y')$ whenever $y \neq y'$, and the obstruction consisting of $D_{B_i^-, B_i^+}^{[a]}(x, y)$ and $D_{B_i^-, B_i^+}^{[b]}(x', y)$ whenever $x \neq x'$. When $a \neq b$, also include the obstruction consisting of $D_{B_i^-, B_i^+}^{[a]}(x, y)$ and $D_{B_i^-, B_i^+}^{[b]}(x, y)$.
- (iii) For every nonempty proper subset $I \subsetneq [5]$, include $\{M(a_1, \dots, a_5), M(b_1, \dots, b_5)\}$ subject to $a_j = b_j$ if and only if $j \in I$.

Thus all possible coordinate overlaps of two distinct tuples are included, not only the case in which exactly one coordinate is shared. Avoiding these matching-conflict obstructions is equivalent to (PM1)–(PM3).

We next treat violations of (Q1)–(Q5). Expand their relation paths into constituent binary-relation statements and enumerate the finitely many cellular-automaton color assignments. Each forbidden alternative then becomes a finite typed relation pattern P with required equalities already imposed, required binary-relation instances, required M -tuples, and finitely many declared inequalities. An *admissible type-preserving homomorphic image* of P is a typed local obstruction Q obtained from a surjection $f: V(P) \rightarrow V(Q)$ with the following properties. The map f sends every vertex to one of the same type, sends every required relation instance and M -tuple of P to the corresponding relation instance or tuple of Q , and satisfies $f(x) \neq f(y)$ whenever P declares $x \neq y$. The relations and tuples of Q are exactly the images of those listed in P .

For each forbidden pattern P , include one representative of every admissible type-preserving homomorphic image of P . There are only finitely many such images, and they can be enumerated effectively, because each is determined by a partition of $V(P)$ into same-type classes that separates every pair declared unequal. An image that already contains a matching-conflict obstruction may be omitted because that conflict is recorded separately. Thus every guarded implication above gives a finite list of patterns by adjoining all its hypotheses to each forbidden alternative; no absence requirement is used. Conversely, every fully present violation in a candidate realization determines a type-preserving homomorphism from one of these patterns. This homomorphism factors through its image as a surjection followed by a type-preserving injective copy. The candidate realization therefore contains a copy of the image, or a copy of a matching-conflict obstruction if that image was omitted.

For example, the violation of (Q2) illustrated in Figure 6 consists of the four fully present paths $D_{\text{hor}}(x, x_E)$, $D_{\text{time}}(x, x_N)$, $D_{\text{time}}(x_E, y)$, and $D_{\text{hor}}(x_N, z)$, together with $y \neq z$. Expanding the four paths and then taking an admissible type-preserving homomorphic image gives a concrete typed local obstruction witnessing the failure of commutation.

These homomorphic images are needed even for short local conditions. Since copies of local obstructions use injective maps, a generic commutation or transition pattern alone would miss a violation in which some of its formal vertices have the same image. Passing to every admissible homomorphic image records

exactly the kernel of every map from the pattern into an actual local configuration. The short-cycle condition in (Q5) separately prevents the displayed input prefix from collapsing.

Let $\mathcal{P}_\beta^{\text{phys}}$ consist of these matching-conflict obstructions and all local obstructions obtained from violations of (Q1)–(Q5). If a candidate realization is not legal, then either one of (PM1)–(PM3) fails, giving a copy of one of the first obstructions, or one of (Q1)–(Q5) fails, giving a copy by the preceding construction. Conversely, every listed obstruction records one of these failures. This proves the asserted equivalence.

The signature and alphabet are fixed. Conditions (Q1)–(Q4) therefore give fixed finite lists, as do the matching-conflict obstructions. Condition (Q5) uses only finitely many paths of length at most $|\beta| + 3$; expanding them and enumerating their color assignments and admissible type-preserving homomorphic images is finite and effective. These paths are the only source of dependence on β , and every resulting obstruction has size $O(|\beta| + 1)$. Finally, no member of $\mathcal{P}_\beta^{\text{phys}}$ imposes an absence condition, so deleting relations, tuples, or vertices cannot create a copy. Hence legality is hereditary under these deletions. ■

3.5 Closed components and tableau recovery

Given a legal realization $\mathfrak{R}$ of $(\mathcal{H}, M)$ for the input β , form an auxiliary multigraph on $V(\mathfrak{R})$ as follows. For every edge e of $\mathcal{H}$ and every directed edge (x, y) of D_e , add an undirected edge xy labeled by e , retaining parallel edges. The M -tuples are ignored. We call the connected components of this multigraph the *relation components* of $\mathfrak{R}$. Because no relation joins two distinct rails, every relation component lies on a unique rail.

Since $\mathcal{H}_i$ is d -regular and the edges of every directed bipartite graph D_e form a matching, every vertex of the auxiliary multigraph has degree at most d . A relation component K of $\mathfrak{R}$ is *closed* if every vertex of K has degree d ; otherwise it is *grounded*. Equivalently, K is grounded if and only if there are a vertex $v \in V(K)$ of type U and an edge e of $\mathcal{H}_i$ incident with U such that no edge of D_e contains v .

If K is closed, then for every edge $e = (x, y)$ or $(x, y; \ell)$, the restriction $D_e \cap ((V_x \cap V(K)) \times (V_y \cap V(K)))$ is a perfect matching between $V_x \cap V(K)$ and $V_y \cap V(K)$. With the orientation from x to y , this perfect matching defines a bijection from the first class onto the second. Indeed, every vertex of K has degree d , and exactly d edges of $\mathcal{H}_i$, counted with multiplicity, are incident with its type. Since each D_e is a matching, every relation prescribed at that type must occur exactly once at the vertex. In particular, if K lies on rail i , the three-step logical paths on that rail induce permutations $D_{\text{hor}}, D_{\text{time}}$ of $V_{C_i} \cap V(K)$. The label path gives exactly one color at every such cell. Totality supplies one colored label pair, and (Q1) forces its return endpoint to be the original cell. Conditions (Q2) and (Q3) then give

$$D_{\text{hor}} D_{\text{time}} = D_{\text{time}} D_{\text{hor}} \quad \text{and} \quad \text{label}(D_{\text{time}} x) = \varphi(\text{label}(D_{\text{hor}}^{-1} x), \text{label}(x), \text{label}(D_{\text{hor}} x)). \quad (3.4)$$

Note that these conclusions use totality only inside an explicitly closed component.

The following schematic separates the two ingredients used in tableau recovery. Closedness makes the logical successor maps total permutations and the local square commute. The anchor then identifies a wall-bounded horizontal compartment, and a finite temporal orbit returns that compartment to its initial row.

We next compare the components formed by the logical-layer relations with the relation components formed by all edges of $\mathcal{H}$. Fix a rail i and a vertex v whose type is one of the eight unhatted types on that rail. In the auxiliary multigraph defining the relation components, delete every edge except those labeled by logical-layer edges of $\mathcal{H}_i$. We call the connected component containing v in the resulting multigraph the *logical relation component* of v . If the full relation component K containing v is closed, then every logical relation prescribed at a vertex of the logical relation component of v is present there. Indeed, every vertex of K has degree d , so each relation prescribed at its type occurs exactly once; the other endpoint of a logical relation lies again in the same logical relation component. Equivalently, if a prescribed logical relation is absent at an unhatted vertex, then every full relation component containing that vertex is grounded.

The converse need not hold because a balancing-layer copy relation or a bridge may be absent at one of its endpoint vertices, grounding a full relation component whose logical relation component is closed. For the equality constructions, however, a realization of the logical-layer signature in which every relation forms a perfect matching between its endpoint classes extends to one with the same property for the full signature. Take isomorphic vertex classes for the hatted balancing copies, transport the logical perfect

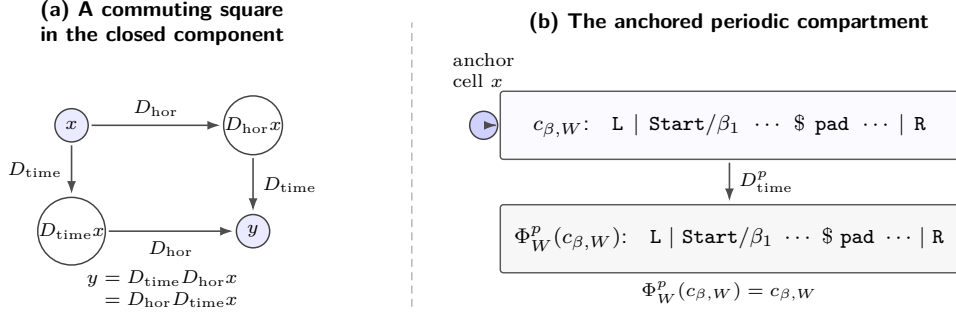


Figure 8: Closed-component recovery schematic. The left panel records only the commuting-square identity forced inside a closed logical component; the ambient component need not be a rectangular torus. The right panel shows the wall-bounded compartment selected by the anchor. Its least temporal return gives a periodic reset tableau.

matchings to the corresponding relations on the balancing layer, and choose a perfect matching for every bridge relation. Thus the balancing layer neither conceals the absence of a prescribed logical relation at a vertex nor obstructs the zero-defect constructions.

We now collect the properties of the resulting finite tableau system.

Lemma 3.5. *The fixed five-rail signature $\mathcal{H}$ and the obstruction lists $\mathcal{P}_\beta^{\text{phys}}$ supplied by Lemma 3.4 have the following additional properties.*

- (i) *There is a finite legal realization of $(\mathcal{H}, \mathbf{M})$ for β in which $\mathbf{M} = \emptyset$ and every directed bipartite graph D_e is a perfect matching between its two vertex classes, with all color relations on the label edge taken together.*
- (ii) *If $\mathbf{U}(\beta)$ halts, there is a finite legal realization of $(\mathcal{H}, \mathbf{M})$ for β in which $\mathbf{M} \neq \emptyset$ and every directed bipartite graph D_e is a perfect matching between its two vertex classes, with all color relations on the label edge taken together.*
- (iii) *If a coordinate a_i of an $\mathbf{M}$ -tuple lies in a closed relation component, let h be the number of vertices of type c_i in the logical relation component of a_i . Then there are integers $W, p \geq 1$ such that $W \geq |\beta| + 2$, $W + 2 \leq h$, $p \leq h$, and $\Phi_W^p(c_{\beta, W}) = c_{\beta, W}$. In particular, if $\mathbf{U}(\beta)$ does not halt, then every coordinate of every $\mathbf{M}$ -tuple belongs to a grounded component.*

In (1) and (2), the five rails of the legal realization may be chosen isomorphic to one another, and all its vertex classes may be chosen to have the same cardinality.

Proof. For (1), choose any $n \geq 1$. On n cells let D_{hor} be an n -cycle, let D_{time} be the identity, and give every cell the fixed background color ($\text{pad}, \square$). Choose perfect matchings for the constituent logical relations so that their compositions give these two maps and every label path returns to its initial cell with the fixed background color. Put $\mathbf{M} = \emptyset$. Conditions (Q1), (Q2), and (Q4) hold by construction, (Q5) is vacuous, and the constant-background rule (3.1) verifies (Q3). Transport these matchings to the balancing layer and choose a perfect matching for every bridge relation. This gives the required realization with every physical vertex class of size n .

Suppose $\mathbf{U}(\beta)$ halts. Choose W, p from Lemma 3.2. On the set $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/(W+2)\mathbb{Z}$, let D_{time} and D_{hor} be the two coordinate successors. Label time row t by $\Phi_W^t(c_{\beta, W})$, including the two wall cells, and identify the right wall's D_{hor} -successor with the left wall. A wall ignores the neighbor on its exterior side, so this is a valid space-time torus. Identify every physical vertex class with a copy of this torus, choose the logical perfect matchings so that each relation path induces its prescribed logical map, and choose arbitrary perfect matchings for the copy relations on the balancing layer and the bridge relations.

Every label path now begins and ends at the same cell, which verifies (Q1); the two coordinate translations commute, which verifies (Q2); and the definition of row $t + 1$ verifies every transition test in (Q3). Since every directed bipartite graph D_e is a perfect matching, all binary matching-conflict obstructions are avoided. To check (Q4), note that the read-only track is always $\beta \$ \text{pad} \dots \text{pad}$ and that the horizontal seam is $R \rightarrow L$. In the displayed phase rows, every **Start** is followed by **Init**, while every **Init** is

followed by **Init** or **R**. Thus (Q4) holds throughout the boot, simulation, and reset sweeps. The identity $\Phi_W^p(c_{\beta,W}) = c_{\beta,W}$ verifies the vertical seam.

On each rail, let $a_i \in V_{A_i}$ be the anchor sent to the copy of cell 1 at time 0, and put the single M -tuple $(a_1, \dots, a_5)$ into M . The coordinate-overlap obstructions are then avoided, and the row at time zero satisfies (Q5). All five rails and all physical vertex classes have the same cardinality, proving part (2).

For (3), let $\mathbf{a} = (a_1, \dots, a_5) \in M$ and suppose that a_i lies in a closed relation component. Let L be the logical relation component of a_i . Since the full relation component containing L is closed, every logical relation prescribed at a vertex of L is present there. In particular, D_{A_i, C_i} sends a_i to a cell $x \in V_{C_i}$. Beyond the legality conditions already imposed, the proof of (3.4) uses only totality of the logical relations, so the same argument applies within L . Thus $D_{\text{hor}}, D_{\text{time}}$ are commuting permutations on its h cell vertices and every cell is labeled.

The anchored conditions give $\text{label}(D_{\text{hor}}^{-1}x) = L$, $\text{label}(x) \in \Sigma_{\text{dyn}}(\mathbf{Start})$, and $\text{label}(D_{\text{hor}}^{j-1}x) \in \Sigma_{\text{inp}}(\beta_j)$ for $1 \leq j \leq \ell_\beta$, as well as $\text{label}(D_{\text{hor}}^{\ell_\beta}x) \in \Sigma_{\text{inp}}(\$)$. We spell out the first-wall argument. By (Q4), the successor of the **Start**-labeled cell x has dynamic symbol **Init**. Thereafter, as long as no right wall has appeared, each successive cell again has dynamic symbol **Init**.

A right wall must eventually appear. Otherwise the finite D_{hor} -orbit of x returns to x ; let $t > 0$ be its least period. If $t = 1$, this is forbidden by (Q5)(d) with $\ell = 1$. If $t \geq 2$, then $D_{\text{hor}}x, \dots, D_{\text{hor}}^{t-1}x$ have dynamic symbol **Init**, so the last adjacency is the forbidden pair **Init** $\rightarrow$ **Start**. Let $W \geq 1$ be least such that $\text{label}(D_{\text{hor}}^Wx) = R$. Thus the dynamic symbols at D_{hor}^jx , $1 \leq j < W$, are all **Init**.

The anchor conditions put the input word $\beta_1 \cdots \beta_{\ell_\beta}$ at $x, \dots, D_{\text{hor}}^{\ell_\beta-1}x$ and put $\$$ at $D_{\text{hor}}^{\ell_\beta}x$. In particular, none of those vertices is the first right wall. Moreover, the $\$ \rightarrow \mathbf{pad}$ clause of (Q4) forces $D_{\text{hor}}^{\ell_\beta+1}x$ to be an interior cell with input symbol **pad**, rather than a wall. Hence $W \geq \ell_\beta + 2$. The **pad** clause and the minimality of W now give

$$\text{label}(D_{\text{hor}}^jx) \in \begin{cases} \Sigma_{\text{inp}}(\beta_{j+1}), & 0 \leq j < \ell_\beta, \\ \Sigma_{\text{inp}}(\$), & j = \ell_\beta, \\ \Sigma_{\text{inp}}(\mathbf{pad}), & \ell_\beta < j < W. \end{cases}$$

All vertices $D_{\text{hor}}^{-1}x, x, D_{\text{hor}}x, \dots, D_{\text{hor}}^Wx$ are distinct. Indeed, suppose $D_{\text{hor}}^i x = D_{\text{hor}}^j x$ for $-1 \leq i < j \leq W$. Since D_{hor} is a permutation, $D_{\text{hor}}^{j-i}x = x$. If $j - i < W$, put $t = j - i$. The case $t = 1$ contradicts (Q5)(d); if $t \geq 2$, the closing adjacency is again the forbidden pair **Init** $\rightarrow$ **Start**. If $j - i = W$, then x carries both the **Start** and **R** colors. The remaining possibility, $j - i = W + 1$, forces $i = -1$ and $j = W$, giving one vertex both wall colors **L** and **R**.

Together with $\text{label}(D_{\text{hor}}^{-1}x) = L$, this proves that the labeled segment from $D_{\text{hor}}^{-1}x$ through D_{hor}^Wx is a copy of the canonical row $c_{\beta,W}$, with W interior cells and the first right wall at its right end. The wall rule isolates this segment from everything outside the two walls.

Let p be the least positive period of x under the permutation D_{time} . Then $p \leq h$ and $D_{\text{time}}^p x = x$. Commutation rules out a twisted return and gives $D_{\text{time}}^p D_{\text{hor}}^j x = D_{\text{hor}}^j D_{\text{time}}^p x = D_{\text{hor}}^j x$ for every integer j . More explicitly, induction on t , using both identities in (3.4), shows that the labels on $D_{\text{time}}^t D_{\text{hor}}^{-1}x, D_{\text{time}}^t x, \dots, D_{\text{time}}^t D_{\text{hor}}^Wx$ are exactly the row $\Phi_W^t(c_{\beta,W})$. At either wall, rule (R1) fixes the wall label independently of its neighbors; the update of an adjacent interior cell reads that wall but never a label beyond it. Hence this induction does not use any cell outside the marked compartment. The equality for $t = p$ therefore shows that the whole marked compartment returns pointwise and $\Phi_W^p(c_{\beta,W}) = c_{\beta,W}$. The $W + 2$ vertices listed above are distinct cell vertices, so $W + 2 \leq h$. This proves the quantitative assertion in part (3). If $U(\beta)$ does not halt, the reset lemma rules out a closed coordinate; hence every coordinate of every M -tuple is grounded. $\blacksquare$

4 The weighted five-rail estimate

For each finite legal realization of $(\mathcal{H}, M)$ for β , we define a homogeneous polynomial of degree five in its vertex weights and maximize it over probability weightings. The construction is calibrated so that incomplete computation records cannot exceed a fixed input-independent baseline, whereas a complete periodic record yields a strict improvement. Thus nonhalting corresponds to equality with the baseline, while halting corresponds to a strictly larger optimum.

We first define this polynomial. Recall from [Section 3](#) the type set $\mathcal{T}$ and its rail vertex sets $\mathcal{T}_i$. By [Fact 3.3](#), the multidigraph $\mathcal{H} = \bigsqcup_{i \in [5]} \mathcal{H}_i$ is fixed, loopless, and d -regular. Here and below, a sum over $e \in \mathcal{H}_i$ runs over its indexed directed edges, so parallel edges are counted separately. The label edge $(\mathbb{B}^-, \mathbb{B}^+)$ is counted once, and all edges on the balancing layer and all bridge edges are included.

Fix a finite legal realization $\mathfrak{R}$ of $(\mathcal{H}, \mathbf{M})$ for β , and let $\mathbf{x} = (x_v)_{v \in V(\mathfrak{R})}$ be a nonnegative weight vector. For each $i \in [5]$, the classes V_s , $s \in \mathcal{T}_i$, partition the vertices on rail i . Define the total weight of that rail by

$$\rho_i := \sum_{s \in \mathcal{T}_i} \sum_{v \in V_s} x_v = \sum_{v \text{ on rail } i} x_v.$$

For an edge $e \in \mathcal{H}_i$ with ordered endpoint types s, t , continue to write D_e for its interpreting matching in $\mathfrak{R}$; for the label edge, D_e means the underlying relation $D_{\mathbb{B}_i^-, \mathbb{B}_i^+}$. Put

$$\mathcal{E}_i := \sum_{\substack{e \in \mathcal{H}_i \\ e: s \rightarrow t}} (\lambda_{\text{rel}}(D_e; \mathbf{x}) + e_2(V_s) + e_2(V_t)) + d \sum_{\{s, t\} \in \binom{\mathcal{T}_i}{2}} e_1(V_s) e_1(V_t).$$

For every ordered pair $(i, j) \in [5]^2$ with $i \neq j$, define the filler factor P_{ij} by

$$P_{ij} := \prod_{h \in [5] \setminus \{i, j\}} \rho_h.$$

The complete unscaled weighted tableau polynomial is

$$\Phi_\beta(\mathfrak{R}, \mathbf{x}) := \sum_{\substack{i, j \in [5] \\ i \neq j}} P_{ij} \mathcal{E}_i + \lambda_{\text{rel}}(\mathbf{M}; \mathbf{x}). \quad (4.1)$$

The factor P_{ij} supplies the total weights of the three rails other than i and j , while $\lambda_{\text{rel}}(\mathbf{M}; \mathbf{x})$ is the contribution of the partial five-matching. Although the coefficients in [\(4.1\)](#) are independent of β , the domain of legal realizations depends on β through the tests in [\(Q5\)](#).

Define the supremal semantic value by

$$\Phi_\beta^* := \sup \left\{ \Phi_\beta(\mathfrak{R}, \mathbf{x}) : \begin{array}{l} \mathfrak{R} \text{ is a finite legal realization of } (\mathcal{H}, \mathbf{M}) \text{ for } \beta, \\ \mathbf{x} \text{ is a probability weighting on } V(\mathfrak{R}) \end{array} \right\}.$$

Define the baseline constant $z_0 := 2d/625$. It is an integer because 625 divides d , and it is independent of β because d was fixed before the input.

The main result of this section is the following exact dichotomy.

Theorem 4.1. *For every input β , every finite legal realization $\mathfrak{R}$ of $(\mathcal{H}, \mathbf{M})$ for β , and every nonnegative weight vector $\mathbf{x}$ of total weight ρ such that every coordinate of every tuple of $\mathbf{M}$ lies in a grounded component of its rail, one has $\Phi_\beta(\mathfrak{R}, \mathbf{x}) \leq z_0 \rho^5$. Moreover, the supremal semantic value satisfies the following two conclusions.*

- (i) $\mathbf{U}(\beta)$ does not halt if and only if $\Phi_\beta^* = z_0 = 2d/625$.
- (ii) $\mathbf{U}(\beta)$ halts if and only if $\Phi_\beta^* > z_0$.

To prove the homogeneous estimate, we first decompose the contribution from each rail into a balanced term and a defect term. We then bound the five-matching contribution in terms of these defects and optimize over the five total rail weights. All weights below are nonnegative.

We begin with the contribution of a single partial matching. Let X and Y be disjoint finite vertex classes, and fix a nonnegative weight vector $\mathbf{x} = (x_v)_{v \in X \sqcup Y}$. For a partial matching $D \subseteq X \times Y$, define its *matching defect* by

$$\text{Def}(D) := \sum_{u \in X \setminus \text{dom } D} x_u^2 + \sum_{v \in Y \setminus \text{im } D} x_v^2 + \sum_{uv \in D} (x_u - x_v)^2.$$

Equivalently, attach a private zero-weight leaf to every vertex in $X \setminus \text{dom } D$ and $Y \setminus \text{im } D$. Then $\text{Def}(D)$ is the sum of the squared weight differences along the matched pairs and these boundary edges. The leaves are used only in this estimate and are not part of the typed matching system. In [Figure 9](#), the operation

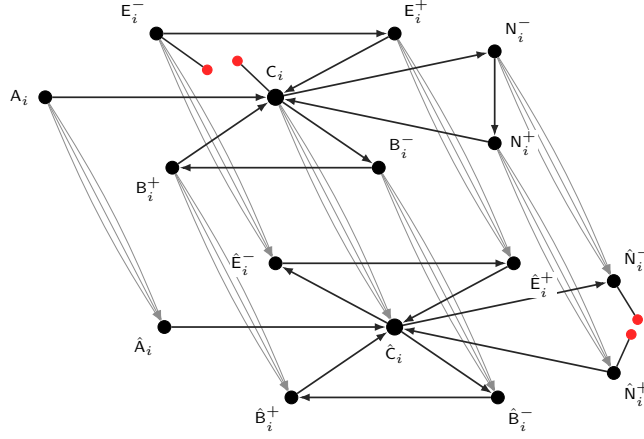


Figure 9: A one-vertex-per-type realization patterned on Figure 4, after deleting two relation instances, one in each layer. Each black point carrying a type label s represents an actual vertex of V_s . The omitted upper instance belongs to D_e for $e = (c_i, E_i^-)$, while the omitted lower instance belongs to $D_{e'}$ for $e' = (\hat{N}_i^-, \hat{N}_i^+)$. Their four exposed endpoints are joined by black edges to distinct formal leaves of weight zero, shown as smaller solid red circles. Each added edge at a vertex w contributes $(x_w - 0)^2 = x_w^2$. The type edges e and e' remain present in $\mathcal{H}_i$; only the two displayed relation instances are missing.

is superimposed on a one-vertex-per-type realization patterned on Figure 4: one realized relation edge in each layer is deleted and the resulting formal leaves are added.

Thus no vertex is added to, and no edge is deleted from, the type multidigraph in Figure 4. In Figure 9, one instance of each of D_e and $D_{e'}$ is deleted, and these two choices are independent. Their four endpoints are unmatched in the corresponding relations, so each receives a private formal leaf. If an actual vertex is unmatched for several incident type edges, it receives one distinct leaf for each pair (v, e) .

For the label edge on rail i , use the relations $D_{B_i^-, B_i^+}^{[a]}$ and $D_{B_i^-, B_i^+}$ defined above. In a legal realization the color relations are pairwise disjoint, and the defect uses $D_{B_i^-, B_i^+}$. Thus $\lambda_{\text{rel}}(D_{B_i^-, B_i^+}; \mathbf{x}) = \sum_a \lambda_{\text{rel}}(D_{B_i^-, B_i^+}^{[a]}; \mathbf{x})$, while the two e_2 terms and the matching defect occur only once for the label edge. Expanding squares gives

$$\lambda_{\text{rel}}(D; \mathbf{x}) + e_2(X) + e_2(Y) = \frac{1}{2}(e_1(X)^2 + e_1(Y)^2) - \frac{1}{2} \text{Def}(D).$$

For each rail $i \in [5]$, define its total defect by

$$\text{Def}_i := \frac{1}{2} \sum_{e \in \mathcal{H}_i} \text{Def}(D_e).$$

Every type has degree d in $\mathcal{H}_i$, with the label edge counted once. Summing the preceding identity therefore gives

$$\sum_{\substack{e \in \mathcal{H}_i \\ e: S \rightarrow T}} (\lambda_{\text{rel}}(D_e; \mathbf{x}) + e_2(V_S) + e_2(V_T)) = \frac{d}{2} \sum_{S \in \mathcal{T}_i} e_1(V_S)^2 - \text{Def}_i.$$

Adding the complete cross terms in the definition of $\mathcal{E}_i$ now completes the square and gives the defect decomposition

$$\mathcal{E}_i = \frac{d}{2} \left(\sum_{S \in \mathcal{T}_i} e_1(V_S) \right)^2 - \text{Def}_i = \frac{d}{2} \rho_i^2 - \text{Def}_i.$$

Fix a rail i . The subgraph of the auxiliary multigraph induced by $\bigcup_{S \in \mathcal{T}_i} V_S$ has one edge labeled e for every $(u, v) \in D_e$ with $e \in \mathcal{H}_i$. If the same ordered vertex pair belongs to both D_e and $D_{e'}$ for two parallel edges e, e' of $\mathcal{H}_i$, both labeled edges are retained and counted in the energy. For a relation component C on rail i , let ∂C be the set of pairs (v, e) such that $v \in C$, the edge $e \in \mathcal{H}_i$ is incident with the type of v , and no edge of D_e contains v . For the estimates below, attach to v a private formal leaf

$\partial_{v,e}$ of weight zero for every $(v,e) \in \partial C$. These leaves are not vertices of $\mathfrak{R}$ and do not alter its relation components. Define

$$\mu_C := \sum_{v \in C} x_v \quad \text{and} \quad \text{Def}_C := \sum_{\substack{e \in \mathcal{H}_i \\ u,v \in C, (u,v) \in D_e}} (x_u - x_v)^2 + \sum_{(v,e) \in \partial C} x_v^2.$$

Here the first sum is over the relation edges, with parallel edges of $\mathcal{H}_i$ counted separately. Then $\text{Def}_i = \frac{1}{2} \sum_C \text{Def}_C$, where the sum is over the relation components on rail i . A relation component is grounded precisely when $\partial C \neq \emptyset$.

The boundary energy controls the largest vertex weight in a grounded component.

Lemma 4.2. *If C is a finite grounded relation component, $x_C^{\max} := \max_{v \in C} x_v$, and $p \geq 4$, then*

$$(x_C^{\max})^3 \leq \frac{9}{4} \mu_C \text{Def}_C, \quad (4.2)$$

and

$$\sum_{v \in C} x_v^p \leq \frac{9}{4} \mu_C^{p-2} \text{Def}_C.$$

Proof. If $x_C^{\max} = 0$, both conclusions are immediate. Assume that $x_C^{\max} > 0$. Choose a vertex $u \in C$ of weight $x_C^{\max}$ and a simple path in C from u to a vertex v for which $(v,e) \in \partial C$ for some edge e . Append the corresponding private zero-weight leaf $\partial_{v,e}$, and write the weights along this path as $a_0 = x_C^{\max}, a_1, \dots, a_\ell, a_{\ell+1} = 0$. Set $b_i = \min_{0 \leq j \leq i} a_j$. Then $b_0 = x_C^{\max}$, the sequence (b_i) is nonincreasing, and

$$\sum_{i=0}^{\ell+1} b_i \leq \sum_{i=0}^{\ell+1} a_i \leq \mu_C, \quad \text{and} \quad \sum_{i=0}^{\ell} (b_i - b_{i+1})^2 \leq \sum_{i=0}^{\ell} (a_i - a_{i+1})^2 \leq \text{Def}_C.$$

The first inequality is immediate. For the second, a nonzero drop of the running minimum at step $i+1$ means $b_{i+1} = a_{i+1} < b_i \leq a_i$, so that $b_i - b_{i+1} \leq a_i - a_{i+1}$; otherwise its contribution is zero.

Linearly interpolate b_i on the unit intervals. Denote the resulting nonincreasing piecewise-linear function by b . By change of variables along its decreasing pieces, we have

$$\int_0^{\ell+1} \sqrt{b(t)} |b'(t)| dt = \int_0^{x_C^{\max}} \sqrt{y} dy = \frac{2}{3} (x_C^{\max})^{3/2}.$$

Moreover, we have

$$\int_0^{\ell+1} b(t) dt = \frac{1}{2} \sum_{i=0}^{\ell} (b_i + b_{i+1}) \leq \mu_C, \quad \text{and} \quad \int_0^{\ell+1} |b'(t)|^2 dt = \sum_{i=0}^{\ell} (b_i - b_{i+1})^2 \leq \text{Def}_C.$$

Cauchy-Schwarz now gives

$$\frac{4}{9} (x_C^{\max})^3 \leq \left(\int b \right) \left(\int |b'|^2 \right) \leq \mu_C \text{Def}_C.$$

This is (4.2). Finally, we have

$$\sum_{v \in C} x_v^p \leq \mu_C (x_C^{\max})^{p-1} = \mu_C (x_C^{\max})^{p-4} (x_C^{\max})^3 \leq \frac{9}{4} \mu_C^{p-2} \text{Def}_C.$$

The first inequality follows from $x_v^{p-1} \leq (x_C^{\max})^{p-1}$ for every $v \in C$. For the second inequality, use (4.2) together with $(x_C^{\max})^{p-4} \leq \mu_C^{p-4}$, which follows from $x_C^{\max} \leq \mu_C$ and $p \geq 4$. $\blacksquare$

We now apply the boundary-moment estimate across the five rails. Write $\boldsymbol{\rho} := (\rho_1, \dots, \rho_5)$ and $\rho := \rho_1 + \dots + \rho_5$. The estimate below needs every coordinate of an $\mathbf{M}$ -tuple to lie in a grounded component; knowing this for only one coordinate would not permit the fivefold Hölder estimate. This is exactly the coordinatewise conclusion of Lemma 3.5(3) in the nonhalting case.

Lemma 4.3. *Assume that every coordinate of every tuple of $\mathbf{M}$ lies in a grounded component of its rail. Then $\lambda_{\text{rel}}(\mathbf{M}; \mathbf{x}) \leq (9/10) \sum_{i \neq j} P_{ij} \text{Def}_i$.*

Proof. Let G_i be the union of the grounded components in rail i . Applying Lemma 4.2 with $p = 5$ component by component gives

$$\sum_{v \in G_i} x_v^5 \leq \frac{9}{4} \sum_C \mu_C^3 \text{Def}_C \leq \frac{9}{2} \rho_i^3 \text{Def}_i.$$

By Hölder's inequality and coordinate-injectivity of $\mathbf{M}$, we have

$$\lambda_{\text{rel}}(\mathbf{M}; \mathbf{x}) \leq \prod_{i=1}^5 \left(\sum_{v \in G_i} x_v^5 \right)^{1/5} \leq \frac{9}{2} \left(\prod_{i=1}^5 \rho_i^3 \text{Def}_i \right)^{1/5}.$$

Fix the 5-cycle $\sigma = (1\ 2\ 3\ 4\ 5)$. For this cycle, each ρ_h is omitted from exactly two of the five factors $P_{i,\sigma(i)}$ and hence occurs in exactly three of them. Therefore $\prod_i P_{i,\sigma(i)} = \prod_i \rho_i^3$, and AM–GM gives

$$\left(\prod_i \rho_i^3 \text{Def}_i \right)^{1/5} = \left(\prod_i P_{i,\sigma(i)} \text{Def}_i \right)^{1/5} \leq \frac{1}{5} \sum_i P_{i,\sigma(i)} \text{Def}_i.$$

Together with the preceding Hölder bound, this proves the stronger bound with the sum over the five cyclic pairs. Enlarging that sum to all ordered pairs gives the bound in the lemma. ■

We finally combine Lemma 4.3 with the rail-defect identity and optimize over the five rail weights. Equation (4.1) gives

$$\Phi_\beta(\mathfrak{R}, \mathbf{x}) = \frac{d}{2} \sum_{i \neq j} P_{ij} \rho_i^2 - \sum_{i \neq j} P_{ij} \text{Def}_i + \lambda_{\text{rel}}(\mathbf{M}; \mathbf{x}).$$

The remaining dependence on $\boldsymbol{\rho}$ is controlled by the following elementary inequality.

Lemma 4.4. *For a nonnegative vector $\boldsymbol{\rho} = (\rho_1, \dots, \rho_5)$ of coordinate sum ρ , we have*

$$\rho e_4(\boldsymbol{\rho}) - 5e_5(\boldsymbol{\rho}) \leq \frac{4}{625} \rho^5.$$

For $\rho > 0$, equality holds if and only if $\rho_1 = \dots = \rho_5 = \rho/5$.

Proof. By homogeneity take $\rho = 1$. Fix three variables z_1, z_2, z_3 and replace the remaining two, x, y , by their average while keeping $x+y$ fixed. If $q_2 = z_1 z_2 + z_1 z_3 + z_2 z_3$ and $q_3 = z_1 z_2 z_3$, then $e_4 = xyq_2 + (x+y)q_3$ and $e_5 = xyq_3$. Consequently the part depending on xy has coefficient $q_2 - 5q_3$. Put $s = z_1 + z_2 + z_3$. Two applications of AM–GM give $s \geq 3q_3^{1/3}$ and $q_2 = z_1 z_2 + z_1 z_3 + z_2 z_3 \geq 3q_3^{2/3}$, and therefore $sq_2 \geq 9q_3$. Since $s \leq 1$, we obtain $q_2 \geq 9q_3 \geq 5q_3$; thus pairwise averaging cannot decrease $e_4 - 5e_5$. Repeatedly average a largest and a smallest coordinate. Each step decreases the variance about $1/5$ by half the square of the current range. If the variance did not tend to zero, then the range, and hence the decrease, would stay bounded away from zero, which is impossible. The resulting vectors therefore converge to the uniform point. Continuity shows that the maximum occurs there, with value $5(1/5)^4 - 5(1/5)^5 = 4/625$. If the three fixed complementary variables contain at least two positive entries, the coefficient of xy is strictly positive, so averaging unequal x, y strictly increases the expression. A boundary point has either at most three positive coordinates, when the expression is zero, or exactly four, when it is their product and is at most $4^{-4} < 4/625$. Hence equality is unique. ■

Expanding $\rho e_4(\boldsymbol{\rho})$ gives $\sum_{i \neq j} \rho_i^2 P_{ij} = \rho e_4(\boldsymbol{\rho}) - 5e_5(\boldsymbol{\rho})$. Indeed, the terms in which the extra factor is the unique variable missing from an e_4 monomial contribute $5e_5$, and all remaining terms are indexed by an ordered pair (i, j) .

We can now prove the theorem stated at the beginning of the section.

Proof of Theorem 4.1. We first prove the homogeneous estimate. Fix a finite legal five-rail realization $\mathfrak{R}$ and a nonnegative weight vector $\mathbf{x}$ of total weight ρ , and suppose that every coordinate of every tuple of $\mathbf{M}$ lies in a grounded component of its rail. The defect decomposition and Lemma 4.3 imply

$$\Phi_\beta(\mathfrak{R}, \mathbf{x}) = \frac{d}{2} \sum_{i \neq j} P_{ij} \rho_i^2 - \sum_{i \neq j} P_{ij} \text{Def}_i + \lambda_{\text{rel}}(\mathbf{M}; \mathbf{x}) \leq \frac{d}{2} (\rho e_4(\boldsymbol{\rho}) - 5e_5(\boldsymbol{\rho})) - \frac{1}{10} \sum_{i \neq j} P_{ij} \text{Def}_i \leq \frac{2d}{625} \rho^5.$$

If $U(\beta)$ does not halt, Lemma 3.5(3) supplies the grounded-coordinate hypothesis for every legal realization. For total weight one, the preceding estimate is therefore at most $2d/625$. Equality is attained by the baseline structure in Lemma 3.5(1). Take isomorphic copies on the five rails, give every vertex within a rail the same positive weight, give each rail total weight $1/5$, and put $M = \emptyset$. Every matching is perfect between equal-weight copies, so all Def_i vanish. The upper bound is then attained when the five rail weights are equal.

If $U(\beta)$ halts, use the periodic reset tableau in Lemma 3.5(2), again with isomorphic equal-weight rails, each of total weight $1/5$. All matchings are perfect between equal-weight copies, so $\text{Def}_i = 0$ for every $i \in [5]$, while the selected tuple of M has strictly positive monomial weight. Therefore $\Phi_\beta(\mathfrak{R}, \mathbf{x}) = 2d/625 + \lambda_{\text{rel}}(M; \mathbf{x}) > 2d/625$. This proves both implications. ■

5 From weighted rails to five-set records

The weighted polynomial constructed in Section 4 is not yet presented as the edge polynomial of a 5-graph. We encode each degree-five term by a tagged five-set whose tag records the term it represents.

5.1 Five-sets with record tags

Fix the five-rail physical type set $\mathcal{T}$ and the d -regular rail signature constructed in Section 3. The sets $\mathcal{T}_1, \dots, \mathcal{T}_5$ of types on the five rails are disjoint.

A *record tag* specifies completely how a five-set is interpreted and counted. Its *kind* is main, reserve, correction, or bonus, but the kind alone is not the tag. Except for the bonus tag, a tag also contains the parameters specified below. Two tags that differ in any parameter are distinct, even if they can be attached to the same underlying five-set.

To motivate the parameters, consider the encoding of one ordinary relation instance $D_e(x, y)$ on rail i . Choose one of the four indices $j \in [5] \setminus \{i\}$ as the omitted rail. To extend the two endpoints x, y to a five-set, use one additional vertex z_h from each of the three rails $h \notin \{i, j\}$. We call these three vertices *fillers*; they occupy the remaining positions in the five-set but carry no part of the projected relation instance. The tag records the edge e , the omitted rail j , and the type of each filler. Once these data are fixed, z_h ranges over every vertex of the recorded type. The tag and the vertex types recover x, y as the two relation endpoints, while the fillers are ignored by the projection.

For every ordered pair $i \neq j$, recall that $P_{ij} = \prod_{h \in [5] \setminus \{i, j\}} \rho_h$. We now expand every term of (4.1) into unit tagged records. There are four kinds.

Main-relation tags. Let $e = (x, y)$ or $(x, y; \ell)$ be an ordinary edge on rail i , including an edge on the balancing layer or a bridge edge. A main-relation tag records its kind, the edge e , an omitted rail $j \neq i$, and one filler type $z_h \in \mathcal{T}_h$ for each $h \notin \{i, j\}$. It is attached to every five-set $\{x, y\} \cup \{z_h : h \in [5] \setminus \{i, j\}\}$ with $(x, y) \in D_e$ and $z_h \in V_{z_h}$ for every $h \in [5] \setminus \{i, j\}$. The endpoint roles are determined by the two distinct types x, y . For the label edge $e = (\mathbf{b}_i^-, \mathbf{b}_i^+)$ on rail i , a main-relation tag additionally records a color $a \in \Sigma_{\text{CA}}$ and uses $(x, y) \in D_{\mathbf{b}_i^-, \mathbf{b}_i^+}^{[a]}$. Thus there is one main-relation tag for each color, and the tags for all color relations project to the same underlying relation while retaining their colors. Accordingly, a main-relation tag has one of the following two forms:

$$\begin{aligned} &(\text{main}, e, j, (z_h)_{h \in [5] \setminus \{i, j\}}) && \text{for an ordinary relation,} \\ &(\text{main}, e, a, j, (z_h)_{h \in [5] \setminus \{i, j\}}) && \text{for a label relation.} \end{aligned}$$

Reserve tags. A reserve tag records its kind, a directed edge e on rail i , a choice $\varepsilon \in \{\text{src}, \text{tgt}\}$ of the source or target type of e , an omitted rail $j \neq i$, and one filler type on each rail $h \notin \{i, j\}$. Its complete form is

$$(\text{reserve}, e, \varepsilon, j, (z_h)_{h \in [5] \setminus \{i, j\}}).$$

The label edge is counted once, without recording a cellular-automaton color. If x is the endpoint type selected by ε , this tag is attached to the five-sets formed from an unordered pair of distinct vertices of V_x and three vertices of the recorded filler types. There is one reserve tag for each endpoint of each edge. These records project to nothing.

Correction tags. A correction tag records its kind, a rail $i \in [5]$, a pair $\{x, y\} \in \binom{\mathcal{T}_i}{2}$, an omitted rail $j \neq i$, one filler type $z_h \in \mathcal{T}_h$ for each $h \notin \{i, j\}$, and a copy index $s \in [d]$. It is attached to every five-set $\{x, y\} \cup \{z_h : h \in [5] \setminus \{i, j\}\}$ with $x \in V_x$, $y \in V_y$, and $z_h \in V_{z_h}$ for every $h \in [5] \setminus \{i, j\}$. Its complete form is

$$(\text{correction}, i, \{x, y\}, j, (z_h)_{h \in [5] \setminus \{i, j\}}, s).$$

These records project to nothing.

Bonus tag. The bonus tag is the single symbol **bonus**, with no additional parameters. It is attached to the five coordinates of every M-tuple. These coordinates lie on distinct rails, so their types recover their order from the underlying five-set. The bonus records project to the relation M.

Take the disjoint union of these four tag families, put c_{tag} equal to its cardinality, and fix a bijection from this union to $[c_{\text{tag}}]$. We identify each tag with its image under this bijection. Thus $\alpha \in [c_{\text{tag}}]$ denotes a complete tag, not merely one of the four kinds. The set of tags is finite and depends only on the fixed machine and rail signature, not on the input β . Every tag prescribes one fixed physical type profile.

A *partially typed tagged 5-graph* on vertex set V is a tuple $\mathfrak{X} = (V, \vartheta; (\mathcal{R}_\alpha)_{\alpha \in [c_{\text{tag}}]})$, where $V(\mathfrak{X}) := V$ is its vertex set, $\vartheta: V \rightarrow \mathcal{T}$ is a partial type assignment, and $\mathcal{R}_\alpha \subseteq \binom{V}{5}$ is the ordinary 5-graph carrying tag α . Membership under different tags is independent, so $A \in \mathcal{R}_\alpha \cap \mathcal{R}_\gamma$ is allowed for $\alpha \neq \gamma$. We call each pair (α, A) with $A \in \mathcal{R}_\alpha$ a *raw record*; it is a tagged five-set before type checking and projection. For a nonnegative weight vector $\mathbf{x}$ put

$$\lambda_\alpha(\mathfrak{X}; \mathbf{x}) := \lambda(\mathcal{R}_\alpha; \mathbf{x}) = \sum_{A \in \mathcal{R}_\alpha} \prod_{v \in A} x_v, \quad \text{and} \quad \lambda_{\text{rec}}(\mathfrak{X}; \mathbf{x}) := \sum_{\alpha \in [c_{\text{tag}}]} \lambda_\alpha(\mathfrak{X}; \mathbf{x}).$$

Thus λ_{rec} is a sum of squarefree degree-five monomials with unit coefficient for each tagged occurrence (α, A) . If the same underlying five-set occurs with several tags, its monomial is counted once for each tag. The system is *fully typed* when $\text{dom } \vartheta = V$.

In the partially typed system used by the uncolored construction, the projection of a raw record (α, A) is defined as follows.

- If some vertex of A is untyped, or if the type profile of A differs from the profile prescribed by α , then (α, A) projects to nothing.
- If the type profile agrees and α is a main-relation tag, then (α, A) projects to the corresponding binary-relation instance. For a color- a label pair, the projection retains the color a .
- If α is the bonus tag, then (α, A) projects to the corresponding M-tuple.
- Reserve and correction tags project to nothing, even when their type profiles agree.

The *projected matching system* of $\mathfrak{X}$ is the candidate realization on $\text{dom } \vartheta$ whose vertex class of type u is $\vartheta^{-1}(u)$. For every ordinary edge e of $\mathcal{H}$, its relation D_e consists of the ordered pairs obtained by projecting all correctly typed main records tagged by e . For the label edge on each rail $i \in [5]$ and each $a \in \Sigma_{CA}$, the relation $D_{\mathbf{B}_i^-, \mathbf{B}_i^+}^{[a]}$ consists of the ordered pairs obtained from the correctly typed color- a main records, and $D_{\mathbf{B}_i^-, \mathbf{B}_i^+}$ is their union over a . Its five-ary relation M consists of the tuples obtained by projecting the correctly typed bonus records. Repeated projections are merged because these are relations rather than multisets.

A partially typed tagged 5-graph $\mathfrak{X}$ is *record-admissible* for the input β if the following two conditions hold.

- Every raw record whose five vertices are typed has the type profile prescribed by its tag.
- The projected matching system is a legal realization of $(\mathcal{H}, \mathbf{M})$ for β in the sense of [Section 3](#). More explicitly, every ordinary edge-indexed binary relation is a partial matching; for every $i \in [5]$, the relation $D_{\mathbf{B}_i^-, \mathbf{B}_i^+} = \bigcup_{a \in \Sigma_{CA}} D_{\mathbf{B}_i^-, \mathbf{B}_i^+}^{[a]}$ is a partial matching and its color relations are pairwise disjoint; M is a partial five-matching; and the tableau conditions (Q1)–(Q5) hold.

Equivalently, the second condition says that the projected matching system contains no copy of any member of $\mathcal{P}_\beta^{\text{phys}}$. It depends only on the projected system and therefore does not inspect which filler triple produced a relation instance or how many raw records produced it. Untyped records project to nothing and hence affect neither condition. Within this section, “admissible” always means record-admissible.

We write RecAdm_β for this record-admissibility predicate. That is, for a partially typed tagged 5-graph $\mathfrak{X}$, the statement $\text{RecAdm}_\beta(\mathfrak{X})$ means that $\mathfrak{X}$ is record-admissible for the input β . Retain the notation Φ_β^* from Section 4, and define

$$\lambda_{\text{rec},\beta}^* := \sup \left\{ \lambda_{\text{rec}}(\mathfrak{X}; \mathbf{x}) : \begin{array}{l} \mathfrak{X} \text{ is a finite fully typed tagged 5-graph such that } \text{RecAdm}_\beta(\mathfrak{X}), \\ \mathbf{x} \text{ is a weighting on } V(\mathfrak{X}) \end{array} \right\}.$$

The main result of this section is that the tagged-record encoding preserves the semantic optimum exactly.

Lemma 5.1. *For every input β , we have $\lambda_{\text{rec},\beta}^* = \Phi_\beta^*$.*

We prove the lemma through three auxiliary results.

We begin with the polynomial identity.

For a finite legal realization $\mathfrak{R}$ of $(\mathcal{H}, \mathbf{M})$ for β , let $\mathfrak{X}_{\mathfrak{R}}$ be the fully typed tagged 5-graph consisting of all main, reserve, correction, and bonus records prescribed by the binary relations, $\mathbf{M}$ -tuples, and vertex classes of $\mathfrak{R}$. Thus, once $\mathfrak{R}$ is fixed, every tagged five-set allowed by one of the four tag definitions is included; none is selected arbitrarily. We call $\mathfrak{X}_{\mathfrak{R}}$ the *saturated tagged-record encoding* of $\mathfrak{R}$.

Lemma 5.2. *Let $\mathfrak{R}$ be a finite legal realization of $(\mathcal{H}, \mathbf{M})$ for β , and let $\mathfrak{X}_{\mathfrak{R}}$ be its saturated tagged-record encoding. Then, for every nonnegative weight vector $\mathbf{x}$, we have $\lambda_{\text{rec}}(\mathfrak{X}_{\mathfrak{R}}; \mathbf{x}) = \Phi_\beta(\mathfrak{R}, \mathbf{x})$.*

Proof. Fix an ordinary edge e on rail i and an index $j \neq i$. Summing the corresponding main records over the filler vertices and then over the filler types gives

$$\lambda_{\text{rel}}(D_e; \mathbf{x}) \prod_{h \notin \{i,j\}} \sum_{v \text{ on rail } h} x_v = P_{ij} \lambda_{\text{rel}}(D_e; \mathbf{x}).$$

For the label edge, the same calculation applies to $D_{\mathbf{B}_i^-, \mathbf{B}_i^+}^{[a]}$ for each $a \in \Sigma_{\text{CA}}$. Summing over a gives the contribution of the underlying label relation. The two reserve families associated with the endpoints of an edge similarly give $P_{ij}e_2(V_x)$ and $P_{ij}e_2(V_y)$. For fixed i, j and $\{x, y\} \in \binom{T_i}{2}$, summing the correction records over all filler profiles and over $s \in [d]$ gives $dP_{ij}e_1(V_x)e_1(V_y)$. Summing over all directed edges of $\mathcal{H}_i$ and all unordered pairs of physical types on rail i , and then over all ordered pairs $i \neq j$, gives $\sum_{i \neq j} P_{ij} \mathcal{E}_i$. The bonus tag gives $\lambda_{\text{rel}}(\mathbf{M}; \mathbf{x})$.

The two relation endpoints occupy distinct physical types, and the three fillers occupy three other rails. Thus all five vertices are distinct and their types recover their roles. The same observation recovers the five coordinates of $\mathbf{M}$. Reserve and correction records project to nothing. Hence every main or bonus projection is unique. $\blacksquare$

For a fully typed tagged 5-graph $\mathfrak{X}$, its *filler saturation* $\mathfrak{X}^+$ is obtained by adding the following records. For every relation instance already present in the projected matching system, add every compatible main record, over all ordered pairs (i, j) and all filler triples of the types prescribed by its tag. Also add every reserve and correction record allowed by the prescribed tag profiles. No bonus record is added, because each projected $\mathbf{M}$ -tuple already determines its unique typed five-set.

Lemma 5.3. *Let $\mathfrak{X}$ be a finite fully typed record-admissible tagged 5-graph. Then its filler saturation $\mathfrak{X}^+$ has the following properties.*

- (i) *The tagged 5-graph $\mathfrak{X}^+$ is fully typed and record-admissible.*
- (ii) *The systems $\mathfrak{X}$ and $\mathfrak{X}^+$ have the same projected matching system.*
- (iii) *For every nonnegative weighting $\mathbf{x}$ on $V(\mathfrak{X})$, one has $\lambda_{\text{rec}}(\mathfrak{X}; \mathbf{x}) \leq \lambda_{\text{rec}}(\mathfrak{X}^+; \mathbf{x})$.*

Proof. All added records use existing typed vertices and have the profiles prescribed by their tags, so full typing is preserved. A new main record projects to a relation instance already present in the projected matching system, and its three filler vertices do not affect that projection. Reserve and correction records project to nothing. Thus the projected matching system is unchanged, and hence remains legal; consequently $\mathfrak{X}^+$ is record-admissible. Finally, filler saturation only adds record monomials with nonnegative coefficients, which proves the last assertion. $\blacksquare$

Thus $\mathfrak{X}^+$ is the saturated tagged-record encoding of the common projected matching system, and its record value is given by Lemma 5.2. Filler saturation is an evaluation operation, not a record-admissibility axiom; requiring saturation would destroy heredity under deleting records.

5.2 Recovering the tableau structure

Let Q be a concrete typed local obstruction, and enlarge it by adjoining a finite set F of new formal vertices disjoint from $V(Q)$. A *relative identification over Q* , also called a *frozen-core identification*, is a type-respecting equivalence relation $\sim$ on $V(Q) \sqcup F$ that identifies no two distinct vertices of Q , respects every displayed inequality and type declaration, and identifies no two positions belonging to the same raw five-set. It may identify new vertices with one another or with an old vertex of the same type. Its equivalence classes form the distinct vertices of the resulting local obstruction. In particular, every equivalence class contains at most one vertex of Q .

Here $V(Q)$ is the frozen core. Its vertices remain distinct, whereas the new vertices in F may coincide whenever the preceding conditions allow it. The two expansions below have this form. Replacing a logical relation by its three-edge relation path adds intermediate vertices, and replacing a physical relation instance by a raw five-set witness adds filler vertices.

If the vertices of Q are mapped injectively and the new vertices are assigned in a type-respecting way, with the five positions of each raw record assigned distinct vertices, equality of their images defines a unique frozen-core identification. Conversely, a copy of the resulting local obstruction recovers such an assignment. Freezing $V(Q)$ ensures that an inequality already certified in Q cannot disappear during either expansion.

For example, start with a concrete local obstruction obtained as an admissible type-preserving homomorphic image of the commutation violation:

$$D_{\text{hor}}(x, x_E), \quad D_{\text{time}}(x, x_N), \quad D_{\text{time}}(x_E, y), \quad D_{\text{hor}}(x_N, z), \quad \text{where } y \neq z.$$

First choose an admissible type-preserving homomorphic image of the displayed five-vertex commutation pattern and regard its vertices as the old names. Expanding the four logical relation instances into their three-step relation paths introduces two intermediate names for each path. Names of the same intermediate type may coincide across different paths if the partial-matching constraints allow it, but no two old names may be identified at this later stage; in particular, the inequality $y \neq z$ survives. Replacing the resulting physical relation instances by raw five-set witnesses introduces the filler names. These fillers may again coincide across different records, or with an old name of the same type, but the five positions within each record remain distinct. The two successive frozen-core identifications therefore record every coincidence that may occur in an actual raw system without losing the original commutation violation.

A *raw local obstruction* is a tuple $Q = (V(Q), \vartheta_Q, \mathcal{R}_Q)$, where $V(Q)$ is a finite vertex set, $\vartheta_Q: V(Q) \rightarrow \mathcal{T}$ is a partial physical-type assignment, and $\mathcal{R}_Q \subseteq [c_{\text{tag}}] \times \binom{V(Q)}{5}$ is a finite set of raw-record requirements. A *requirement* of Q is either a type requirement $\vartheta_Q(w) = U$, with $w \in \text{dom } \vartheta_Q$, or a raw-record requirement $(\alpha, A) \in \mathcal{R}_Q$. We say that $\mathfrak{X}$ *contains a copy* of Q if there is an injective map $f: V(Q) \rightarrow V(\mathfrak{X})$ such that $\vartheta_{\mathfrak{X}}(f(w)) = \vartheta_Q(w)$ for every $w \in \text{dom } \vartheta_Q$ and $f(A) \in \mathcal{R}_{\alpha}(\mathfrak{X})$ for every $(\alpha, A) \in \mathcal{R}_Q$. Thus deleting a vertex, a type assignment, or a raw record cannot create a copy.

When a local obstruction is first written with names that may coincide, we enumerate its admissible type-preserving homomorphic images, omitting those that identify two positions of one five-set or assign two types to one name, and use copies of the resulting local obstructions. This is the preceding homomorphic-image construction with no frozen core. When new names are introduced while expanding an already concrete local obstruction, we instead use only frozen-core identifications. The kernel observation then turns every assignment extending a copy of the concrete obstruction into exactly one copy of an obstruction obtained by a relative identification.

The next lemma expands the tableau tests and gives the certificate bound required by the rooted construction.

Lemma 5.4. *Given β , one can compute a finite list $\mathcal{P}_{\beta}^{\text{raw}}$ of raw local obstructions and an integer b_{β} with the following properties.*

- (i) A partially typed tagged 5-graph is record-admissible precisely when it contains no copy of any member of $\mathcal{P}_\beta^{\text{raw}}$.
- (ii) Every $Q \in \mathcal{P}_\beta^{\text{raw}}$ satisfies $|V(Q)| \leq b_\beta$ and $|\text{dom } \vartheta_Q| + |\mathcal{R}_Q| \leq b_\beta$.
- (iii) Every $Q \in \mathcal{P}_\beta^{\text{raw}}$ satisfies $\text{dom } \vartheta_Q = V(Q)$, and for every $w \in V(Q)$ there is some $(\alpha, A) \in \mathcal{R}_Q$ with $w \in A$.
- (iv) Let $\mathfrak{X}$ be record-admissible, let $v \in V(\mathfrak{X}) \setminus \text{dom } \vartheta_{\mathfrak{X}}$, and fix $u \in \mathcal{T}$. Define $\mathfrak{X}'$ by $\vartheta_{\mathfrak{X}'} = \vartheta_{\mathfrak{X}} \cup \{(v, u)\}$ and $\mathcal{R}_\alpha(\mathfrak{X}') = \{A \in \mathcal{R}_\alpha(\mathfrak{X}) : v \notin A\}$ for every $\alpha \in [c_{\text{tag}}]$. Then $\mathfrak{X}'$ is record-admissible.

Proof. We construct $\mathcal{P}_\beta^{\text{raw}}$ in four stages.

- (i) *Logical cores.* We first pass from the schematic tests to concrete logical cores. Let $\mathcal{P}_\beta^{\text{log}}$ be the following finite list. For each of the schemas (Q1)–(Q5), enumerate every admissible type-preserving homomorphic image allowed by its required equalities, inequalities, and types. The resulting objects are concrete logical obstructions with distinct named positions. This preliminary image step has no frozen core and records every equality pattern that the logical names may have in an actual violation.
- (ii) *Physical relation paths.* We now carry out the first of two frozen-core expansions. Fix $Q \in \mathcal{P}_\beta^{\text{log}}$ and declare its vertices old. Expand every logical successor, label, or anchor relation along the fixed physical relation path from Section 3; the internal vertices of the relation paths are new names, while each M-tuple is left unchanged. Enumerate all local obstructions obtained by relative identifications over Q , rather than allowing identifications that merge two vertices of Q . Denote the resulting finite list of concrete physical-relation obstructions by $\mathcal{P}_\beta^{\text{path}}$. This is exactly the sublist of the earlier physical list $\mathcal{P}_\beta^{\text{phys}}$ that arises from (Q1)–(Q5), now with the logical core recorded explicitly. The matching-conflict members of $\mathcal{P}_\beta^{\text{phys}}$ are handled separately by the elementary obstructions below. Put $v_\beta := \max \left\{ 1, \max_{Q \in \mathcal{P}_\beta^{\text{path}}} |V(Q)| \right\}$. Let a_β be the maximum of 1 and, over all $Q \in \mathcal{P}_\beta^{\text{path}}$, the number of required physical relation instances and M-tuples in Q . A maximum over an empty list is understood to be zero.
- (iii) *Raw witnesses.* Each physical binary-relation instance has a finite set of raw witness schemes. To obtain one, choose its main-relation tag, place its two endpoints in the prescribed positions, and add three typed filler names on the other rails. A color- a label pair uses the tag carrying color a , while an M-tuple uses the bonus tag and its five coordinate names.

For the second expansion, fix a concrete $Q \in \mathcal{P}_\beta^{\text{path}}$ and declare all its vertices old. Choose one raw witness scheme independently for each required physical relation instance or M-tuple; its filler names are new. Enumerate all local obstructions obtained by relative identifications over Q . Distinct vertices of Q remain distinct, whereas filler names may coincide with one another or with a same-type vertex of Q , provided that every raw record still has five distinct vertices. Call the resulting list $\mathcal{P}_\beta^{\text{tab}}$. It is finite and effective because the relation paths, type set, tag set, and witness schemes are fixed.

- (iv) *Elementary obstructions.* Complete the list by the fixed elementary obstructions which express the following conditions.
 - (a) a raw record whose five vertices are all typed but whose type profile is not the one prescribed by its tag;
 - (b) two projected records violating the partial-matching property of an edge-indexed binary relation, including a conflict between two colors on the label edge (whether or not the two colored endpoint pairs coincide); and
 - (c) two projected bonus records whose identification kernel on their five coordinates is a nonempty proper subset of $[5]$.

More explicitly, for every ordinary edge e we use the two concrete cores $D_e(x, y), D_e(x, z)$ with $y \neq z$, and $D_e(x, y), D_e(z, y)$ with $x \neq z$. For the label edge $(\mathbf{b}^-, \mathbf{b}^+)$, the two required pairs may lie in any two color relations; if the colors differ, the case in which both endpoints coincide is included as well. For $\mathbf{M}$, for each $\emptyset \neq I \subsetneq [5]$ we include the concrete two-tuple core $\mathbf{M}(x_1, \dots, x_5)$ and $\mathbf{M}(y_1, \dots, y_5)$, where $x_i = y_i$ precisely for $i \in I$. Thus all thirty possible nontrivial overlap kernels, not merely the

generic one-coordinate overlap, are present. The wrong-profile obstructions in part (a) are already raw local obstructions and require no projection. Each projected elementary core in parts (b) and (c) is expanded to raw witnesses by the frozen-core rule used to define $\mathcal{P}_\beta^{\text{path}}$. In particular, the distinct targets or sources in a matching conflict can never be collapsed by its filler expansion.

There are finitely many such obstructions because the type and tag sets are fixed. Let $\mathcal{P}_\beta^{\text{raw}}$ be their union with $\mathcal{P}_\beta^{\text{tab}}$.

We record the asserted support invariant. In every schematic test (Q1)–(Q5), each logical name lies in a successor, label, anchor relation, or M-tuple. Expanding a relation path places each old logical name and each new intermediate name in a physical relation instance, while an M-name remains in its M-tuple. Replacing a physical relation instance by a raw witness places both endpoint names (or all five M-names) and every new filler name in a raw-record requirement. All these names receive their physical types.

In a wrong-profile obstruction, the displayed raw record itself contains and supports all five typed names. In a matching-conflict obstruction for an edge-indexed binary relation or for M, every core name lies in a projected relation instance or tuple. Expanding each projected requirement by a raw witness therefore supports every core name, and the added filler names lie in those same witnesses. Finally, these identifications preserve displayed types and cannot create an isolated equivalence class. Hence every name in every member of $\mathcal{P}_\beta^{\text{raw}}$ is typed and raw-record-supported.

We verify exactness in both directions. First suppose that a raw system contains a copy of a member of $\mathcal{P}_\beta^{\text{tab}}$. Projecting the raw witness records in this copy recovers every requirement of its concrete physical core $Q \in \mathcal{P}_\beta^{\text{path}}$. The frozen-core condition ensures that the restriction to $V(Q)$ is still injective. Consequently an inequality in the forbidden tableau obstruction cannot have vanished in the raw expansion, and the projected matching system contains a copy of the corresponding typed local obstruction. For a wrong-profile elementary obstruction, its displayed raw record directly witnesses a fully typed record whose type profile is not the one prescribed by its tag, with no projection involved. For a matching conflict involving an edge-indexed binary relation or M, projecting the chosen raw witnesses recovers the elementary conflict core, while the frozen-core condition preserves every required distinction among its old vertices.

Conversely, suppose that the projected matching system violates a tableau test. Stage (ii) supplies a copy of some $Q \in \mathcal{P}_\beta^{\text{path}}$. For every required relation instance or tuple in this copy, choose a raw record that projects to it. The images of the old vertices are distinct, so equality among the chosen intermediate and filler vertices defines a unique frozen-core identification. Since a raw record is a five-element set, this identification never merges two positions of one record. The chosen records therefore give a copy of a member of $\mathcal{P}_\beta^{\text{tab}}$.

The same frozen-core argument lifts every matching conflict involving an edge-indexed binary relation or M in the projected matching system to an elementary raw obstruction; a fully typed record whose type profile is not the one prescribed by its tag is already such an obstruction. Hence avoidance of $\mathcal{P}_\beta^{\text{raw}}$ is equivalent to record admissibility.

Replacing one physical relation instance introduces at most three new filler names and one raw-record requirement. After types are displayed, a tableau expansion therefore uses at most $v_\beta + 3a_\beta$ vertices and at most $v_\beta + 4a_\beta$ requirements. The elementary profile and matching-conflict obstructions use at most ten vertices and at most twelve requirements. Thus the explicit computable choice

$$b_\beta := \max\{12, v_\beta + 4a_\beta\} \quad (5.1)$$

has the required property. Identifying positions can only decrease both quantities.

Finally, let $\mathfrak{X}$, v , U , and $\mathfrak{X}'$ be as in part (4). If $\mathfrak{X}'$ were not record-admissible, part (1) would give a copy of some $Q \in \mathcal{P}_\beta^{\text{raw}}$. Deleting records cannot create such a copy, so it must use the new assignment $\vartheta_{\mathfrak{X}'}(v) = U$. Thus some $w \in \text{dom } \vartheta_Q$ is mapped to v . By part (3), there is an $(\alpha, A) \in \mathcal{R}_Q$ with $w \in A$. Its image would be a raw record of $\mathfrak{X}'$ containing v , contrary to the definition of $\mathcal{R}_\alpha(\mathfrak{X}')$. This proves part (4). $\blacksquare$

By Lemma 5.4, the predicate RecAdm_β is equivalent to avoiding the finite raw local-obstruction list $\mathcal{P}_\beta^{\text{raw}}$. The tag set and physical type set are fixed; the finite list of marked input tests, and hence the computable number b_β , carries the dependence on β .

The following lemma gives the two conversions between record-admissible tagged systems and legal five-rail realizations that will be used below.

Lemma 5.5. *For every input β , the following statements hold.*

- (i) *The projected matching system of every fully typed RecAdm_β system is a legal realization of $(\mathcal{H}, \mathbf{M})$ for β .*
- (ii) *Conversely, for every finite legal realization $\mathfrak{R}$ of $(\mathcal{H}, \mathbf{M})$ for β and every nonnegative weight vector $\mathbf{x}$, there is a finite fully typed record-admissible tagged 5-graph $\mathfrak{X}_\mathfrak{R}$ such that $\lambda_{\text{rec}}(\mathfrak{X}_\mathfrak{R}; \mathbf{x}) = \Phi_\beta(\mathfrak{R}, \mathbf{x})$. Any isolated vertices added to make type classes nonempty have weight zero.*

Proof. The endpoint types give a unique projection for every main relation instance. The matching-conflict obstructions make each edge-indexed binary relation functional and injective, so composition along a relation path gives the required logical partial map. Every forbidden logical obstruction has been expanded using all raw-witness and identification choices; hence a copy of it would contradict record admissibility. This proves the first assertion.

Conversely, fix a finite legal five-rail realization $\mathfrak{R}$ and a nonnegative weight vector $\mathbf{x}$. Let $\tilde{\mathfrak{R}}$ be obtained by adjoining to every empty physical type one isolated vertex, and extend $\mathbf{x}$ by giving every such vertex weight zero. Every name in every member of $\mathcal{P}_\beta^{\text{phys}}$ belongs to a required relation instance, so adjoining these isolated vertices cannot create a copy of a typed local obstruction. Thus $\tilde{\mathfrak{R}}$ is legal, every vertex class is nonempty, and $\Phi_\beta(\tilde{\mathfrak{R}}, \mathbf{x}) = \Phi_\beta(\mathfrak{R}, \mathbf{x})$.

Construct a fully typed raw seed $\mathfrak{X}_0$ as follows. For every ordinary relation instance $D_e(u, v)$ of $\tilde{\mathfrak{R}}$ on rail i , choose any $j \neq i$, any filler types $z_h \in \mathcal{T}_h$ for $h \notin \{i, j\}$, and vertices $z_h \in V_{z_h}(\tilde{\mathfrak{R}})$, and insert $\{u, v, z_h : h \notin \{i, j\}\}$ with the corresponding main-relation tag. For a color- a label pair $D_{\mathbf{b}_i^-, \mathbf{b}_i^+}^{[a]}(u, v)$, use the tag carrying its given color a . For every $\mathbf{M}$ -tuple $(u_1, \dots, u_5)$ insert $\{u_1, \dots, u_5\}$ with the bonus tag. The five vertices of each main record are distinct because its two endpoint types are distinct and its fillers lie on three distinct other rails; the five vertices of each bonus record lie on the five distinct rails and are likewise distinct.

Every seed record therefore has the type profile prescribed by its tag, and the projected matching system of $\mathfrak{X}_0$ is exactly $\tilde{\mathfrak{R}}$. Hence $\mathfrak{X}_0$ is record-admissible; equivalently, by Lemma 5.4, it avoids $\mathcal{P}_\beta^{\text{raw}}$. In particular, equality among filler vertices chosen for different relation instances cannot create a new violation.

Let $\mathfrak{X}_\mathfrak{R}$ be the filler saturation of $\mathfrak{X}_0$. Parts (1)–(2) of Lemma 5.3 show that $\mathfrak{X}_\mathfrak{R}$ remains record-admissible and has the same projected matching system. It contains all main, reserve, and correction records prescribed by that system, while all bonus records were already inserted in the seed. Therefore Lemma 5.2 gives $\lambda_{\text{rec}}(\mathfrak{X}_\mathfrak{R}; \mathbf{x}) = \Phi_\beta(\tilde{\mathfrak{R}}, \mathbf{x}) = \Phi_\beta(\mathfrak{R}, \mathbf{x})$. ■

The projection and filler-saturation lemmas now give the two required inequalities.

Proof of Lemma 5.1. We prove the two inequalities separately. Let $(\mathfrak{X}, \mathbf{x})$ be any finite fully typed record-admissible system with total weight one, let $\mathfrak{R}$ be its projected matching system, and let $\mathfrak{X}^+$ be its filler saturation. Part (i) of Lemma 5.5 makes $\mathfrak{R}$ a legal five-rail realization, and part (3) of Lemma 5.3 gives $\lambda_{\text{rec}}(\mathfrak{X}; \mathbf{x}) \leq \lambda_{\text{rec}}(\mathfrak{X}^+; \mathbf{x})$. The saturated system contains every main, reserve, and correction record prescribed by the relation instances and vertex classes of $\mathfrak{R}$, while it creates no new projected relation instance. Lemma 5.2 therefore gives $\lambda_{\text{rec}}(\mathfrak{X}^+; \mathbf{x}) = \Phi_\beta(\mathfrak{R}, \mathbf{x}) \leq \Phi_\beta^*$. Taking the supremum over $(\mathfrak{X}, \mathbf{x})$ proves $\lambda_{\text{rec}, \beta}^* \leq \Phi_\beta^*$.

Conversely, let $\mathfrak{R}$ be any finite legal realization of $(\mathcal{H}, \mathbf{M})$ for β , equipped with a probability weighting $\mathbf{x}$. Part (ii) of Lemma 5.5 produces a fully typed record-admissible tagged 5-graph $\mathfrak{X}_\mathfrak{R}$ satisfying $\lambda_{\text{rec}}(\mathfrak{X}_\mathfrak{R}; \mathbf{x}) = \Phi_\beta(\mathfrak{R}, \mathbf{x})$. Taking the supremum first over $(\mathfrak{R}, \mathbf{x})$ gives $\lambda_{\text{rec}, \beta}^* \geq \Phi_\beta^*$. This proves the equality. ■

6 From typed records to uncolored hypergraphs

Section 5 replaces the computation by a finite system of typed, tagged five-set records with optimum Φ_β^* . These types and record tags are useful for describing the construction, but they are not part of an ordinary hypergraph. This section removes them by encoding all auxiliary information in the intersection

of an edge with a fixed set of root vertices. This encoding defines a monotone class $\mathcal{C}_\beta$ of simple uncolored r -graphs whose class Lagrangian is an explicit function of Φ_β^* .

From this point on, N denotes the number of root labels. Set $r_{\text{typ}} := r - 1$ and $r_{\text{rec}} := r - 5$. The encoding uses N distinguished root vertices. An edge containing r roots is a background edge, an edge containing r_{typ} roots declares the type of its one data vertex, and an edge containing r_{rec} roots represents a tagged record on its five data vertices. Distinct perfect matchings from Baranyai decompositions provide disjoint root supports for the different roles, types, and tags. In particular, two occurrences of the same five-set with different tags become distinct simple r -edges rather than parallel edges.

The passage to uncolored hypergraphs uses only five properties of the record model. We record them before beginning the construction.

Put $m := |\mathcal{T}|$ and fix a bijection $\mathcal{T} \rightarrow [m]$. Via this bijection, regard the partially typed tagged 5-graphs of Section 5 as *typed c_{tag} -record systems*. We retain the notation $V(\mathfrak{X})$, $\vartheta_{\mathfrak{X}}$, $\mathcal{R}_\alpha(\mathfrak{X})$, $\lambda_\alpha(\mathfrak{X}; \mathbf{x})$, and $\lambda_{\text{rec}}(\mathfrak{X}; \mathbf{x})$. The tag indices and the admissibility predicate include the projected relation data and the reserve and correction records constructed in that section.

Properties (S1) and (S2) allow the uncolored construction to complete and repair type assignments without losing admissibility. Properties (S3) and (S4) ensure that this passage preserves the weighted optimum. Property (S5) then transfers the halting versus nonhalting distinction to that optimum.

- (S1) **Finite local obstructions.** For each machine input β there is an effective admissibility predicate RecAdm_β on finite typed record systems that depends only on their isomorphism types. Failure is witnessed by a copy of one of finitely many raw local obstructions. There is an effectively computable bound b_β for both the number of vertices and the number of requirements in any such obstruction. Deleting records, or deleting data vertices and restricting their types, cannot create a failure.
- (S2) **Typing isolated vertices.** If v is untyped and every raw record containing v is deleted, then one may give v an arbitrary type without destroying admissibility.
- (S3) **Weighted upper bound.** One has $\lambda_{\text{rec}}(\mathfrak{X}; \mathbf{x}) \leq \Phi_\beta^* \rho^5$ for every fully typed admissible system $\mathfrak{X}$ and every nonnegative weight vector of total weight ρ .
- (S4) **Approximation from below.** For every $\varepsilon > 0$ there is a finite fully typed admissible system $\mathfrak{X}$ and a probability weighting $\mathbf{x}$ for which $\lambda_{\text{rec}}(\mathfrak{X}; \mathbf{x}) > \Phi_\beta^* - \varepsilon$.
- (S5) **Fixed baseline dichotomy.** The signature (m, c_{tag}) is independent of β , and there is a fixed integer $z_0 \geq 0$ such that $\text{U}(\beta) \uparrow$ exactly when $\Phi_\beta^* = z_0$, and $\text{U}(\beta) \downarrow$ exactly when $\Phi_\beta^* > z_0$.

The record systems constructed in Section 5 satisfy (S1)–(S5). More precisely, their fixed parameters are $m = 80$, c_{tag} , and $z_0 = 2d/625 \in \mathbb{Z}$, where 625 divides the regular degree d . These parameters are independent of β . Parts (1) and (2) of Lemma 5.4 give (S1), and part (4) gives (S2). Homogeneity gives the factor ρ^5 in (S3), while the definition of $\lambda_{\text{rec}, \beta}^*$ gives (S4). Finally, Lemma 5.1 identifies this record optimum with Φ_β^* , and Theorem 4.1 gives (S5).

For a nonnegative vector $\mathbf{x}$ of total weight ρ , we shall use the elementary record bounds

$$0 \leq \lambda_\alpha(\mathfrak{X}; \mathbf{x}) \leq e_5(\mathbf{x}) \leq \frac{\rho^5}{5!} = \frac{\rho^5}{120}. \quad (6.1)$$

6.1 Root gadgets from Baranyai decompositions

For a finite set V , a *perfect matching* in an h -graph on V is a set of pairwise disjoint edges whose union is V . We use the following form of Baranyai's factorization theorem.

Theorem 6.1 ([4]). *If h and N are positive integers with $h \mid N$, then the complete h -graph $\binom{[N]}{h}$ can be partitioned into perfect matchings.*

For fixed h and N , such a factorization can be found by finite exhaustive search. Thus every factorization and every selected matching below can be constructed effectively once the parameters have been fixed.

Fix $r \geq 8$. Let $\ell_{\text{typ}}, N \geq 1$ and $n_{\text{hole}} \geq 0$ be integers, with N divisible by $r, r_{\text{typ}}, r_{\text{rec}}$, whose calibrated values will be fixed in the next subsection. Assume that the Baranyai decompositions below contain the requested numbers of perfect matchings. Lemma 6.9 will choose these parameters effectively and then freeze the selected matchings. For each $h \in \{r, r_{\text{typ}}, r_{\text{rec}}\}$, fix a Baranyai decomposition $\mathfrak{B}_h$ of $\binom{[N]}{h}$ into

perfect matchings. Thus $\mathfrak{B}_h$ has exactly $\binom{N-1}{h-1}$ members, and distinct members of the same $\mathfrak{B}_h$ have disjoint edge sets.

Each edge of a selected perfect matching is used as a possible root support. Since every $\mathcal{P} \in \mathfrak{B}_h$ consists of N/h edges, at the uniform root weighting $\bar{\mathbf{u}} = (s/N)\mathbf{1}$ we have $\lambda(\mathcal{P}; \bar{\mathbf{u}}) = (N/h)(s/N)^h = s^h/(hN^{h-1})$. Thus all selected matchings of uniformity h make the same contribution at the uniform root weighting. This symmetry is the reason for choosing the root supports from Baranyai factorizations. We select the following distinct members of the corresponding fixed decompositions.

- (i) The matchings $\mathcal{H}_1, \dots, \mathcal{H}_{n_{\text{hole}}}$ are selected from $\mathfrak{B}_r$.
- (ii) For every $i \in [m]$ and $j \in [\ell_{\text{typ}}]$, the matching $\mathcal{P}_{i,j}^{\text{type}}$ is selected from $\mathfrak{B}_{r_{\text{typ}}}$.
- (iii) For every tag $\alpha \in [c_{\text{tag}}]$, the matching $\mathcal{S}_\alpha$ is selected from $\mathfrak{B}_{r_{\text{rec}}}$.

The successive record and root-support encodings are shown in Figure 10. The picture uses one main-relation record; reserve, correction, and bonus records use the same tagged-five-set interface before the root supports are added.

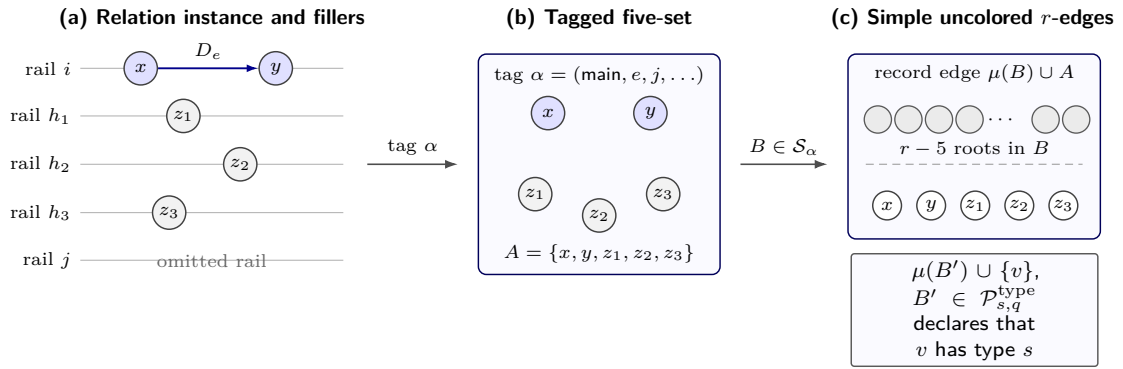


Figure 10: The two encodings that remove the rail relations, types, and tags. (a) A main-relation instance on rail i is completed to a five-set by one filler on each of three other rails; rail j is omitted. The fillers carry no projected relation data. (b) The complete tag α records the relation edge, the omitted rail, and the filler types. (c) A support $B \in \mathcal{S}_\alpha$ of size $r - 5$ turns the tagged five-set into the simple uncolored r -edge $\mu(B) \cup A$. Likewise, a selected $(r - 1)$ -support $B' \in \mathcal{P}_{s,q}^{\text{type}}$ makes the edge $\mu(B') \cup \{v\}$ declare the type s of one data vertex.

6.2 The uncolored admissible class

Let G be an r -graph and let μ be a partial $[N]$ -rooting of G in the sense of Section 2. A vertex outside $\text{im } \mu$ is a *data vertex*. The *root support* of an edge E of G is the set $\{a \in \text{dom } \mu : \mu(a) \in E\}$.

An edge is *permitted* if it has one of the following profiles.

Background edges. An edge is a background edge if it has r roots, no data vertices, and its root support is not an edge of any matching $\mathcal{H}_t$ with $t \in [n_{\text{hole}}]$.

Type edges. A type edge has $r_{\text{typ}} = r - 1$ roots and one data vertex.

Record edges. A record edge has $r_{\text{rec}} = r - 5$ roots and five data vertices, and its root support is an edge of $\mathcal{S}_\alpha$ for some $\alpha \in [c_{\text{tag}}]$.

A type edge with data vertex v is *generic* if its root support belongs to none of the matchings $\mathcal{P}_{i,j}^{\text{type}}$; if its root support is an edge of $\mathcal{P}_{i,j}^{\text{type}}$, then it declares that v has type i . A record edge with data set A and root support in $\mathcal{S}_\alpha$ declares the raw record $A \in \mathcal{R}_\alpha$.

If every edge of G is permitted and no data vertex receives two distinct type declarations, define $\vartheta_{G,\mu}(v) = i$ precisely when a type edge declares that v has type i . Let $\mathfrak{X}_{G,\mu}$ be the partially typed tagged 5-graph whose vertex set is the set of data vertices, whose partial type assignment is $\vartheta_{G,\mu}$, and whose family $\mathcal{R}_\alpha(\mathfrak{X}_{G,\mu})$ consists of the five-sets declared by record edges with tag α .

The pair (G, μ) is *admissible* if every edge of G is permitted, no data vertex receives two distinct type declarations, and $\text{RecAdm}_\beta(\mathfrak{X}_{G,\mu})$ holds.

Define

$$\mathcal{C}_\beta := \left\{ G: \begin{array}{l} G \text{ is a finite } r\text{-graph and there exists a partial } [N]\text{-rooting } \mu \text{ of } G \\ \text{such that } (G, \mu) \text{ is admissible} \end{array} \right\}. \quad (6.2)$$

At the rooted r -graph level, *bundle completion* repeats an existing raw-record declaration over its prescribed root supports. More precisely, once an edge declares a raw α -record A , bundle completion adds $\mu(B) \cup A$ for every edge $B \in \mathcal{S}_\alpha$. It introduces no new raw record. This operation is distinct from filler saturation, which adds tagged five-set records before the root supports are introduced.

Note that admissibility allows missing root labels, untyped data vertices, and incomplete record bundles. Requiring completion would destroy the closure of $\mathcal{C}_\beta$ under taking subgraphs. The following lemma supplies the missing roots and edges only when the Lagrangian is evaluated, combining bundle completion with the analogous root-only and type-edge completions. These operations expose the polynomial represented by the three edge profiles without changing the declared types or raw records.

Let (G, μ) be an admissible partially $[N]$ -rooted r -graph. An *admissible numerical-completion* of (G, μ) is an admissible totally $[N]$ -rooted r -graph $(\hat{G}, \hat{\mu})$ satisfying the following conditions.

- (i) The r -graph $\hat{G}$ contains G as a subgraph, $\hat{\mu}$ extends μ , and

$$V(\hat{G}) \setminus V(G) = \hat{\mu}([N] \setminus \text{dom } \mu).$$

Thus the only new vertices are those carrying the previously missing root labels.

- (ii) For every $B \in \binom{[N]}{r}$ that is not an edge of any $\mathcal{H}_t$ with $t \in [n_{\text{hole}}]$, the root-only edge $\hat{\mu}(B)$ belongs to $\hat{G}$.
- (iii) For every data vertex v of (G, μ) and every $B \in \binom{[N]}{r_{\text{typ}}}$ that is not an edge of any $\mathcal{P}_{i,j}^{\text{type}}$ with $i \in [m]$ and $j \in [\ell_{\text{typ}}]$, the generic type edge $\hat{\mu}(B) \cup \{v\}$ belongs to $\hat{G}$.
- (iv) If a data vertex v is declared to have type i , then $\hat{\mu}(B) \cup \{v\}$ belongs to $\hat{G}$ for every $j \in [\ell_{\text{typ}}]$ and every edge $B \in \mathcal{P}_{i,j}^{\text{type}}$. If a raw α -record A is declared, then $\hat{\mu}(B) \cup A$ belongs to $\hat{G}$ for every edge $B \in \mathcal{S}_\alpha$.
- (v) The declared record system is unchanged, that is, $\mathfrak{X}_{\hat{G}, \hat{\mu}} = \mathfrak{X}_{G, \mu}$.

Throughout this part, we abbreviate *admissible numerical-completion* to *admissible completion*.

If $\mathbf{x}$ is a nonnegative weighting on G , its *zero extension* to an admissible completion agrees with $\mathbf{x}$ on $V(G)$ and assigns weight zero to every vertex of $V(\hat{G}) \setminus V(G)$.

Lemma 6.2. *Every admissible partially $[N]$ -rooted r -graph has an admissible completion.*

Proof. Let (G, μ) be an admissible partially $[N]$ -rooted r -graph. For every label in $[N] \setminus \text{dom } \mu$, add a distinct new vertex and map the label to it. This defines a total rooting $\hat{\mu}$. Because the new root vertices lie in no edge of G , every old edge retains its root support and data set.

Starting from G , add all edges required in parts (ii)–(iv) of the definition. Every added edge is permitted. The edges in parts (ii) and (iii) create no declaration, while those in part (iv) repeat existing type or raw-record declarations. Hence $\mathfrak{X}_{\hat{G}, \hat{\mu}} = \mathfrak{X}_{G, \mu}$. In particular, $\text{RecAdm}_\beta(\mathfrak{X}_{\hat{G}, \hat{\mu}})$ holds, so $(\hat{G}, \hat{\mu})$ is an admissible completion of (G, μ) . $\blacksquare$

For an admissible completion $(\hat{G}, \hat{\mu})$ equipped with the zero extension $\hat{\mathbf{x}}$ of a weighting on G , we now suppress the hats. Let $\mathfrak{X} = \mathfrak{X}_{G, \mu}$ be its partially typed record system. Put $\mathbf{u} := (u_1, \dots, u_N)$, $u_a := x_{\mu(a)}$, $s := \sum_{a=1}^N u_a$, and $\rho := 1 - s$. We also use $\mathbf{x}$ for the restriction of the weighting to the data vertices. Let η_i be the total weight of the vertices declared to have type i , and put $y_\alpha := \lambda_\alpha(\mathfrak{X}; \mathbf{x})$. For arbitrary vectors $\boldsymbol{\eta} = (\eta_i)_{i \in [m]}$ and $\mathbf{y} = (y_\alpha)_{\alpha \in [c_{\text{tag}}]}$, define the master polynomial

$$\Psi_N(\mathbf{u}, \rho, \boldsymbol{\eta}, \mathbf{y}) := e_r(\mathbf{u}) + \rho e_{r_{\text{typ}}}(\mathbf{u}) - \sum_{t=1}^{n_{\text{hole}}} \lambda(\mathcal{H}_t; \mathbf{u}) - \sum_{i=1}^m (\rho - \eta_i) \sum_{j=1}^{\ell_{\text{typ}}} \lambda(\mathcal{P}_{i,j}^{\text{type}}; \mathbf{u}) + \sum_{\alpha=1}^{c_{\text{tag}}} y_\alpha \lambda(\mathcal{S}_\alpha; \mathbf{u}). \quad (6.3)$$

The Lagrangian value of the completed rooted r -graph is $\Psi_N(\mathbf{u}, \rho, \boldsymbol{\eta}, \mathbf{y})$ with the η_i and y_α just defined. The first two terms form the complete background. The sum over the matchings $\mathcal{H}_t$ removes the root-only edges belonging to those matchings, the double sum over the matchings $\mathcal{P}_{i,j}^{\text{type}}$ penalizes missing type

declarations, and the final sum rewards the declared records. A typed vertex of type i is joined to every r_{typ} -set except the selected supports belonging to types other than i , while an untyped vertex is joined to no selected support. Formula (6.3) is consequently valid even before every data vertex is typed. The one-variable polynomial below is obtained from this exact expression after the root weights have been balanced and all data vertices have been typed.

6.3 The calibrated polynomial and the main statement

The estimates below will show that, after completing the admissible rooted r -graph, balancing the root weights, and typing all data vertices, only the total root weight remains to be optimized. We record this one-variable polynomial here so that the exact output of the compiler can be stated before its technical proof.

For the calibration, choose $\ell_{\text{typ}} \geq 32c_{\text{tag}}r_{\text{typ}}/(3r_{\text{rec}})$ such that also

$$\frac{(m-1)\ell_{\text{typ}}}{r_{\text{typ}}} > \frac{5z_0}{r_{\text{rec}}}. \quad (6.4)$$

Call N *admissible* if it is a positive multiple of $\text{lcm}(r, r_{\text{typ}}, r_{\text{rec}})$. For an admissible N , put

$$n_{\text{hole}} := \frac{(m-1)\ell_{\text{typ}}(N - r_{\text{typ}})}{r_{\text{typ}}} - \frac{z_0(5N - r_{\text{rec}})}{r_{\text{rec}}}. \quad (6.5)$$

Because $r_{\text{typ}} \mid N$, $r_{\text{rec}} \mid N$, and $z_0 \in \mathbb{Z}$, we have

$$\frac{N - r_{\text{typ}}}{r_{\text{typ}}} = \frac{N}{r_{\text{typ}}} - 1 \in \mathbb{Z}, \quad \frac{5N - r_{\text{rec}}}{r_{\text{rec}}} = \frac{5N}{r_{\text{rec}}} - 1 \in \mathbb{Z}.$$

Thus both terms in (6.5) are integers. The coefficient of N in that formula is positive by (6.4); hence, for all sufficiently large admissible N , the number n_{hole} is a nonnegative integer and satisfies $n_{\text{hole}} = O(N)$. The effective parameter-selection lemma below chooses such an N and verifies that the three Baranyai decompositions contain the requested numbers of matchings.

Define

$$C_{\text{typ}} := \binom{N}{r_{\text{typ}}} - \frac{(m-1)\ell_{\text{typ}}N}{r_{\text{typ}}}, \quad \text{and} \quad a_{\text{bg}} := \frac{\binom{N}{r} - n_{\text{hole}}N/r}{N^r},$$

and, for $z \in \mathbb{R}$,

$$\Upsilon_z(s) := a_{\text{bg}}s^r + \frac{C_{\text{typ}}}{N^{r_{\text{typ}}}}(1-s)s^{r_{\text{typ}}} + \frac{z}{r_{\text{rec}}N^{r_{\text{rec}}-1}}(1-s)^5s^{r_{\text{rec}}}, \quad \text{where } 0 \leq s \leq 1. \quad (6.6)$$

Set $\Upsilon_z^* := \max_{0 \leq s \leq 1} \Upsilon_z(s)$. For each fixed $s \in [0, 1]$, the coefficient of z in $\Upsilon_z(s)$ is nonnegative. Thus $\Upsilon_z(s)$ is nondecreasing in z , and hence the map $z \mapsto \Upsilon_z^*$ is nondecreasing.

Finally, put

$$J_0 := \binom{N}{r} - \frac{n_{\text{hole}}N}{r} + C_{\text{typ}} + \frac{z_0N}{r_{\text{rec}}}, \quad \text{and} \quad \tau_r := \frac{r!J_0}{(N+1)^r}. \quad (6.7)$$

Since N is divisible by $r, r_{\text{typ}}, r_{\text{rec}}$, the number J_0 is an integer. We also require $J_0 > 0$; this holds for all sufficiently large admissible N , since the first term in the definition of J_0 has degree r , while every negative term outside C_{typ} has degree at most two and $C_{\text{typ}} = \binom{N}{r_{\text{typ}}} - O(N)$.

The main result of this section is the following exact compiler identity. It includes an effective choice of all remaining parameters and root-support matchings.

Theorem 6.3. *There is an effectively computable integer r_{num} such that, for every $r \geq r_{\text{num}}$, the remaining parameters and root-support matchings can be selected effectively so that $\Lambda(\mathcal{C}_\beta) = \Upsilon_{\Phi_\beta}^*$ for every input β .*

We prove the theorem in four steps. First, we use the master polynomial Ψ_N from (6.3) and control its root-support matching terms. Second, a quantitative Maclaurin estimate localizes the root weights, after which the perturbation bound makes them equal. Third, we assign types to all remaining data vertices without decreasing the Lagrangian value. Finally, we calibrate this polynomial at z_0 and make all parameter choices effective.

The next elementary estimate is used twice, globally to localize a possible maximizer and locally to make all root weights equal.

Lemma 6.4. *Let $h \geq 2$, let $\mathcal{P}$ be a perfect matching in $\binom{[N]}{h}$, and let $\mathbf{u} \in \mathbb{R}_{\geq 0}^N$ have coordinate sum s . Write $\bar{\mathbf{u}} = (s/N, \dots, s/N)$ and $V_{\mathbf{u}} := \sum_j (u_j - s/N)^2$. Then*

$$|\lambda(\mathcal{P}; \mathbf{u}) - \lambda(\mathcal{P}; \bar{\mathbf{u}})| \leq B_h s^{h-2} V_{\mathbf{u}}, \quad \text{where} \quad B_h := \begin{cases} \frac{1}{2}, & h = 2, \\ \frac{h-1}{2(h-2)^{h-2}}, & h \geq 3. \end{cases}$$

If, in addition, $u_j \leq 2s/N$ for every j , then

$$|\lambda(\mathcal{P}; \mathbf{u}) - \lambda(\mathcal{P}; \bar{\mathbf{u}})| \leq \frac{h-1}{2} \left(\frac{2s}{N} \right)^{h-2} V_{\mathbf{u}}. \quad (6.8)$$

Proof. For $h = 2$, the Hessian entries corresponding to two vertices in the same edge of $\mathcal{P}$ are 1, so Taylor's theorem gives both bounds with $B_2 = 1/2$. Suppose therefore that $h \geq 3$. The gradient of $\lambda(\mathcal{P}; \cdot)$ at $\bar{\mathbf{u}}$ is constant, and hence its linear term vanishes on the hyperplane $\sum_j u_j = s$.

The nonzero Hessian entries join two vertices in the same edge of $\mathcal{P}$; each is the product of the other $h-2$ coordinates in that edge. Along the segment from $\bar{\mathbf{u}}$ to $\mathbf{u}$, AM-GM bounds this product by $(s/(h-2))^{h-2}$. Thus every absolute Hessian row sum is at most $(h-1)(s/(h-2))^{h-2}$. Taylor's theorem and the operator-norm bound by the maximum absolute row sum prove the first bound. Under the additional coordinate bound, the same argument gives $(h-1)(2s/N)^{h-2}$ and proves (6.8). ■

6.4 Balancing the root weights and typing the data

The next two steps reduce the optimization to one variable. First, every weighting above the calibrated baseline is localized near the complete-frame optimum, where averaging makes all root weights equal. Once the root weights are equal, deleting the records incident with an untyped vertex costs no more than the type edges gained by assigning it a type. The remaining parameter is the total root weight s .

Put $n := N + 1$ and $\mathbf{z} := (u_1, \dots, u_N, \rho)$. Define

$$\alpha_{r,n} := \frac{\binom{n}{r}}{n^r}, \quad \text{and} \quad V_{\mathbf{z}} := \sum_{a=1}^N \left(u_a - \frac{1}{n} \right)^2 + \left(\rho - \frac{1}{n} \right)^2.$$

Since $\bar{\mathbf{u}} = s\mathbf{1}/N$, we have

$$V_{\mathbf{z}} = V_{\mathbf{u}} + \frac{n}{N} \left(\rho - \frac{1}{n} \right)^2.$$

We use the following quantitative consequence of Maclaurin's inequalities; see Hardy, Littlewood, and Pólya [27, Theorem 52] for the classical form.

Lemma 6.5. *For every $\mathbf{z} \in \mathbb{R}_{\geq 0}^n$ with $\sum_j z_j = 1$, we have*

$$e_r(\mathbf{z}) \leq \alpha_{r,n} - \alpha_{r,n} \sum_{j=1}^n \left(z_j - \frac{1}{n} \right)^2.$$

Proof. Write $V = \sum_j (z_j - 1/n)^2$. Then

$$e_2(\mathbf{z}) = \frac{1}{2} \left(1 - \sum_j z_j^2 \right) = \binom{n}{2} n^{-2} - \frac{V}{2}.$$

Maclaurin's inequalities give

$$\frac{e_r(\mathbf{z})}{\binom{n}{r}} \leq \left(\frac{e_2(\mathbf{z})}{\binom{n}{2}} \right)^{r/2} = n^{-r} \left(1 - \frac{n}{n-1} V \right)^{r/2}.$$

Because $r/2 \geq 1$ and $0 \leq nV/(n-1) \leq 1$, the last factor is at most $1 - nV/(n-1) \leq 1 - V$. This proves the claim. ■

The identity $e_r(\mathbf{u}) + \rho e_{r_{\text{typ}}}(\mathbf{u}) = e_r(\mathbf{z})$ and the nonnegativity of $\sum_{t=1}^{n_{\text{hole}}} \lambda(\mathcal{H}_t; \mathbf{u})$ and of the type penalty, together with Lemmas 6.4 and 6.5, imply

$$\Psi_N(\mathbf{u}, \rho, \boldsymbol{\eta}, \mathbf{y}) \leq \alpha_{r,n} - \alpha_{r,n} V_{\mathbf{z}} + \frac{c_{\text{tag}} \rho^5}{120} \left(\frac{s^{r_{\text{rec}}}}{r_{\text{rec}} N^{r_{\text{rec}}-1}} + B_{r_{\text{rec}}} s^{r_{\text{rec}}-2} V_{\mathbf{u}} \right).$$

We require r to be large enough that

$$\frac{c_{\text{tag}} B_{r_{\text{rec}}}}{120} \leq \frac{1}{8r!}. \quad (6.9)$$

For all sufficiently large N , we have $\alpha_{r,n} \geq 1/(2r!)$. Hence (6.9), together with $\rho, s \leq 1$ and $V_{\mathbf{u}} \leq V_{\mathbf{z}}$, gives

$$\frac{c_{\text{tag}} B_{r_{\text{rec}}}}{120} \rho^5 s^{r_{\text{rec}}-2} V_{\mathbf{u}} \leq \frac{1}{8r!} \rho^5 s^{r_{\text{rec}}-2} V_{\mathbf{u}} \leq \frac{1}{4} \alpha_{r,n} V_{\mathbf{z}},$$

and therefore

$$\Psi_N(\mathbf{u}, \rho, \boldsymbol{\eta}, \mathbf{y}) \leq \alpha_{r,n} - \frac{3}{4} \alpha_{r,n} V_{\mathbf{z}} + \varepsilon_N \rho^5 s^{r_{\text{rec}}}, \quad \text{where } \varepsilon_N := \frac{c_{\text{tag}}}{120 r_{\text{rec}} N^{r_{\text{rec}}-1}}. \quad (6.10)$$

We next compare the polynomial at an arbitrary root-weight vector $\mathbf{u}$ with its value at the averaged vector $\bar{\mathbf{u}}$. This is the local estimate needed to prove uniformization. It is not enough to know where the polynomial associated with one perfect matching is maximized because a single matching can strongly favor an unbalanced root vector. The complete frame gives a uniform quadratic gain, while the perturbations from all selected matchings have vanishing linear term at the uniform vector.

Lemma 6.6. *Fix $m, c_{\text{tag}}, r, \ell_{\text{typ}}$ with $r \geq 8$, and fix constants $K_{\text{hole}}, K_V > 0$. For all sufficiently large N , the following holds. If $n_{\text{hole}} \leq K_{\text{hole}} N$, $\rho < 2/(N+1)$, $V_{\mathbf{u}} \leq K_V N^{2-r}$, $0 \leq \eta_i \leq \rho$ for every $i \in [m]$, and $0 \leq y_\alpha \leq \rho^5/120$ for every $\alpha \in [c_{\text{tag}}]$, then $\Psi_N(\bar{\mathbf{u}}, \rho, \boldsymbol{\eta}, \mathbf{y}) \geq \Psi_N(\mathbf{u}, \rho, \boldsymbol{\eta}, \mathbf{y})$, with strict inequality unless $\mathbf{u} = \bar{\mathbf{u}}$.*

Proof. Since $s = 1 - \rho$ and $\rho < 2/(N+1)$, we have $s > 1/2$ for large N . Moreover, for every $a \in [N]$, we have

$$\left| u_a - \frac{s}{N} \right| \leq \sqrt{V_{\mathbf{u}}} \leq \sqrt{K_V} N^{1-r/2} = o(N^{-1}),$$

where the last estimate uses $r > 4$. Since $s > 1/2$, this proves that $s/(2N) \leq u_a \leq 2s/N$ for every $a \in [N]$ and all sufficiently large N . Repeatedly average two root coordinates. If a, b are averaged, the gain in e_r is

$$\frac{(u_a - u_b)^2}{4} e_{r-2}([N] \setminus \{a, b\}; \mathbf{u}).$$

Every averaging step preserves the interval $s/(2N) \leq u_a \leq 2s/N$. For sufficiently large N , we have

$$e_{r-2}([N] \setminus \{a, b\}; \mathbf{u}) \geq \binom{N-2}{r-2} \left(\frac{s}{2N} \right)^{r-2} \geq \frac{s^{r-2}}{2^{r-1}(r-2)!}.$$

For example, repeatedly average coordinates of largest and smallest weight. If $\mathbf{u}'$ is obtained from $\mathbf{u}$ by replacing u_a, u_b by their average, then the total weight is unchanged and the exact variance identity is

$$V_{\mathbf{u}} - V_{\mathbf{u}'} = \frac{(u_a - u_b)^2}{2}.$$

If the variance did not tend to zero, then the range, and hence the decrease in this display, would stay bounded away from zero. Thus the sequence converges to $\bar{\mathbf{u}}$, and the variance decrements telescope to $V_{\mathbf{u}}$. Summing the preceding gain along this sequence gives

$$[e_r(\bar{\mathbf{u}}) + \rho e_{r_{\text{typ}}}(\bar{\mathbf{u}})] - [e_r(\mathbf{u}) + \rho e_{r_{\text{typ}}}(\mathbf{u})] \geq \frac{s^{r-2}}{2^r(r-2)!} V_{\mathbf{u}};$$

we used only the e_r gain, since $e_{r_{\text{typ}}}$ also increases under averaging.

Every Lagrangian polynomial $\lambda(\mathcal{P}; \cdot)$ associated with a perfect matching has constant gradient at $\bar{\mathbf{u}}$. Applying the local estimate (6.8), the total absolute perturbation caused by the $\mathcal{H}_t$ -terms is at most

$$n_{\text{hole}} \frac{r-1}{2} \left(\frac{2s}{N} \right)^{r-2} V_{\mathbf{u}} = O_{r, K_{\text{hole}}} (N^{3-r}) s^{r-2} V_{\mathbf{u}}.$$

The type term has $m\ell_{\text{typ}}$ selected matchings, each with coefficient at most $\rho < 2/n$. Also, $\lambda_\alpha(\mathfrak{X}; \mathbf{x}) \leq \rho^5/120 = O(N^{-5})$ and $r_{\text{rec}} - 2 = r - 7$. The type and record perturbations are therefore bounded, respectively, by $O_{m, \ell_{\text{typ}}, r}(N^{2-r})s^{r-2}V_{\mathbf{u}}$ and $O_{c_{\text{tag}}, r}(N^{2-r})s^{r-2}V_{\mathbf{u}}$. Here the root-domain bound $s > 1/2$ absorbs the missing fixed powers of s into the constants.

If $V_{\mathbf{u}} > 0$, divide the background, type, and record error bounds by the complete-frame gain above. Their three ratios are, respectively, $O_{r, K_{\text{hole}}}(N^{3-r})$, $O_{m, \ell_{\text{typ}}, r}(N^{2-r})$, and $O_{c_{\text{tag}}, r}(N^{2-r})$, and hence tend to zero. For large N their sum is less than half the complete-frame gain. If $V_{\mathbf{u}} = 0$, then $\mathbf{u} = \bar{\mathbf{u}}$ already. This proves the asserted inequality and its strict form. The type and record perturbation estimates use only $0 \leq \rho - \eta_i \leq \rho$ and $0 \leq y_\alpha \leq \rho^5/120$, so the same argument proves the asserted coefficient-bound extension. $\blacksquare$

We next type the remaining data vertices. This typing repair comes after root balancing because its loss estimate uses the uniform value associated with every selected root matching.

Lemma 6.7. *Assume $\mathbf{u} = \bar{\mathbf{u}}$, $\rho < 2/n$, $s > 1/2$, and $\ell_{\text{typ}} \geq 32c_{\text{tag}}r_{\text{typ}}/(3r_{\text{rec}})$. Then all data vertices can be typed without decreasing the Lagrangian value.*

Proof. Let v be untyped and have weight x_v . Delete every raw record containing v , including all copies in its root bundle. If $x_v = 0$, this deletion and the subsequent assignment of an arbitrary type do not change the Lagrangian value, so suppose $x_v > 0$. For each tag, the sum of the remaining fourfold data monomials is at most $e_4(\mathbf{x}) \leq \rho^4/24$. Since $\lambda(\mathcal{S}_\alpha; \bar{\mathbf{u}}) = s^{r_{\text{rec}}}/(r_{\text{rec}}N^{r_{\text{rec}}-1})$, the loss is at most $c_{\text{tag}}x_v\rho^4s^{r_{\text{rec}}}/(24r_{\text{rec}}N^{r_{\text{rec}}-1})$.

Give v an arbitrary type and add its ℓ_{typ} type bundles. The type assignment is admissible by (S2), and the gain is $\ell_{\text{typ}}x_v s^{r_{\text{typ}}}/(r_{\text{typ}}N^{r_{\text{typ}}-1})$.

The ratio of this gain to the loss bound is at least

$$\frac{24r_{\text{rec}}\ell_{\text{typ}}}{c_{\text{tag}}r_{\text{typ}}} \left(\frac{s}{N\rho} \right)^4 > \frac{3r_{\text{rec}}\ell_{\text{typ}}}{32c_{\text{tag}}r_{\text{typ}}} \geq 1,$$

because $s > 1/2$ and $N\rho < 2$. Hence the gain is at least the actual loss. Repeating the typing repair types every data vertex. Records deleted at an earlier step are not charged again. $\blacksquare$

After this repair, $\sum_{i=1}^m(\rho - \eta_i) = (m-1)\rho$. Before repair the left side exceeds $(m-1)\rho$ by exactly the total weight of the untyped vertices. This additional term explains why typing is completed before the one-variable evaluation.

6.5 Calibration and effective parameter choice

At uniform root weights and after typing the data vertices, every selected perfect matching has the Lagrangian value used in the definition of Υ_z . We now prove the calibration and effectivity assertions needed to finish [Theorem 6.3](#).

We first verify that the definition of J_0 calibrates the nonhalting value z_0 at the balanced point $s_0 = N/(N+1)$.

Lemma 6.8. *For all sufficiently large admissible N , the following statements hold.*

- (i) *The unique maximizer of Υ_{z_0} is $s_0 := N/(N+1)$.*
- (ii) *The calibrated optimum satisfies $\Upsilon_{z_0}^* = \Upsilon_{z_0}(s_0) = J_0/(N+1)^r$.*
- (iii) *For every $z > z_0$, we have $\Upsilon_z^* > \Upsilon_{z_0}^*$.*
- (iv) *We have $0 < J_0 < \binom{N+1}{r}$ and $\tau_r \in \mathbb{Q} \cap (0, 1)$.*

Proof. Let $\rho_0 = 1/(N+1)$. The unperturbed first two terms

$$b_N(s) := \frac{\binom{N}{r}}{N^r} s^r + \frac{\binom{N}{r_{\text{typ}}}}{N^{r_{\text{typ}}}} (1-s)s^{r_{\text{typ}}}$$

are exactly e_r evaluated at the $(N+1)$ -vector $(s/N, \dots, s/N, 1-s)$. Hence b_N is uniquely maximized at s_0 . By (6.5), differentiating the three corrections to b_N and using $s_0 = N\rho_0$ gives

$$\Upsilon'_{z_0}(s_0) = \rho_0^{r-1} \left[-n_{\text{hole}} + \frac{(m-1)\ell_{\text{typ}}(N-r_{\text{typ}})}{r_{\text{typ}}} - \frac{z_0(5N-r_{\text{rec}})}{r_{\text{rec}}} \right] = 0.$$

We verify that stationarity is global and unique. By Lemma 6.5, we have

$$\alpha_{r,n} - b_N(s) \geq \alpha_{r,n} \frac{N+1}{N} ((1-s) - \rho_0)^2.$$

Write $\rho := 1-s$ and define

$$\delta_N := \alpha_{r,n} - \frac{J_0}{(N+1)^r} = \left(\frac{(m-1)\ell_{\text{typ}}}{r_{\text{typ}}} - \frac{5z_0}{r_{\text{rec}}} \right) \frac{N}{r(N+1)^{r-1}} > 0. \quad (6.11)$$

The equality follows by substituting (6.5) and (6.7), and the strict inequality is (6.4). Since $b_N(s_0) = \alpha_{r,n}$ and $\Upsilon_{z_0}(s_0) = J_0/(N+1)^r$, while the background and type corrections to $b_N(s)$ are nonpositive, the preceding stability bound gives, for every $s \in [0, 1]$,

$$\Upsilon_{z_0}(s) - \Upsilon_{z_0}(s_0) \leq -\alpha_{r,n} \frac{N+1}{N} (\rho - \rho_0)^2 + \delta_N + \frac{z_0}{r_{\text{rec}} N^{r_{\text{rec}}-1}} \rho^5 s^{r_{\text{rec}}}. \quad (6.12)$$

In particular, $\delta_N = \Theta(N^{2-r})$, and $\alpha_{r,n}(N+1)/N \geq 1/(2r!)$ for all sufficiently large admissible N .

Suppose that $\Upsilon_{z_0}(s) \geq \Upsilon_{z_0}(s_0)$ and $\rho \geq 2\rho_0$. Then $|\rho - \rho_0| \geq \rho/2$. Since $r \geq 8$ and $0 \leq \rho \leq 1$, for all sufficiently large N the last term in (6.12) is at most $\rho^2/(16r!)$. Moreover, $\rho \geq 2/(N+1)$ and $\delta_N = o(\rho^2)$ uniformly on this range, so we have

$$\Upsilon_{z_0}(s) - \Upsilon_{z_0}(s_0) \leq -\frac{\rho^2}{16r!} + \delta_N < 0,$$

a contradiction. Hence every global maximizer satisfies $\rho < 2\rho_0$. On this range the record term in (6.12) is $O(N^{1-r})$; the same inequality then shows that there is a constant $K > 0$, independent of N , such that every global maximizer belongs to

$$I_N := \left\{ s \in [0, 1] : |(1-s) - \rho_0| \leq K N^{1-r/2} \right\}.$$

On I_N , $b''_N(s)$ is bounded above by a negative constant depending only on r . To see this explicitly, b''_N converges uniformly on $[3/4, 1]$ to the second derivative of $s^r/r! + (1-s)s^{r-1}/(r-1)!$, whose value at $s = 1$ is $-1/(r-2)! < 0$, and I_N shrinks to $s = 1$. The second derivatives of the background and type corrections are $O(N^{2-r})$, while that of the record correction is $O(N^{3-r})$ on I_N . Consequently $\Upsilon''_{z_0}(s) < 0$ throughout I_N for large N . Since $\Upsilon'_{z_0}(s_0) = 0$, s_0 is the unique global maximizer.

Substitution of $s_0 = N/(N+1)$ into (6.6) gives the asserted calibrated identity; integrality follows from $r, r_{\text{typ}}, r_{\text{rec}} \mid N$. If $z > z_0$, then, because $0 < s_0 < 1$, we have

$$\Upsilon_z^* \geq \Upsilon_z(s_0) > \Upsilon_{z_0}(s_0) = \Upsilon_{z_0}^*,$$

Positivity of J_0 is one of the largeness conditions imposed when N is selected. Equation (6.11) gives $J_0 < \binom{N+1}{r}$. Hence $\tau_r > 0$, while $\tau_r < r! \binom{N+1}{r} / (N+1)^r < 1$. ■

The proof of Lemma 6.8 establishes the required estimates for all sufficiently large uniformities and root frames. To make the compiler effective, however, we need a terminating procedure that chooses $r_{\text{num}}, \ell_{\text{typ}}, N$, and the required perfect matchings without using the input β or the unknown value Φ_β^* . The next lemma provides this procedure and records the two uniform conclusions needed later, namely localization above the baseline and nondecrease when the root weights are averaged.

Lemma 6.9. *There is an effectively computable integer $r_{\text{num}} \geq 16$ such that, for every $r \geq r_{\text{num}}$, parameters ℓ_{typ}, N and the required systems of perfect matchings can be selected. For the selected N and systems of perfect matchings, set $n = N+1$. For $\mathbf{u} \in \mathbb{R}_{\geq 0}^N$, $\rho \geq 0$, $\boldsymbol{\eta} \in \mathbb{R}_{\geq 0}^m$, and $\mathbf{y} \in \mathbb{R}_{\geq 0}^{\text{ctag}}$, use the master polynomial $\Psi_N(\mathbf{u}, \rho, \boldsymbol{\eta}, \mathbf{y})$ from (6.3) and write $s = \sum_a u_a$. The following statements hold.*

- (i) The selection is effective. First compute ℓ_{typ} from $m, c_{\text{tag}}, z_0, r$, then compute N from these quantities and ℓ_{typ} , and finally construct the required systems of perfect matchings.
- (ii) Whenever $\sum_a u_a + \rho = 1$, $\sum_i \eta_i \leq \rho$, $0 \leq y_\alpha \leq \rho^5/120$ for every $\alpha \in [c_{\text{tag}}]$, and $\Psi_N(\mathbf{u}, \rho, \boldsymbol{\eta}, \mathbf{y}) > J_0/n^r$, we have $\rho < 2/n$ and $s > 1/2$.
- (iii) Under the hypotheses in the preceding item, we have

$$\Psi_N(\bar{\mathbf{u}}, \rho, \boldsymbol{\eta}, \mathbf{y}) \geq \Psi_N(\mathbf{u}, \rho, \boldsymbol{\eta}, \mathbf{y}), \quad \text{where } \bar{\mathbf{u}} = (s/N)\mathbf{1}, \quad (6.13)$$

with equality only when $\mathbf{u} = \bar{\mathbf{u}}$.

Proof. For $r \geq 8$, put $a_r := r!B_{r-5}$. Stirling's estimate gives $a_r \sim (\sqrt{2\pi} e^7/2)r^{17/2}e^{-r} \rightarrow 0$. This convergence can be turned into an effective bound valid for every sufficiently large r . For $r \geq 16$, set $t = r - 6 \geq 10$. The definition of B_h gives

$$\frac{a_{r+1}}{a_r} = \left(1 + \frac{8}{t} + \frac{7}{t^2}\right) \left(1 - \frac{1}{t}\right)^{t-1}.$$

Using $\log(1+x) \leq x$ and $\log(1-x) \leq -x$, we obtain

$$\log \frac{a_{r+1}}{a_r} \leq -1 + \frac{9}{t} + \frac{7}{t^2} < 0.$$

Thus $(a_r)_{r \geq 16}$ is strictly decreasing. Enumerate $r = 16, 17, \dots$ until (6.9) holds, and let r_{num} be the first accepted value. The tested inequality is rational, the search terminates because $a_r \rightarrow 0$, and monotonicity shows that (6.9) holds for every $r \geq r_{\text{num}}$.

Now fix an arbitrary $r \geq r_{\text{num}}$. Having set $r_{\text{typ}} = r - 1$ and $r_{\text{rec}} = r - 5$, choose an integer $\ell_{\text{typ}} \geq 32c_{\text{tag}}r_{\text{typ}}/(3r_{\text{rec}})$ satisfying (6.4).

The asymptotic estimates in the proof of Lemma 6.8 show that every sufficiently large admissible N works, but they do not provide an explicit lower bound on N . To obtain a computable choice, enumerate the positive multiples N of $\text{lcm}(r, r_{\text{typ}}, r_{\text{rec}})$. For each candidate, compute n_{hole} and J_0 from (6.5) and (6.7), and discard it unless $n_{\text{hole}} \geq 0$, $J_0 > 0$, $n_{\text{hole}} < \binom{N-1}{r-1}$, $m\ell_{\text{typ}} \leq \binom{N-1}{r_{\text{typ}}-1}$, and $c_{\text{tag}} \leq \binom{N-1}{r_{\text{rec}}-1}$. For every remaining candidate, find, by exhaustive search, a complete Baranyai decomposition $\mathfrak{B}_h$ for each $h \in \{r, r_{\text{typ}}, r_{\text{rec}}\}$, and select the required numbers of distinct perfect matchings from the three decompositions. A candidate is a finite partition of $\binom{[N]}{h}$; checking that every part consists of disjoint h -sets covering $[N]$ and that every h -set occurs exactly once is a finite decidable test. Baranyai's theorem guarantees that this exhaustive search succeeds. Since the first matching-count inequality is strict, at least one member of $\mathfrak{B}_r$ is not among $\mathcal{H}_1, \dots, \mathcal{H}_{n_{\text{hole}}}$. Every edge of that member is therefore a permitted root-only edge.

For each surviving candidate, perform two exact tests. The first is

$$\forall t \in [0, 1] \quad t \neq \frac{N}{N+1} \implies \Upsilon_{z_0}(t) < \Upsilon_{z_0}\left(\frac{N}{N+1}\right). \quad (6.14)$$

Writing $s = \sum_a u_a$ and $\bar{\mathbf{u}} = (s/N)\mathbf{1}$, let $\mathbf{A}_N$ denote the conjunction

$$\begin{aligned} \mathbf{u} \in \mathbb{R}_{\geq 0}^N, \quad \rho \in \mathbb{R}_{\geq 0}, \quad \boldsymbol{\eta} \in \mathbb{R}_{\geq 0}^m, \quad \mathbf{y} \in \mathbb{R}_{\geq 0}^{c_{\text{tag}}}, \quad s + \rho = 1, \quad \sum_i \eta_i \leq \rho, \\ 0 \leq y_\alpha \leq \rho^5/120 \quad \text{for every } \alpha \in [c_{\text{tag}}], \quad \text{and} \quad \Psi_N(\mathbf{u}, \rho, \boldsymbol{\eta}, \mathbf{y}) > J_0/n^r. \end{aligned}$$

The second test is the universal closure of

$$\begin{aligned} \mathbf{A}_N \implies \rho < 2/n \quad \text{and} \quad s > 1/2, \\ \mathbf{A}_N \implies \Psi_N(\bar{\mathbf{u}}, \rho, \boldsymbol{\eta}, \mathbf{y}) \geq \Psi_N(\mathbf{u}, \rho, \boldsymbol{\eta}, \mathbf{y}), \\ \mathbf{A}_N \wedge \Psi_N(\bar{\mathbf{u}}, \rho, \boldsymbol{\eta}, \mathbf{y}) = \Psi_N(\mathbf{u}, \rho, \boldsymbol{\eta}, \mathbf{y}) \implies \sum_a (u_a - s/N)^2 = 0. \end{aligned}$$

For fixed N , the polynomials associated with the selected perfect matchings are explicit finite polynomials with rational coefficients. After clearing positive integer denominators, both tests are first-order sentences

over the ordered field of the reals; for example, $t \neq N/(N+1)$ can be written as $(nt - N)^2 > 0$. Tarski [69] proved that such sentences are decidable, and Basu, Pollack, and Roy [6] give an algorithmic treatment. Hence one can determine whether both tests hold.

The search terminates. Indeed, Lemma 6.8 proves (6.14) for every sufficiently large admissible N . The derivation of (6.10) uses only $\sum_{t=1}^{n_{\text{hole}}} \lambda(\mathcal{H}_t; \mathbf{u}) \geq 0$, $\rho - \eta_i \geq 0$, and $0 \leq y_\alpha \leq \rho^5/120$, so it applies uniformly to all $\boldsymbol{\eta}, \mathbf{y}$ satisfying the hypotheses in part (2). It gives the required localization as follows. Recall from (6.11) that $\delta_N = \alpha_{r,n} - J_0/n^r = \Theta(N^{2-r})$. If $\Psi_N(\mathbf{u}, \rho, \boldsymbol{\eta}, \mathbf{y}) > J_0/n^r$, then

$$\frac{3}{4}\alpha_{r,n}V_{\mathbf{z}} < \delta_N + \varepsilon_N \rho^5 s^{r_{\text{rec}}}. \quad (6.15)$$

Suppose that $\rho \geq 2/n$. Then $V_{\mathbf{z}} \geq (\rho - 1/n)^2 \geq \rho^2/4$, whereas $\delta_N/\rho^2 = O(N^{4-r})$ and $\varepsilon_N \rho^5/\rho^2 = O(N^{6-r})$. Since $r \geq 8$ and $\alpha_{r,n}$ is bounded away from zero, this contradicts (6.15) for all sufficiently large N . Hence $\rho < 2/n$.

It follows that the last term in (6.15) is $O(N^{1-r})$ and hence $V_{\mathbf{z}} = O(N^{2-r})$. In particular, $V_{\mathbf{u}} = O(N^{2-r})$ and $s = 1 - \rho > 1/2$. The implicit constants depend only on $(m, c_{\text{tag}}, z_0, r, \ell_{\text{typ}})$. Moreover, $\eta_i \geq 0$ and $\sum_i \eta_i \leq \rho$ imply $0 \leq \eta_i \leq \rho$. The coefficient-bound extension in Lemma 6.6 therefore applies uniformly throughout this domain and proves (6.13), including its strict form.

Finally, $n_{\text{hole}} = O(N)$, the three Baranyai decompositions contain arbitrarily many perfect matchings, and $J_0 > 0$ eventually. Thus every candidate test holds for all sufficiently large divisible N . The first accepted candidate supplies N , the three decompositions, and all selected perfect matchings. Every datum used by the search depends only on $(m, c_{\text{tag}}, z_0, r, \ell_{\text{typ}})$, never on β or on the unknown value Φ_β^* . ■

For reference, Table 3 separates the finite data fixed once and for all from the choices made after the uniformity and the input word are supplied. In particular, the real-closed-field tests select root-frame data before β is known; they do not query or approximate the weighted optimum Φ_β^* .

Fix an arbitrary $r \geq r_{\text{num}}$ and the first accepted remaining parameters and systems of perfect matchings for this r . For the proof, specialize the class $\mathcal{C}_\beta$ in (6.2) to these choices. We suppress its dependence on r ; all choices are independent of β .

We now prove the theorem stated near the beginning of the section. The upper bound completes an admissibly rooted r -graph, balances its root weights, types its data vertices, and then applies (S3). The reverse bound uses (S4) to realize record values approaching Φ_β^* .

Proof of Theorem 6.3. We prove the upper bound. Fix $G \in \mathcal{C}_\beta$ and an arbitrary probability weighting $\mathbf{x}$ of G . Choose an admissible partial rooting μ witnessing that $G \in \mathcal{C}_\beta$, and apply Lemma 6.2 to (G, μ) . Let $(\hat{G}, \hat{\mu})$ be the resulting admissible completion and let $\hat{\mathbf{x}}$ be the zero extension of $\mathbf{x}$. Put $\text{Val}_{\text{comp}} := \lambda(\hat{G}; \hat{\mathbf{x}})$. Since completion only adds edges and assigns weight zero to its new vertices, $\text{Val}_{\text{comp}} \geq \lambda(G; \mathbf{x})$. For the remainder of the upper-bound proof, suppress the hats, and let $\mathfrak{X} = \mathfrak{X}_{G,\mu}$ be the record system declared by the completed rooted r -graph. If $\text{Val}_{\text{comp}} \leq \Upsilon_{z_0}^*$, the desired bound follows because $\Phi_\beta^* \geq z_0$ and the map $z \mapsto \Upsilon_z^*$ is nondecreasing. Suppose instead that $\text{Val}_{\text{comp}} > \Upsilon_{z_0}^* = J_0/n^r$.

Let η_i be the total type- i weight, and put $y_\alpha := \lambda_\alpha(\mathfrak{X}; \mathbf{x})$. Distinct types have disjoint vertex sets, so $\sum_i \eta_i \leq \rho$, and (6.1) gives $0 \leq y_\alpha \leq \rho^5/120$. Parts (2)–(3) of Lemma 6.9 therefore apply to the root weights $\mathbf{u}$, the total data weight ρ , the type weights $\boldsymbol{\eta}$, and the record values $\mathbf{y}$. They give $\rho < 2/n$ and $s > 1/2$, and make the root weights uniform without decreasing the Lagrangian value.

Lemma 6.7 now makes every data vertex typed, again without decreasing the Lagrangian value. By (S3), the identity $\sum_i (\rho - \eta_i) = (m - 1)\rho$, and the uniform values associated with the perfect matchings, the resulting Lagrangian value is at most $\Upsilon_{\Phi_\beta^*}(s) \leq \Upsilon_{\Phi_\beta^*}^*$. This proves the upper bound for every $G \in \mathcal{C}_\beta$ and every probability weighting on G .

For the reverse inequality, fix $s \in [0, 1]$ and $0 < \varepsilon < \Phi_\beta^*$. By (S4), take a finite fully typed record system satisfying RecAdm_β with probability weights and record value exceeding $\Phi_\beta^* - \varepsilon$. Scale its weights by $1 - s$, add a total root frame with every root weight s/N , and include all completion edges. The resulting uncolored r -graph belongs to $\mathcal{C}_\beta$, and its displayed probability weighting has Lagrangian value at least $\Upsilon_{\Phi_\beta^* - \varepsilon}(s)$. Letting $\varepsilon \downarrow 0$ and then maximizing over s proves $\Lambda(\mathcal{C}_\beta) \geq \Upsilon_{\Phi_\beta^*}^*$. ■

Object	Depends on	Effective construction or status
$U, d, \mathcal{T}, m, c_{\text{tag}}, z_0$	Fixed universal machine and construction	Finite normalization, relation-path and d -regular rail construction, and record-tag enumeration. Fixed once and independent of both r and β .
r_{num}	Fixed data	Search the explicit rational inequality (6.9); monotonicity supplies a certified bound valid for every larger r .
$r_{\text{typ}} = r - 1, r_{\text{rec}} = r - 5$, and ℓ_{typ}	Fixed data and the chosen $r \geq r_{\text{num}}$	The first two are explicit. Choose the first integer ℓ_{typ} satisfying the finitely many displayed rational inequalities.
N, n_{hole}, J_0 and the selected Baranyai matchings	Fixed data, r , and ℓ_{typ}	Enumerate admissible divisible N . For each candidate, find the finite Baranyai decompositions by exhaustive search and decide the two polynomial certificates by quantifier elimination over real closed fields.
$\tau_r = r!J_0/(N+1)^r$	The preceding r -dependent data	An exactly computable rational; it is independent of β .
$K_{*,r}$	The preceding r -dependent data	Complete the fixed empty-mark baseline record system and apply the finite rooted compiler to obtain a preliminary template. Compute its fixed positive rational optimal weighting and take the corresponding integer blowup in canonical edge-list form. This construction is effective from r and independent of β .
$\mathcal{P}_{\beta}^{\text{raw}}, b_{\beta}, \mathcal{A}_{\beta}, \mathcal{F}_{r,\beta}$	β and the already fixed r -dependent data	Finite local-obstruction expansion, bounded-root elimination, and enumeration of pair-covering extensions. These are the input-dependent outputs.
Φ_{β}^*	β	Weighted supremum used to prove correctness. It is not an input to any parameter search, and no algorithm for computing it is assumed.

Table 3: Dependencies and effective choices in the direct construction.

By Theorem 4.1 and Lemma 6.8, the exact identity implies that $\Lambda(\mathcal{C}_{\beta}) = \tau_r/r!$ when U does not halt on β , and $\Lambda(\mathcal{C}_{\beta}) > \tau_r/r!$ when U halts on β .

7 From rooted classes to Turán density

The construction in Section 6 encodes the computation in the class Lagrangian of a monotone class $\mathcal{C}_{\beta}$ of uncolored r -graphs. Two tasks remain before this becomes an ordinary Turán problem. Membership in $\mathcal{C}_{\beta}$ is defined by the existence of a suitable partial placement of the root labels, and the class Lagrangian is not yet the Turán density of an explicitly given finite forbidden family.

We handle these tasks in two steps. First, every unsuccessful partial rooting has a bounded obstruction that remains an obstruction until a new root is placed inside it. Since there are only finitely many root labels, a finite search tree gives a uniform bound on minimal unrooted obstructions. Enumerating all hypergraphs up to this bound produces a finite family $\mathcal{A}_{\beta}$ with $\mathcal{C}_{\beta} = \text{Forb}(\mathcal{A}_{\beta})$.

Second, following Mubayi’s extension method [52], we replace each member of $\mathcal{A}_{\beta}$ by all of its bounded pair-covering extensions. Pair covering prevents vertices of a forbidden configuration from collapsing into one vertex class of a blowup. Conversely, the support of a minimum-support optimal weighting covers pairs. These two facts give an exact identity between the class Lagrangian and the Turán density of the resulting finite family.

Accordingly, once $\mathcal{A}_{\beta}$ has been constructed, put

$$\mathcal{F}_{r,\beta} := \bigcup_{J \in \mathcal{A}_{\beta}} \text{Ext}(J).$$

The two steps below establish the complete conversion chain

$$\mathcal{C}_\beta = \text{Forb}(\mathcal{A}_\beta), \quad \text{and} \quad \pi(\mathcal{F}_{r,\beta}) = \Lambda_r(\mathcal{C}_\beta) = r! \Upsilon_{\Phi_\beta}^*.$$

We next verify the bounded-witness hypothesis of [Proposition 2.8](#) for the resulting class.

Lemma 7.1. *For a partial $[N]$ -rooting μ of an r -graph G , let $\text{Adm}_\beta(G, \mu)$ mean that (G, μ) is admissible in the sense of [Section 6.2](#). With the explicit local-obstruction bound b_β from [\(5.1\)](#), the predicate Adm_β has the w_β -bounded-witness property for $w_\beta := r \max\{2, b_\beta\}$. It is decidable uniformly in β .*

Proof. Fix an inadmissible partial rooting (G, μ) . By the definition of admissibility, one of the following three alternatives occurs.

- (1) An edge A is not permitted. Take F_μ to be the one-edge subgraph with edge A .
- (2) Two permitted unary edges declare different types for the same data vertex. Take their union for F_μ .
- (3) The typed record system contains a copy of some Q in the finite list $\mathcal{P}_\beta^{\text{raw}}$ from [Lemma 5.4](#). For every type or raw-record requirement in this copy of Q , choose one edge of G that supplies it, and let F_μ be the union of the chosen edges. By the support assertion in [Lemma 5.4](#), every vertex of Q is typed and occurs in a raw-record requirement. The chosen edges therefore contain every vertex of the copy.

In case (3), the obstruction has at most b_β requirements. The witnesses in cases (1), (2), and (3) use at most r , $2r$, and rb_β vertices, respectively, and hence always at most w_β vertices.

We prove persistence. Suppose $F_\mu \subseteq H \subseteq G$ and $\nu \supseteq \mu$, and no newly used root label is mapped into $V(F_\mu) \setminus \text{im } \mu$. For every witness edge $A \in F_\mu$, each vertex which was rooted under μ retains the same root label under ν , and no vertex which was data under μ becomes rooted under ν . Consequently, A has the same root support under ν as under μ , and its data set satisfies

$$A \setminus \text{im } \nu = A \setminus \text{im } \mu.$$

Whether A is permitted and any declaration it makes depend only on its root support, its data set, and the fixed lists of perfect matchings. Since these data are unchanged, every witness edge has the same permitted status and declaration under ν as under μ .

It follows in case (1) that A is still not permitted, and in case (2) that the two conflicting type declarations remain. In case (3), the support assertion just cited ensures that every vertex of the copy lies in $V(F_\mu)$. Hence none becomes rooted under ν , and the displayed persistence condition preserves every declaration selected for the copy. The declared record system of (H, ν) therefore still contains that copy of Q . Thus (H, ν) is inadmissible. Taking the contrapositive gives precisely the persistent witness condition in [Definition 2.7](#).

Finally, whether an edge is permitted is decided by the fixed finite lists of perfect matchings, and $\mathcal{P}_\beta^{\text{raw}}$ is computed by [Lemma 5.4](#). On a finite input there are only finitely many edges and injective maps from these obstructions, so Adm_β is decidable uniformly in β . $\blacksquare$

Combining [Lemma 7.1](#) with [Proposition 2.8](#) gives an effective forbidden family.

Lemma 7.2. *There is an algorithm which, on input β , produces a finite family $\mathcal{A}_\beta$ of r -graphs such that $\mathcal{C}_\beta = \text{Forb}(\mathcal{A}_\beta)$. Every member of $\mathcal{A}_\beta$ has an edge, and $\mathcal{C}_\beta$ contains an r -graph with an edge.*

Proof. The class $\mathcal{C}_\beta$ is monotone. Indeed, if $H \subseteq G$ and μ witnesses $G \in \mathcal{C}_\beta$, restrict μ to the root labels whose image vertices lie in $V(H)$. Every root of a surviving edge also survives, so its root support, permitted status, and declaration are unchanged. Deleting other edges only deletes declarations, and avoidance of the raw local-obstruction list $\mathcal{P}_\beta^{\text{raw}}$ is preserved under taking subgraphs.

[Lemma 7.1](#) and [Proposition 2.8](#), applied with the root-label set $[N]$ and $w = w_\beta$, now give the asserted finite obstruction family. Explicitly, every minimal obstruction has a subobstruction on at most $B_\beta := w_\beta \sum_{j=0}^N (Nw_\beta)^j$ vertices. Enumerating all r -graphs through order B_β , checking every partial injective $[N]$ -rooting, and retaining those r -graphs with no admissible rooting computes $\mathcal{A}_\beta$.

Every edgeless r -graph belongs to $\mathcal{C}_\beta$: use the empty rooting, for which there are no edges that are not permitted, declarations, or local obstructions. Hence every r -graph retained in $\mathcal{A}_\beta$ has an edge.

It remains to note that the class is not edgeless. The matchings $\mathcal{H}_1, \dots, \mathcal{H}_{n_{\text{hole}}}$ were selected from the fixed Baranyai decomposition $\mathfrak{B}_r$. The strict inequality $n_{\text{hole}} < \binom{N-1}{r-1}$ imposed in the proof of Lemma 6.9 leaves a member of $\mathfrak{B}_r$ unselected. Every edge of that member belongs to none of the selected matchings. Root one such edge by its r labels and use it as the sole edge. It is a permitted root-only edge, creates no type or raw-record declaration, and hence is admissible. ■

The finite-root elimination in Lemma 7.2 and the pair-covering transfer in Lemma 2.9 now complete the proof of the main theorem.

Proof of Theorem 1.1. Compute r_{num} as in Lemma 6.9, and fix an arbitrary input $r \geq r_{\text{num}}$. Select the remaining parameters for this r by that lemma; they do not depend on β . Lemma 6.8 computes $\tau_r \in \mathbb{Q} \cap (0, 1)$. Given β , compute the finite obstruction family $\mathcal{A}_\beta$ by Lemma 7.2, and then compute $\mathcal{F}_{r,\beta} = \bigcup_{J \in \mathcal{A}_\beta} \text{Ext}(J)$. The pair-covering transfer in Lemma 2.9 and Theorem 6.3 give $\pi(\mathcal{F}_{r,\beta}) = \Lambda_r(\mathcal{C}_\beta) = r! \Upsilon_{\Phi_\beta}^*$, while Theorem 4.1 gives $\Phi_\beta^* = z_0$ exactly in the nonhalting case and $\Phi_\beta^* > z_0$ in the halting case. The calibration satisfies $\Upsilon_{z_0}^* = J_0/(N+1)^r$, and Lemma 6.8 gives $\Upsilon_z^* > \Upsilon_{z_0}^*$ for every $z > z_0$. Thus the two alternatives have the stated values. ■

Part II

Structural universality

By Theorem 1.1, $\pi(\mathcal{F}_\beta) = \tau_{\text{base}}$ when $\mathcal{U}(\beta)$ does not halt, whereas $\pi(\mathcal{F}_\beta) > \tau_{\text{base}}$ when it halts. This part converts these two density alternatives into a classification of the full extremal limit space of a different finite-family Turán problem. We first set up the limit-space and stability framework and describe the combinatorial gadget that will carry the two possible phases. We then isolate the weighted one-phase/two-phase mechanism, realize this mechanism by a finite ordinary forbidden family, and prove the required structural and stability properties.

The diagram below summarizes the five stages of the structural construction.

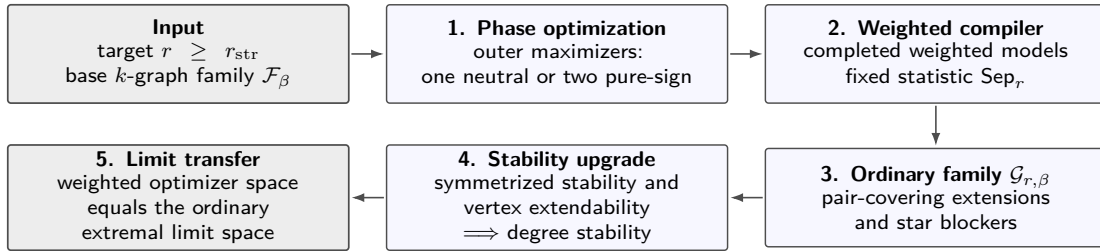


Figure 11: The structural proof chain for $r \geq r_{\text{str}}$. The ordinary forbidden family $\mathcal{G}_{r,\beta}$ is computed from the target uniformity r and the edge list of the base family $\mathcal{F}_\beta = \mathcal{F}_{k,\beta}$ of k -graphs. The value $z_\beta = \pi(\mathcal{F}_\beta)$ enters only the optimizer analysis and is never queried by the compiler.

8 Structural framework and weighted optimizer spaces

We begin by fixing the limit-space and stability terminology used throughout this part. We use the homomorphism-density formulation of dense hypergraph limits. For $r = 2$, Lovász and Szegedy [48] constructed graphon limit objects, and Borgs, Chayes, Lovász, Sós, and Vesztegombi [9, Definition 2.1] called convergence of all homomorphism densities *left convergence*. For general r , Elek and Szegedy [14] developed an ultraproduct construction of hypergraph limits, Austin [1] gave an exchangeability-based account, and Zhao [75, Section 1.4] gave a regularity-based construction and recorded compactness in the homomorphism-density topology.

Fix an integer $r \geq 2$. For finite r -graphs F and G with $v(G) > 0$, let

$$t(F, G) := \frac{\text{hom}(F, G)}{v(G)^{v(F)}}.$$

A sequence (G_n) of finite r -graphs with $v(G_n) \rightarrow \infty$ is *convergent* if $t(F, G_n)$ converges for every finite r -graph F . Identifying sequences with the same limiting homomorphism densities gives a compact metrizable space $\mathfrak{W}_r$. We equip $\mathfrak{W}_r$ with the *homomorphism-density topology*, in which $W_n \rightarrow W$ exactly when $t(F, W_n) \rightarrow t(F, W)$ for every finite r -graph F . For $W \in \mathfrak{W}_r$, put $\text{dens}_r(W) := t(K_r^{(r)}, W)$. If $G_n \rightarrow W$, then $\text{dens}_r(W) = \lim_n |G_n| / \binom{v(G_n)}{r}$.

For a finite family $\mathcal{F}$ of r -graphs, its *extremal limit space* is

$$\mathcal{E}(\mathcal{F}) := \{W \in \mathfrak{W}_r : t(F, W) = 0 \text{ for every } F \in \mathcal{F} \text{ and } \text{dens}_r(W) = \pi(\mathcal{F})\}.$$

This is a nonempty compact set. A sequence (G_n) of $\mathcal{F}$ -free r -graphs is *asymptotically extremal* if $v(G_n) \rightarrow \infty$ and $|G_n| / \binom{v(G_n)}{r} \rightarrow \pi(\mathcal{F})$. Conversely, every $W \in \mathcal{E}(\mathcal{F})$ is the limit of an asymptotically extremal $\mathcal{F}$ -free sequence. Indeed, approximate W by finite r -graphs. The homomorphism density of every $F \in \mathcal{F}$ in these approximants tends to zero, so the hypergraph removal lemma (see Gowers [25] and Rödl and Schacht [61, Theorem 3]) allows us to remove all copies of the finitely many members of $\mathcal{F}$ by deleting $o(n^r)$ edges. This does not change the limit.

For two r -graphs on the same vertex set, put $\text{edit}(G, H) := |G \Delta H|$. If $\mathfrak{C}_n$ is a class of n -vertex r -graphs, write $\text{edit}(G, \mathfrak{C}_n) := \min_{H \in \mathfrak{C}_n} \text{edit}(G, H)$. A finite family $\mathcal{F}$ of r -graphs is *edit stable* with respect to a construction class $\mathfrak{C} = (\mathfrak{C}_n)$ if, for every $\varepsilon > 0$, there are $\delta > 0$ and n_0 such that every $\mathcal{F}$ -free G on $n \geq n_0$ vertices with $|G| \geq \text{ex}(n, \mathcal{F}) - \delta n^r$ satisfies $\text{edit}(G, \mathfrak{C}_n) \leq \varepsilon n^r$.

Following the graph-algebra terminology in Lovász [47, Chapter 6], a *quantum r -graph statistic* is a finite real linear combination of homomorphism densities. Such a statistic defines a continuous function on $\mathfrak{W}_r$. Induced densities are also quantum-graph statistics, by Boolean Möbius inversion.

We now define finite profiles and give precise formulations of the five notions used in Corollary 1.3. We then introduce the weighted optimizer spaces used in the construction.

A finite r -profile is a pair $\mathbf{P} = (\mathbf{a}, \mathcal{M}_{\mathbf{P}})$, where $\mathbf{a} = (a_1, \dots, a_s)$ satisfies $a_i \geq 0$ and $\sum_{i=1}^s a_i = 1$. In an n -vertex realization, type i receives $a_i n + O(1)$ vertices. The set $\mathcal{M}_{\mathbf{P}}$ is a collection of multisets of size r on the type set $[s]$.

For each n , first take $n_i = \lfloor a_i n \rfloor$, and then add one to the coordinates with the largest fractional parts, breaking ties by type index, until $\sum_i n_i = n$. For every labeled n -element vertex set V , let $\mathfrak{B}_{\mathbf{P}}(V)$ contain all r -graphs obtained from ordered partitions $V = V_1 \cup \dots \cup V_s$ with $|V_i| = n_i$. An r -set is an edge exactly when the multiset of the types of its vertices belongs to $\mathcal{M}_{\mathbf{P}}$. We write $\mathfrak{B}_{\mathbf{P}}(n)$ when the ground set is clear. Thus all valid labeled type partitions are included, and the construction class does not depend on a distinguished labeling.

Every sequence $B_n \in \mathfrak{B}_{\mathbf{P}}(n)$ has one common homomorphism-density limit, denoted by $W_{\mathbf{P}}$. Indeed, for each fixed r -graph F , expand a uniformly random map $V(F) \rightarrow V(B_n)$ according to the types of its images. Assignments involving a zero-proportion type have total weight $o(1)$, while, conditional on any assignment to positive-proportion types, the probability of a collision among the finitely many sampled vertices is $O_{F, \mathbf{P}}(n^{-1})$. On the remaining maps, every edge indicator is determined solely by $\mathcal{M}_{\mathbf{P}}$. Thus $t(F, B_n)$ converges to a quantity depending only on $\mathbf{P}$. Since all ordered partitions with the prescribed class sizes give isomorphic r -graphs, the limit is independent of all labeling choices.

Fix $r \geq 2$ and a continuous quantum r -graph statistic Sep_r . For a finite family $\mathcal{F}$ of r -graphs, we define the following properties of its extremal problem.

- (i) It has a *unique extremal limit* if $|\mathcal{E}(\mathcal{F})| = 1$.
- (ii) It has a *connected extremal space* if $\mathcal{E}(\mathcal{F})$ is connected as a subspace of $\mathfrak{W}_r$.
- (iii) It exhibits *Sep_r -symmetry breaking* if there are $W_+, W_- \in \mathcal{E}(\mathcal{F})$ with $\text{Sep}_r(W_+) > 0 > \text{Sep}_r(W_-)$.
- (iv) It has a *Sep_r -separated two-phase decomposition* if, for some $\eta > 0$, every $W \in \mathcal{E}(\mathcal{F})$ satisfies either $\text{Sep}_r(W) \geq \eta$ or $\text{Sep}_r(W) \leq -\eta$, and both alternatives occur.
- (v) It has *Erdős–Simonovits stability* if there is one finite r -profile $\mathbf{P}$, with limit $W_{\mathbf{P}} \in \mathcal{E}(\mathcal{F})$, such that every asymptotically extremal $\mathcal{F}$ -free sequence satisfies $\text{edit}(G_n, \mathfrak{B}_{\mathbf{P}}(v(G_n))) = o(v(G_n)^r)$.

We next introduce the weighted models used in the construction. The optimization is first carried out on finite weighted r -graphs, whereas the structural conclusions concern the limit space $\mathfrak{W}_r$. The following definitions explain how to pass between the two settings.

Fix $r \geq 2$. Finite weighted r -graphs form a convenient dense family of representatives in $\mathfrak{W}_r$. A *weighted r -graph* is a pair $\mathbf{G} = (G, \mathbf{x})$, where G is a finite r -graph and $\mathbf{x} = (x_v)_{v \in V(G)}$ is a probability vector. For a finite r -graph F , define

$$t(F, \mathbf{G}) := \sum_{\phi: V(F) \rightarrow V(G)} \prod_{u \in V(F)} x_{\phi(u)} \prod_{e \in F} \mathbf{1}_{\{\phi(e) \in G\}}.$$

Here, the sum is over all maps $\phi: V(F) \rightarrow V(G)$, but the product of indicators is one precisely when ϕ is a homomorphism. If two vertices of an edge of F receive the same image, the corresponding indicator is zero because G has no repeated-vertex edge. For the one-edge r -graph $K_r^{(r)}$, define

$$\text{dens}_r(\mathbf{G}) := t(K_r^{(r)}, \mathbf{G}) = r! \lambda(G; \mathbf{x}).$$

When the weighted r -graph is displayed through its two components, we abbreviate

$$t(F, G, \mathbf{x}) := t(F, (G, \mathbf{x})), \quad \text{and} \quad \text{dens}_r(G, \mathbf{x}) := \text{dens}_r((G, \mathbf{x})).$$

Thus $\lambda_r(G) = \max_{\mathbf{x}} \text{dens}_r(G, \mathbf{x})$, where the maximum is over all probability weightings of G .

The following elementary invariance will be used whenever a weighted vertex is split into clones.

Lemma 8.1. *Let $(G, \mathbf{x})$ be a weighted r -graph, and let $(G', \mathbf{x}')$ be obtained by cloning $v \in V(G)$ into a clone class C . Suppose that $x'_u = x_u$ for every $u \in V(G) \setminus \{v\}$ and that the vertices in C have nonnegative weights summing to x_v . Then $t(F, G', \mathbf{x}') = t(F, G, \mathbf{x})$ for every finite r -graph F . The same conclusion holds for every iterated cloning when weights are split in this way at each step.*

Proof. Project every member of C back to v and group the weighted maps $\phi: V(F) \rightarrow V(G')$ by their projected map $\bar{\phi}: V(F) \rightarrow V(G)$. If an edge of F contains two vertices projected to v , then its image is not an edge in G ; after cloning it is also not an edge, even when the two vertices choose different clones, because no host edge contains two members of C . Otherwise each edge uses at most one preimage of v , so all clone choices have the same edge indicator. Summing their weights replaces every factor x'_v by $(\sum_{y \in C} x'_y)^a = x_v^a$. Thus the total contribution of every projected map is unchanged. Iteration proves the final assertion. $\blacksquare$

For completeness, every finite weighted r -graph represents a well-defined point of $\mathfrak{W}_r$. Fix $\mathbf{G} = (G, \mathbf{x})$, choose rational probability weightings $\mathbf{x}^{(j)} \rightarrow \mathbf{x}$, and let D_j be a common positive denominator of the coordinates of $\mathbf{x}^{(j)}$. Choose integers $m_j \rightarrow \infty$, and form the blowup B_j of G whose vertex class over $v \in V(G)$ has size $m_j D_j x_v^{(j)}$, omitting a class when this number is zero. Then $v(B_j) = m_j D_j \rightarrow \infty$. Grouping homomorphisms to B_j by their projection to G gives, for every finite r -graph F ,

$$t(F, B_j) = t(F, G, \mathbf{x}^{(j)}) \longrightarrow t(F, G, \mathbf{x}).$$

Thus (B_j) is convergent and represents a point determined by the weighted densities of $\mathbf{G}$, independently of the rational approximations and multipliers. Taking $F = K_r^{(r)}$ shows that its one-edge density is $\text{dens}_r(\mathbf{G}) = r! \lambda(G; \mathbf{x})$, with the normalization used above.

Enumerate the finite r -graphs as $F_1, F_2, \dots$ and put

$$d_{\text{hom}}(W, W') := \sum_{i \geq 1} 2^{-i} |t(F_i, W) - t(F_i, W')|.$$

The density results of Elek and Szegedy [14] and Zhao [75] show that this metric induces the topology already fixed on $\mathfrak{W}_r$. Since the ordinary graphs in any defining sequence, equipped with their uniform weightings, converge to the same limit, finite weighted r -graphs are dense in $\mathfrak{W}_r$. We use dens_r for the one-edge density of both weighted r -graphs and limit objects.

For a class $\mathcal{C}$ of finite r -graphs containing an r -graph with nonempty vertex set, let $\mathfrak{W}(\mathcal{C})$ be the closure of all weighted members $(G, \mathbf{x})$ with $G \in \mathcal{C}$. By the definitions of Λ_r and dens_r , we have

$$\Lambda_r(\mathcal{C}) = \max \{ \text{dens}_r(W) : W \in \mathfrak{W}(\mathcal{C}) \} = r! \sup_{G \in \mathcal{C}} \lambda(G),$$

and define the optimizer space

$$\mathfrak{L}(\mathcal{C}) := \{W \in \mathfrak{W}(\mathcal{C}) : \text{dens}_r(W) = \Lambda_r(\mathcal{C})\}.$$

Compactness gives the maximum and shows that $\mathfrak{L}(\mathcal{C})$ is nonempty and compact whenever $\mathcal{C}$ contains an r -graph with at least one edge.

We will use the following compactness observation several times to pass from a classification of the exact maximizers to a corresponding statement for near-maximizers.

Lemma 8.2. *Let X be a compact metric space, let $f: X \rightarrow \mathbb{R}$ be continuous, and put $f^* := \max_X f$ and $X_{\max} := f^{-1}(f^*)$. Then, for every $\varepsilon > 0$, there is $\delta > 0$ such that every $x \in X$ satisfying $f(x) \geq f^* - \delta$ also satisfies $\text{dist}(x, X_{\max}) < \varepsilon$. Consequently, every sequence with $f(x_n) \rightarrow f^*$ has distance tending to zero from $X_{\max}$.*

Indeed, the compact set $\{x : \text{dist}(x, X_{\max}) \geq \varepsilon\}$ is disjoint from $X_{\max}$, so the maximum of f on it is strictly smaller than f^* .

Applied to $X = \mathfrak{W}(\mathcal{C})$ and $f = \text{dens}_r$, this gives the near-optimal assertions in [Theorem 11.1](#) once the optimizer set has been classified.

9 The phase gadget and its analytic mechanism

Recall the fixed notation $k = r_{\text{num}}$, $\mathcal{F}_\beta = \mathcal{F}_{k,\beta}$, and $\tau_{\text{base}} = \tau_k$ from the Introduction.

Every member of $\mathcal{F}_\beta$ has no isolated vertices, by the definition of the pair-covering extensions used to construct $\mathcal{F}_\beta$. Hence $\mathcal{F}_\beta^\circ = \mathcal{F}_\beta$. Apply the finite homomorphic closure from [Lemma 2.4](#) and set

$$\mathcal{C}_\beta^{\text{in}} := \text{Forb}(\mathcal{Q}(\mathcal{F}_\beta)), \quad \text{and} \quad z_\beta = \Lambda_k(\mathcal{C}_\beta^{\text{in}}) = \pi(\mathcal{F}_\beta). \quad (9.1)$$

Then $z_\beta = \tau_{\text{base}}$ in the nonhalting case and $z_\beta > \tau_{\text{base}}$ in the halting case. The inner class $\mathcal{C}_\beta^{\text{in}}$ is blowup closed; this property is used when data vertices are split in [Section 10](#).

The rooted construction will encode two signed copies of the inner problem and reduce their interaction to a symmetric three-variable optimization. We first analyze that optimization abstractly in [Section 9.1](#); its finite rooted realization is constructed in [Section 10](#).

9.1 The abstract phase objective

We now isolate the real-variable optimization underlying the gadget of [Section 9](#). Here u is the total data weight, a and b are the total weights of the two sign classes, and z is the optimum of the inner Turán problem.

Fix an integer $k \geq 3$, a real number $\tau \in [0, 1)$, and a nonempty compact interval $I \subset (0, 1)$. Let $\Phi_{\text{base}}, \alpha, \gamma: I \rightarrow \mathbb{R}$ extend to C^2 functions on an open interval containing I , with $\alpha(u), \gamma(u) > 0$ there. Put

$$\Omega_I := \{(u, a, b) : u \in I, a, b \geq 0, \text{ and } a + b \leq u\}.$$

Thus the two sign classes together use at most the total data weight. For $z \geq \tau$, define $\Phi_z: \Omega_I \rightarrow \mathbb{R}$ by

$$\Phi_z(u, a, b) := \Phi_{\text{base}}(u) + \alpha(u)(z - \tau)(a^k + b^k) - \gamma(u)(a + b)^{k+1}. \quad (9.2)$$

Since Ω_I is compact and Φ_z is continuous, its maximum exists; write

$$\Phi_z^* := \max_{(u,a,b) \in \Omega_I} \Phi_z(u, a, b).$$

For $z \in [\tau, 1]$, define

$$J_z(u) := \frac{k^k}{(k+1)^{k+1}} (z - \tau)^{k+1} \frac{\alpha(u)^{k+1}}{\gamma(u)^k}.$$

Lemma 9.1. *Suppose that Φ_{base} has a unique maximizer u_0 on I . Let $U \subsetneq I$ be a proper closed interval that is a relative neighborhood of u_0 in I , and assume that the following conditions hold.*

(A1) *There is $\mu > 0$ such that $\Phi_{\text{base}}''(u) \leq -\mu$ for every $u \in U$.*

(A2) We have $\sup_{z \in [\tau, 1]} \sup_{u \in I} J_z(u) < \frac{1}{2}(\Phi_{\text{base}}(u_0) - \max_{I \setminus \text{int}_I U} \Phi_{\text{base}})$.

(A3) We have $\sup_{z \in [\tau, 1]} \sup_{u \in U} |J_z''(u)| < \mu/2$.

(A4) We have $k\alpha(u)(1 - \tau)/((k + 1)\gamma(u)) < u$ for every $u \in U$.

Then, for every $z \in [\tau, 1]$, the maximizers of Φ_z on Ω_I are as follows.

- (i) If $z = \tau$, the unique maximizer is $(u_0, 0, 0)$.
- (ii) If $z > \tau$, there is a unique $u_z \in \text{int}_I U$, and the maximizers are exactly $(u_z, t_z, 0)$ and $(u_z, 0, t_z)$, where $t_z := k\alpha(u_z)(z - \tau)/((k + 1)\gamma(u_z)) > 0$.

Proof. Put $t = a + b$. Strict convexity of $x \mapsto x^k$ gives $a^k + b^k \leq t^k$, with equality for $t > 0$ exactly at $(a, b) = (t, 0)$ or $(0, t)$. For fixed u , the scalar function

$$f_{u,z}(t) := \alpha(u)(z - \tau)t^k - \gamma(u)t^{k+1}$$

is uniquely maximized at $t = 0$ when $z = \tau$. When $z > \tau$, its unique positive critical point on $[0, \infty)$ is $t_z(u) := k\alpha(u)(z - \tau)/((k + 1)\gamma(u))$. Thus the maximum over the constrained interval $[0, u]$ is at most $J_z(u)$ for every $u \in I$. If $u \in U$, then condition (A4) makes $t_z(u)$ feasible, so the constrained maximum equals $J_z(u)$.

We localize the maximizers by comparing Φ_z outside $\text{int}_I U$ with its value at the neutral point $(u_0, 0, 0)$. If $u \in I \setminus \text{int}_I U$, then condition (A2) gives

$$\max_{\substack{a, b \geq 0 \\ a+b \leq u}} \Phi_z(u, a, b) \leq \Phi_{\text{base}}(u) + J_z(u) \leq \max_{I \setminus \text{int}_I U} \Phi_{\text{base}} + \sup_{v \in I} J_z(v) < \Phi_{\text{base}}(u_0) = \Phi_z(u_0, 0, 0).$$

Hence every maximizer lies in $\text{int}_I U$. On U , conditions (A1) and (A3) give $(\Phi_{\text{base}} + J_z)'' \leq -\mu/2$, so $\Phi_{\text{base}} + J_z$ is strictly concave there. If $z > \tau$, it has the unique maximizer u_z , and $t_z = t_z(u_z)$; the equality case of the convexity bound above gives the two sign choices in part (ii). When $z = \tau$, $J_z = 0$, so the outer maximizer is u_0 and $t = 0$. $\blacksquare$

The two alternatives in Lemma 9.1 are illustrated schematically below.

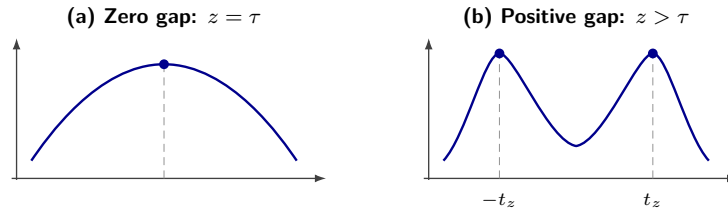


Figure 12: The one-phase/two-phase transition for $\Phi_z(u, a, b) = \Phi_{\text{base}}(u) + \alpha(u)(z - \tau)(a^k + b^k) - \gamma(u)(a + b)^{k+1}$. Each curve gives the largest value of $\Phi_z(u, a, b)$ when the difference $d = a - b$ is fixed. (a) At the baseline $z = \tau$, the unique maximizer is the neutral point $(u_0, 0, 0)$. (b) When $z > \tau$, this maximizer splits into two symmetric choices, $(u_z, t_z, 0)$ and $(u_z, 0, t_z)$, corresponding to the positive and negative phases.

By Lemma 8.2, for each fixed $z \in [\tau, 1]$, every sequence in Ω_I whose Φ_z -values tend to Φ_z^* has distance tending to the corresponding set of maximizers described in the lemma.

For later edit estimates we record a quantitative consequence for each fixed positive gap.

Lemma 9.2. *Under the hypotheses of Lemma 9.1, for every fixed $z > \tau$ there are $c_z, \eta_z > 0$ such that every $(u, a, b) \in \Omega_I$ satisfying $\Phi_z(u, a, b) \geq \Phi_z^* - \eta_z$ also satisfies*

$$\Phi_z^* - \Phi_z(u, a, b) \geq c_z((u - u_z)^2 + (a + b - t_z)^2 + ab).$$

Proof. By Lemma 8.2 and the phase classification, after reducing η_z every point under consideration lies in a fixed small neighborhood of one of the two pure maximizers. In particular, u lies in a compact interval $U' \subseteq \text{int}_I U$ containing u_z , and $t = a + b$ lies near t_z . On U' the positive scalar critical point is feasible by condition (A4). Recall that

$$f_{u,z}(t) = \alpha(u)(z - \tau)t^k - \gamma(u)t^{k+1}, \quad \text{and} \quad J_z(u) = \max_{0 \leq s \leq u} f_{u,z}(s).$$

For every such feasible (u, a, b) , we have

$$\begin{aligned}\Phi_z^* - \Phi_z(u, a, b) &= ((\Phi_{\text{base}} + J_z)(u_z) - (\Phi_{\text{base}} + J_z)(u)) \\ &\quad + (J_z(u) - f_{u,z}(t)) + \alpha(u)(z - \tau)(t^k - a^k - b^k).\end{aligned}$$

All three terms are nonnegative. Strong concavity of $\Phi_{\text{base}} + J_z$ gives a positive quadratic lower bound for the first term near u_z . The positive maximizer $t_z(u) = k\alpha(u)(z - \tau)/((k+1)\gamma(u))$ is C^1 on U' , and $f''_{u,z}(t_z(u)) = -(k+1)\gamma(u)t_z(u)^{k-1} < 0$. Consequently, in a sufficiently small neighborhood of (u_z, t_z) , the second term is bounded below by a positive multiple of $(t - t_z(u))^2$. Since $t_z(\cdot)$ is C^1 on the compact interval U' , there is $L_z < \infty$ such that $|t_z(u) - t_z| \leq L_z|u - u_z|$. Let $a_z > 0$ be a quadratic lower-bound coefficient for the first term and let $b_z > 0$ be one for the second term. Decrease b_z , if necessary, so that $b_z L_z^2 \leq a_z/2$. Using $(A - B)^2 \geq A^2/2 - B^2$ with $A = t - t_z$ and $B = t_z(u) - t_z$, we obtain

$$(t - t_z(u))^2 \geq \frac{1}{2}(t - t_z)^2 - L_z^2(u - u_z)^2.$$

It follows that the sum of the first two terms is at least $(a_z/2)(u - u_z)^2 + (b_z/2)(t - t_z)^2$.

Finally, we have

$$t^k - a^k - b^k = \sum_{j=1}^{k-1} \binom{k}{j} a^j b^{k-j} \geq k ab (a^{k-2} + b^{k-2}).$$

When t is close to $t_z > 0$, at least one of a, b is at least $t_z/4$; hence $a^{k-2} + b^{k-2}$ is bounded below by a positive constant depending only on z . Combining the three estimates gives the asserted bound locally. The feasible domain is compact and the maximizer set consists of the two pure points, so the loss has a positive minimum on the complement of this local neighborhood. Decreasing c_z and η_z completes the proof. $\blacksquare$

10 Finite rooted realization of the phase function

We now realize the phase function by a finite rooted r -graph construction. The goal is to preserve its optimizer space while making all parameters effective. Write the rational threshold in lowest terms as $\tau_{\text{base}} = p/q$, where $p, q \in \mathbb{N}$, $0 < p < q$, and $\gcd(p, q) = 1$. We fix the number of translation orbits in the penalty design by

$$d_P := 8kq. \tag{10.1}$$

Once k and the reduced fraction p/q are fixed, so is d_P , independently of β . The remaining structural data will depend effectively only on these parameters and the target uniformity r .

10.1 The root frame, support designs, and rooted edge types

The frame and the support designs have different roles. Asymmetry will later recover the individual root labels, while pair coverage prevents edgewise-injective homomorphisms from identifying vertices that must remain distinct. Regularity makes the equal-root vector stationary, and the orbit counts give the coefficients α and γ in the scalar phase function.

The padding parameter $c \geq 4$ will be chosen sufficiently large in terms of k, p, q , and we put $r := k + c$. For a uniform hypergraph $\mathcal{S}$ on a finite root-label set M , we use $\lambda(\mathcal{S}; \mathbf{y})$ for the same edge polynomial as in Section 2. The uniformity of $\mathcal{S}$ will always be clear from context.

For positive integers h, m , an h -set $A \subseteq \mathbb{Z}_m$, and $t \in \mathbb{Z}_m$, put $A + t := \{a + t \pmod{m} : a \in A\}$. The translation orbit of A is $\{A + t : t \in \mathbb{Z}_m\}$. We call this orbit *full* if $A + t = A$ only for $t = 0$, or equivalently if it contains exactly m distinct h -sets. Every full orbit is h -regular. Indeed, translation symmetry gives every vertex the same degree, and counting incidences shows that this degree is h .

Lemma 10.1. *One can effectively compute an integer $c_{\text{des}} = c_{\text{des}}(k, p, q)$ such that, for every $c \geq c_{\text{des}}$ (with $r = k + c$), one can effectively construct a root set M of order m , an r -graph $\mathcal{J} \subseteq \binom{M}{r}$, pairwise disjoint c -graphs $\mathcal{Z}_+, \mathcal{Z}_-, \mathcal{T}_+, \mathcal{T}_-, \mathcal{D}_{\text{disp}} \subseteq \binom{M}{c}$, and a $(c-1)$ -graph $\mathcal{P} \subseteq \binom{M}{c-1}$ with the following properties.*

- (i) *The hypergraph $\mathcal{J}$ is 2-regular and asymmetric, and every pair of root labels is contained in an r -set outside $\mathcal{J}$. On the box $0 \leq y_i \leq 2/m$, we have $\|D^2\lambda(\mathcal{J}; \mathbf{y})\|_{\text{op}} \leq 2(r-1)(2/m)^{r-2}$. At every constant root vector $\mathbf{y} = t\mathbf{1}$, we have $\langle \nabla_{\mathbf{y}}\lambda(\mathcal{J}; t\mathbf{1}), \mathbf{h} \rangle = 0$ for every $\mathbf{h} \in \mathbf{1}^\perp$.*

- (ii) One has $r(r+4)/2 \leq m \leq r(r+5)/2$.
- (iii) Each of the five c -graphs is a union of full translation orbits on $\mathbb{Z}_m$, while $\mathcal{P}$ is a union of d_P full $(c-1)$ -orbits. The numbers of c -orbits assigned to $\mathcal{Z}_+$, $\mathcal{Z}_-$, $\mathcal{T}_+$, $\mathcal{T}_-$, and $\mathcal{D}_{\text{disp}}$ are q , q , p , p , and 1 , respectively. Consequently, all these designs are regular, and $|\mathcal{Z}_+| = |\mathcal{Z}_-| = qm$, $|\mathcal{T}_+| = |\mathcal{T}_-| = pm$, $|\mathcal{D}_{\text{disp}}| = m$, and $|\mathcal{P}| = d_P m$.
- (iv) The labeled neighborhood designs $\mathcal{B}_+ := \mathcal{D}_{\text{disp}} \cup \mathcal{T}_- \cup \mathcal{Z}_+$ and $\mathcal{B}_- := \mathcal{D}_{\text{disp}} \cup \mathcal{T}_+ \cup \mathcal{Z}_-$ are distinct.

We call the members of $\mathcal{J}$ the *frame holes*, because the root-only frame used below has precisely the complementary edge set $\binom{M}{r} \setminus \mathcal{J}$.

Proof. We first choose the asymmetric frame with a controlled number of roots. Baron and Imrich [5, Satz 2, p. 140] proved that, for every integer $r > 6$, the minimum order of an asymmetric r -regular graph is $r+4$ when r is even and $r+5$ when r is odd. Put

$$n := \begin{cases} r+4, & r \text{ even,} \\ r+5, & r \text{ odd,} \end{cases}$$

and choose an asymmetric r -regular graph G on $[n]$. This choice is effective. Enumerate the finitely many labeled graphs on $[n]$, test r -regularity, and retain a graph precisely when the identity is its only automorphism, as can be checked by testing all $n!$ vertex permutations; take the lexicographically first retained graph. The cited theorem guarantees that this finite search succeeds for every $r \geq 7$.

Let $\mathcal{J}$ be the hypergraph dual of G . Put $M := G$, viewed as the edge set of G , and, for each $v \in V(G)$, let the set of the r graph-edges incident with v be an edge of $\mathcal{J}$. Thus $\mathcal{J}$ is r -uniform and every root has degree two.

The hyperedges of $\mathcal{J}$ are the distinct vertex stars of G . Two such hyperedges intersect if and only if the corresponding graph vertices are adjacent. Hence the intersection graph of the hyperedges of $\mathcal{J}$ is exactly G , and every automorphism of $\mathcal{J}$ induces an automorphism of G . Since G is asymmetric, every hyperedge of $\mathcal{J}$ is fixed. Each root is the unique intersection of the two hyperedges corresponding to its endpoints in G , so every root is fixed as well. Therefore $\mathcal{J}$ is asymmetric. Moreover, we have $m = |M| = nr/2$, which gives the stated interval. Two distinct roots, viewed as two edges of G , lie in at most one common vertex star. Since $\binom{m-2}{r-2} > 1$ for $r \geq 7$, every pair is contained in an r -set outside $\mathcal{J}$.

On the box $0 \leq y_i \leq 2/m$, every row of $D^2\lambda(\mathcal{J}; \mathbf{y})$ has nonnegative entries and row sum at most $2(r-1)(2/m)^{r-2}$, because every root has degree two in $\mathcal{J}$. The operator norm is at most the maximum row sum. Moreover, 2-regularity makes all coordinate derivatives equal at a constant root vector, so the gradient is parallel to $\mathbf{1}$. This completes the proof of part (i).

Identify M with $\mathbb{Z}_m$. We give an explicit orbit system. For $h \in \{c, c-1\}$ let $s_c := 2q + 2p + 1$ and $s_{c-1} := d_P$, and, for $1 \leq j \leq s_h$, define the integer h -set $A_{h,j} := \{0, 1, \dots, h-2\} \cup \{2h+j\}$. After increasing c we may assume $m > 4c + 2 \max\{s_c, s_{c-1}\} + 10$. Hence every $A_{h,j}$ embeds in $\mathbb{Z}_m$ without wrap-around. Its cyclic gap sequence has $h-2$ consecutive unit gaps, one gap of length $h+j+2$, and one gap of length $m-(2h+j)$; by the preceding inequality, the last gap is the unique largest gap. A translation preserving $A_{h,j}$ must therefore fix the successor of this largest gap, namely 0, and is the identity. The same gap sequence also shows that $A_{h,j}$ and $A_{h,j'}$ are translates only when $j = j'$. Consequently the full translation orbit $\{A_{h,j} + t : t \in \mathbb{Z}_m\}$ has m distinct edges, and these orbits are pairwise disjoint for fixed h . Every such orbit is h -regular, because each of its m edges has size h and the translation action is transitive.

Allocate disjoint ranges of the s_c many c -orbits, with q assigned to each of $\mathcal{Z}_+$ and $\mathcal{Z}_-$, p to each of $\mathcal{T}_+$ and $\mathcal{T}_-$, and one to $\mathcal{D}_{\text{disp}}$. Allocate the d_P many $(c-1)$ -orbits to $\mathcal{P}$. The five c -uniform families are pairwise disjoint by construction, and the two neighborhood designs are distinct because, for example, $\mathcal{Z}_+ \subseteq \mathcal{B}_+ \setminus \mathcal{B}_-$. This proves parts (iii)–(iv).

Claim 10.2. *Let $\mathcal{S}$ be the union of s full translation orbits of h -sets. Then every root has degree sh in $\mathcal{S}$, and on the box $0 \leq y_i \leq 2/m$ we have $\|D^2\lambda(\mathcal{S}; \mathbf{y})\|_{\text{op}} \leq sh(h-1)(2/m)^{h-2}$. In particular, at every constant root vector $\mathbf{y} = t\mathbf{1}$, the gradient of $\lambda(\mathcal{S}; \mathbf{y})$ is parallel to $\mathbf{1}$, so its restriction to the tangent space $\mathbf{1}^\perp$ is zero.*

Proof of the claim. Every root has degree sh in $\mathcal{S}$. The row of $D^2\lambda(\mathcal{S}; \mathbf{y})$ indexed by i has nonnegative entries, and

$$\sum_{j \neq i} \partial_{ij} \lambda(\mathcal{S}; \mathbf{y}) \leq (2/m)^{h-2} \sum_{S \in \mathcal{S}: i \in S} |S \setminus \{i\}| = sh(h-1)(2/m)^{h-2}.$$

The operator norm is at most the maximum row sum. Finally, regularity makes all coordinate derivatives equal at a constant vector, so the gradient is parallel to $\mathbf{1}$. $\blacksquare$

It remains to obtain one lower bound that works for every larger c . The frame construction works for every $r \geq 7$, and the orbit construction works whenever $r(r+4)/2 > 4c + 2 \max\{2q + 2p + 1, d_P\} + 10$. The left side is a quadratic polynomial in c because $r = k + c$, whereas the right side is linear. Thus one can compute c_{des} by increasing $c \geq \max\{4, 7 - k\}$ until this inequality holds and its difference has positive forward difference; it then holds for every larger c . For each such input c , the finite graph search above followed by the displayed orbit formulas constructs all the required objects. This proves both the uniform lower bound and its effectivity. $\blacksquare$

Fix the root-label set M , the frame-hole r -graph $\mathcal{J}$, and the support designs $\mathcal{D}_{\text{disp}}, \mathcal{Z}_+, \mathcal{Z}_-, \mathcal{T}_+, \mathcal{T}_-$, and $\mathcal{P}$ supplied by Lemma 10.1. Fix an r -graph G together with a partial M -rooting μ , and put $U := V(G) \setminus \text{im } \mu$. We call the vertices of U *data vertices*. Relative to μ , the *rooted edge type* of an edge is its category in the following list, with the sign σ included for raw record and threshold edges. An edge is *permitted* if it has one of these rooted edge types.

Frame edges.

A root-only r -set is permitted exactly when its root support does not belong to $\mathcal{J}$. Every edge consisting of one data vertex and $r - 1$ distinct roots is permitted.

Dispersion edges.

For $D \in \mathcal{D}_{\text{disp}}$, every edge $X \cup D$ with $X \in \binom{U}{k}$ is permitted.

Raw record edges.

For $\sigma \in \{+, -\}$ and $Z \in \mathcal{Z}_\sigma$, every edge $X \cup Z$ with $X \in \binom{U}{k}$ is a raw σ -record edge. Its data set X is a σ -record.

Threshold edges.

For $\sigma \in \{+, -\}$ and $T \in \mathcal{T}_\sigma$, every edge $X \cup T$ with $X \in \binom{U}{k}$ is permitted.

Penalty edges.

For $P \in \mathcal{P}$, every edge $Y \cup P$ with $Y \in \binom{U}{k+1}$ is permitted.

Note that the rooted edge type of every permitted edge is unique. Indeed, the numbers of data vertices are respectively 0, 1, k , and $k + 1$ for a root-only frame edge, a one-data frame edge, a k -data edge, and a penalty edge. These numbers are distinct because $k \geq 3$. Among the k -data edges, the pairwise disjoint support designs $\mathcal{D}_{\text{disp}}, \mathcal{Z}_+, \mathcal{Z}_-, \mathcal{T}_+$, and $\mathcal{T}_-$ determine the edge type uniquely.

For $\sigma \in \{+, -\}$, the σ -*projection* is the k -graph G_σ on the data vertex set U whose edges are the data sets of the raw σ -record edges. Equivalently,

$$G_\sigma := \left\{ X \in \binom{U}{k} : X \cup Z \in G \text{ for some } Z \in \mathcal{Z}_\sigma \right\}.$$

Put $U_\sigma := \bigcup_{X \in G_\sigma} X$. Thus U_σ is the set of data vertices that occur in raw σ -record edges. A rooted r -graph whose edges are all permitted is *admissible* when it contains none of the following forbidden configurations.

- (V1) A data vertex in $U_+ \cap U_-$.
- (V2) A threshold edge $X \cup T$ with $T \in \mathcal{T}_\sigma$ and $X \subseteq U_\sigma$.
- (V3) A penalty edge $Y \cup P$ with $P \in \mathcal{P}$ and $Y \subseteq U_+ \cup U_-$.
- (V4) A copy of a member of $\mathcal{Q}(\mathcal{F}_\beta)$ in G_σ , for either $\sigma \in \{+, -\}$.

Here membership of a vertex in U_σ is witnessed by one raw σ -record edge containing that vertex, and a copy in (V4) is vertex-injective. More explicitly, a witness for (V1) consists of two raw record edges of opposite signs meeting in a designated data vertex. A witness for (V2) consists of the offending threshold edge together with one raw σ -record edge through each of its data vertices. A witness for (V3) consists of

the offending penalty edge together with, for each of its data vertices y , a choice $\sigma_y \in \{+, -\}$ and one raw σ_y -record edge covering y . A witness for (V4) consists of a vertex-injective projected copy and, for every edge of that copy, one raw record edge of the relevant sign realizing the projected edge. There are only finitely many overlap types among the auxiliary data vertices in these witnesses, and the frozen-root identifications described below form a finite effective list. Thus each of (V1)–(V4) has a finite edge witness, each sign projection is $\mathcal{Q}(\mathcal{F}_\beta)$ -free, and the resulting witness catalogue is finite and effective because the support systems and $\mathcal{Q}(\mathcal{F}_\beta)$ are finite.

Define

$$\mathcal{D}_\beta := \{G : G \text{ is an } r\text{-graph admitting an admissible partial } M\text{-rooting}\}.$$

To pass from these rooted admissibility conditions to a finite unrooted obstruction family for $\mathcal{D}_\beta$, we must also detect violations when some auxiliary vertices coincide in the host. A *rooted configuration* consists of a finite r -graph on named vertices, specified root labels on some of these vertices, and finitely many requirements $x \neq y$ between named vertices that must remain distinct even when they do not lie in a common edge. A *frozen-root identification* is an equivalence relation on the named vertices which fixes all root names, never identifies a pair required to be distinct, and never identifies two vertices of one edge. Applying it replaces each equivalence class by one named vertex and merges repeated image edges. There are finitely many such identifications, and they can be enumerated. We include every resulting configuration that remains injective on each displayed edge and preserves every required distinction.

Let (G, μ) be an admissible partially M -rooted r -graph, and put $U := V(G) \setminus \text{im } \mu$. An *admissible structural-completion* of (G, μ) is an admissible totally M -rooted r -graph $(\hat{G}, \hat{\mu})$ satisfying the following conditions.

- (i) The graph $\hat{G}$ contains G as a subgraph, $\hat{\mu}$ extends μ , and $V(\hat{G}) \setminus V(G) = \hat{\mu}(M \setminus \text{dom } \mu)$. Thus the only new vertices are those carrying the previously missing root labels.
- (ii) For every $B \in \binom{M}{r} \setminus \mathcal{J}$, one has $\hat{\mu}(B) \in \hat{G}$. For every $v \in U$ and $B \in \binom{M}{r-1}$, one has $\hat{\mu}(B) \cup \{v\} \in \hat{G}$. For every $D \in \mathcal{D}_{\text{disp}}$ and $X \in \binom{U}{k}$, one has $\hat{\mu}(D) \cup X \in \hat{G}$.
- (iii) For every $\sigma \in \{+, -\}$, $T \in \mathcal{T}_\sigma$, and $X \in \binom{U}{k}$ with $X \not\subseteq U_\sigma$, one has $\hat{\mu}(T) \cup X \in \hat{G}$. For every $P \in \mathcal{P}$ and $Y \in \binom{U}{k+1}$ with $Y \not\subseteq U_+ \cup U_-$, one has $\hat{\mu}(P) \cup Y \in \hat{G}$.
- (iv) For every $\sigma \in \{+, -\}$, $X \in G_\sigma$, and $Z \in \mathcal{Z}_\sigma$, one has $\hat{\mu}(Z) \cup X \in \hat{G}$.
- (v) If $\hat{G}_\sigma$ and $\hat{U}_\sigma$ denote the σ -projection and its covered vertex set determined by $(\hat{G}, \hat{\mu})$, then $\hat{G}_\sigma = G_\sigma$ and $\hat{U}_\sigma = U_\sigma$ for every $\sigma \in \{+, -\}$.

Throughout this part, we abbreviate *admissible structural-completion* to *admissible completion*. Whenever a weighting on $V(G)$ is carried to an admissible completion, we extend it by zero on the new vertices $\hat{\mu}(M \setminus \text{dom } \mu)$; in particular, the extended vector is again a probability weighting.

The next lemma passes from these finite rooted conditions to the unrooted class and records the completion used in the subsequent optimization.

Lemma 10.3. *The class $\mathcal{D}_\beta$ is monotone and has an effectively computable finite obstruction family $\mathcal{A}_\beta^{\text{root}}$. Moreover, every admissible partially M -rooted r -graph has an admissible completion.*

Proof. We first prove monotonicity and the bounded-witness property. If a partial rooting (G, μ) is inadmissible, then either some edge of G has no permitted rooted edge type or one of (V1)–(V4) fails. In the first case, take that edge as the witness. In the second case, take the union of the edges in one of the listed forbidden rooted configurations witnessing the failure. The list contains all configurations obtained by frozen-root identifications and is finite and effective, so the orders of these witnesses are bounded by an effectively computable constant w .

Suppose that F_μ is such a witness, $F_\mu \subseteq H \subseteq G$, and $\nu \supseteq \mu$ assigns no new root label to a vertex of F_μ that was unrooted under μ . Every vertex of the witness then retains its root or data status, every witnessing edge retains its rooted edge type, and the same forbidden configuration remains present in (H, ν) . Thus (H, ν) is inadmissible, and the admissibility predicate has the w -bounded-witness property of Definition 2.7.

To prove that $\mathcal{D}_\beta$ is monotone, let $H \subseteq G$ with $G \in \mathcal{D}_\beta$, and restrict an admissible rooting μ of G to the root labels whose image vertices lie in $V(H)$. If this restricted rooting made H inadmissible, choose

a witness F as above. Restoring the omitted roots places no new root on F , since their image vertices lie outside $V(H)$. The same witness would therefore make (G, μ) inadmissible, a contradiction. Hence $H \in \mathcal{D}_\beta$.

Admissibility is decidable from the finite support tables, the finite family $\mathcal{Q}(\mathcal{F}_\beta)$, and the finite tests (V1)–(V4). Applying Proposition 2.8 gives the finite effective obstruction family $\mathcal{A}_\beta^{\text{root}}$.

For completion, fix an admissible partially M -rooted r -graph (G, μ) . For every label in $M \setminus \text{dom } \mu$, add a fresh vertex and map the label to it; this defines the total rooting $\hat{\mu}$. Starting with all edges of G , for each sign σ complete every record in G_σ over all supports in $\mathcal{Z}_\sigma$. Next add all frame and dispersion edges, every threshold edge $\hat{\mu}(T) \cup X$ with $X \not\subseteq U_\sigma$, and every penalty edge $\hat{\mu}(P) \cup Y$ with $Y \not\subseteq U_+ \cup U_-$. Denote the resulting r -graph by $\hat{G}$.

The added record edges use only data sets already in G_σ , and none of the other added edges is a raw record edge. Hence $\hat{G}_\sigma = G_\sigma$ and $\hat{U}_\sigma = U_\sigma$ for each sign σ . It follows that (V1) and (V4) are unchanged, while the subset conditions used to add the threshold and penalty edges ensure that (V2) and (V3) remain false. Thus $(\hat{G}, \hat{\mu})$ is admissible and satisfies all five conditions in the definition of an admissible completion. ■

A *completed weighted model* is a triple $(D, \mu, \mathbf{x})$, where (D, μ) is an admissible completion and $\mathbf{x}$ is a probability weighting on $V(D)$. We write $\lambda(D; \mathbf{x})$ for its Lagrangian value. The rooting remains part of the model because it determines the root vector, the data vertices, and the two sign projections.

For a completed weighted model $(D, \mu, \mathbf{x})$, let G_+ and G_- be its two sign projections and let U_+ and U_- be their covered vertex sets. We use the following notation throughout this part:

$$\begin{aligned} m &:= |M|, & U &:= V(D) \setminus \text{im } \mu, & u &:= \sum_{v \in U} x_v, & a &:= \sum_{v \in U_+} x_v, & b &:= \sum_{v \in U_-} x_v, \\ y_i &:= x_{\mu(i)} \text{ for every } i \in M, & \mathbf{y} &:= (y_i)_{i \in M}, & \varrho(u) &:= \frac{1-u}{m}, & \bar{\mathbf{y}}(u) &:= \varrho(u) \mathbf{1}_M. \end{aligned} \quad (10.2)$$

Here U is the data vertex set, u is its total weight, and a and b are the positive-sign and negative-sign weights. By (V1), U_+ and U_- are disjoint. The vector $\bar{\mathbf{y}}(u)$ is the equal root-weight vector of total weight $1-u$. The projected k -graphs inherit the data weights from $\mathbf{x}$, and $e_j(S)$ denotes the elementary symmetric polynomial evaluated at the inherited weights on a data vertex set S .

We can now state the optimizer classification for the rooted class.

Theorem 10.4. *Let z_β be as in (9.1). The following statements hold.*

- (i) *If $z_\beta = \tau_{\text{base}}$, the optimizer space of $\mathcal{D}_\beta$ consists of one fixed finite-step limit W_0 .*
- (ii) *If $z_\beta > \tau_{\text{base}}$, there are unique parameters u_β and $t_\beta > 0$ such that the optimizer space of $\mathcal{D}_\beta$ is the disjoint union of two nonempty phases. In one phase, the total data, positive-sign, and negative-sign weights of representing completed weighted models converge to $(u_\beta, t_\beta, 0)$; in the other, they converge to $(u_\beta, 0, t_\beta)$.*

The classification also describes the internal structure of the two phases. Each optimizer in either phase admits a representing sequence of completed weighted models whose root vectors converge to $\bar{\mathbf{y}}(u_\beta)$ and whose normalized active projection converges to an optimizer of the inner class $\mathcal{C}_\beta^{\text{in}}$. Conversely, every inner optimizer is represented in both sign phases. Once the sign and the limiting inner optimizer are fixed, the resulting limit represented by these models, including the limiting contributions of all edge classes imposed by completion, is uniquely determined by the fixed edge-class tables.

We prove the theorem in four steps.

- (i) Data cloning makes the individual data weights negligible without changing the represented limit.
- (ii) Global localization and root averaging reduce the weighted problem to maximizing Φ_{z_β} .
- (iii) The effective parameter hierarchy verifies the hypotheses of the phase-roof lemma.
- (iv) The reduced-realization lemma identifies the scalar maximizers with all optimizer limits.

We shall optimize over completed weighted models, so we first record that completion does not change an extremal limit.

If $\widehat{\mathbf{G}}$ is obtained from a weighted r -graph $\mathbf{G}$ by adjoining zero-weight vertices, if necessary, and then adding edges, and if $\text{dens}_r(\widehat{\mathbf{G}}) - \text{dens}_r(\mathbf{G}) = \eta$, then for every fixed r -graph F , we have

$$0 \leq t(F, \widehat{\mathbf{G}}) - t(F, \mathbf{G}) \leq |F|\eta. \quad (10.3)$$

Here η is the gain in normalized one-edge density, equivalently $r!$ times the gain in $\lambda(G; \mathbf{x})$. Indeed, under a random weighted map of $V(F)$, use a union bound over the edges of F . More explicitly, let $(\mathbf{G}_n)$ be any weighted sequence from $\mathcal{D}_\beta$ converging to an optimizer limit. For each n , choose an admissible partial rooting and let $\widehat{\mathbf{G}}_n$ be the resulting completion, with the weighting extended by zero on the newly added roots. Since $\text{dens}_r(\mathbf{G}_n) \rightarrow \Lambda_r(\mathcal{D}_\beta)$ and $\text{dens}_r(\widehat{\mathbf{G}}_n) \leq \Lambda_r(\mathcal{D}_\beta)$, the completion gains tend to zero. Equation (10.3) then shows that $t(F, \widehat{\mathbf{G}}_n) - t(F, \mathbf{G}_n) \rightarrow 0$ for every fixed F . Thus completion changes every fixed homomorphism density by $o(1)$ along an optimizing sequence. Every optimizer limit therefore has a representing sequence of completed weighted models, and the scalar optimization below describes the full optimizer space rather than merely its maximum value.

10.2 Data splitting and evaluation

The limiting scalar expression obtained after data splitting depends only on the total weights of the data classes, whereas a finite weighted model may contain vertices of comparatively large weight. By Lemma 8.1, cloning data vertices splits those weights without changing any homomorphism density. The following lemma shows that cloning also preserves admissibility in the present rooted construction. When a data vertex is cloned, the partial rooting is unchanged and every clone belongs to exactly the same sets among U_+ and U_- as the original vertex.

Lemma 10.5. *Cloning one data vertex of an admissible rooted r -graph preserves admissibility.*

Proof. Let x be the data vertex being cloned. Every copied edge has the same rooted support and rooted edge type as the old edge. All clones have the same membership in U_+ and U_- as x . Hence a new violation of (V1), (V2), or (V3) would collapse to the same violation before cloning.

Suppose that the cloned model contained a forbidden projected copy. Collapse all clones of x back to x . No projected edge contains two clones, so the collapse is injective on each projected edge. The collapsed copy is therefore an edgewise-injective homomorphic image of a member of $\mathcal{Q}(\mathcal{F}_\beta)$. Since $\mathcal{Q}(\mathcal{F}_\beta)$ is closed under further edgewise-injective images, the old projection would already have contained a forbidden member. ■

For a completed weighted model $(D, \mu, \mathbf{x})$, use the notation from (10.2). The completion conditions give the following decomposition:

$$\begin{aligned} \lambda(D; \mathbf{x}) = & e_r(\mathbf{y}) - \lambda(\mathcal{J}; \mathbf{y}) + u e_{r-1}(\mathbf{y}) + \lambda(\mathcal{D}_{\text{disp}}; \mathbf{y}) e_k(U) + \sum_{\sigma \in \{+, -\}} \lambda(\mathcal{T}_\sigma; \mathbf{y}) (e_k(U) - e_k(U_\sigma)) \\ & + \lambda(\mathcal{P}; \mathbf{y}) (e_{k+1}(U) - e_{k+1}(U_+ \cup U_-)) + \lambda(\mathcal{Z}_+; \mathbf{y}) \lambda(G_+; \mathbf{x}) + \lambda(\mathcal{Z}_-; \mathbf{y}) \lambda(G_-; \mathbf{x}). \end{aligned} \quad (10.4)$$

The four groups of terms correspond, in order, to the frame and dispersion edge classes, the two threshold edge classes, the penalty edge class, and the two sign projections. This decomposition is chosen so that splitting data vertices and equalizing the root weights reduce it to the scalar upper envelope in (10.10).

The completed weighted models have the same optimum as $\mathcal{D}_\beta$. More explicitly, the supremum ranges over all admissible completions (D, μ) and all probability weightings $\mathbf{x}$ on $V(D)$:

$$\Lambda(\mathcal{D}_\beta) = \sup \{ \lambda(D; \mathbf{x}) : (D, \mu) \text{ is an admissible completion and } \mathbf{x} \text{ is a probability weighting on } V(D) \}.$$

Lemma 10.6. *The following statements hold.*

- (i) *Replacing a data vertex by two clones having the same membership in U_+ and U_- , distributing its weight between them, and taking the admissible completion supplied by Lemma 10.3 cannot decrease the Lagrangian and does not change either projection value.*
- (ii) *If $\lambda(\mathcal{D}_{\text{disp}}; \mathbf{y}) > 0$, replacing a data vertex of weight $x > 0$ by k equal clones and re-completing increases the Lagrangian by at least $\lambda(\mathcal{D}_{\text{disp}}; \mathbf{y})(x/k)^k$.*

- (iii) If there is no data vertex, take the constant sequence. Otherwise, repeatedly replacing a data vertex of maximum weight by k equal clones and re-completing produces a sequence whose largest data-vertex weight tends to zero. For every data set S carried along by these clonings, we have $e_j(S) \rightarrow (\sum_{v \in S} x_v)^j / j!$ for $j \in \{k, k+1\}$.
- (iv) Let $(D, \mu, \mathbf{x})$ be a completed weighted model in which every root has weight at least $1/(2(m+1))$, and let ξ be its largest data-vertex weight, with $\xi = 0$ when D has no data vertex. Then

$$m \left(\frac{1}{2(m+1)} \right)^c \left(\frac{\xi}{k} \right)^k \leq \Lambda(\mathcal{D}_\beta) - \lambda(D; \mathbf{x}).$$

Moreover, if $(D', \mu', \mathbf{x}')$ is obtained from $(D, \mu, \mathbf{x})$ by finitely many clonings of data vertices and re-completions, then, for every fixed r -graph F , we have

$$0 \leq t(F, D', \mathbf{x}') - t(F, D, \mathbf{x}) \leq |F|r! (\Lambda(\mathcal{D}_\beta) - \lambda(D; \mathbf{x})).$$

Proof. Clone a data vertex as in Lemma 10.5, give the clones the prescribed weights, and then take the admissible completion. The projected inner class $\mathcal{C}_\beta^{\text{in}} = \text{Forb}(\mathcal{Q}(\mathcal{F}_\beta))$ is blowup closed by Lemma 2.4. Hence each sign projection is the corresponding blowup of the old one, so both projection polynomials are unchanged. Completion deletes no edge and adds no new edge to either sign projection; it only restores the unconditional bundles and the threshold and penalty bundles permitted by (V2) and (V3).

If a variable of weight x is split into θx and $(1-\theta)x$, the increase of an elementary symmetric polynomial in the re-completed model is $\theta(1-\theta)x^2 e_{j-2}$ (the remaining variables). Every edge class in the first three lines of (10.4) contributes a difference $e_j(U) - e_j(S)$ with $S \subseteq U$, except for the positive dispersion term. Because all clones have the same membership in U_+ and U_- as their ancestor, the splitting formula shows that these differences do not decrease. The re-completion is essential here because it adds every unconditional dispersion edge on a k -set of distinct clones. Thus replacing a vertex of weight $x > 0$ by k equal clones creates the new dispersion monomial obtained by selecting all k clones, and the gain is at least $\lambda(\mathcal{D}_{\text{disp}}; \mathbf{y})(x/k)^k$. If there is no data vertex, there is nothing to split. Otherwise, repeatedly splitting a data vertex of maximum weight into k equal clones and re-completing preserves the total weights and the sign projections and drives the largest data-vertex weight to zero. Indeed, if Q denotes the sum of the squares of the data weights, splitting the chosen weight x into k equal parts decreases Q by $(1-1/k)x^2$; if the successive maximum weights failed to tend to zero, then Q would decrease by a fixed positive amount infinitely often, which is impossible because $Q \geq 0$.

We now make the gap estimate explicit. If D has no data vertex, then $\xi = 0$ and the first bound in part (iv) is immediate. Otherwise, every root has weight at least $1/(2(m+1))$, and $|\mathcal{D}_{\text{disp}}| = m$, so $\lambda(\mathcal{D}_{\text{disp}}; \mathbf{y}) \geq m(1/(2(m+1)))^c$. Split a data vertex of maximum weight ξ into k equal clones and re-complete. The new model is admissible, so its Lagrangian is at most $\Lambda(\mathcal{D}_\beta)$, whereas the preceding dispersion monomial increases the Lagrangian by at least the left side of the first bound in part (iv). This proves the bound. Consequently, the largest data-vertex weight tends to zero along any sequence of such models whose optimality gaps tend to zero.

Before re-completion, cloning preserves every homomorphism density by Lemma 8.1. We expand the common-refinement argument used to control all of the intervening completions. Write the finite chain as $D = D_0 \rightarrow D_1 \rightarrow \dots \rightarrow D_s = D'$, where each arrow either clones a data vertex or performs the subsequent completion. Take a common refinement of all clone partitions occurring in the chain, and pull every D_i forward to a weighted rooted r -graph $\widehat{D}_i$ on this one refined vertex set. The weights of the final clones sum to the weights of their ancestors. Pure cloning therefore preserves both the Lagrangian value and every homomorphism density by Lemma 8.1.

These refined hypergraphs are monotone: $\widehat{D}_0 \subseteq \widehat{D}_1 \subseteq \dots \subseteq \widehat{D}_s$. Indeed, every clone has the same membership in U_+ and U_- as its ancestor, and every clone-copy of an edge of the previously completed model remains permitted. A cloning step gives equality after passage to the common refinement, whereas a completion changes neither sign projection nor the sets U_+, U_- ; it adds edges and deletes none, by Lemma 10.3. Consequently, if $\widehat{\mathbf{x}}$ is the common refined weighting, then

$$0 \leq \lambda(\widehat{D}_s; \widehat{\mathbf{x}}) - \lambda(\widehat{D}_0; \widehat{\mathbf{x}}) = \lambda(D'; \mathbf{x}') - \lambda(D; \mathbf{x}) \leq \Lambda(\mathcal{D}_\beta) - \lambda(D; \mathbf{x}).$$

Merge all new edges into the single set $\widehat{D}_s \setminus \widehat{D}_0$. The corresponding increase in normalized one-edge density is at most $r! (\Lambda(\mathcal{D}_\beta) - \lambda(D; \mathbf{x}))$. Since cloning also preserves the two endpoint densities, one

application of the union bound (10.3) gives

$$0 \leq t(F, D', \mathbf{x}') - t(F, D, \mathbf{x}) = t(F, \widehat{D}_s, \widehat{\mathbf{x}}) - t(F, \widehat{D}_0, \widehat{\mathbf{x}}) \leq |F|r!(\Lambda(\mathcal{D}_\beta) - \lambda(D; \mathbf{x})),$$

which proves the second bound in part (iv).

Finally, if the variable family is empty, then the convergence in part (iii) is immediate. Otherwise, put $s = \sum_v x_v$ and $\xi = \max_v x_v$. A union bound over a repeated pair in an ordered j -tuple gives the explicit estimate $0 \leq s^j - j!e_j(\mathbf{x}) \leq \binom{j}{2}\xi s^{j-1}$. This proves part (iii). $\blacksquare$

10.3 Global localization and root balancing

Before reducing the weighted optimization to one variable, we must exclude weight vectors near the boundary of the simplex. The dense frame term uniquely favors the balanced $(m+1)$ -part vector, and the remaining support classes are too sparse to overturn that preference. The next lemma gives the required uniform estimate.

For the lower comparison used below, let $(G_{\text{bal}}, \mu_{\text{bal}})$ be the edgeless totally M -rooted r -graph whose vertex set consists of the m root vertices $\mu_{\text{bal}}(i)$, for $i \in M$, and one additional data vertex v . Since G_{bal} has no raw record edges, $U_+ = U_- = \emptyset$. Let $(D_{\text{bal}}, \mu_{\text{bal}})$ be its admissible completion supplied by Lemma 10.3, and let $\mathbf{x}_{\text{bal}}$ assign weight $1/(m+1)$ to every vertex.

Lemma 10.7. *If c , and hence $r = k + c$, is sufficiently large in terms of k, p, q , the following statements hold. For completed weighted models $(D, \mu, \mathbf{x})$, we use the notation from (10.2).*

(i) *There is $\varepsilon_r = o(m^{-2})$ such that all positive terms in (10.4) other than $e_r(\mathbf{y}) + ue_{r-1}(\mathbf{y})$ satisfy*

$$\begin{aligned} & \lambda(\mathcal{D}_{\text{disp}}; \mathbf{y})e_k(U) + \sum_{\sigma \in \{+, -\}} \lambda(\mathcal{T}_\sigma; \mathbf{y})(e_k(U) - e_k(U_\sigma)) \\ & + \lambda(\mathcal{P}; \mathbf{y})(e_{k+1}(U) - e_{k+1}(U_+ \cup U_-)) + \sum_{\sigma \in \{+, -\}} \lambda(\mathcal{Z}_\sigma; \mathbf{y})\lambda(G_\sigma; \mathbf{x}) \leq \varepsilon_r \binom{m+1}{r} (m+1)^{-r}, \end{aligned}$$

(ii) *Put $\delta_r := (2m/r)/\binom{m+1}{r} = o(m^{-2})$. For the completed weighted model $(D_{\text{bal}}, \mu_{\text{bal}}, \mathbf{x}_{\text{bal}})$ defined above, we have $(\mathbf{y}, u) = (m+1)^{-1}\mathbf{1}_{m+1}$ and*

$$\lambda(D_{\text{bal}}; \mathbf{x}_{\text{bal}}) = \left(\binom{m+1}{r} - \frac{2m}{r} \right) (m+1)^{-r} = (1 - \delta_r) \binom{m+1}{r} (m+1)^{-r}.$$

(iii) *If $\lambda(D; \mathbf{x}) = \Lambda(\mathcal{D}_\beta)$, then*

$$\|(\mathbf{y}, u) - (m+1)^{-1}\mathbf{1}_{m+1}\|_2^2 \leq \varepsilon_r + \delta_r < \frac{1}{50m^2}.$$

(iv) *There is $\eta_r > 0$ such that if $\lambda(D; \mathbf{x}) \geq \Lambda(\mathcal{D}_\beta) - \eta_r$, then*

$$\|(\mathbf{y}, u) - (m+1)^{-1}\mathbf{1}_{m+1}\|_2^2 < \frac{1}{25m^2}.$$

Consequently, $1/(2(m+1)) < y_i < 2/(m+1)$ for every $i \in M$, and $1/(2(m+1)) < u < 2/(m+1)$.

Proof. We use a bounded-degree estimate. Let $\mathcal{S}$ be an h -graph on the roots, let its maximum vertex degree be Δ , put $y_{\text{root}} := \sum_i y_i$, and let $d := r - h$. For an edge S write $w_S := \sum_{i \in S} y_i$. Weighted AM-GM and $\sum_{S \in \mathcal{S}} w_S \leq \Delta y_{\text{root}}$ give

$$\lambda(\mathcal{S}; \mathbf{y}) \leq h^{-h} \sum_{S \in \mathcal{S}} w_S^h \leq h^{-h} (\max_{S \in \mathcal{S}} w_S)^{h-1} \sum_{S \in \mathcal{S}} w_S \leq \frac{\Delta}{h^h} y_{\text{root}}^h.$$

Also $e_d(U) \leq u^d/d!$. Maximizing the product in the preceding bound under $y_{\text{root}} + u = 1$ yields

$$\lambda(\mathcal{S}; \mathbf{y})e_d(U) \leq \frac{\Delta d^d}{d!r^r}. \quad (10.5)$$

The same estimate applies to either projection polynomial, which is at most $e_k(U)$, and to every difference of elementary symmetric polynomials by discarding the negative term.

The c -uniform designs have maximum degree $O_{k,p,q}(c)$ and the $(c-1)$ -uniform design $\mathcal{P}$ has maximum degree $d_P(c-1)$. Thus (10.5), summed over the fixed list of positive edge classes, is at most $C_{k,p,q}r/r^r$. The controlled frame construction in Lemma 10.1 gives $r(r+4)/2 \leq m \leq r(r+5)/2$. Since $m+1 \asymp r^2$, the product formula for $\binom{m+1}{r}(m+1)^{-r}$ gives $\binom{m+1}{r}(m+1)^{-r} \geq c_0/r!$ for an absolute $c_0 > 0$. Stirling's inequality then shows that the preceding sparse-term bound, divided by $\binom{m+1}{r}(m+1)^{-r}$, is $O_{k,p,q}(r^{3/2}e^{-r}) = o(m^{-2})$. This proves the first assertion. For the finite reference model D_{bal} , the data set consists of one neutral vertex, so $e_k(U) = e_{k+1}(U) = 0$ and both projection polynomials vanish. Hence every sparse term in (10.4) vanishes except the frame hole. Since $|\mathcal{J}| = 2m/r$, its contribution is $(2m/r)(m+1)^{-r}$, which gives the stated value of D_{bal} and the exact formula for δ_r . The same product estimate shows that $\delta_r = o(m^{-2})$.

It remains to localize. For $(\mathbf{y}, u) \in \Delta_{m+1}$, put $V := \|(\mathbf{y}, u) - (m+1)^{-1}\mathbf{1}_{m+1}\|_2^2$. Maclaurin's inequality between e_2 and e_r gives

$$\frac{e_r(\mathbf{y}) + ue_{r-1}(\mathbf{y})}{\binom{m+1}{r}(m+1)^{-r}} \leq \left(1 - \frac{m+1}{m}V\right)^{r/2} \leq 1 - V,$$

because $r \geq 4$. We have $\lambda(D_{\text{bal}}; \mathbf{x}_{\text{bal}}) = (1 - \delta_r)\binom{m+1}{r}(m+1)^{-r}$. On the other hand, every completed weighted model $(D, \mu, \mathbf{x})$ satisfies $\lambda(D; \mathbf{x}) \leq e_r(\mathbf{y}) + ue_{r-1}(\mathbf{y}) + \varepsilon_r \binom{m+1}{r}(m+1)^{-r}$. Therefore an optimizer satisfies $V \leq \varepsilon_r + \delta_r$. Taking c large makes this smaller than $1/(50m^2)$. The near-optimal assertion is uniform and does not require a compactness argument over the varying finite data structures. Indeed, if a completed weighted model $(D, \mu, \mathbf{x})$ has $V \geq 1/(25m^2)$, then the displayed Maclaurin bound and the positive-term estimate give

$$\lambda(D; \mathbf{x}) \leq \left(1 - \frac{1}{25m^2} + \varepsilon_r\right) \binom{m+1}{r}(m+1)^{-r},$$

whereas the optimum is at least $(1 - \delta_r)\binom{m+1}{r}(m+1)^{-r}$. Since $\varepsilon_r + \delta_r < 1/(50m^2)$, every such point lies at least $\binom{m+1}{r}(m+1)^{-r}/(50m^2)$ below the optimum. Thus one may take $\eta_r := \binom{m+1}{r}(m+1)^{-r}/(100m^2)$. Every completed weighted model whose value is within η_r of the optimum has $V < 1/(25m^2)$, uniformly over all admissible completions and probability weightings. The coordinate bounds follow. $\blacksquare$

Global localization places every optimizer in an interior box. We now show that, inside this box, the root weights are exactly equal. For fixed u , the dense frame contribution satisfies $e_r(\mathbf{y}) + ue_{r-1}(\mathbf{y}) = e_r(y_1, \dots, y_{|M|}, u)$. Pair averaging gives a uniform quadratic gain unless the root weights are equal. The sparse support polynomials have zero tangent gradient at the equal-root vector and a much smaller Hessian, so they cannot offset this gain.

For the remainder of this part, put $I := [1/(2(m+1)), 2/(m+1)]$.

Lemma 10.8. *If c , and hence $r = k + c$, is sufficiently large in terms of k, p, q , the following statements hold for all admissible completions (D, μ) and all probability weightings $\mathbf{x}$ on $V(D)$.*

- (i) *Every optimizer of (10.4), and every sufficiently near-optimal point, has $u \in I$ and $\varrho(u)/2 \leq y_i \leq 2\varrho(u)$ for every $i \in M$.*
- (ii) *Under the notation of (10.2), let $(D, \mu, \mathbf{x})$ be a completed weighted model with $u \in I$ whose root vector $\mathbf{y}$ satisfies the bounds in part (i). Define the probability weighting $\mathbf{x}^{\text{eq}}$ on $V(D)$ by $x_v^{\text{eq}} = x_v$ for every data vertex $v \in U$ and $x_{\mu(i)}^{\text{eq}} = \varrho(u)$ for every $i \in M$. Then $\lambda(D; \mathbf{x}^{\text{eq}}) \geq \lambda(D; \mathbf{x})$, with equality if and only if $\mathbf{y} = \bar{\mathbf{y}}(u)$.*

Proof. Part (i) follows from Lemma 10.7. Indeed, its near-optimal estimate gives $|u - (m+1)^{-1}|, |y_i - (m+1)^{-1}| < 1/(5m)$; for $m \geq 3$ these inequalities imply $u \in I$ and the asserted bounds on the root coordinates. For part (ii), fix a vector $\mathbf{y}$ satisfying those bounds. We prove the asserted comparison quantitatively. Average two coordinates y_i, y_j while keeping their sum fixed. The coefficient of $y_i y_j$ in $e_r(\mathbf{y}) + ue_{r-1}(\mathbf{y})$ is

$$e_{r-2}(\mathbf{y}_{M \setminus \{i,j\}}) + ue_{r-3}(\mathbf{y}_{M \setminus \{i,j\}}) \geq \binom{m-2}{r-2} \left(\frac{1-u}{2m}\right)^{r-2}.$$

Thus the increase in the dense frame term $e_r(\mathbf{y}) + ue_{r-1}(\mathbf{y})$ produced by averaging y_i and y_j is at least $\binom{m-2}{r-2}(\varrho(u)/2)^{r-2}(y_i - y_j)^2/4$, whereas the squared distance to $\bar{\mathbf{y}}(u)$ decreases by $(y_i - y_j)^2/2$. Repeated pair averaging preserves the bounds in part (i) and converges to $\bar{\mathbf{y}}(u)$. Telescoping therefore gives

$$e_r(\bar{\mathbf{y}}(u)) + ue_{r-1}(\bar{\mathbf{y}}(u)) - e_r(\mathbf{y}) - ue_{r-1}(\mathbf{y}) \geq \nu_r(u) \|\mathbf{y} - \bar{\mathbf{y}}(u)\|_2^2, \quad \nu_r(u) := \frac{1}{2} \binom{m-2}{r-2} \left(\frac{1-u}{2m} \right)^{r-2}. \quad (10.6)$$

Let $E_u(\mathbf{y})$ be the sum of all remaining root-dependent terms in (10.4), with the data weights fixed. The absolute values of their data coefficients are bounded by $u^k/k!$ for the c -supports and by $u^{k+1}/(k+1)!$ for the $(c-1)$ -supports. The derivative bound for $\mathcal{J}$ in Lemma 10.1, together with Claim 10.2 for the orbit supports, gives

$$\|D^2 E_u(\mathbf{y})|_{\mathbf{1}^\perp}\|_{\text{op}} \leq \Xi_r,$$

where

$$\Xi_r := 2^{r-2} m^{2-r} \left(2(r-1) + \frac{(2q+2p+1)c(c-1)}{k!} + \frac{d_P(c-1)(c-2)}{(k+1)!} \right). \quad (10.7)$$

The 2-regularity of $\mathcal{J}$ and the regularity of every orbit support make the gradient of each summand of E_u parallel to $\mathbf{1}$ at $\bar{\mathbf{y}}(u)$. Hence the tangent gradient of E_u vanishes there. Taylor's formula along the segment from $\bar{\mathbf{y}}(u)$ to $\mathbf{y}$ now yields

$$|E_u(\mathbf{y}) - E_u(\bar{\mathbf{y}}(u))| \leq \frac{\Xi_r}{2} \|\mathbf{y} - \bar{\mathbf{y}}(u)\|_2^2. \quad (10.8)$$

The comparison is uniform in $u \in I$. Indeed, $1-u \geq 1/2$ and $\binom{m-2}{r-2} \geq ((m-2)/(r-2))^{r-2}$. Together with $m \geq r(r+4)/2$, this gives

$$\frac{\Xi_r}{\inf_{u \in I} \nu_r(u)} \leq C_{k,p,q} r^4 \left(\frac{32}{r} \right)^{r-2} = o(1).$$

Choose c so large that $\Xi_r < \inf_I \nu_r$. Combining (10.6) and (10.8) proves strict improvement unless $\mathbf{y} = \bar{\mathbf{y}}(u)$. $\blacksquare$

At the equalized root vector, regularity and part (iii) of Lemma 10.1 give

$$\lambda(\mathcal{Z}_+; \bar{\mathbf{y}}(u)) = \lambda(\mathcal{Z}_-; \bar{\mathbf{y}}(u)) = qm\varrho(u)^c, \quad \text{and} \quad \lambda(\mathcal{T}_+; \bar{\mathbf{y}}(u)) = \lambda(\mathcal{T}_-; \bar{\mathbf{y}}(u)) = pm\varrho(u)^c. \quad (10.9)$$

By (V4), both sign projections G_+ and G_- belong to $\mathcal{C}_\beta^{\text{in}}$. The restrictions of $\mathbf{x}$ to U_+ and U_- have total weights a and b , respectively. Hence Lemma 2.4 and the k -homogeneity of the Lagrangian give $\lambda(G_+; \mathbf{x}) \leq (z_\beta/k!)a^k$ and $\lambda(G_-; \mathbf{x}) \leq (z_\beta/k!)b^k$. Conversely, since $z_\beta = \Lambda_k(\mathcal{C}_\beta^{\text{in}})$, there are weighted k -graphs $(H_n, \mathbf{w}^{(n)})$ whose underlying k -graphs belong to $\mathcal{C}_\beta^{\text{in}}$, whose weightings are probability vectors, and for which $k!\lambda(H_n; \mathbf{w}^{(n)}) \rightarrow z_\beta$. Combining these bounds and approximating models with Lemma 10.6, we obtain the following reduced upper envelope at fixed (u, a, b) .

$$\Phi_{\text{base}}(u) + \alpha(u)(z_\beta - \tau_{\text{base}})(a^k + b^k) - \gamma(u)(a+b)^{k+1}, \quad (10.10)$$

where

$$\alpha(u) := \frac{qm\varrho(u)^c}{k!}, \quad \text{and} \quad \gamma(u) := \frac{d_P m \varrho(u)^{c-1}}{(k+1)!}, \quad (10.11)$$

and the fixed zero-active polynomial is

$$\begin{aligned} \Phi_{\text{base}}(u) &:= \binom{m}{r} \varrho(u)^r + u \binom{m}{r-1} \varrho(u)^{r-1} - \frac{2m}{r} \varrho(u)^r \\ &\quad + \frac{(1+2p)m}{k!} \varrho(u)^c u^k + \frac{d_P m}{(k+1)!} \varrho(u)^{c-1} u^{k+1}. \end{aligned} \quad (10.12)$$

Here $|\mathcal{J}| = 2m/r$ because $\mathcal{J}$ is 2-regular. In particular, Φ_{base} is a fixed rational polynomial independent of β . The finite reference model D_{bal} used in Lemma 10.7 should not be confused with the limiting value $\Phi_{\text{base}}(1/(m+1))$. In D_{bal} the neutral data weight is carried by one vertex, so all positive sparse terms vanish. By contrast, $\Phi_{\text{base}}(1/(m+1))$ is the closed limiting value approached by splitting the same total

data weight among arbitrarily many neutral vertices, and this splitting creates the positive correction terms displayed above. The realization direction, including the placement of every positive-weight active vertex in the appropriate set U_σ , is proved in [Lemma 10.10](#); it shows that (10.10) is the exact closed supremum.

The four groups of edge terms that produce (10.10) are summarized below.

Contribution group	Combinatorial role	Contribution to the reduced upper envelope
Input-independent terms	The frame and dispersion edges, together with the all-data parts of the threshold and penalty bundles, determine the baseline value of the neutral construction.	The baseline $\Phi_{\text{base}}(u)$.
Signed records	Encode one inner k -graph on each of the positive and negative data classes.	$+\alpha(u)z_\beta(a^k + b^k)$.
Missing threshold edges	Remove the calibrated threshold contribution from a same-sign k -set contained in U_σ .	$-\alpha(u)\tau_{\text{base}}(a^k + b^k)$.
Missing penalty edges	Remove the penalty contribution from a $(k+1)$ -set contained in $U_+ \cup U_-$, thereby penalizing the total weight of this union.	$-\gamma(u)(a+b)^{k+1}$.

Table 4: Combinatorial origin of the reduced one-phase/two-phase expression in (10.10). The functions α and γ are positive and are determined by the regular root-support designs.

With the penalty multiplicity fixed by (10.1), it remains to choose the root-padding parameter $c = r - k$ sufficiently large. We require this choice to make Φ_{base} strongly concave, to make the active correction a smaller perturbation of the baseline, and to keep the active maximizer within the feasible region. The following hierarchy makes all three requirements effective and independent of β .

Proposition 10.9. *There is an effectively computable integer r_{str} such that, for every integer $r \geq r_{\text{str}}$, putting $c = r - k$ and using the designs from [Lemma 10.1](#), the conclusions of [Lemmas 10.7](#) and [10.8](#) and the following statements all hold.*

- (i) *There exist $u_0 \in \text{int } I$ and $\mu_r > 0$ such that u_0 is the unique maximizer of Φ_{base} on I and $\Phi_{\text{base}}''(u) \leq -\mu_r/2$ for every $u \in I$.*
- (ii) *For every $z \in [\tau_{\text{base}}, 1]$, the maximized phase gain*

$$J_z(u) := \frac{k^k}{(k+1)^{k+1}} (z - \tau_{\text{base}})^{k+1} \frac{\alpha(u)^{k+1}}{\gamma(u)^k}$$

satisfies $\|J_z''\|_{L^\infty(I)} < \mu_r/4$. Consequently $\Phi_{\text{base}} + J_z$ has a unique maximizer $u_z \in \text{int } I$.

- (iii) *For every $u \in I$ and $z \in [\tau_{\text{base}}, 1]$, the optimal total active weight*

$$t_z(u) := \frac{k\alpha(u)(z - \tau_{\text{base}})}{(k+1)\gamma(u)}$$

lies in the feasible interval $[0, u]$; in fact, $t_z(u) \leq u/2$.

The threshold r_{str} depends only on k and $\tau_{\text{base}} = p/q$. Once $r \geq r_{\text{str}}$ is fixed, all auxiliary parameters and root-support designs can be computed from k , τ_{base} , and r , without using β or z_β .

Proof. We prove the proposition directly from explicit estimates that hold for every sufficiently large r . All constants below are rational and computable from k, p, q . We first obtain uniform lower bounds for global localization and root equalization, then control the curvature and phase gain, and finally take the maximum of these bounds.

First consider global localization. The product inequality $\prod_i (1 - x_i) \geq 1 - \sum_i x_i$ for $x_i \in [0, 1]$ and the lower bound on m give

$$\binom{m+1}{r} (m+1)^{-r} = \frac{1}{r!} \prod_{j=0}^{r-1} \left(1 - \frac{j}{m+1}\right) \geq \frac{5}{(r+4)r!}. \quad (10.13)$$

Put

$$C_{\text{loc}} := (1 + 2p + 2q) \frac{k^k}{k!} + d_P \frac{(k+1)^{k+1}}{(k+1)!}.$$

Summing (10.5) as in the proof of Lemma 10.7, and using $r!/r^r \leq 2^{-\lfloor r/2 \rfloor}$, gives the following rational bound for ε_r . The exact evaluation of D_{bal} in that lemma gives the bound for δ_r ; thus

$$\varepsilon_r \leq \frac{C_{\text{loc}} r(r+4)}{5} 2^{-\lfloor r/2 \rfloor}, \quad \text{and} \quad \delta_r = \frac{2m/r}{\binom{m+1}{r}} \leq 2r \left(\frac{2}{r} \right)^r.$$

For the last inequality, use $\binom{m+1}{r} \geq ((m-r+2)/r)^r$, $m-r+2 \geq r^2/2$, and $2m/r \leq r+5 \leq 2r$. Since $m \leq r(r+5)/2$, it follows that

$$50m^2(\varepsilon_r + \delta_r) \leq \Phi_{\text{loc}}(r),$$

where

$$\Phi_{\text{loc}}(r) := \frac{5}{2} C_{\text{loc}} r^3 (r+4)(r+5)^2 2^{-\lfloor r/2 \rfloor} + 100r^5 \left(\frac{2}{r} \right)^r. \quad (10.14)$$

Both parity subsequences of the first summand are strictly decreasing for $r \geq 32$, since its ratio under $r \mapsto r+2$ is at most $(1+2/r)^6/2 < 1$. The second summand is decreasing there, with successive ratio at most $2/r$, and both terms tend effectively to zero. Let r_{loc} be the first integer $s \geq 32$ for which $\Phi_{\text{loc}}(s) < 1$ and $\Phi_{\text{loc}}(s+1) < 1$. The two parity subsequences then give $\Phi_{\text{loc}}(r) < 1$ for every $r \geq r_{\text{loc}}$. This is precisely the uniform estimate required in Lemma 10.7.

Next consider root equalization. Since $\varrho(u)/2 \geq (4m)^{-1}$ for every $u \in I$, while $m-2 \geq r^2/2$ and $c \leq r$, define Φ_{eq} and obtain

$$\begin{aligned} \Phi_{\text{eq}}(r) &:= 2 \left(2 + \frac{2q+2p+1}{k!} + \frac{d_P}{(k+1)!} \right) r^2 \left(\frac{16}{r} \right)^{r-2}, \\ \frac{\Xi_r}{\inf_{u \in I} \nu_r(u)} &\leq \Phi_{\text{eq}}(r), \end{aligned} \quad (10.15)$$

where the inequality follows from (10.6)–(10.7). The successive ratio of the right side is at most $16/r$ for $r \geq 3$. Thus the first $r \geq 32$ for which $\Phi_{\text{eq}}(r) < 1$ gives an effective lower bound r_{eq} for Lemma 10.8.

It remains to make the concavity and phase-gain estimates effective. Write

$$B_0(u) := \binom{m}{r} \varrho(u)^r + u \binom{m}{r-1} \varrho(u)^{r-1}, \quad C_r := \binom{m}{r} m^{-r}, \quad \text{and} \quad b_r := \frac{(r-1)(m+1)}{m-r+1}.$$

Then $B_0(u) = C_r(1-u)^{r-1}(1+b_ru)$, and direct differentiation gives

$$-B_0''(u) = C_r(r-1)(1-u)^{r-3}(2b_r(1-u) - (r-2)(1+b_ru)). \quad (10.16)$$

Define

$$L_r := \frac{r(r-1)}{(r+4)r!}, \quad C_\Delta := 2 + \frac{2^k(1+2p)}{k!} + \frac{2^{k+1}d_P}{(k+1)!}, \quad \text{and} \quad C_J := \frac{k^k}{(k+1)^{k+1}} \frac{((k+1)!)^k q^{k+1}}{(k!)^{k+1} d_P^k}.$$

Using (10.11), the phase gain has the exact form

$$J_z(u) = C_J(z - \tau_{\text{base}})^{k+1} m \varrho(u)^r. \quad (10.17)$$

For $r \geq 8$, formula (10.16) gives

$$-B_0''(u) \geq L_r \quad \text{for every } u \in I. \quad (10.18)$$

Indeed, $C_r \geq r!^{-1}(1 - r(r-1)/(2m)) \geq 5/((r+4)r!)$, and Bernoulli's inequality gives $(1-u)^{r-3} \geq 1 - 2(r-3)/(m+1) \geq 1/2$. Moreover, $b_r r u < 4$, so the final bracket in (10.16) is at least $r-4 \geq r/2$. Multiplying these three bounds gives the slightly stronger lower bound $5r(r-1)/(4(r+4)r!)$.

For $j = 0, 1, 2$, direct differentiation of $m \varrho(u)^h u^d$, where $h+d=r$, gives

$$\|(\Phi_{\text{base}} - B_0)^{(j)}\|_{L^\infty(I)} \leq D_j := C_\Delta r^2 m^{j+1-r}, \quad \text{and} \quad \sup_{z \in [\tau_{\text{base}}, 1]} \|J_z^{(j)}\|_{L^\infty(I)} \leq Q_j := C_J r^2 m^{1-r}.$$

The required Leibniz expansion is

$$\left| \frac{d^j}{du^j} (m \varrho(u)^h u^d) \right| \leq m \sum_{\substack{0 \leq a \leq j \\ a \leq h, j-a \leq d}} \binom{j}{a} (h)_a m^{-a} \varrho(u)^{h-a} (d)_{j-a} u^{d-j+a}.$$

Substitute $\varrho(u) \leq 1/m$, $u \leq 2/m$, $m \geq r^2/2$, and $j \leq 2$. For $j = 2$, the three surviving summands, after division by $2^d r^2 m^{3-r}$, are at most $1/4$, $1/m$, and $1/m^2$, respectively; the cases $j = 0, 1$ are smaller. The frame-hole term and its first two derivatives are bounded by $2r^2 m^{j+1-r}$. This proves the first displayed bound with the stated value of C_Δ . The formula (10.17) gives the second bound directly.

Put $u_* := 1/(m+1)$ and $U_{\text{roof}} := [u_* - 1/(8(m+1)), u_* + 1/(8(m+1))]$. Then $U_{\text{roof}} \subsetneq I$. The following four rational inequalities imply the required concavity, localization, and phase-gain bounds.

$$D_2 < \frac{L_r}{4}, \quad Q_2 < \frac{L_r}{4}, \quad D_1 + Q_1 < \frac{L_r}{64(m+1)}, \quad \text{and} \quad Q_0 < \frac{L_r}{2048(m+1)^2}. \quad (10.19)$$

They give $\Phi''_{\text{base}} \leq -3L_r/4$ and $(\Phi_{\text{base}} + J_z)'' \leq -L_r/2$. At $u_* = 1/(m+1)$, the identity $B'_0(u_*) = 0$ gives $|\Phi'_{\text{base}}(u_*)| \leq D_1$ and $|(\Phi_{\text{base}} + J_z)'(u_*)| \leq D_1 + Q_1$. The third inequality in (10.19), together with the two curvature bounds, gives opposite derivative signs at $u_* \pm 1/(16(m+1))$. Thus Φ_{base} and $\Phi_{\text{base}} + J_z$ each have a unique critical point within $1/(16(m+1))$ of u_* and the required derivative signs at the endpoints of I . In particular, $u_0 \in \text{int}_I U_{\text{roof}}$. Therefore, for every $u \in I$ with $|u - u_*| \geq 1/(8(m+1))$, we have

$$\Phi_{\text{base}}(u_0) - \Phi_{\text{base}}(u) \geq \frac{3L_r}{2048(m+1)^2}.$$

The last inequality in (10.19) shows that $\sup J_z$ is less than one third of this gap, and hence less than half of it, while the second gives the Hessian bound. Thus conditions (A1)–(A3) hold on U_{roof} with the parameter $\mu = L_r/2$. Equivalently, parts (i) and (ii) hold with the rational choice $\mu_r = L_r$.

After multiplying the first two inequalities in (10.19) by $4/L_r$, the third by $64(m+1)/L_r$, and the fourth by $2048(m+1)^2/L_r$, the bounds for D_j and Q_j , together with $m+1 \leq 2m$ and $1/L_r \leq 3r!$, show that every resulting left side is at most $24576(C_\Delta + C_J)r^2 r! m^{3-r}$. Thus all four inequalities follow when this quantity is less than 1. Using $r! \leq r^r$ and $m \geq r^2/2$, it is at most

$$\Phi_{\text{roof}}(r) := 3072(C_\Delta + C_J)r^8 \left(\frac{2}{r}\right)^r. \quad (10.20)$$

Its successive ratio is $(2/r)(1+1/r)^{7-r} \leq 2/r$ for $r \geq 8$. Hence the first $r \geq 8$ with $\Phi_{\text{roof}}(r) < 1$ gives an effective lower bound r_{roof} for the phase-roof estimates.

Finally, feasibility needs no additional lower bound on r . By (10.1) and (10.11), we obtain

$$t_z(u) \leq \frac{1-u}{8m} \leq \frac{1}{8m} \leq \frac{1}{4(m+1)} \leq \frac{u}{2}.$$

This proves part (iii). It also verifies condition (A4) on U_{roof} , so all four hypotheses of Lemma 9.1 are instantiated with $U = U_{\text{roof}}$ and $\mu = L_r/2$. Let $r_{\text{des}} = k + c_{\text{des}}$ be the effective lower bound for the designs from Lemma 10.1, and define

$$r_{\text{str}} := \max\{r_{\text{des}}, r_{\text{loc}}, r_{\text{eq}}, r_{\text{roof}}, 32\}. \quad (10.21)$$

Every comparison used to compute this integer is between rational numbers, and the monotonicity just proved shows that every $r \geq r_{\text{str}}$ passes the same tests. This proves the asserted effective lower bound and completes the proof. $\blacksquare$

To compute r_{str} , obtain $k = r_{\text{num}}$ and $\tau_{\text{base}} = p/q$ from Theorem 1.1 using the fixed input β_∞ , compute the four lower bounds listed below, and take their maximum as in (10.21).

Note that none of the lower bounds in Table 5 depends on β .

For the remainder of the structural construction, fix an arbitrary target uniformity $r \geq r_{\text{str}}$, put $c = r - k$, and choose the lexicographically first frame in the finite search of Lemma 10.1, together with the displayed orbit supports. Since the target uniformity r is now fixed, we omit it from the notation for objects constructed at that uniformity, writing $\mathcal{A}_\beta^{\text{root}}$, $\mathcal{D}_\beta$, $\mathcal{G}_\beta$, W_0 , Sep , $\mathbf{P}_0$, and $\mathfrak{P}_0$ for their r -indexed versions.

Lower bound	Definition and source	Dependence	Conclusion
r_{des}	$k + c_{\text{des}}$, where c_{des} is computed in Lemma 10.1 .	k, p, q	For every $r \geq r_{\text{des}}$, the asymmetric frame and all support designs can be constructed effectively.
r_{loc}	The first $s \geq 32$ with $\Phi_{\text{loc}}(s) < 1$ and $\Phi_{\text{loc}}(s+1) < 1$; see (10.13) – (10.14) .	k, p, q	For every $r \geq \max\{r_{\text{des}}, r_{\text{loc}}\}$, both parity classes are controlled and Lemma 10.7 holds.
r_{eq}	The first $s \geq 32$ with $\Phi_{\text{eq}}(s) < 1$; see (10.15) .	k, p, q	For every $r \geq \max\{r_{\text{des}}, r_{\text{loc}}, r_{\text{eq}}\}$, the sparse Hessian is smaller than the root-averaging gain and Lemma 10.8 holds.
r_{roof}	The first $s \geq 8$ with $\Phi_{\text{roof}}(s) < 1$; see (10.18) – (10.20) .	k, p, q	For every $r \geq \max\{r_{\text{des}}, r_{\text{roof}}\}$, the four inequalities in (10.19) give parts (i) and (ii), with $\mu_r = L_r$, and conditions (A1) – (A3) .
r_{str}	$\max\{r_{\text{des}}, r_{\text{loc}}, r_{\text{eq}}, r_{\text{roof}}, 32\}$; see (10.21) .	The four preceding lower bounds	For every $r \geq r_{\text{str}}$, the designs, global localization, root equalization, and all three conclusions of Proposition 10.9 hold simultaneously. The lower bound 32 also covers the later size inequalities involving the uniformity.

Table 5: Effective lower bounds in the structural construction.

The r -index is restored in the final statements. By contrast, $\mathcal{F}_\beta = \mathcal{F}_{k,\beta}$, $\mathcal{C}_\beta^{\text{in}}$, z_β , and τ_{base} are determined by the fixed base k -uniform problem and do not depend on the target uniformity r .

The scalar optimization in [Lemma 9.1](#) and [Proposition 10.9](#) determines the extremal value, but the main theorem requires a classification of all extremal limits. We must therefore prove two directions: every extremal limit yields a maximizer (u, a, b) of Φ_{z_β} together with the corresponding inner extremal limits, and every such choice can be realized by a sequence of completed weighted models. The next lemma establishes these two directions. In particular, it removes positive-weight isolated vertices from an inner optimizing sequence before raw record edges are added to place the remaining active vertices in U_+ or U_- . With $\nu_r(u)$ and Ξ_r defined in [\(10.6\)](#) and [\(10.7\)](#), respectively, put

$$\underline{\nu}_r := \inf_{u \in I} (\nu_r(u) - \Xi_r/2) > 0.$$

Lemma 10.10. *For $u \in I$ and $a, b \geq 0$ with $a+b \leq u$, use the phase function $\Phi_{z_\beta}(u, a, b)$ from [\(9.2\)](#), with $\tau = \tau_{\text{base}}$, with α and γ defined in [\(10.11\)](#), and with Φ_{base} defined in [\(10.12\)](#). The following statements hold.*

- (i) *Every optimizer limit of $\mathcal{D}_\beta$ has a representing sequence of completed weighted models for which the root vector is $o(1)$ from $\bar{\mathbf{y}}(u_n)$ and the largest data-vertex weight is $o(1)$. Along every subsequence on which its total data, positive-sign, and negative-sign weights (u_n, a_n, b_n) converge to (u, a, b) , this limit maximizes Φ_{z_β} . If, for example, $a > 0$, then, along that same subsequence, $\text{dist}_{d_{\text{hom}}}((G_{+,n}, \mathbf{x}^{(n)}/a_n), \mathfrak{L}(\mathcal{C}_\beta^{\text{in}})) \rightarrow 0$, and the analogous assertion holds for the negative sign when $b > 0$.*
- (ii) *Conversely, let $W_+, W_- \in \mathfrak{L}(\mathcal{C}_\beta^{\text{in}})$ and let (u, a, b) be as above. There is a sequence of completed weighted models whose roots have weights $\bar{\mathbf{y}}(u)$, whose positive, negative, and neutral data classes have total weights $a, b, u-a-b$, whose largest data-vertex weight tends to zero, and whose normalized positive and negative sign projections tend to W_+ and W_- , respectively, whenever the corresponding weights a and b are positive. The sequence converges in every r -graph homomorphism density, and its Lagrangian values tend to $\Phi_{z_\beta}(u, a, b)$.*
- (iii) *Once (u, a, b) and the normalized limit of each sign class of positive weight are fixed, the completion rules uniquely determine the resulting limit. If $a = b = 0$, neither W_+ nor W_- contributes, and the result is the same finite-step limit for every β . If $a > 0$ and $b = 0$, only W_+ remains relevant; if $a = 0$ and $b > 0$, only W_- remains relevant.*

Proof. Start with an optimizing sequence of completed weighted models $(D_n, \mu_n, \mathbf{x}^{(n)})$. For the model indexed by n , use the notation from (10.2) with subscript n . By Lemma 10.7, all sufficiently large terms lie in the localized box. Let $\mathbf{x}_{\text{eq}}^{(n)}$ be obtained from $\mathbf{x}^{(n)}$ by leaving the data weights unchanged and replacing the root weights by the equal vector $\bar{\mathbf{y}}(u_n)$. The comparison in (10.6)–(10.8) and the fact that no completed weighted model has Lagrangian exceeding $\Lambda(\mathcal{D}_\beta)$ give

$$0 \leq \nu_r \|\mathbf{y}_n - \bar{\mathbf{y}}(u_n)\|_2^2 \leq \lambda(D_n; \mathbf{x}_{\text{eq}}^{(n)}) - \lambda(D_n; \mathbf{x}^{(n)}) \leq \Lambda(\mathcal{D}_\beta) - \lambda(D_n; \mathbf{x}^{(n)}) = o(1).$$

Hence the root vectors are $o(1)$ apart in ℓ^2 and, because the fixed number of roots is m , also in ℓ^1 . For two weightings $\mathbf{x}, \mathbf{x}'$ of the same finite model, maximal coupling of the sampled vertices gives

$$|t(F, D, \mathbf{x}) - t(F, D, \mathbf{x}')| \leq v(F) \|\mathbf{x} - \mathbf{x}'\|_1.$$

Thus root equalization changes every fixed homomorphism density by $o(1)$ and preserves the represented limit. By Lemma 10.6, every completed near-maximizing sequence in the localized box may then be split and re-completed so that its largest data-vertex weight tends to zero, without changing its represented optimizer limit; relabel the resulting sequence as $(D_n, \mu_n, \mathbf{x}^{(n)})$.

If $a_n > 0$, put $z_{+,n} := k! \lambda(G_{+,n}; \mathbf{x}^{(n)}) / a_n^k \leq z_\beta$; equivalently, this is $k!$ times the value obtained after normalizing the inherited weights on U_+ to sum to one. The inequality follows from Lemma 2.4. Put $z_{+,n} = 0$ when $a_n = 0$, and define $z_{-,n}$ in the same way, with value zero when $b_n = 0$. Part (iii) and the equal-root identities (10.9) show that $\lambda(D_n; \mathbf{x}^{(n)})$ differs by $o(1)$ from

$$\Phi_{\text{base}}(u_n) + \alpha(u_n) \left((z_{+,n} - \tau_{\text{base}}) a_n^k + (z_{-,n} - \tau_{\text{base}}) b_n^k \right) - \gamma(u_n) (a_n + b_n)^{k+1},$$

where a term corresponding to a class of weight zero is read as zero. Replacing $z_{\sigma,n}$ by z_β can only increase this expression. It follows that every subsequential parameter limit is bounded above by the corresponding value of Φ_{z_β} .

We next prove the converse. Together with the limiting expressions in part (iii), it shows that the preceding upper envelope is the exact closed supremum at fixed (u, a, b) . Choose weighted k -graphs $H_{\sigma,n}$ whose underlying k -graphs belong to $\mathcal{C}_\beta^{\text{in}}$, which converge to W_σ , and which satisfy $\text{dens}_k(H_{\sigma,n}) \rightarrow z_\beta$. Such sequences exist by the definition of $\mathfrak{L}(\mathcal{C}_\beta^{\text{in}})$. They may be assumed to have no isolated vertex. Indeed, if s_n is the total weight of the isolated vertices and $H_{\sigma,n}^\circ$ is obtained by deleting them and renormalizing, then

$$\text{dens}_k(H_{\sigma,n}) = (1 - s_n)^k \text{dens}_k(H_{\sigma,n}^\circ) \leq (1 - s_n)^k z_\beta.$$

Here $z_\beta \geq \tau_{\text{base}} > 0$, so $s_n \rightarrow 0$. Coupling the original and conditioned vertex samples shows that, for every fixed k -graph Q ,

$$|t(Q, H_{\sigma,n}) - t(Q, H_{\sigma,n}^\circ)| \leq v(Q) s_n = o(1).$$

Thus deleting the isolated vertices preserves W_σ . Apply iterated cloning until the largest normalized vertex weight tends to zero; by Lemma 8.1, this preserves all inner homomorphism densities exactly.

Scale the positive and negative inner weights to sum to a and b , respectively, and split the remaining weight $u - a - b$ among neutral vertices whose largest weight tends to zero. Give the roots the weights $\bar{\mathbf{y}}(u)$. For each $\sigma \in \{+, -\}$, insert the raw σ -record edges prescribed by $H_{\sigma,n}$, and then perform the admissible completion of Lemma 10.3. Because the inner k -graphs have no isolated vertex, every positive vertex belongs to U_+ and every negative vertex belongs to U_- . The sets U_+ and U_- are disjoint, their projections are $\mathcal{Q}(\mathcal{F}_\beta)$ -free, and completion adds threshold and penalty edges according to the subset conditions in (V2) and (V3). Hence the resulting rooted models are admissible.

Part (iii), the convergence $\text{dens}_k(H_{\sigma,n}) \rightarrow z_\beta$, and (10.9) now give convergence of their Lagrangian values to $\Phi_{z_\beta}(u, a, b)$. They also converge in every homomorphism density. To see this directly, fix an r -graph F and expand $t(F, \cdot)$ over the finitely many role assignments

$$\varpi: V(F) \longrightarrow M \cup \{+, -, 0\},$$

where 0 denotes the neutral data class. For a fixed ϖ , the root weights and the frame, dispersion, threshold, and penalty indicators are determined by the class weights and the fixed edge-class tables. For each sign σ , collect the k -sets of formal vertices which occur as the data part of a raw σ -record edge under ϖ . These sets form a finite projected k -graph $F_{\varpi,\sigma}$ on the formal vertices assigned sign σ ; vertices which occur in no such set are retained as isolated vertices.

Ignoring maps in which two formal data vertices are sent to the same data vertex, the contribution of ϖ therefore has the form $c_\varpi t(F_{\varpi,+}, H_{+,n})t(F_{\varpi,-}, H_{-,n})$, where c_ϖ is the product of the relevant root and class weights, multiplied by a zero–one compatibility factor determined by the fixed edge-class tables. A factor belonging to a sign class of weight zero is omitted; assignments using that sign already have coefficient zero. Let ξ_n be the largest data-vertex weight. The total weight of maps with a repeated data vertex is at most $\binom{v(F)}{2} \sum_{v \in U} x_v^2 \leq \binom{v(F)}{2} \xi_n$. Thus the factorization has an $O_F(\xi_n)$ error, uniformly over the finitely many role assignments. Every term consequently converges, and its limit depends only on (u, a, b, W_+, W_-) and the fixed tables.

The upper bound already proved and this converse imply

$$\Lambda(\mathcal{D}_\beta) = \Phi_{z_\beta}^* = \max_{\substack{u \in I, \\ a, b \geq 0 \\ a+b \leq u}} \Phi_{z_\beta}(u, a, b).$$

Fix a subsequence on which $(u_n, a_n, b_n) \rightarrow (u, a, b)$, and put

$$R_n := \Phi_{\text{base}}(u_n) + \alpha(u_n)((z_{+,n} - \tau_{\text{base}})a_n^k + (z_{-,n} - \tau_{\text{base}})b_n^k) - \gamma(u_n)(a_n + b_n)^{k+1}.$$

The estimate $\lambda(D_n; \mathbf{x}^{(n)}) = R_n + o(1)$ and the optimizing assumption give

$$0 \leq \Lambda(\mathcal{D}_\beta) - R_n = o(1), \quad \text{and} \quad R_n \leq \Phi_{z_\beta}(u_n, a_n, b_n) \leq \Lambda(\mathcal{D}_\beta).$$

Consequently, we have

$$0 \leq \alpha(u_n)(a_n^k(z_\beta - z_{+,n}) + b_n^k(z_\beta - z_{-,n})) = \Phi_{z_\beta}(u_n, a_n, b_n) - R_n = o(1).$$

If $a > 0$, then eventually $a_n > 0$, and the positive continuous function α is bounded away from zero on I . Hence $\alpha(u_n)a_n^k(z_\beta - z_{+,n}) = o(1)$ implies $z_{+,n} \rightarrow z_\beta$. The normalized positive projections are weighted members of $\mathcal{C}_\beta^{\text{in}}$, and their normalized one-edge densities are exactly $z_{+,n}$. Compactness stability in $\mathfrak{W}(\mathcal{C}_\beta^{\text{in}})$ therefore proves the asserted convergence. The same argument, along this same subsequence, applies to the negative projection whenever $b > 0$. This proves parts (i) and (ii). The finite expansion in the preceding paragraph proves part (iii). When $a = b = 0$, precisely the m root roles and one neutral data role remain, whereas a pure-sign limit retains precisely the corresponding inner optimizer. ■

11 The order parameter and weighted phase construction

The two optimizer phases are defined using rooted signs, but the final extremal limits carry no root labels. We therefore need one quantum-graph statistic, fixed before the input is known, that records the sign after the labels have been forgotten. For every finite r -graph F , the map $W \mapsto t(F, W)$ is continuous. These functions distinguish any two distinct r -graph limits, include the constant function 1, and satisfy $t(F_1, W)t(F_2, W) = t(F_1 \sqcup F_2, W)$. Their finite linear span, which consists of the quantum r -graph statistics, is therefore an algebra of continuous functions that contains the constants and separates points. The Stone–Weierstrass theorem shows that any two disjoint compact subsets of $\mathfrak{W}_r$ can be separated by such a statistic. This abstract existence statement is insufficient here because the separator must be independent of the input. We instead construct an explicit induced statistic: the sparse asymmetric root frame pins the labeled root roles, after which the statistic reads the two fixed distinct regular support designs.

We construct, for each sign, a partial induced pattern that recognizes the k -sets belonging to that sign projection after the root labels have been forgotten. Fix $\sigma \in \{+, -\}$ and take the formal vertex set $A \cup M$, where $A = \{a_1, \dots, a_k\}$. On the root vertices prescribe the exact frame, so an r -subset of M is an edge precisely when it does not belong to $\mathcal{J}$. For every $S \in \binom{M}{r}$, prescribe $A \cup S$ to be an edge precisely when $S \in \mathcal{B}_+$ for $\sigma = +$, and precisely when $S \in \mathcal{B}_-$ for $\sigma = -$. No condition is imposed on the remaining r -sets.

Boolean Möbius inversion over the finitely many prescribed nonedges turns this partial induced pattern into a quantum r -graph, denoted by Q_σ . If $Q = \sum_H c_H H$ is a quantum r -graph, define its value on a weighted model by $t(Q; D, \mathbf{x}) := \sum_H c_H t(H; D, \mathbf{x})$. This definition is independent of the chosen presentation because it is the linear extension of the homomorphism-density functional. Equivalently, $t(Q_\sigma; D, \mathbf{x})$ is the weighted density of maps realizing the prescribed frame and the corresponding exact sign-neighborhood pattern.

Define the fixed quantum-graph statistic

$$\text{Sep}(W) := t(Q_+, W) - t(Q_-, W). \quad (11.1)$$

The main result of this section is the following weighted compiler.

Theorem 11.1. *There is a total algorithm which, on input $r \geq 2$ and β , outputs a finite family $\mathcal{A}_{r,\beta}^{\text{root}}$ of r -graphs. For every $r \geq r_{\text{str}}$ there are a fixed finite-step limit $W_{0,r}$ and a fixed continuous statistic Sep_r , independent of β , with $\text{Sep}_r(W_{0,r}) = 0$, such that, with $\mathcal{D}_{r,\beta} := \text{Forb}(\mathcal{A}_{r,\beta}^{\text{root}})$, the following hold.*

- (i) *If $\mathcal{U}$ does not halt on β , then $\mathfrak{L}(\mathcal{D}_{r,\beta}) = \{W_{0,r}\}$.*
- (ii) *If $\mathcal{U}$ halts on β , then $\mathfrak{L}(\mathcal{D}_{r,\beta}) = \mathfrak{L}_{r,+}(\beta) \dot{\cup} \mathfrak{L}_{r,-}(\beta)$, where the two sets are nonempty and compact and, for some $c_{r,\beta} > 0$, $\text{Sep}_r = c_{r,\beta}$ on $\mathfrak{L}_{r,+}(\beta)$ and $\text{Sep}_r = -c_{r,\beta}$ on $\mathfrak{L}_{r,-}(\beta)$.*
- (iii) *Let $\mathbf{G}_n = (G_n, \mathbf{x}^{(n)})$ with $G_n \in \mathcal{D}_{r,\beta}$. If $r! \lambda(G_n; \mathbf{x}^{(n)}) \rightarrow \Lambda_r(\mathcal{D}_{r,\beta})$, then the d_{hom} -distance from $\mathbf{G}_n$ to $\mathfrak{L}(\mathcal{D}_{r,\beta})$ tends to zero.*

The proof has two ingredients.

- (i) The first lemma evaluates Q_+ and Q_- exactly on every completed weighted model.
- (ii) The second uses this formula to show that Sep vanishes on the neutral optimizer and has opposite signs, bounded away from zero, on the two active phases.

Lemma 11.2. *Let $(D, \mu, \mathbf{x})$ be a completed weighted model, and use the notation from (10.2). Then*

$$t(Q_+; D, \mathbf{x}) = \left(\prod_{i \in M} y_i \right) k! \lambda(G_+; \mathbf{x}), \quad \text{and} \quad t(Q_-; D, \mathbf{x}) = \left(\prod_{i \in M} y_i \right) k! \lambda(G_-; \mathbf{x}). \quad (11.2)$$

Proof. The proof has three steps.

- (i) The required edges force injectivity on the formal vertices.
- (ii) The asymmetric frame recovers every root role.
- (iii) The neighborhood of the active k -set identifies its sign projection.

We first check that the required-edge system covers every pair of formal vertices. Two formal roots lie together in a required frame edge because $\binom{M}{r} \setminus \mathcal{J}$ covers pairs. Given a formal root i and an active vertex a_j , regularity and nonemptiness of $\mathcal{D}_{\text{disp}}$ provide a set $D \in \mathcal{D}_{\text{disp}}$ containing i , and the required edge $A \cup D$ contains both. Finally, any one set $D \in \mathcal{D}_{\text{disp}}$ gives a required edge $A \cup D$ containing every pair of active vertices. Consequently every nonzero occurrence of Q_σ is injective on the full formal vertex set. Identifying any two formal vertices would collapse a required edge in the host r -graph. In particular, the active images form a genuine k -set.

We next recover the roles in such an occurrence. No two formal roots can occupy the same root role. Each root role consists of one distinguished root vertex, whereas the occurrence is injective.

Let $S \subseteq M$ be the set of formal roots whose images are data vertices, and write $s := |S|$ and $m_{\text{root}} := m - s$. We claim that $s = 0$. Suppose first that $s \geq 2$ and $m_{\text{root}} \geq r - 2$. Consider the r -sets containing exactly two vertices of S . Their number is $\binom{s}{2} \binom{m_{\text{root}}}{r-2}$. This number is larger than $|\mathcal{J}| = 2m/r$. Indeed, $2m/r \leq r + 7$. If $s \geq r$, then $\binom{s}{2} \geq \binom{r}{2} > r + 7 \geq 2m/r$ for $r \geq 32$; if $2 \leq s < r$, then $m_{\text{root}} \geq m - r + 1 > r + 7 \geq 2m/r$ and $\binom{m_{\text{root}}}{r-2} \geq m_{\text{root}}$. Thus one of these r -sets lies outside $\mathcal{J}$. It is a prescribed edge whose image has exactly two data vertices, contrary to the fact that the available data multiplicities are $0, 1, k, k + 1$ and $k \geq 3$.

If $s \geq 2$ and $m_{\text{root}} < r - 2$, then $s > m - r + 2 > r + 7 \geq 2m/r$. Since $s > r$ and $\binom{s}{r} \geq s$, we have $\binom{s}{r} > |\mathcal{J}|$, so some r -subset of S lies outside $\mathcal{J}$. Its image has r data vertices, again an unavailable multiplicity because $r = k + c \geq k + 4$. We have proved $s \leq 1$.

If $s = 1$, let i be the unique formal root mapped to data and choose a hole $J \in \mathcal{J}$ containing i , which exists because $\mathcal{J}$ is 2-regular. The formal set J is prescribed to be a nonedge, whereas its image has one data vertex and $r - 1$ vertices in distinct root roles; it is therefore a one-data frame edge required by the completion. This contradiction proves $s = 0$.

Consequently the formal roots occupy all m root roles and induce a permutation of M . The prescribed edges and nonedges on the root set say that this permutation preserves $\mathcal{J}$. As $\mathcal{J}$ is asymmetric, the permutation is the identity.

Every active formal vertex is data. Indeed, $\mathcal{D}_{\text{disp}}$ is regular of positive degree. If a_j occupied root role i , choose $S \in \mathcal{D}_{\text{disp}}$ containing i . The required edge $A \cup S$ would then contain two formal vertices with the same image, contradicting the injectivity of the occurrence.

The root roles are now fixed. Suppose first that the data k -set A_0 is an edge of the positive projection. The completion rule supplies $A_0 \cup Z$ for every $Z \in \mathcal{Z}_+$. Every vertex of A_0 belongs to U_+ , so no $A_0 \cup T$ with $T \in \mathcal{T}_+$ is present. By (V1), none of these vertices belongs to U_- , and hence every negative-threshold edge on A_0 is present. Dispersion edges are always present, and no other c -support defines a permitted k -data edge. Thus the exact root neighborhood of A_0 is $\mathcal{B}_+$. Conversely, the occurrence of every $\mathcal{Z}_+$ -support in that neighborhood says exactly that A_0 belongs to the positive projection. The negative statement is symmetric.

It follows that, after the root images have been chosen, the active ordered k -tuple contributes exactly when its underlying set is an edge of G_σ . The root choices have total weight $\prod_i y_i$, and the ordered active tuples contribute $k!\lambda(G_\sigma; \mathbf{x})$. This proves (11.2). ■

Lemma 11.3. *The statistic Sep is continuous, independent of β , and satisfies $\text{Sep}(W_0) = 0$. Suppose that $z_\beta > \tau_{\text{base}}$. Every optimizer W in the positive phase, represented by a sequence of completed weighted models with $(u_n, a_n, b_n) \rightarrow (u_\beta, t_\beta, 0)$, satisfies $\text{Sep}(W) = \varrho(u_\beta)^m z_\beta t_\beta^k > 0$. Every optimizer in the negative phase, represented by a sequence with $(u_n, a_n, b_n) \rightarrow (u_\beta, 0, t_\beta)$, satisfies $\text{Sep}(W) = -\varrho(u_\beta)^m z_\beta t_\beta^k < 0$. Consequently, for every fixed halting input β , $|\text{Sep}|$ is bounded away from zero on both optimizer phases.*

Proof. Continuity follows because $Q_+ - Q_-$ is a finite quantum graph. At W_0 both sign projections have zero weight, so (11.2) gives $\text{Sep}(W_0) = 0$. Let W lie in the positive phase and choose a representing sequence of completed weighted models with $(u_n, a_n, b_n) \rightarrow (u_\beta, t_\beta, 0)$. The root vectors converge to $\bar{\mathbf{y}}(u_\beta)$, the normalized positive projections converge to an optimizer of $\mathcal{C}_\beta^{\text{in}}$, and the negative projection has limiting total weight zero. Hence $k!\lambda(G_{+,n}; \mathbf{x}^{(n)}) \rightarrow z_\beta t_\beta^k$ and $k!\lambda(G_{-,n}; \mathbf{x}^{(n)}) \rightarrow 0$. Substitution in (11.2), followed by continuity, gives the asserted positive formula. The negative case is symmetric. Since $\varrho(u_\beta)$, z_β , and t_β are positive for every fixed halting input, the final assertion follows. ■

The sign formula in Lemma 11.3 now completes the rooted phase classification stated in Theorem 10.4.

Proof of Theorem 10.4. Fix $W \in \mathcal{L}(\mathcal{D}_\beta)$ and choose an optimizing sequence of completed weighted models converging to W . For the model indexed by n , use the notation from (10.2) with subscript n . By part (i), every subsequential limit of (u_n, a_n, b_n) maximizes Φ_{z_β} . The conclusions and effective estimates in Proposition 10.9 verify all hypotheses of Lemma 9.1.

If $z_\beta = \tau_{\text{base}}$, the phase function $\Phi_{\tau_{\text{base}}}$ has the unique maximizer $(u_0, 0, 0)$. Part (iii) therefore shows that every optimizing sequence has the same finite-step limit W_0 , independent of β . Part (ii) realizes this point, so the optimizer space is exactly $\{W_0\}$. This proves part (i).

Suppose that $z_\beta > \tau_{\text{base}}$. The one-phase/two-phase lemma gives exactly the two maximizers

$$(u_\beta, t_\beta, 0) \quad \text{and} \quad (u_\beta, 0, t_\beta), \quad \text{where} \quad t_\beta = \frac{k\alpha(u_\beta)(z_\beta - \tau_{\text{base}})}{(k+1)\gamma(u_\beta)} > 0,$$

where u_β is the unique maximizer of $\Phi_{\text{base}} + J_{z_\beta}$. Choose a subsequence on which (u_n, a_n, b_n) converges. Its limit is one of the two displayed triples, so this subsequence is pure-sign in the limit. The root vectors converge to $\bar{\mathbf{y}}(u_\beta)$ by part (i). The convergence asserted in that part and compactness give an inner limit in $\mathcal{L}(\mathcal{C}_\beta^{\text{in}})$ along a further subsequence. This further subsequence still converges to W , and its normalized active projections converge to the chosen inner limit. Thus it is a representing sequence of the form asserted in the theorem.

Conversely, part (ii) realizes every member of $\mathcal{L}(\mathcal{C}_\beta^{\text{in}})$ in either sign. The two displayed triples and the limiting root vector are unique. Part (iii), proved by the finite homomorphism-density expansion, shows that the sign and inner optimizer determine the resulting limit represented by completed weighted models, including the limiting contributions of all edge classes imposed by completion. Lemmas 11.2 and 11.3 show that a positive representation and a negative representation cannot determine the same unlabeled

limit. This proves part (ii). The phase-sensitive edit-stability argument in [Proposition 15.3](#) uses the quantitative estimate in [Lemma 9.2](#). ■

The two lemmas now supply the fixed sign detector required in the weighted compiler.

Proof of Theorem 11.1. For $2 \leq r < r_{\text{str}}$, output the empty forbidden family. Given (r, β) with $r \geq r_{\text{str}}$, put $c = r - k$, compute the canonical frame and orbit supports fixed in [Proposition 10.9](#), and compute $\mathcal{F}_\beta = \mathcal{F}_{k,\beta}$ from [Theorem 1.1](#). Enumerate $\mathcal{Q}(\mathcal{F}_\beta)$ and construct the finite rooted conditions of [Section 10](#). The algorithm in [Lemma 10.3](#) outputs $\mathcal{A}_{r,\beta}^{\text{root}}$. Every step is a finite operation uniform in (r, β) , so the map is total computable. The canonical choice of frame and supports shows that $W_{0,r}$ and Sep_r depend only on r , not on β , and [Lemma 11.3](#) gives $\text{Sep}_r(W_{0,r}) = 0$.

By [Lemma 2.4](#), the phase function is Φ_{z_β} , where $z_\beta = \pi(\mathcal{F}_\beta)$. In the nonhalting case this parameter equals τ_{base} , and [Theorem 10.4](#) gives the singleton optimizer W_0 . In the halting case it is larger than τ_{base} , so [Theorem 10.4](#) gives two pure-sign optimizer types with total data weight u_β and active-sign weight $t_\beta > 0$. Put $c_{r,\beta} := \varrho(u_\beta)^m z_\beta t_\beta^k > 0$, the common magnitude given by [Lemma 11.3](#), and define

$$\mathfrak{L}_{r,+}(\beta) := \mathfrak{L}(\mathcal{D}_{r,\beta}) \cap \text{Sep}_r^{-1}(\{c_{r,\beta}\}), \quad \text{and} \quad \mathfrak{L}_{r,-}(\beta) := \mathfrak{L}(\mathcal{D}_{r,\beta}) \cap \text{Sep}_r^{-1}(\{-c_{r,\beta}\}).$$

The rooted phase classification and [Lemma 11.3](#) show that these sets are nonempty, disjoint, and cover the optimizer space. Since Sep_r is continuous, they are closed subsets of the compact optimizer space and are therefore compact. The near-optimal assertion is [Lemma 8.2](#). ■

12 Canonical models and finite star blockers

We now transfer the weighted optimizer space of $\mathcal{D}_\beta$ to an ordinary finite-family Turán problem. We first define the ordinary models generated by the rooted construction and then construct a finite obstruction family controlling which vertices can be added to them.

We begin with the ordinary models obtained by completing the rooted constructions and then blowing up their root vertices.

Definition 12.1. Start with a totally rooted admissible r -graph, clone data vertices arbitrarily many times, and complete the resulting rooted r -graph as in [Lemma 10.3](#). Replace each root $i \in M$ by an arbitrary vertex class V_i . For every edge e of the completed rooted r -graph and every choice of one vertex $v_i \in V_i$ for each $i \in e \cap M$, include the ordinary edge $(e \setminus M) \cup \{v_i : i \in e \cap M\}$; each data vertex remains a singleton. The resulting ordinary r -graph is a *canonical completed model*. A *canonical model* is any subgraph of a canonical completed model. Empty root classes are allowed. Let $\mathfrak{C}_\beta$ denote the class of all canonical models.

Lemma 12.2. *The class $\mathfrak{C}_\beta$ defined in Definition 12.1 is monotone. It contains every ordinary blowup of every member of $\mathcal{D}_\beta$.*

Proof. The first assertion is built into the definition. For the second, choose an admissible partial rooting, add the missing roots as new vertices, delete any data vertices whose desired blowup multiplicity is zero, and clone every remaining data vertex according to its desired multiplicity. Deleting data vertices preserves admissibility, and [Lemma 10.5](#) shows that the subsequent clonings do as well. Complete the resulting rooted r -graph and blow up its root vertices. The original ordinary blowup is a subgraph of the resulting canonical completed model. ■

To recognize these models inside a host with a specified role partition, we use the following rooted compression.

Suppose an r -graph H has a partition $\mathcal{V} = (U, (V_i)_{i \in M})$ with $V(H) = U \dot{\cup} \bigcup_{i \in M} V_i$. An edge is *root-transversal* if it meets every V_i in at most one vertex. For each root-transversal edge e , form $(e \cap U) \cup \{i \in M : e \cap V_i \neq \emptyset\}$. The *rooted compression* $\text{Comp}_\mathcal{V}(H)$ is the rooted r -graph on $U \dot{\cup} M$ formed by these edges. Every $i \in M$ is retained as a distinguished root, even when V_i is empty or no edge of H meets V_i . We say that H belongs to $\mathfrak{C}_\beta$ *through* $\mathcal{V}$ when H is a subgraph of a canonical completed model whose canonical partition is $\mathcal{V}$.

This compression exactly detects membership through the fixed partition. If H is a subgraph of a canonical completed model with partition $\mathcal{V}$, then compressing its root classes shows that every edge of H

is root-transversal and that $\text{Comp}_{\mathcal{V}}(H)$ is admissible. Conversely, if these two conditions hold, complete the admissible rooted compression, replace each root label i by the class V_i , and retain every data vertex as a singleton. Every edge of H then expands from an edge of its rooted compression, so H is a subgraph of the resulting canonical completed model. We record this equivalence for later use.

Fact 12.3. *For a fixed partition $\mathcal{V} = (U, (V_i)_{i \in M})$, the following are equivalent.*

- (i) H belongs to $\mathfrak{C}_{\beta}$ through $\mathcal{V}$.
- (ii) Every edge of H is root-transversal and $\text{Comp}_{\mathcal{V}}(H)$ is an admissible totally rooted r -graph.

Canonical models come with a role partition, whereas an ordinary forbidden family cannot refer to that partition. We therefore build finite configurations that recover the roles and certify the failure of every possible placement of one new vertex. The asymmetric frame fixes the root roles in the host partition, the pins fix the witness roles, and the star blockers combine one failed extension for each tentative role.

Introduce one formal data role $\star$ and set $\mathcal{R}_{\text{role}} := M \cup \{\star\}$. Define the role frame

$$\mathcal{R} := \left(\binom{M}{r} \setminus \mathcal{J} \right) \cup \left\{ \{\star\} \cup S : S \in \binom{M}{r-1} \right\}.$$

Lemma 12.4. *Let $\phi: \mathcal{R}_{\text{role}} \rightarrow M \cup \{\text{data}\}$ be a role assignment, where $\phi(x) = i$ means that x is assigned root role i , and $\phi(x) = \text{data}$ means that x is assigned the data role. Suppose that, for every edge of $\mathcal{R}$, the roles assigned to its vertices can occur on a permitted rooted edge. Then $\phi(\star) = \text{data}$ and $\phi(i) = i$ for every $i \in M$.*

Proof. The frame covers pairs. Hence two formal frame vertices cannot map to the same root role, since an edge containing them would repeat that root role. Let $S := \phi^{-1}(\text{data})$. Since $|\mathcal{R}_{\text{role}}| = m + 1$ and there are only m root roles, S is nonempty.

Suppose $|S| \geq 2$. If $|\mathcal{R}_{\text{role}} \setminus S| \geq r - 2$, choose an edge containing formal $\star$ and exactly two members of S . If $\star \in S$, use one further member of S and $r - 2$ vertices outside S ; if $\star \notin S$, use two members of S and $r - 3$ further vertices outside S . Every r -set containing formal $\star$ is an edge of $\mathcal{R}$. Its image has exactly two data vertices, impossible because the available data multiplicities are $0, 1, k, k + 1$ and $k \geq 3$.

If $|\mathcal{R}_{\text{role}} \setminus S| < r - 2$, then the bound $m \geq r(r + 4)/2$ implies $|S| \geq r$. If $\star \in S$, choose an r -set in S containing $\star$; if $\star \notin S$, choose formal $\star$ and $r - 1$ members of S . The image has respectively r or $r - 1$ data vertices. Since $r = k + c$ with $c \geq 4$, neither multiplicity is available. Thus $|S| = 1$, and ϕ is a permutation of the $m + 1$ roles. After representing the data role by $\star$, every permitted rooted edge with zero or one data vertex has a role set belonging to $\mathcal{R}$, so the hypothesis gives $\phi(\mathcal{R}) \subseteq \mathcal{R}$; equality follows because ϕ is a permutation.

The degree of $\star$ in $\mathcal{R}$ is $\binom{m}{r-1}$. Since $\mathcal{J}$ is 2-regular, every root role has degree $\binom{m-1}{r-2} + \binom{m-1}{r-1} - 2 = \binom{m}{r-1} - 2$. Hence $\star$ is fixed. The induced permutation of M preserves $\binom{M}{r} \setminus \mathcal{J}$, and hence preserves $\mathcal{J}$. As $\mathcal{J}$ is asymmetric, it is the identity. $\blacksquare$

Fix formal frame vertices r_i for $i \in M$ and a distinguished formal data vertex d . For another vertex y , put

$$\text{Pin}_{\star}(y) := \left\{ \{y\} \cup \{r_i : i \in S\} : S \in \binom{M}{r-1} \right\}, \quad (12.1)$$

and, for $i \in M$,

$$\text{Pin}_i(y) := \left\{ \{y, d\} \cup \{r_j : j \in S\} : S \in \binom{M \setminus \{i\}}{r-2} \right\}. \quad (12.2)$$

Once the frame roles are fixed by Lemma 12.4, these edge systems determine the role of y . If $\text{Pin}_{\star}(y)$ is present and y has root role j , choosing $S \ni j$ in (12.1) produces an edge that repeats root role j . If $\text{Pin}_i(y)$ is present and y has the data role, each edge in (12.2) contains the two data vertices y and d . If instead y has a root role $j \neq i$, choosing $S \ni j$ in (12.2) again produces an edge with a repeated root role. The intended assignments are possible because the corresponding edges have one data vertex and $r - 1$ distinct root roles and are present in every canonical completed model. We record the conclusion for later use.

Fact 12.5. *Once the frame roles are fixed by Lemma 12.4, the edge system $\text{Pin}_{\star}(y)$ forces y to have data role $\star$, while $\text{Pin}_i(y)$ forces y to have root role i for every $i \in M$.*

12.1 Centered one-vertex certificates

We now convert a rooted extension failure into an ordinary finite configuration. A *role-labeled centered configuration* is a triple (C, v_C°, ϖ_C) , where C is a finite r -graph, $v_C^\circ \in V(C)$ is its distinguished center, and $\varpi_C: V(C) \rightarrow \mathcal{R}_{\text{role}}$ assigns a role to every vertex. A map from C to a canonically partitioned ordinary r -graph is *role respecting* if role i is mapped into the root class V_i and role $\star$ is mapped into the data class U . We always require the map to be edgewise injective.

Fix a tentative role t for a new vertex v . We separate invalid edge profiles from the four admissibility violations. Let E be an r -set of distinct formal vertices. For a role assignment $\varpi: E \rightarrow \mathcal{R}_{\text{role}}$, put $\ell(\varpi) := |\varpi^{-1}(\star)|$ and call $\varpi(E) \cap M$ its *root support*. We call the role profile specified by ϖ *permitted* when no root role is repeated and one of the following holds.

- (a) $\ell(\varpi) = 0$ and the root support lies in $\binom{M}{r} \setminus \mathcal{J}$;
- (b) $\ell(\varpi) = 1$;
- (c) $\ell(\varpi) = k$ and the root support lies in $\mathcal{D}_{\text{disp}} \cup \mathcal{Z}_+ \cup \mathcal{Z}_- \cup \mathcal{T}_+ \cup \mathcal{T}_-$;
- (d) $\ell(\varpi) = k + 1$ and the root support lies in $\mathcal{P}$.

All other role assignments are invalid. For every invalid assignment with a distinguished formal vertex of role t , regard its r formal vertices as one r -edge, designate the distinguished vertex as the center, and retain the full role assignment. This is an *invalid one-edge t -certificate*. In particular, if a through- v edge contains another vertex in the same tentative root class, the corresponding certificate has two distinct formal vertices carrying the same root role; the repeated role is not collapsed in the formal edge.

Suppose now that every edge through v is root-transversal and has a permitted role profile. Compressing these edges under the tentative role partition gives the finite rooted conditions of Section 10. Since the previous rooted compression is admissible, failure of the extension has one of the following forms.

- (F2) a (V1) double-sign certificate;
- (F3) a (V2) threshold certificate;
- (F4) a (V3) penalty certificate;
- (F5) a (V4) realization of some $Q \in \mathcal{Q}(\mathcal{F}_\beta)$ in one sign projection.

For uniformity of notation we call the invalid one-edge certificates (F1). In each of (F2)–(F5), a witnessing rooted configuration uses at least one edge of the extended compression that is not present in the previous admissible compression; otherwise the same violation would already occur there. Every such new compression edge has an ordinary preimage through v under the tentative partition.

Every member of $\mathcal{Q}(\mathcal{F}_\beta)$ has no isolated vertices, so every data-variable class in an (F5) certificate occurs in a displayed raw record edge. Hence a new data class represented by the center in an (F5) witness gives a nonempty center incidence.

The five labels (F1)–(F5) give an exhaustive classification, but several violations may occur simultaneously. We make the classification disjoint by using the displayed order. For a tentative role t whose placement fails, first test whether a through-center edge has an invalid role profile. If not, test (V1), (V2), (V3), and (V4), in that order, and assign the failed placement to the first violated condition. If none of these five tests fails, then every edge is root-transversal with a permitted role profile and the extended rooted compression is admissible. By Fact 12.3, the placement succeeds, a contradiction. Thus every failed placement has a uniquely determined first failed test; the particular witness for that test need not be unique, and the catalogue retains every possible witness.

Fix a rooted violation certificate Q and a tentative role t . A *legal center-incidence choice* for (Q, t) is defined as follows. If $t = \star$, it is one data-variable class of Q that occurs in at least one displayed edge. If $t = i \in M$, it is a nonempty set of the displayed occurrences of root label i in Q , containing at most one occurrence from each displayed edge.

For each rooted violation certificate of type (F2)–(F5), tentative role t , and legal center-incidence choice, define its *maximal ordinary role lift* as follows. The center is assigned role t . Every data-variable class of the certificate is represented by one role- $\star$ vertex, with the selected class represented by the center when $t = \star$. For every displayed edge e and every root-label occurrence i in e , use the center when $t = i$ and this occurrence was selected; otherwise use a fresh role- i vertex private to the pair (e, i) . No additional

identifications are made. A frozen-root identification never identifies two vertices of one edge, and a legal center-incidence choice uses the center for at most one vertex of any displayed edge. Hence every displayed edge of the lift has r distinct formal vertices.

Case	First failed condition	Catalogue member	Property retained once the frame, pins, and center role are fixed
(F1)	A through-center edge is not root-transversal or its role profile is not permitted.	The invalid one-edge certificate with the offending role assignment on its r distinct formal vertices.	Edge injectivity preserves these distinct vertices, while the frame and pins preserve their roles.
(F2)	No (F1) failure occurs, but (V1) fails at a data vertex in $U_+ \cap U_-$.	A maximal lift of the two record-edge witnesses of opposite signs.	The two disjoint sign-support families and their common displayed data vertex remain visible.
(F3)	Neither earlier case occurs, but (V2) fails at a threshold edge.	A maximal lift of the threshold edge and one same-sign record-edge witness for each of its k data vertices.	The threshold support and the witnesses placing all k data vertices in the same set U_σ remain visible.
(F4)	No earlier case occurs, but (V3) fails at a penalty edge.	A maximal lift of the penalty edge and one signed-record witness for each of its $k + 1$ data vertices.	The penalty support and the witnesses placing all $k + 1$ data vertices in $U_+ \cup U_-$ remain visible.
(F5)	No earlier case occurs, but (V4) fails in one sign projection.	A maximal lift of a realization of some $Q \in \mathcal{Q}(\mathcal{F}_\beta)$ in one sign projection.	The projected image is a further edgewise-injective image of Q and hence remains in $\mathcal{Q}(\mathcal{F}_\beta)$.

Table 6: The ordered catalogue of one-vertex extension failures.

The frozen-root identifications and maximal lifts divide the possible coincidences as follows. Coincidences between different host edges are handled by an edgewise-injective homomorphism, whereas vertices within one invalid role profile are never identified. An actual invalid through-center edge is therefore the role-respecting image of its invalid one-edge certificate.

Lemma 12.6. *For every $t \in \mathcal{R}_{\text{role}}$ there is an effectively computable finite family $\mathcal{C}_{\beta,t}$ of role-labeled centered configurations, consisting of invalid one-edge certificates and maximal role lifts. There is also a computable constant B_β^{wit} with the following property.*

Suppose that $H - v$ belongs to $\mathfrak{C}_\beta$ through a canonical partition $\mathcal{V} = (U, (V_i)_{i \in M})$, but $H \notin \mathfrak{C}_\beta$. For $t \in \mathcal{R}_{\text{role}}$, let $\mathcal{V}^t$ be the tentative partition obtained by placing v in U when $t = \star$ and in V_t when $t \in M$. Simultaneously for all $t \in \mathcal{R}_{\text{role}}$, one can choose $C_t \in \mathcal{C}_{\beta,t}$ and an edgewise-injective homomorphism $\psi_t: C_t \rightarrow H$, role respecting with respect to $\mathcal{V}^t$, such that $\psi_t(v_{C_t}^\circ) = v$ and $\psi_t(V(C_t) \setminus \{v_{C_t}^\circ\}) \subseteq V(H) \setminus \{v\}$. Moreover, these choices satisfy $|\bigcup_{t \in \mathcal{R}_{\text{role}}} \psi_t(V(C_t) \setminus \{v_{C_t}^\circ\})| \leq B_\beta^{\text{wit}}$.

Proof. Fix any canonical partition $\mathcal{V}$ through which $H - v$ belongs to $\mathfrak{C}_\beta$. No tentative partition $\mathcal{V}^t$ succeeds, for otherwise [Fact 12.3](#) would put H in $\mathfrak{C}_\beta$. Fix $t \in \mathcal{R}_{\text{role}}$ and apply the ordered tests in [Table 6](#). If (F1) is the first failure, the actual offending edge, with the roles supplied by $\mathcal{V}^t$, is the role-respecting edgewise-injective image of its invalid one-edge certificate.

Suppose that (F1) does not occur. Then all edges through v are root-transversal with permitted role profiles, so $\mathcal{V}^t$ has a well-defined extended rooted compression. Since the placement does not succeed, at least one of (V1)–(V4) fails. Choose a witnessing configuration for the first failed condition. This configuration contains at least one displayed edge of the extended compression that is not present in the previous compression, since otherwise it would already contradict the admissibility of $H - v$. For each displayed edge of the previous compression choose an ordinary preimage in $H - v$, and for each displayed new compression edge choose a preimage through v . Recording every data class or root occurrence represented by v selects a nonempty retained center-incidence choice. The maximal-lift construction then gives the required role-respecting edgewise-injective map to H , and all noncentral vertices map into $H - v$.

It remains to verify finiteness and effectivity. For (F1), there are only finitely many role assignments on r formal vertices, and invalidity is decidable from the fixed support tables. For (F2)–(F4), the violation

templates, frozen-root identifications, support choices, and center-incidence choices are finite. For (F5), the extra choice is a member of the finite family $\mathcal{Q}(\mathcal{F}_\beta)$. Include every invalid one-edge certificate and every maximal ordinary lift in the enumeration.

Each catalogue is nonempty. Since $r = k + c > k + 1$, the all-data assignment is an invalid (F1) profile for $t = \star$, while an assignment repeating root role t is invalid for $t \in M$. Since r is fixed throughout this part, we continue to suppress it from the notation and define the computable bound

$$B_\beta^{\text{wit}} := \sum_{t \in \mathcal{R}_{\text{role}}} \max_{C \in \mathcal{C}_{\beta,t}} (|V(C)| - 1).$$

For any simultaneous choice of the maps ψ_t , the union of their images outside v has order at most B_β^{wit} . Every list and test is computable from the edge lists of $\mathcal{Q}(\mathcal{F}_\beta)$ and of the support systems. ■

12.2 Star blockers and image persistence

We now combine the failure certificates for all tentative roles at one common center. A shared role frame and complete pins will ensure that an edgewise-injective homomorphic image preserves the role contradiction. After defining this construction, we delete isolated vertices and take the homomorphic closure $\mathcal{Q}$, as in Section 2.

For every tuple $(C_t)_{t \in \mathcal{R}_{\text{role}}}$ with $C_t \in \mathcal{C}_{\beta,t}$, identify their centers to one vertex v and keep every other formal vertex distinct. Add a fresh disjoint copy of the role frame, with frame vertices r_i for $i \in M$ and a distinguished data vertex d . For every noncentral vertex y of every C_t , add the complete pin $\text{Pin}_{\varpi_{C_t}(y)}(y)$. The resulting finite r -graph is a *base star blocker*.

Put $\mathcal{K}_\beta := \mathcal{Q}(\{B^\circ : B \text{ is a base star blocker}\})$.

The role-selection mechanism behind the shared-center construction is shown in Figure 13. The blocker contains every branch simultaneously; the highlighted branch is the one indexed by the role eventually assigned to the image of the center.

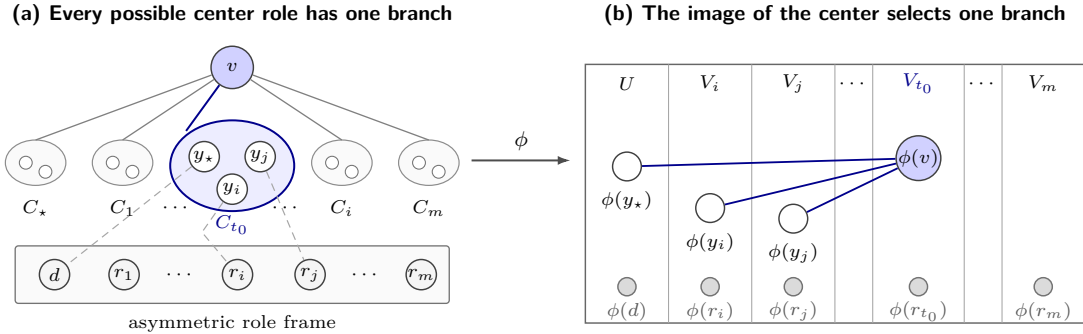


Figure 13: Why a base star blocker has no edgewise-injective homomorphic image in a canonical completed model. (a) The configurations C_t , one for each tentative role $t \in M \cup \{\star\}$, share only the center v . The highlighted branch C_{t_0} shows three representative noncentral vertices; the dashed connectors stand for complete pin systems, not individual edges, and every omitted noncentral vertex is pinned similarly. (b) The diagram depicts the case $t_0 \in M$; the gray bottom points are the displayed frame images. If $\phi(v) \in V_{t_0}$, the role frame and pins force the vertices of C_{t_0} into their prescribed classes, so the rooted violation survives. If $t_0 = \star$, replace V_{t_0} by U . The blue segments group the displayed vertices into $\phi(C_{t_0})$ and do not represent individual r -edges.

Lemma 12.7. *There is no edgewise-injective homomorphism from a base star blocker to a canonical completed model. Consequently no member of $\mathcal{K}_\beta$ embeds in a canonical completed model.*

Proof. Suppose that $\phi: B \rightarrow M_0$ is an edgewise-injective homomorphism from a base star blocker to a canonical completed model. Assign each frame vertex the role of its image under ϕ in the canonical partition of M_0 . Since every formal frame edge maps to an actual edge of M_0 , the roles assigned to that edge can occur on a permitted rooted edge. Thus this role assignment satisfies the hypothesis of Lemma 12.4; the pair-covering property of the frame and edge injectivity also make the formal frame

vertices distinct in the image. Hence r_i has role i for every $i \in M$, and d has role $\star$. By [Fact 12.5](#), every noncentral witness vertex has its prescribed role. The center has some role $t_0 \in \mathcal{R}_{\text{role}}$ in the canonical partition. If it is identified with any other blocker vertex, that vertex necessarily has the same role t_0 ; an identification with a differently pinned or framed role is already impossible. Consider the selected certificate C_{t_0} .

We check the five failure types.

- (F1) The catalogue member is one edge on r distinct formal vertices. Edge injectivity preserves the r distinct image vertices, while the frame roles identified above and the pins preserve the complete role assignment. The repeated root role, or the other reason that the role profile is not permitted, therefore remains.
- (F2) The images of the two record edges retain root supports belonging to the two disjoint typed support families $\mathcal{Z}_+$ and $\mathcal{Z}_-$ and still share the displayed data vertex; hence that vertex belongs to both U_+ and U_- .
- (F3) The threshold edge keeps its k distinct data vertices, and the image of each designated record-edge witness places its displayed vertex in the same set U_σ . Thus (V2) persists.
- (F4) The identical argument with $k + 1$ data vertices proves persistence; here the chosen sign may vary from one data vertex to another, exactly as in (V3).
- (F5) Let $Q \in \mathcal{Q}(\mathcal{F}_\beta)$ be the displayed projected copy. The restriction of ϕ to every record edge is injective, so the induced map from Q to the corresponding sign projection of M_0 is edgewise injective. Its image Q' is therefore an edgewise-injective homomorphic image of Q . Since $\mathcal{Q}(\mathcal{F}_\beta)$ is closed under further edgewise-injective images, $Q' \in \mathcal{Q}(\mathcal{F}_\beta)$. That sign projection of the canonical completed model M_0 would contain Q' , contradicting (V4).

Thus the image of C_{t_0} always gives a rooted violation in the rooted compression of M_0 , a contradiction. If a member of $\mathcal{K}_\beta$ embedded in M_0 , composition with the edgewise-injective homomorphism defining it would give precisely the excluded edgewise-injective homomorphism from a base star blocker. ■

13 The final ordinary forbidden family

The final family has two complementary components. The pair-covering extensions $\text{Ext}(A)$ and their fixed distinguished cores were defined in [Section 2](#); they force the projected base to avoid the rooted obstruction family. The star blockers exclude a vertex that cannot be assigned any canonical role.

Define $\mathcal{G}_\beta := \mathcal{Q}\left(\left(\bigcup_{A \in \mathcal{A}_\beta^{\text{root}}} \text{Ext}(A)\right)^\circ\right) \cup \mathcal{K}_\beta$. For the arbitrary fixed target uniformity in force, the global notation for this family is $\mathcal{G}_{r,\beta} := \mathcal{G}_\beta$.

The point of this construction is the following exact transfer from the weighted rooted class to the ordinary forbidden family.

Proposition 13.1. *One has $\pi(\mathcal{G}_\beta) = \Lambda_r(\mathcal{D}_\beta)$.*

Its proof uses the following two properties of the final forbidden family.

Lemma 13.2. (i) *The family $\mathcal{G}_\beta$ is finite and effectively computable, and $\text{Forb}(\mathcal{G}_\beta)$ is closed under ordinary blowups.*

(ii) *Every r -graph in $\mathfrak{C}_\beta$ is $\mathcal{G}_\beta$ -free.*

Proof. For part (i), a finite hypergraph has finitely many edgewise-injective homomorphic images, so the family is finite and effective. If a blowup of H contains $F \in \mathcal{G}_\beta$, projection to the blowup vertex classes is edgewise injective and produces an edgewise-injective image F' . Since the family is closed under such images, $F' \in \mathcal{G}_\beta$ and F' embeds in H .

For part (ii), star blockers are excluded by [Lemma 12.7](#). Suppose a canonical completed model contained an edgewise-injective image of some $Q \in \text{Ext}(A)$. Every pair of vertices in the fixed distinguished A -core $\iota_{A,Q}(V(A))$ lies in an edge of Q . Hence neither the map defining this image nor projection to the canonical root vertex classes can identify two core vertices. Their composition with $\iota_{A,Q}$ would embed A in the admissible base r -graph, contradicting $A \in \mathcal{A}_\beta^{\text{root}}$. The assertion for subgraphs follows. ■

The two parts of the lemma now prove the density identity.

Proof of Proposition 13.1. For each $j \geq 1$, choose a weighted member of $\mathcal{D}_\beta$ whose normalized value is at least $\Lambda_r(\mathcal{D}_\beta) - 1/j$, fix an admissible partial rooting, and add zero-weight vertices for the missing root labels. Perturb its probability vector by a sufficiently small rational amount so that every existing data coordinate is strictly positive, every root coordinate is nonnegative, and the normalized value remains at least $\Lambda_r(\mathcal{D}_\beta) - 2/j$. Such vectors are dense in the finite probability simplex. The perturbation changes neither the rooted edge set nor the sets U_+ and U_- . A zero root weight is allowed and will give an empty root vertex class.

For this rational weighting, clone each data vertex according to the positive numerator of its weight, complete the resulting cloned rooted r -graph, and realize the root weights by vertex-class sizes. Choose a sufficiently large common multiple of the denominators so that the resulting order is at least j and the difference between $r!|N|/v(N)^r$ and $|N|/\binom{v(N)}{r}$ is at most $1/j$. By Lemma 8.1, the cloning preserves every weighted homomorphism density, in particular the original Lagrangian value, while completion only increases the Lagrangian value and cannot exceed the class optimum. The resulting canonical completed model is free by part (ii) of Lemma 13.2. Taking these models for $j = 1, 2, \dots$ gives orders tending to infinity and ordinary densities tending to $\Lambda_r(\mathcal{D}_\beta)$. Thus $\pi(\mathcal{G}_\beta) \geq \Lambda_r(\mathcal{D}_\beta)$.

For the reverse inequality, the identity map is edgewise injective and every member of $\text{Ext}(A)$ has no isolated vertices, so $\bigcup_A \text{Ext}(A) \subseteq \mathcal{Q}((\bigcup_A \text{Ext}(A))^\circ) \subseteq \mathcal{G}_\beta$. Following Mubayi's extension method [52], we use the minimum-support argument. If H is $\mathcal{G}_\beta$ -free, choose a minimum-support Lagrangian-maximizing weighting and let D be its positive support. Then D covers pairs. Otherwise the weights on an uncovered pair can be moved to one endpoint without changing the value, contradicting minimal support.

If D contained some $A \in \mathcal{A}_\beta^{\text{root}}$, take the union of the A -copy and one covering edge for every pair of its vertices, retaining only vertices which occur in this union. Every edgeless r -graph belongs to $\mathcal{D}_\beta$, so each obstruction A has an edge and in particular at least two vertices. The resulting extension has no isolated vertex, has order at most the bound in the definition of $\text{Ext}(A)$, and is therefore a forbidden member of $\text{Ext}(A)$. Hence $D \in \mathcal{D}_\beta$ and $\lambda(H) \leq \Lambda(\mathcal{D}_\beta)$. Applying the uniform weighting to H and letting its order tend to infinity gives $\pi(\mathcal{G}_\beta) \leq \Lambda_r(\mathcal{D}_\beta)$. ■

14 Symmetrized stability and vertex extendability

The symmetrization method goes back to Zykov [76]; for hypergraphs we use the terminology of Liu, Mubayi, and Reiher [42, pp. 38–39]. For $v \in V(H)$, its *link* is

$$L_H(v) := \{A \in \binom{V(H) \setminus \{v\}}{r-1} : A \cup \{v\} \in H\}.$$

Two vertices $u, v \in V(H)$ are *equivalent*, written $u \sim_H v$, if $L_H(u) = L_H(v)$. An r -graph H is *symmetrized* if every pair of non-equivalent vertices is contained in a common edge.

The following standard observation is recorded, for example, by Liu, Mubayi, and Reiher [42, p. 49, proof of Lemma 4.2].

Fact 14.1. *Every symmetrized r -graph H is a blowup of an r -graph D that covers pairs; the blowup classes are precisely the equivalence classes of $\sim_H$.*

The degree-stability criterion applied in the next section requires two concrete inputs: symmetrized stability and vertex extendability. We verify them here. The first proposition gives the symmetrized statement. The general marginal estimate that follows turns near-optimal weighted value into the vertex-degree control needed for the extension argument.

Proposition 14.2. *Every symmetrized $\mathcal{G}_\beta$ -free r -graph belongs to $\mathfrak{C}_\beta$.*

Proof. By Fact 14.1, write H as a blowup of an r -graph D that covers pairs. If $D \notin \mathcal{D}_\beta$, then D contains some $A \in \mathcal{A}_\beta^{\text{root}}$. For every pair of vertices in this copy choose an edge of D containing the pair, and take the union on exactly the vertices occurring in the copy and the chosen edges. As in the reverse-inequality proof of Proposition 13.1, this is a no-isolated A -pair-covering extension of the bounded order used in the definition of $\text{Ext}(A)$, so it embeds in H , contradicting $\mathcal{G}_\beta$ -freeness. Thus $D \in \mathcal{D}_\beta$. By Lemma 12.2, every blowup of D belongs to $\mathfrak{C}_\beta$. ■

The second main conclusion of this section is the required one-vertex extension property.

Proposition 14.3. *There are $\zeta_\beta > 0$ and n_β such that the following holds. If H is a $\mathcal{G}_\beta$ -free r -graph on $n \geq n_\beta$ vertices with $\delta(H) \geq (\Lambda_r(\mathcal{D}_\beta)/(r-1)! - \zeta_\beta)n^{r-1}$ and $H - v \in \mathfrak{C}_\beta$ for some v , then $H \in \mathfrak{C}_\beta$.*

Its proof has three ingredients.

- (i) We first bound every coordinate derivative at a near-optimal weighting.
- (ii) Exact compression then converts this bound into an ordinary vertex-degree estimate.
- (iii) Finally, the witness catalogue turns any failed extension into a forbidden star blocker.

The following estimate is independent of the construction. After compression, its coordinate derivatives become normalized ordinary vertex degrees.

Lemma 14.4. *Let $\mathcal{C}$ be a class of r -graphs. If $G \in \mathcal{C}$ and a probability weighting $\mathbf{x}$ satisfies $\lambda(G; \mathbf{x}) \geq \Lambda(\mathcal{C}) - \eta$, then for every $i \in V(G)$, we have*

$$\partial_i \lambda(G; \mathbf{x}) \leq r\Lambda(\mathcal{C}) + \sqrt{\frac{8\eta}{(r-2)!}}. \quad (14.1)$$

Proof. Put $\mathbf{h} = \mathbf{e}_i - \mathbf{x}$ and $g(t) = \lambda(G; (1-t)\mathbf{x} + t\mathbf{e}_i)$. Since $(1-t)\mathbf{x} + t\mathbf{e}_i$ is a probability weighting on the same r -graph, $g(t) \leq \Lambda(\mathcal{C})$ for $0 \leq t \leq 1$. Euler's identity gives

$$g'(0) = \partial_i \lambda(G; \mathbf{x}) - r\lambda(G; \mathbf{x}). \quad (14.2)$$

For every probability vector $\mathbf{z}$, we have

$$|D_{\mathbf{z}}^2 \lambda(G; \mathbf{z})[\mathbf{h}, \mathbf{h}]| \leq \frac{1}{(r-2)!} \left(\sum_j |h_j| \right)^2 \leq \frac{4}{(r-2)!} =: B.$$

Let $A = g'(0)$. If $A \leq 0$, (14.1) follows from (14.2). If $A > 0$, then $A \leq \partial_i \lambda(G; \mathbf{x}) \leq 1/(r-1)! \leq B$, so $t = A/B \in [0, 1]$. Taylor's inequality gives

$$\Lambda(\mathcal{C}) \geq g(t) \geq g(0) + At - \frac{B}{2}t^2 = g(0) + \frac{A^2}{2B}.$$

Thus $A \leq \sqrt{2B\eta}$, and (14.1) follows from (14.2) and $\lambda(G; \mathbf{x}) \leq \Lambda(\mathcal{C})$. ■

Let M_0 be a canonical completed model on $N \geq 1$ ordinary vertices, and fix one presentation of M_0 by a completed rooted graph D and root classes $(V_i)_{i \in M}$. Give root i weight $|V_i|/N$ and every data vertex weight $1/N$; denote this probability weighting by $\mathbf{x}$. The presentation map $\pi: V(M_0) \rightarrow V(D)$ sends every vertex of V_i to root i and fixes every singleton data vertex. The full graph D remains part of the presentation: when $V_i = \emptyset$, its edges containing i need not be recoverable from M_0 , but every such edge has zero weighted contribution.

Lemma 14.5. *For every finite r -graph F , we have*

$$t(F, M_0) = t(F, D, \mathbf{x}), \quad (14.3)$$

where M_0 carries its uniform vertex weighting. In particular, $\lambda(D; \mathbf{x}) = |M_0|/N^r$. For every ordinary vertex $y \in V(M_0)$, we have

$$\partial_{\pi(y)} \lambda(D; \mathbf{x}) = \frac{\deg_{M_0}(y)}{N^{r-1}}. \quad (14.4)$$

Proof. Group the maps $\phi: V(F) \rightarrow V(M_0)$ by the presented map $\bar{\phi} = \pi \circ \phi$. If some edge of F has two vertices sent by $\bar{\phi}$ to the same root, its edge indicator is zero in D ; every lift also has indicator zero because canonical edges are root-transversal. Otherwise the edge indicators of all lifts agree with those of $\bar{\phi}$. There are $|V_i|^{\bar{\phi}^{-1}(i)}$ independent lifts over root i , while a data coordinate has one lift. If $V_i = \emptyset$ and $\bar{\phi}^{-1}(i) \neq \emptyset$, both this lift count and the corresponding weight are zero. After normalization by $N^{v(F)}$, the total lift weight is therefore

$$\prod_{i \in M} \left(\frac{|V_i|}{N} \right)^{|\bar{\phi}^{-1}(i)|} \prod_{v \in U} \left(\frac{1}{N} \right)^{|\bar{\phi}^{-1}(v)|},$$

which is exactly the weight of $\bar{\phi}$ in $(D, \mathbf{x})$. Summing over $\bar{\phi}$ proves (14.3). Taking F to be one r -edge gives the second assertion of the lemma. Differentiating the corresponding edge monomials deletes the coordinate of y and counts precisely the ordinary edges containing y , proving (14.4). ■

Combining (14.3) with Lemma 11.2 shows that the identities in (11.2) also hold for every canonical completed ordinary model after compression, and hence, by continuity, on limits of such models.

We can now prove vertex extendability. Suppose that adjoining one vertex destroys canonicity. The witness catalogue then supplies a bounded failure for every tentative role. We complete the old canonical r -graph and choose frame representatives at random; the minimum-degree hypothesis makes it possible to realize all required frame and pin edges simultaneously, producing a forbidden star blocker.

Proof of Proposition 14.3. Put $H_0 := H - v$ and $N := n - 1$. Choose a canonical partition of H_0 and complete it, on the same ordinary vertex set, to a canonical completed model $M_0 \supseteq H_0$. For every old vertex y , we have

$$\deg_{H_0}(y) \geq \deg_H(y) - \binom{n-2}{r-2} \geq \left(\frac{\Lambda_r(\mathcal{D}_\beta)}{(r-1)!} - \zeta_\beta - o(1) \right) N^{r-1}. \quad (14.5)$$

Summing (14.5) over all N old vertices and using $\sum_y \deg_{H_0}(y) = r|H_0|$ gives $|H_0|/N^r \geq \Lambda(\mathcal{D}_\beta) - \zeta_\beta/r - o(1)$. Compression makes M_0 a weighted member of $\mathcal{D}_\beta$, so $|M_0|/N^r \leq \Lambda(\mathcal{D}_\beta)$. Therefore, with $\Delta := M_0 \setminus H_0$, we have $|\Delta| \leq (\zeta_\beta/r + o(1))N^r$. The compressed completed model is near optimal. By Lemma 10.7, after ζ_β is chosen sufficiently small there is a constant $c_{\text{cls}} > 0$, depending only on the fixed construction, such that

$$|V_i| \geq c_{\text{cls}}N \quad \text{for every } i \in M, \quad \text{and} \quad |U| \geq c_{\text{cls}}N, \quad (14.6)$$

where U is the data class of the canonical partition.

Apply Lemmas 14.4 and 14.5 to $\mathcal{C} = \mathcal{D}_\beta$ with $\eta = \zeta_\beta/r + o(1)$. Equations (14.1) and (14.4) give

$$\frac{\deg_{M_0}(y)}{N^{r-1}} \leq \frac{\Lambda_r(\mathcal{D}_\beta)}{(r-1)!} + \sqrt{\frac{8(\zeta_\beta/r + o(1))}{(r-2)!}}.$$

Subtracting the lower bound (14.5), and decreasing ζ_β so that $\zeta_\beta \leq 1$, gives, for every old vertex y ,

$$\deg_{M_0}(y) - \deg_{H_0}(y) \leq \left(\left(1 + \sqrt{\frac{8}{r(r-2)!}} \right) \sqrt{\zeta_\beta} + o(1) \right) N^{r-1}. \quad (14.7)$$

Assume for a contradiction that $H \notin \mathfrak{C}_\beta$. For every role $t \in \mathcal{R}_{\text{role}}$, choose $C_t \in \mathcal{C}_{\beta,t}$ and an edgewise-injective map ψ_t as in Lemma 12.6. Let V_{exc} be the union of their old image vertices, so $|V_{\text{exc}}| \leq B_\beta^{\text{wit}}$. The common center is $v \notin V(H_0)$, while $V_{\text{exc}} \subseteq V(H_0)$ contains every other vertex used by any witness map, including coincidences between different witness images.

The vertex classes $(V_i)_{i \in M}$ and U are pairwise disjoint. We choose one new representative from each class outside V_{exc} , so the frame representatives and the distinguished data representative are mutually distinct and avoid all noncentral witness images. Independently choose

$$r_i \in V_i \setminus V_{\text{exc}} \quad \text{for every } i \in M, \quad \text{and} \quad d \in U \setminus V_{\text{exc}}, \quad (14.8)$$

uniformly. By (14.6), these choices are well defined for all sufficiently large N .

Let X_{fr} count required edges of the role frame which are missing from H_0 . Every such edge is present in M_0 . A selected frame edge uses r prescribed vertex classes, each of size at least $c_{\text{cls}}N - B_\beta^{\text{wit}}$. Hence every fixed edge of Δ is selected with probability at most $(c_{\text{cls}}N - B_\beta^{\text{wit}})^{-r}$. Its role profile determines at most one formal frame edge, both for a root-only frame edge and for a one-data frame edge. Consequently, we have

$$\mathbb{E}X_{\text{fr}} \leq |\Delta|(c_{\text{cls}}N - B_\beta^{\text{wit}})^{-r} \leq C_{\beta,1}\zeta_\beta + o(1).$$

Now fix a noncentral formal vertex y in one of the chosen C_t , and write $\bar{y} := \psi_t(y)$. Let X_y count missing edges in its complete pin. Every pin edge is present in M_0 and contains $\bar{y}$. Once $\bar{y}$ is fixed, the remaining

$r - 1$ vertices are chosen from prescribed vertex classes of size at least $c_{\text{cls}}N - B_{\beta}^{\text{wit}}$. An edge of M_0 incident with $\bar{y}$ but missing from H_0 realizes at most one formal pin edge, because its role profile uniquely determines the subset of frame roles used in (12.1) or (12.2). Thus (14.7) gives

$$\mathbb{E}X_y \leq (\deg_{M_0}(\bar{y}) - \deg_{H_0}(\bar{y}))(c_{\text{cls}}N - B_{\beta}^{\text{wit}})^{-(r-1)} \leq C_{\beta,2}\sqrt{\zeta_{\beta}} + o(1).$$

The total number of formal noncentral vertices over all chosen catalogues is bounded by a constant depending on β . Therefore we have

$$\mathbb{E} \left[X_{\text{fr}} + \sum_y X_y \right] \leq C_{\beta,1}\zeta_{\beta} + C_{\beta,3}\sqrt{\zeta_{\beta}} + o(1).$$

Choose $\zeta_{\beta} > 0$ so small that the right side is eventually below one. The random variable is a nonnegative integer, so some choice in (14.8) makes it zero.

Use the maps ψ_t on the witness lifts, send the formal frame to the selected representatives, and send every pin to the corresponding selected vertices. The maps agree at the common center v and are already edgewise injective on every witness edge.

The selected frame vertices and the distinguished data representative lie outside the bounded witness core and in pairwise disjoint prescribed vertex classes. A root-only frame edge uses distinct root roles, a one-data frame edge uses the distinguished data representative and distinct root roles, a $\text{Pin}_{\star}$ -edge uses its pinned data vertex and distinct roots, and a Pin_i -edge uses its role- i vertex, the distinguished data representative, and roots whose roles avoid i . Hence no frame or pin edge suffers an identification. We therefore obtain an edgewise-injective homomorphism from a base star blocker into H . Its image is a member of $\mathcal{K}_{\beta} \subseteq \mathcal{G}_{\beta}$, contradicting the assumed freeness of H . ■

15 Degree stability and ordinary edit stability

We first recall the degree-stability criterion to which Propositions 14.2 and 14.3 will be applied. Recall that a family $\mathcal{F}$ of r -graphs is nondegenerate when $\pi(\mathcal{F}) > 0$, and call it *blowup invariant* if every blowup of an $\mathcal{F}$ -free r -graph remains $\mathcal{F}$ -free. By Lemma 2.1, blowup invariance is equivalent to the formulation in Liu–Mubayi–Reiher in which every $\mathcal{F}$ -free r -graph is $\mathcal{F}$ -hom-free.

The hereditary-class form used below also appears explicitly in work of Hou, Li, Liu, Mubayi, and Zhang [30, Theorem 6.2], Zhang, Hou, and Li [74, Theorem 18], and Liu [39, Theorem 2.3]. Subsequent work strengthened and extended this framework. Hou, Liu, and Zhao [31, Theorem 1.1] showed that edge stability together with vertex extendability implies degree stability, thereby covering families beyond the blowup-invariant or vertex-stable settings of the original framework. Chen and Liu [11, Theorems 1.6 and 1.7] extended vertex extendability to general extremal problems on hypergraphs and gave axiomatic criteria for degree stability based on symmetrization or edge stability.

The stability theorem is stated for a hereditary construction family $\mathfrak{H}$, with “contained in $\mathfrak{H}$ ” meaning “is a subgraph of a member of $\mathfrak{H}$ ”. Every target class used in this work is monotone, so containment is equivalent to membership. We record the resulting specialization used here.

Let $\mathfrak{C}$ be a monotone family of $\mathcal{F}$ -free r -graphs closed under isomorphisms. We say that $\mathcal{F}$ is *strongly symmetrized-stable* with respect to $\mathfrak{C}$ if every symmetrized $\mathcal{F}$ -free r -graph belongs to $\mathfrak{C}$. This condition implies the symmetrized-stability hypothesis of Liu–Mubayi–Reiher. The family $\mathcal{F}$ is *vertex-extendable* with respect to $\mathfrak{C}$ if there are $\eta > 0$ and n_0 such that every $\mathcal{F}$ -free H on $n \geq n_0$ vertices satisfying $\delta(H) \geq (\pi(\mathcal{F})/(r-1)! - \eta)n^{r-1}$ has the property that $H - v \in \mathfrak{C}$ for some vertex v implies $H \in \mathfrak{C}$. It is *degree-stable* if the same minimum-degree condition alone implies $H \in \mathfrak{C}$.

Degree stability is supplied by the following hereditary-class specialization of the theorem of Liu, Mubayi, and Reiher.

Theorem 15.1 ([42, Theorem 1.7]). *Let $\mathcal{F}$ be a nondegenerate blowup-invariant family of r -graphs, and let $\mathfrak{C}$ be a family of $\mathcal{F}$ -free r -graphs closed under isomorphisms and taking arbitrary subgraphs. If $\mathcal{F}$ is both strongly symmetrized-stable and vertex-extendable with respect to $\mathfrak{C}$, then $\mathcal{F}$ is degree-stable with respect to $\mathfrak{C}$.*

The construction-specific hypotheses of Theorem 15.1 are supplied by Propositions 14.2 and 14.3; the remaining basic hypotheses are recorded in the proof below. Thus degree stability follows, and we

then use the standard implication from degree stability to edit stability. Only the further passage to a near-extremal completed canonical model requires an argument specific to our construction.

Theorem 15.2. *The following statements hold.*

- (i) *There are $\varepsilon_\beta > 0$ and $n_0(\beta)$ such that every $\mathcal{G}_\beta$ -free r -graph H on $n \geq n_0(\beta)$ vertices with $\delta(H) \geq (\Lambda_r(\mathcal{D}_\beta)/(r-1)! - \varepsilon_\beta)n^{r-1}$ belongs to $\mathfrak{C}_\beta$.*
- (ii) *For every $\varepsilon, \eta > 0$ there are $\delta > 0$ and n_0 such that every $\mathcal{G}_\beta$ -free r -graph G on $n \geq n_0$ vertices with $|G| \geq \text{ex}(n, \mathcal{G}_\beta) - \delta n^r$ is within εn^r edges of a canonical completed model N on the same vertex set satisfying $|N| \geq \text{ex}(n, \mathcal{G}_\beta) - \eta n^r$. In particular, if (G_j) is an asymptotically extremal sequence and $n_j := v(G_j)$, then its edit distance from the canonical completed model class is $o(n_j^r)$.*

Proof. For part (i), the hypotheses of [Theorem 15.1](#) hold as follows.

- (a) By part (i) of [Lemma 13.2](#), the forbidden family is blowup invariant.
- (b) The finite reference model D_{bal} from [Lemma 10.7](#) has positive frame density, and hence $\Lambda_r(\mathcal{D}_\beta) > 0$. By [Proposition 13.1](#), $\pi(\mathcal{G}_\beta) = \Lambda_r(\mathcal{D}_\beta)$, so the family is nondegenerate.
- (c) By the definition of $\mathfrak{C}_\beta$, [Lemma 12.2](#), and part (ii) of [Lemma 13.2](#), the target class is closed under isomorphisms and taking subgraphs and consists of $\mathcal{G}_\beta$ -free r -graphs.
- (d) Symmetrized stability is [Proposition 14.2](#).
- (e) Vertex extendability is [Proposition 14.3](#).

Therefore [Theorem 15.1](#) applies and proves part (i).

For part (ii), fix $\varepsilon, \eta > 0$. For completeness, we prove the standard consequence of degree stability noted by Liu, Mubayi, and Reiher [[42](#), Section 1.1 and Fact 2.5]. We claim that for every $\xi > 0$ there are $\delta_0 > 0$ and n_0 such that every $\mathcal{G}_\beta$ -free G on $n \geq n_0$ vertices with $|G| \geq \text{ex}(n, \mathcal{G}_\beta) - \delta_0 n^r$ has a set $Z \subseteq V(G)$ satisfying $|Z| \leq \xi n$ and $G - Z \in \mathfrak{C}_\beta$.

To prove the claim, put $a := \pi(\mathcal{G}_\beta)/(r-1)! > 0$. By [Proposition 13.1](#), the threshold in part (i) is $(a - \varepsilon_\beta)q^{r-1}$. Set $\gamma := \min\{\varepsilon_\beta/2, a/2\}$ and let $n_1 := n_0(\beta)$ be as in that part. Since $a - \gamma \geq a - \varepsilon_\beta$, every $\mathcal{G}_\beta$ -free r -graph J on $q \geq n_1$ vertices with $\delta(J) \geq (a - \gamma)q^{r-1}$ belongs to $\mathfrak{C}_\beta$. Fix $\xi > 0$, put $\rho := \min\{\xi, 1/2\}$, and choose

$$\delta_0 < \frac{\gamma}{2r} (1 - (1 - \rho)^r).$$

Starting from G , whenever the current graph has order q and contains a vertex of degree less than $(a - \gamma)q^{r-1}$, delete such a vertex. Suppose that the process makes $k := \lfloor \rho n \rfloor$ deletions, and let G' be the remaining graph and $m := n - k$. Since $\text{ex}(q, \mathcal{G}_\beta) = aq^r/r + o(q^r)$, $m/n = 1 - \rho + o(1)$, and $\sum_{j=m+1}^n j^{r-1} = (n^r - m^r)/r + O(n^{r-1})$, we have

$$\begin{aligned} |G'| - \text{ex}(m, \mathcal{G}_\beta) &> \text{ex}(n, \mathcal{G}_\beta) - \text{ex}(m, \mathcal{G}_\beta) - \delta_0 n^r - (a - \gamma) \sum_{j=m+1}^n j^{r-1} \\ &= \left(\frac{\gamma}{r} (1 - (1 - \rho)^r) - \delta_0 + o(1) \right) n^r > 0 \end{aligned}$$

for all sufficiently large n , contradicting the definition of $\text{ex}(m, \mathcal{G}_\beta)$. Thus the process stops after fewer than k deletions. If Z is the deleted set and $q = n - |Z|$, then $q \geq (1 - \rho)n$, the graph $G - Z$ is still $\mathcal{G}_\beta$ -free, and $\delta(G - Z) \geq (a - \gamma)q^{r-1}$. After increasing n_0 so that $q \geq n_1$, part (i) gives $G - Z \in \mathfrak{C}_\beta$, while $|Z| < k \leq \rho n \leq \xi n$. This proves the claim.

Choose $\xi > 0$ and $0 < \delta \leq \delta_0$ so that $\delta + 2\xi < \varepsilon$ and $\delta + \xi < \eta$. Let G satisfy the hypotheses of part (ii). Adjoin the vertices of Z to $G - Z$ as isolated data vertices, obtaining an n -vertex canonical model M . Indeed, choose a canonical completed model witnessing $G - Z \in \mathfrak{C}_\beta$. In its rooted construction, adjoin the vertices of Z as neutral data vertices before completion, and retain only the edges of $G - Z$ after completing the enlarged rooted model. This realizes M as a subgraph of a canonical completed model. Then $|G| - |M|$ is at most the number of edges meeting Z , and hence $|G| - |M| \leq |Z| \binom{n-1}{r-1} \leq \xi n^r$. Complete M to a canonical completed model N on the same vertex set. By part (ii) of [Lemma 13.2](#), the r -graph N is $\mathcal{G}_\beta$ -free, so $|N| - |M| \leq \text{ex}(n, \mathcal{G}_\beta) - |M| \leq (\delta + \xi)n^r$. It follows that $\text{edit}(G, N) \leq |G| - |M| + |N| - |M| < \varepsilon n^r$, while $|N| \geq |M| \geq \text{ex}(n, \mathcal{G}_\beta) - (\delta + \xi)n^r > \text{ex}(n, \mathcal{G}_\beta) - \eta n^r$. The final assertion follows by taking ε and η to zero along an asymptotically extremal sequence. $\blacksquare$

Ordinary edit stability places a near-extremal r -graph close to the whole canonical class. We now refine this conclusion by distinguishing the zero-active construction from the two pure-sign constructions. We begin by introducing the scalar parameters and completed phase classes needed to state this refinement.

The same ordinary canonical completed model may admit more than one rooted presentation, and empty root classes need not be recoverable from its edge set. In this section, a *presented canonical completed model* consists of the ordinary graph together with one witnessing completed rooted graph and its canonical partition. All compression and scalar notation below is relative to this chosen presentation, and every estimate is valid for every such presentation. When a phase class is regarded as a class of ordinary graphs, membership means that the graph admits a presentation satisfying the displayed conditions.

If N is an n -vertex presented canonical completed model, let $(D_N, \mathbf{x}_N)$ be its presented rooted graph with the weighting from Lemma 14.5. Edges of D_N meeting an empty root class have zero weight, so both exact identities in that lemma remain valid for the complete presented rooted graph. Write $u(N)$ for its total data proportion and $y_i(N)$ for its root proportions. Let $G_\sigma(N)$ be the σ -projection of D_N , let $U_\sigma(N)$ be its covered vertex set, and write $a_\sigma(N)$ for the total proportion of $U_\sigma(N)$, for $\sigma \in \{+, -\}$. If $a_\sigma(N) > 0$, put $z_\sigma(N) := k! \lambda(G_\sigma(N); \mathbf{x}_N) / a_\sigma(N)^k$, and put $z_\sigma(N) := 0$ when $a_\sigma(N) = 0$. By Lemma 2.4, $0 \leq z_\sigma(N) \leq z_\beta$.

Let $\bar{\mathbf{x}}_N$ be obtained from $\mathbf{x}_N$ by replacing its root weights by the equal vector $\bar{\mathbf{y}}(u(N))$ and leaving its data weights unchanged.

We now define the completed phase classes. In the zero-gap case $z_\beta = \tau_{\text{base}}$, let u_0 be the unique maximizer of Φ_{base} on I supplied by part (i) of Proposition 10.9. The derivative-sign argument based on (10.19) in the proof of that part isolates u_0 in the rational interval

$$\left(\frac{15}{16(m+1)}, \frac{17}{16(m+1)} \right) \subset I.$$

Hence u_0 is algebraic and effectively specified as the unique zero of the rational polynomial Φ'_{base} in this computable interval. Fix an ordering of the type set $M \cup \{\star\}$, where $\star$ denotes a neutral data vertex, and define $a_\star := u_0$ and $a_i := \varrho(u_0)$ for every $i \in M$. For a set $S \subseteq M$, write $\star^j \uplus S$ for the multiset consisting of j copies of $\star$ and one copy of each element of S . Let

$$\mathcal{M}_0 := \{S: S \in \binom{M}{r} \setminus \mathcal{J}\} \cup \{\star \uplus S: S \in \binom{M}{r-1}\} \cup \{\star^k \uplus S: S \in \mathcal{D}_{\text{disp}} \cup \mathcal{T}_+ \cup \mathcal{T}_-\} \cup \{\star^{k+1} \uplus S: S \in \mathcal{P}\}.$$

The finite profile $\mathbf{P}_0 := ((a_t)_{t \in M \cup \{\star\}}, \mathcal{M}_0)$ therefore encodes exactly the completed zero-active model; with the uniformity index restored it is denoted $\mathbf{P}_{0,r}$. For every n , use the largest-fractional-parts rounding fixed in Section 8 simultaneously on all $m+1$ types. Let $\mathfrak{P}_0$ be the fixed finite-profile construction class whose n -vertex members form $\mathfrak{B}_{\mathbf{P}_0}(n)$, including all labeled realizations of its prescribed type partition. With the suppressed index restored, write this class as $\mathfrak{P}_{0,r}$.

Suppose now that $z_\beta > \tau_{\text{base}}$. Let u_β and $t_\beta > 0$ be the parameters supplied by Theorem 10.4. Call a presented canonical completed model N *pure+* if $U_-(N) = \emptyset$, and *pure-* if $U_+(N) = \emptyset$. For $\sigma \in \{+, -\}$ and $\eta > 0$, let $\mathfrak{P}_{\beta,\sigma}(n, \eta)$ consist of the ordinary underlying graphs of pure- σ presented canonical completed models N on n vertices satisfying

$$|u(N) - u_\beta| + \sum_{i \in M} |y_i(N) - \varrho(u_\beta)| + |a_\sigma(N) - t_\beta| \leq \eta, \quad \text{and} \quad z_\sigma(N) \geq z_\beta - \eta. \quad (15.1)$$

The completed σ -phase is represented by these classes as $n \rightarrow \infty$ and $\eta \downarrow 0$.

The phase classes defined in (15.1) yield the following phase-sensitive refinement of part (ii) of Theorem 15.2.

Proposition 15.3. *Let (G_j) be an asymptotically extremal $\mathcal{G}_\beta$ -free sequence, and put $n_j := v(G_j)$.*

- (i) *If $z_\beta = \tau_{\text{base}}$, then $\text{edit}(G_j, \mathfrak{P}_0(n_j)) = o(n_j^r)$.*
- (ii) *If $z_\beta > \tau_{\text{base}}$, then there are signs $\sigma_j \in \{+, -\}$, numbers $\eta_j \downarrow 0$, and $P_j \in \mathfrak{P}_{\beta,\sigma_j}(n_j, \eta_j)$ such that $\text{edit}(G_j, P_j) = o(n_j^r)$.*

Its proof uses two estimates. The first compares a finite canonical completed model with the closed scalar envelope Φ_{z_β} ; the only error comes from repeated data vertices and is $O(n^{-1})$.

Lemma 15.4. *Uniformly over all n -vertex presented canonical completed models N with $u(N) \in I$ and root vector satisfying the bounds in part (i) of Lemma 10.8, write $u := u(N)$. The following statements hold.*

(i) *We have*

$$\begin{aligned} \lambda(D_N; \bar{\mathbf{x}}_N) &= \Phi_{\text{base}}(u) + \alpha(u) \sum_{\sigma \in \{+, -\}} (z_\sigma(N) - \tau_{\text{base}}) a_\sigma(N)^k \\ &\quad - \gamma(u)(a_+(N) + a_-(N))^{k+1} + O_{k,p,q,r}(n^{-1}). \end{aligned} \quad (15.2)$$

(ii) *In particular, we have*

$$\lambda(D_N; \bar{\mathbf{x}}_N) \leq \Phi_{z_\beta}(u(N), a_+(N), a_-(N)) + O_{k,p,q,r}(n^{-1}), \quad (15.3)$$

where Φ_{z_β} is the function defined in (9.2) with $z = z_\beta$.

(iii) *Moreover, comparing the equalized and original compressed weightings gives*

$$\lambda(D_N; \bar{\mathbf{x}}_N) - \lambda(D_N; \mathbf{x}_N) \geq \underline{\nu}_r \sum_{i \in M} (y_i(N) - \varrho(u(N)))^2, \quad (15.4)$$

where $\underline{\nu}_r > 0$ is the constant fixed before Lemma 10.10.

Proof. Every compressed data vertex has weight $1/n$. If S is any set of data vertices of total weight s and $j \in \{k, k+1\}$, then the total weight of ordered j -tuples with a repeated coordinate is at most

$$\binom{j}{2} \left(\sum_{v \in S} x_v^2 \right) s^{j-2} \leq \binom{j}{2} \frac{s^{j-1}}{n}.$$

Consequently, $0 \leq s^j - j!e_j(S) \leq \binom{j}{2}n^{-1}$. At the equal root vector, all support polynomials have the exact values in (10.9), while $\lambda(G_\sigma(N); \mathbf{x}_N) = z_\sigma(N)a_\sigma(N)^k/k!$ by definition. Substituting the preceding bound into the Lagrangian decomposition (10.4) gives (15.2); the error is uniform because the number of edge classes and their coefficients are fixed. The upper bound (15.3) follows from $z_\sigma(N) \leq z_\beta$. Finally, (15.4) is exactly the difference estimate obtained by combining (10.6) and (10.8). ■

The second estimate is the following local form of the completion rule.

Lemma 15.5. *Let N_1 and N_2 be presented canonical completed models on the same vertex set with the same canonical partition $\mathcal{V} = (U, (V_i)_{i \in M})$. For $q \in \{1, 2\}$ and $\sigma \in \{+, -\}$, let $G_\sigma^{(q)}$ and $U_\sigma^{(q)}$ be the σ -projection and its covered vertex set in the completed rooted graph belonging to the chosen presentation of N_q . Suppose that there is a set $S \subseteq U$ such that $G_\sigma^{(1)}[U \setminus S] = G_\sigma^{(2)}[U \setminus S]$ and $U_\sigma^{(1)} \setminus S = U_\sigma^{(2)} \setminus S$ for both signs σ . Then an r -set has different edge status in the ordinary underlying graphs N_1 and N_2 only if it meets S .*

Proof. Fix an ordinary r -set disjoint from S . Its vertices have the same role profile in the common partition for both presentations. If this profile is not permitted, the set is absent from both graphs. Root-only and one-data frame profiles, as well as dispersion profiles, are unconditional. For a raw σ -record profile with data set X , the set is an edge of N_q exactly when $X \in G_\sigma^{(q)}$, and this membership is independent of q . For a threshold profile with data set X , it is an edge exactly when $X \not\subseteq U_\sigma^{(q)}$; for a penalty profile with data set Y , it is an edge exactly when $Y \not\subseteq U_+^{(q)} \cup U_-^{(q)}$. Since the data set is disjoint from S , these subset conditions are also independent of q . These are all the permitted profiles, so the fixed r -set has the same edge status in N_1 and N_2 . ■

Proof of Proposition 15.3. By part (ii) of Theorem 15.2, replace G_j at a cost $o(n_j^r)$ by a canonical completed model N_j whose extremal gap is $o(n_j^r)$, and retain the witnessing presentation produced by that completion. Let $(D_{N_j}, \mathbf{x}_{N_j})$ be its presented weighted rooted graph. Then $\lambda(D_{N_j}; \mathbf{x}_{N_j}) \rightarrow \Lambda(\mathcal{D}_\beta)$. By Lemma 10.7, for all large j its total data weight belongs to I and its root vector satisfies the bounds in part (i) of Lemma 10.8, so Lemma 15.4 applies.

Since root equalization does not decrease the Lagrangian value and no weighted member of $\mathcal{D}_\beta$ can exceed $\Lambda(\mathcal{D}_\beta)$, we have

$$\Lambda(\mathcal{D}_\beta) - o(1) = \lambda(D_{N_j}; \mathbf{x}_{N_j}) \leq \lambda(D_{N_j}; \bar{\mathbf{x}}_{N_j}) \leq \Lambda(\mathcal{D}_\beta). \quad (15.5)$$

By (15.4), the root imbalance satisfies $\sum_{i \in M} (y_i(N_j) - \varrho(u(N_j)))^2 \rightarrow 0$.

Assume first that $z_\beta = \tau_{\text{base}}$. Formula (15.3) and (15.5) show that the scalar parameters form a maximizing sequence for $\Phi_{\tau_{\text{base}}}$. The zero-gap part of Lemma 9.1, or equivalently compactness plus the strict concavity of $\Phi_{\tau_{\text{base}}}$ and the negative penalty term, therefore gives $u(N_j) \rightarrow u_0$ and $a_+(N_j) + a_-(N_j) \rightarrow 0$. Together with the root-imbalance convergence in (15.4), this yields $y_i(N_j) \rightarrow \varrho(u_0)$ for every $i \in M$. Let $S_j := U_+(N_j) \cup U_-(N_j)$ be the set of active vertices, so $|S_j| = o(n_j)$. Reclassify the vertices of S_j as neutral by deleting their raw signed-record declarations. Let $\mathbf{a}^{(j)}$ be the resulting vector of root and neutral proportions, and let $\mathbf{a}^0$ be the fixed target vector defining $\mathfrak{P}_0$. These limits give $\|\mathbf{a}^{(j)} - \mathbf{a}^0\|_1 = o(1)$.

The largest-fractional-parts rounding changes each prescribed class size by at most one, so the vertices can be repartitioned by moving a set T_j with

$$|T_j| \leq \frac{n_j}{2} \|\mathbf{a}^{(j)} - \mathbf{a}^0\|_1 + O_r(1) = o(n_j).$$

Every raw σ -record edge has all its data vertices in U_σ . Delete all raw signed-record edges from the presented rooted graph of N_j , complete through the same partition, and denote the resulting ordinary graph by N'_j . The old and new sign projections and covered sets agree outside S_j , so Lemma 15.5 shows that N_j and N'_j can differ only on edges meeting S_j .

Now move the vertices in T_j and complete through the resulting target partition, obtaining B_j . Both N'_j and B_j have empty sign projections. If an r -set is disjoint from T_j , every one of its vertices has the same root or data role in the two partitions. A direct check of the fixed rooted edge types then gives the same edge status in both completions: frame and dispersion edges are unconditional, threshold and penalty edges see the same empty covered sets, and no raw record edge occurs. Hence N'_j and B_j can differ only on edges meeting T_j . It follows that

$$\text{edit}(N_j, B_j) \leq |S_j \cup T_j| \binom{n_j}{r-1} = o(n_j^r).$$

The graph B_j belongs to $\mathfrak{P}_0(n_j)$, proving part (i).

Now suppose $z_\beta > \tau_{\text{base}}$. Again, (15.3) and (15.5) make the scalar parameters a maximizing sequence for Φ_{z_β} . Apply the quantitative phase-separation estimate in Lemma 9.2. For each j choose $\sigma_j \in \{+, -\}$ so that the corresponding pure maximizer is the closer one. Then $u(N_j) \rightarrow u_\beta$, $y_i(N_j) \rightarrow \varrho(u_\beta)$ for every $i \in M$, $a_{\sigma_j}(N_j) \rightarrow t_\beta$, and $a_{-\sigma_j}(N_j) \rightarrow 0$.

It remains to check the inner condition (15.1). If, along a subsequence, $z_{\sigma_j}(N_j) \leq z_\beta - \varepsilon$ for some fixed $\varepsilon > 0$, then (15.2), the preceding convergences, and $z_{-\sigma_j}(N_j) \leq z_\beta$ give

$$\lambda(D_{N_j}; \bar{\mathbf{x}}_{N_j}) \leq \Phi_{z_\beta}^* - \alpha(u_\beta)\varepsilon \left(\frac{t_\beta}{2}\right)^k + o(1),$$

a contradiction to (15.5). Hence $z_{\sigma_j}(N_j) \rightarrow z_\beta$.

Finally, let $S_j := U_{-\sigma_j}(N_j)$ be the set of minority-sign vertices. Since $a_{-\sigma_j}(N_j) \rightarrow 0$, we have $|S_j| = o(n_j)$. Reclassify S_j as neutral by deleting the minority-sign raw record declarations, and complete the resulting pure- σ_j model to obtain P_j .

By (V1), no dominant-sign record edge meets S_j , whereas every minority-sign record edge has all its data vertices in S_j . The dominant sign projection, and hence its value z_{σ_j} , is therefore unchanged, and the two assignments agree outside S_j . By Lemma 15.5, every edge whose status changes meets S_j . Quantitatively, $\text{edit}(N_j, P_j) \leq |S_j| \binom{n_j}{r-1} = o(n_j^r)$. Choose $\eta_j \downarrow 0$ to dominate the errors in the preceding convergences and in $z_{\sigma_j}(N_j) \rightarrow z_\beta$. Then $P_j \in \mathfrak{P}_{\beta, \sigma_j}(n_j, \eta_j)$ and $\text{edit}(N_j, P_j) = o(n_j^r)$. Together with the initial replacement of G_j by N_j , this proves part (ii). $\blacksquare$

16 Extremal limit transfer and the structural theorem

It remains to show that the weighted and ordinary constructions have the same extremal limits. Edit stability and compression give one inclusion; rational approximation, cloning data vertices, and root blowups give the reverse inclusion.

Theorem 16.1. *The extremal limit space of the final family equals the weighted optimizer space. In symbols, $\mathcal{E}(\mathcal{G}_\beta) = \mathcal{L}(\mathcal{D}_\beta)$.*

Proof. Fix first $W \in \mathcal{E}(\mathcal{G}_\beta)$. By the removal observation in [Section 8](#), choose an asymptotically extremal $\mathcal{G}_\beta$ -free sequence (G_ℓ) converging to W , and put $n_\ell := v(G_\ell) \rightarrow \infty$. By part (ii) of [Theorem 15.2](#), there are canonical completed models N_ℓ on the same vertex sets such that $\text{edit}(G_\ell, N_\ell) = o(n_\ell^r)$. Fix one witnessing presentation for each N_ℓ . For every fixed r -graph F , a union bound over the edges of F gives

$$|t(F, G_\ell) - t(F, N_\ell)| \leq |F| \frac{r! \text{edit}(G_\ell, N_\ell)}{n_\ell^r} = o(1).$$

Thus (G_ℓ) and (N_ℓ) have the same limit points. Give the presented rooted graph of N_ℓ the weights determined by its root-class sizes and retain every data vertex as a singleton. By [Lemma 14.5](#), including its zero-weight observation for empty root classes, the resulting weighted members $(D_\ell, \mathbf{x}_\ell)$ satisfy

$$r! \lambda(D_\ell; \mathbf{x}_\ell) = \frac{r! |N_\ell|}{n_\ell^r} \rightarrow \Lambda_r(\mathcal{D}_\beta).$$

Equation (14.3) shows that compression preserves every homomorphism density exactly. Hence $(D_\ell, \mathbf{x}_\ell) \rightarrow W$. Each compressed model is a weighted member of $\mathcal{D}_\beta$, so the definition of the closed space $\mathfrak{W}(\mathcal{D}_\beta)$ gives $W \in \mathfrak{W}(\mathcal{D}_\beta)$. Moreover, continuity of the one-edge density and the preceding display give $\text{dens}_r(W) = \Lambda_r(\mathcal{D}_\beta)$. Thus $W \in \mathcal{L}(\mathcal{D}_\beta)$, proving the first inclusion in the asserted equality.

Conversely, fix $W \in \mathcal{L}(\mathcal{D}_\beta)$ and use the enumeration $F_1, F_2, \dots$ fixed in [Section 8](#). Put

$$\varepsilon_j := \frac{1}{j(1 + r! \max_{1 \leq i \leq j} |F_i|)}.$$

Because W is an optimizer limit, for each j we may choose a weighted member $(D_j, \mathbf{x}^{(j)})$ of $\mathcal{D}_\beta$, together with an admissible partial rooting witnessing this membership, such that simultaneously

$$\begin{aligned} d_{\text{hom}}((D_j, \mathbf{x}^{(j)}), W) &< \varepsilon_j, \\ \max_{1 \leq i \leq j} |t(F_i, D_j, \mathbf{x}^{(j)}) - t(F_i, W)| &< \varepsilon_j, \end{aligned} \tag{16.1}$$

$$r! \lambda(D_j; \mathbf{x}^{(j)}) > \Lambda_r(\mathcal{D}_\beta) - \varepsilon_j. \tag{16.2}$$

Indeed, take a weighted sequence converging to W and use the continuity of the one-edge density and of the first j density coordinates. We include (16.1) explicitly because the weighted-metric condition alone would introduce a factor depending on i .

Let $\overline{D}_j$ be the completion relative to this chosen rooting, with $\mathbf{x}^{(j)}$ extended by zero on every newly added root vertex. Its gain in normalized one-edge density is at most $\Lambda_r(\mathcal{D}_\beta) - r! \lambda(D_j; \mathbf{x}^{(j)}) < \varepsilon_j$. Thus, by (10.3), for every $i \leq j$, we have

$$0 \leq t(F_i, \overline{D}_j, \mathbf{x}^{(j)}) - t(F_i, D_j, \mathbf{x}^{(j)}) \leq |F_i| \varepsilon_j. \tag{16.3}$$

On this fixed completed rooted r -graph, choose a rational probability vector $\mathbf{x}_{\text{rat}}^{(j)}$ for which every data coordinate is strictly positive and every root coordinate is nonnegative, and such that

$$\begin{aligned} |r! \lambda(\overline{D}_j; \mathbf{x}_{\text{rat}}^{(j)}) - r! \lambda(\overline{D}_j; \mathbf{x}^{(j)})| &< \varepsilon_j, \\ \max_{1 \leq i \leq j} |t(F_i, \overline{D}_j, \mathbf{x}_{\text{rat}}^{(j)}) - t(F_i, \overline{D}_j, \mathbf{x}^{(j)})| &< \varepsilon_j. \end{aligned} \tag{16.4}$$

Rational vectors with all data coordinates positive are dense in the finite probability simplex, so continuity gives this perturbation. It changes only the weighting, not the underlying admissible rooted r -graph, its sign projections, or the sets U_+ and U_- . Combining (16.2) with the Lagrangian perturbation bound gives

$$0 \leq \Lambda_r(\mathcal{D}_\beta) - r! \lambda(\overline{D}_j; \mathbf{x}_{\text{rat}}^{(j)}) < 2\varepsilon_j. \tag{16.5}$$

Take a common denominator for $\mathbf{x}_{\text{rat}}^{(j)}$. Multiplying that denominator and all numerators by an arbitrary positive integer does not change the vector, so we may choose a denominator $n_j \geq j$; in particular,

$n_j \rightarrow \infty$. Write each data weight as a_v/n_j and each root weight as b_i/n_j . Replace the data vertex v by a_v singleton data clones of weight $1/n_j$. Data cloning preserves every density by [Lemma 8.1](#). Complete the cloned rooted r -graph before the root blowup. By [\(16.5\)](#), its gain in normalized one-edge density is at most $2\varepsilon_j$, so, for every $1 \leq i \leq j$, the corresponding increase in $t(F_i, \cdot)$ lies between 0 and $2|F_i|\varepsilon_j$.

Finally replace root i by a vertex class of size b_i . If $b_i = 0$, this is an empty root vertex class, which is allowed in [Definition 12.1](#). The root blowup preserves every weighted homomorphism density exactly, by the exact compression identity [\(14.3\)](#). We obtain an ordinary canonical completed model N_j of order n_j , free by part (ii) of [Lemma 13.2](#). Combining [\(16.1\)](#), [\(16.3\)](#), [\(16.4\)](#), and the bound $2|F_i|\varepsilon_j$ for completing the cloned rooted graph, for every $i \leq j$ we have the explicit bound

$$|t(F_i, N_j) - t(F_i, W)| \leq (2 + 3|F_i|)\varepsilon_j.$$

Moreover, [\(16.5\)](#) and the fact that the second completion only adds edges give

$$\Lambda_r(\mathcal{D}_\beta) - 2\varepsilon_j < \frac{r!|N_j|}{n_j^r} \leq \Lambda_r(\mathcal{D}_\beta).$$

Since $n_j \rightarrow \infty$, it follows that $|N_j|/\binom{n_j}{r} \rightarrow \Lambda_r(\mathcal{D}_\beta)$. For each fixed i , the right side of the preceding explicit bound tends to zero once $j \geq i$. Thus $N_j \rightarrow W$, and every weighted optimizer limit is an ordinary extremal limit. $\blacksquare$

With the extremal limit spaces identified, we can now complete the proofs of the structural theorem and its undecidability corollary.

Proof of Theorem 1.2. The integer r_{str} is effectively computable by [Proposition 10.9](#). For $2 \leq r < r_{\text{str}}$, define $\mathcal{G}_{r,\beta}$ to be the empty family. Given (r, β) with $r \geq r_{\text{str}}$, the weighted construction computes $\mathcal{A}_{r,\beta}^{\text{root}}$. The witness catalogues, star blockers, pair-covering extensions, and edgewise-injective closures in [Sections 12](#) and [13](#) are finite operations uniform in r . Hence $(r, \beta) \mapsto \mathcal{G}_{r,\beta}$ is total computable.

Now fix an arbitrary $r \geq r_{\text{str}}$ and suppress its index as in the structural construction. The objects $W_{0,r}$ and Sep_r are the locally denoted W_0 and Sep ; they depend only on r and are independent of β . The finite quantum graph in [\(11.1\)](#) defines Sep_r , and [Lemma 11.3](#) gives $\text{Sep}_r(W_{0,r}) = 0$.

By [Theorem 16.1](#), the extremal limit space is exactly the weighted optimizer space classified in [Theorem 11.1](#). This proves the singleton and two-phase statements, including compactness and separation by Sep .

The edit assertions are exactly [Proposition 15.3](#). For an asymptotically extremal sequence (G_j) , the proposition gives $o(v(G_j)^r)$ -closeness in the nonhalting case to the integer realizations of the fixed profile $\mathbf{P}_{0,r}$, namely the construction class $\mathfrak{P}_{0,r}$, whose members converge to $W_{0,r}$. In the halting case it gives $o(v(G_j)^r)$ -closeness to the union of the two completed pure-sign phase classes. The selected sign σ_j may depend on j , while the total data and sign-class proportions converge to the unique phase optimum and the dominant inner projection converges to its extremal value. $\blacksquare$

Proof of Corollary 1.3. Fix an arbitrary $r \geq r_{\text{str}}$ and map the input word β to the finite family $\mathcal{G}_{r,\beta}$. Write $W_0 = W_{0,r}$ and $\text{Sep} = \text{Sep}_r$ locally. If $\mathbf{U}(\beta) \uparrow$, then the extremal space is the singleton $\{W_0\}$ and $\text{Sep}(W_0) = 0$. Hence uniqueness, connectedness, and Erdős–Simonovits stability hold, while the two sign properties fail.

If $\mathbf{U}(\beta) \downarrow$, then the two nonempty compact phase sets are disjoint and are separated by Sep . Thus uniqueness and connectedness fail, while both sign properties hold. Erdős–Simonovits stability also fails. If both phases were edit-close to integer realizations of one fixed profile $\mathbf{P}$, then sequences from both phases would converge to the uniquely defined profile limit $W_{\mathbf{P}}$ and therefore have the same value of the continuous statistic Sep , a contradiction.

An algorithm deciding any one of the listed properties would therefore decide whether $\mathbf{U}(\beta)$ halts. $\blacksquare$

Part III

Gödel–Rosser incompleteness

The proofs in Parts I and II are ordinary mathematical proofs carried out in the usual ZFC foundations. The purpose of this part is not to repeat those arguments. Instead, we isolate the finite certificates carried by the numerical compiler and record the uniform set-theoretic equivalences needed for the Gödel–Rosser substitution.

Fix $r \geq r_{\text{num}}$ and the corresponding input-independent compiler data. The numerical target, whose components are defined below and in Appendix A, is the uniform formula

$$\text{ZFC} \vdash \forall d \forall e \forall f \left(\text{Valid}_{\text{PR}}(d) \wedge \text{CodeFam}_r^{\text{num}}(d, e, f) \longrightarrow \left([\text{All}_{\text{PR}}(d, e) \iff \text{TuranEq}_{\tau_r}(f)] \wedge \text{Base}_r(f) \right) \right).$$

Once this formula is available, numerical incompleteness follows by the standard Rosser substitution. The structural argument uses the same input together with an analogous coded truth table for the five properties in Corollary 1.3.

The detailed arithmetizations, proof codes, typed compiler ledgers, and structural limit-space formulas are given in Appendices A and B. The main text retains the certificate argument and the logical composition.

Formalization scope. All assertions of ZFC-provability below are conventional metamathematical formalizability statements. The appendices specify the formulas and finite proof transformations; no proof-assistant formalization is claimed.

Throughout this part, fix $r \geq r_{\text{num}}$ and abbreviate $\mathcal{F}_\beta := \mathcal{F}_{r,\beta}$ and $\tau_* := \tau_r$. These are the data supplied by Theorem 1.1. For the fixed compiler interface, also used in Part IV and Appendix A, put $r_{\text{rec}} := r - 5$, write N_{rt} for the number of root labels, and put $s_{\text{phys}} := 16$, the number of physical types on one rail. We retain $\mathbf{M}$ for the marked five-ary relation of the direct compiler. All input-independent data are computable from r . Claims below that a compiler is primitive recursive are made after this r -dependent setup has been fixed.

17 Finite certificate normal form

The main result of this section is the fixed-template normal form in Theorem 17.6. Its technical core is Theorem 17.5: part (i) turns a finite halting trace into a balanced improvement witness, while part (ii) recovers a bounded halting trace from every such witness. Thus the certificate mechanism runs in both directions:

$$\begin{aligned} & \text{finite halting trace} \longrightarrow \text{periodic reset tableau} \longrightarrow \text{balanced improvement witness,} \\ & \text{balanced improvement witness} \longrightarrow \text{closed relation component} \longrightarrow \text{bounded halting trace.} \end{aligned}$$

17.1 Balanced improvement witnesses

We first record the input-independent baseline template supplied by the direct construction.

Lemma 17.1. *There is a total algorithm which, on input $r \geq r_{\text{num}}$, constructs a finite r -graph $K_{*,r}$ whose uniform weighting is optimal and whose balanced blowups have limiting density τ_r . In particular, $\lambda_r(K_{*,r}) = \tau_r$. For every input word β :*

- (i) $K_{*,r}$ is $\mathcal{F}_{r,\beta}$ -hom-free, equivalently every blowup of $K_{*,r}$ is $\mathcal{F}_{r,\beta}$ -free;
- (ii) If $\mathbf{U}$ halts on β , then there exists a finite $\mathcal{F}_{r,\beta}$ -hom-free r -graph G with $\lambda_r(G) > \lambda_r(K_{*,r})$.

Proof. Take the input-independent baseline realization from Lemma 3.5 and Theorem 4.1, with $\mathbf{M} = \emptyset$, equal rail weights, and all relation matchings perfect. The exact record and rooted translations in Lemma 5.2 and Theorem 6.3 give a finite template K^0 with a positive rational weighting of normalized value τ_r . The same realization is admissible for every input β , so every blowup of K^0 is $\mathcal{F}_{r,\beta}$ -free. Applying the nonhalting equality to the fixed word $\beta_\infty = 0$ gives

$$\lambda_r(K^0) \leq \pi(\mathcal{F}_{r,\beta_\infty}) = \tau_r.$$

The displayed rational weighting attains τ_r , so K^0 is optimal. Clearing the rational weights produces $K_{*,r}$, whose uniform weighting attains τ_r . If $U(\beta)$ halts, the positive branch of the same construction gives a finite $\mathcal{F}_{r,\beta}$ -hom-free template with strictly larger value. The effective choices and the canonical edge-list construction are recorded in [Section A.3](#). ■

For an $\mathcal{F}$ -hom-free finite template K , we say that K is *asymptotically optimal for $\mathcal{F}$* if $\lambda_r(K) = \pi(\mathcal{F})$. Thus $K_{*,r}$ is asymptotically optimal for $\mathcal{F}_\beta$ exactly when $U(\beta)$ does not halt, while halting is equivalent to the existence of a better finite blowup template.

We next replace optimized real weightings by uniformly weighted finite templates. For a finite r -graph H with $v(H) > 0$, allowing $|H| = 0$, define its *balanced blowup value* by $\lambda_{\text{bal}}(H) := r!|H|/v(H)^r$. This is the value at the uniform weighting and the limiting edge density of the balanced blowups of H .

Write the fixed rational threshold in lowest terms as $\tau_* = a_*/b_*$, where $a_*, b_* \in \mathbb{N}$ and $b_* > 0$. For a finite family $\mathcal{F}$ of r -graphs and a finite r -graph H , let $\text{Improve}_{\tau_*}(\mathcal{F}, H)$ be the assertion that H is $\mathcal{F}$ -hom-free and

$$b_* r!|H| > a_* v(H)^r. \quad (17.1)$$

Define $\text{BalSharp}_{\tau_*}(\mathcal{F})$ to mean $\forall H \neg \text{Improve}_{\tau_*}(\mathcal{F}, H)$. The predicate inside the universal quantifier is primitive recursive because hom-freeness is checked by enumerating finitely many vertex maps and (17.1) is an integer inequality. Thus this is a literal Π_1^0 sentence, with no quantification over real weights and no appeal to quantifier elimination. The following lemma both replaces optimized real weights by one uniformly weighted finite hypergraph and identifies the absence of such an improvement witness with sharpness of the fixed baseline.

Lemma 17.2. *Let $\mathcal{F}$ be a finite family of r -graphs and let $q \in \mathbb{Q}$. Then $\pi(\mathcal{F}) > q$ if and only if there is a finite $\mathcal{F}$ -hom-free r -graph H with $\lambda_{\text{bal}}(H) > q$. In particular, $\text{BalSharp}_{\tau_*}(\mathcal{F})$ holds if and only if $\pi(\mathcal{F}) \leq \tau_*$. If, in addition, $K_{*,r}$ is $\mathcal{F}$ -hom-free, then these conditions are equivalent to $\pi(\mathcal{F}) = \tau_*$ and to the asymptotic optimality of $K_{*,r}$ for $\mathcal{F}$.*

Proof. Suppose first that such an H exists. Every blowup of H is $\mathcal{F}$ -free by [Lemma 2.1](#), and the balanced blowups have edge density tending to $\lambda_{\text{bal}}(H) > q$. Hence $\pi(\mathcal{F}) > q$.

Conversely, suppose that $\pi(\mathcal{F}) > q$. By [Corollary 2.5](#), there is a finite $\mathcal{F}$ -hom-free r -graph G with $\lambda_r(G) > q$. Choose a probability weighting $\mathbf{x}$ of G such that $r!\lambda(G; \mathbf{x}) > q$. By continuity and density of the rational points in the probability simplex, after deleting zero-weight vertices if necessary we may take $x_i = c_i/D$, where $c_i \in \mathbb{N}_{>0}$ and $\sum_i c_i = D$, and still have $r!\lambda(G; \mathbf{x}) > q$. Replace vertex i of G by a vertex class of size c_i , and let H be the resulting blowup. Projection to the vertex classes shows that H is $\mathcal{F}$ -hom-free. Moreover, $|H| = \sum_{e \in G} \prod_{i \in e} c_i$ and $\lambda_{\text{bal}}(H) = r!|H|/D^r = r!\lambda(G; \mathbf{x}) > q$.

Taking $q = \tau_*$ shows that $\text{BalSharp}_{\tau_*}(\mathcal{F})$ is equivalent to $\pi(\mathcal{F}) \leq \tau_*$. If $K_{*,r}$ is $\mathcal{F}$ -hom-free, its blowups give $\pi(\mathcal{F}) \geq \lambda_r(K_{*,r}) = \tau_*$. The remaining equivalences follow. ■

17.2 The certificate-preserving compiler

For a binary word β and $t \in \mathbb{N}$, define $\text{Trace}_U(\beta, t)$ to hold if the computation of U on β halts within t steps. This is a primitive recursive predicate, and a complete halting transcript can be recovered by bounded simulation. We separate the total computation of the r -dependent setup from the primitive recursive maps obtained after that setup is fixed.

Proposition 17.3. *The compiler has the following two uniformity properties.*

- (i) *The map $(r, \beta) \mapsto \langle \mathcal{F}_{r,\beta} \rangle$, extended by a fixed default value on invalid inputs, is total computable. The corresponding map from (r, d, e) to the canonical code of the family associated with a primitive recursive predicate code d and a parameter e is also total computable.*
- (ii) *After r and all input-independent compiler data have been fixed, the maps $\beta \mapsto \langle \mathcal{F}_{r,\beta} \rangle$ and $(d, e) \mapsto \langle \mathcal{F}_{r,d,e} \rangle$ are primitive recursive.*

Proof. As r varies, the terminating parameter searches in [Lemma 6.9](#) compute the finite Baranyai decompositions and all other input-independent data. The remaining compiler operations are finite, so the resulting maps are total computable. Once r and these data are fixed, the retained bounds on local obstructions, root elimination, and pair-covering extensions make every enumeration primitive recursively

bounded; canonicalization and specialization are bounded searches as well. The explicit bounds and off-domain conventions are given in [Section A.3](#). ■

Lemma 17.4. *Let G be a finite $\mathcal{F}_\beta$ -hom-free r -graph with $\lambda_r(G) > \tau_*$. Then there are integers W, p such that $\Phi_W^p(c_{\beta,W}) = c_{\beta,W}$, $W + 2 \leq v(G)$, and $p \leq v(G)$. In particular, $U(\beta)$ halts within $v(G)$ steps.*

Proof. Put $n := v(G)$ and choose a minimum-support Lagrangian-maximizing weighting of G . Since G is $\mathcal{F}_\beta$ -hom-free, it is in particular $\mathcal{F}_\beta$ -free. The certificate clause of [Lemma 2.9](#) therefore gives a support core $D \in \mathcal{C}_\beta$ with $\lambda(D) = \lambda(G) > \tau_*/r! = \Upsilon_{z_0}^*$.

Choose an admissible partial rooting of D and apply the completion, root-balancing, and typing-repair operations from the proof of [Theorem 6.3](#). None of these operations decreases the Lagrangian value. They add only the fixed root set and introduce no new nonisolated data vertices. Let C be the resulting completed rooted graph, let $\mathbf{y}$ be its weighting, and let $\mathfrak{X}$ be the corresponding fully typed record system. Let s and $\rho = 1 - s$ be the total root and data weights, and let $\mathbf{x}$ be the data weighting. Parts (2)–(3) of [Lemma 6.9](#) give $s > 1/2$.

At the balanced root vector, the exact completed-polynomial identity, with the baseline record contribution subtracted, gives

$$\lambda(C; \mathbf{y}) - \Upsilon_{z_0}(s) = \frac{s^{r_{\text{rec}}}}{r_{\text{rec}} N_{\text{rt}}^{r_{\text{rec}}-1}} (\lambda_{\text{rec}}(\mathfrak{X}; \mathbf{x}) - z_0 \rho^5).$$

The left side is positive because $\lambda(C; \mathbf{y}) > \Upsilon_{z_0}^* \geq \Upsilon_{z_0}(s)$, and the coefficient on the right is positive because $s > 1/2$. Thus the expression in parentheses is positive. If $\rho = 0$, then all data weights vanish and $\lambda_{\text{rec}}(\mathfrak{X}; \mathbf{x}) = 0$, a contradiction. Therefore $\rho > 0$.

Put $z_{\mathfrak{X}} := \lambda_{\text{rec}}(\mathfrak{X}; \mathbf{x})/\rho^5$. Since $\lambda(C; \mathbf{y}) = \Upsilon_{z_{\mathfrak{X}}}(s) > \Upsilon_{z_0}(s)$ and the coefficient of z in $\Upsilon_z(s)$ is positive, we have $z_{\mathfrak{X}} > z_0$.

Let $\mathfrak{X}^+$ be the filler saturation of $\mathfrak{X}$, and let $\mathfrak{R}$ be its projected matching system. This operation adds no vertices, and [Lemma 5.3](#) shows that it preserves record admissibility and does not decrease λ_{rec} . Part (i) of [Lemma 5.5](#) makes $\mathfrak{R}$ legal, while [Lemma 5.2](#) gives $\Phi_\beta(\mathfrak{R}, \mathbf{x}) = \lambda_{\text{rec}}(\mathfrak{X}^+; \mathbf{x}) \geq \lambda_{\text{rec}}(\mathfrak{X}; \mathbf{x}) > z_0 \rho^5$. If every coordinate of every M-tuple belonged to a grounded relation component, the homogeneous estimate in [Theorem 4.1](#) would give $\Phi_\beta(\mathfrak{R}, \mathbf{x}) \leq z_0 \rho^5$, a contradiction. Thus some marked coordinate belongs to a closed relation component.

Let h be the number of cell vertices in the corresponding logical relation component. The admissible numerical-completion in [Lemma 6.2](#) adds only root vertices and edges, and its zero extension assigns weight zero to every new root vertex. Root balancing changes only weights, typing repair uses the existing data vertices, and filler saturation adds records on those same vertices. Thus the cell vertices of $\mathfrak{R}$ lie in the original support of D , and $h \leq n$. Part (iii) of [Lemma 3.5](#) recovers integers W, p with $\Phi_W^p(c_{\beta,W}) = c_{\beta,W}$, $W + 2 \leq h \leq n$, and $p \leq h \leq n$.

By the converse direction of [Lemma 3.2](#), a return to $c_{\beta,W}$ can occur only after the simulated computation has halted. Each temporal CA step simulates at most one step of U , so the genuine computation before reset has length at most $p \leq n$. ■

The next theorem gives certificate transformations in both directions. Its first part turns a finite halting trace into a finite improvement witness, while its second part recovers a halting-time bound from any such witness.

Theorem 17.5. *Using the fixed canonical codes for binary words and finite r -graphs, there are total primitive recursive maps*

$$\text{Build}_r: \mathbb{N}^2 \rightarrow \mathbb{N} \quad \text{and} \quad \text{Bound}_r: \mathbb{N}^2 \rightarrow \mathbb{N}.$$

The first map takes a binary-word code and a halting-time bound and returns a canonical r -graph code. The second takes a binary-word code and a canonical r -graph code and returns a halting-time bound. These maps have the following properties.

- (i) *If $\text{Trace}_U(\beta, t)$, then $\text{Improve}_{\tau_*}(\mathcal{F}_\beta, \text{Build}_r(\beta, t))$.*
- (ii) *If $\text{Improve}_{\tau_*}(\mathcal{F}_\beta, H)$, then $U(\beta)$ halts within $v(H)$ steps, and $\text{Bound}_r(\beta, H)$ is a valid halting time bound.*

Consequently, U does not halt on β exactly when $\text{BalSharp}_{\tau_*}(\mathcal{F}_\beta)$.

Proof. Assume $\text{Trace}_U(\beta, t)$. Bounded simulation up to t recovers the complete finite halting transcript, and the reset construction computes a periodic tableau. The positive construction in Part I turns it into a rationally weighted admissible rooted r -graph of value strictly above the baseline; clearing denominators gives the balanced witness $\text{Build}_r(\beta, t)$. After r is fixed, every step is primitive recursive. The typed implementation and its off-domain behavior are recorded in [Section A.3](#).

Conversely, an improvement witness H satisfies $\lambda_r(H) \geq \lambda_{\text{bal}}(H) > \tau_*$. The bounded-recovery lemma therefore shows that $U(\beta)$ halts within $v(H)$ steps. Put $\text{Bound}_r(\beta, H) = v(H)$ on legal graph codes and 0 otherwise, and use the fixed one-edge r -graph as the default value of Build_r off the trace domain. Bounded simulation recovers the corresponding transcript. Taking negations gives the final equivalence. ■

The semantic alternatives and the two finite certificate transformations are collected in [Figure 14](#). The upper row records equivalent properties. The lower row records the explicit finite objects used to pass between the two sides of the positive branch.

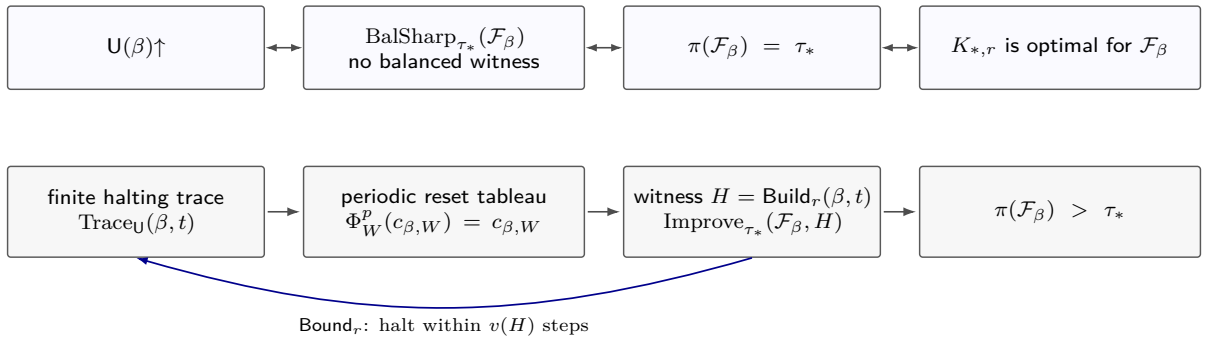


Figure 14: The finite certificate normal form for the fixed numerical compiler. The double arrows in the upper row are semantic equivalences, with the fixed baseline template supplying the implication from sharpness to equality. In the lower row, a finite halting trace builds a periodic reset tableau and then a balanced improvement witness. Conversely, every such witness yields, through a closed relation component, a halting-time bound of at most its order. Neither construction requires deciding whether U halts on β .

We finally pass from the halting predicate of U to an arbitrary Π_1^0 predicate. Fix a decidable effective syntax for primitive recursive predicates. Its decoder is total, with syntactically invalid strings interpreted as the constant true predicate. We fix a set-theoretic formula $\text{Valid}_{\text{PR}}(d)$ for validity. ZFC proves that a valid code has a unique Boolean value on every input. This is a uniform definability convention, not an assertion that there is a primitive recursive universal evaluator. For a binary primitive recursive predicate $R(e, n)$ presented by a chosen expression in this syntax, write $\ulcorner R \urcorner$ for the natural-number Gödel code of that expression. For a fixed valid code $d = \ulcorner R \urcorner$, its evaluation is the primitive recursive predicate $R(e, n)$.

The parameter theorem, also called the s - m - n theorem, says that fixed parameters can be compiled effectively into a program index. It uniformly produces an index $\text{Idx}_{\text{test}}(d, e)$ for a machine which successively tests $R(e, 0), R(e, 1), \dots$ and halts at the first failure. Put $\beta_{d,e} := \text{tr}_U(\text{Idx}_{\text{test}}(d, e), 0)$ and $\mathcal{F}_{r,d,e} := \mathcal{F}_{r,\beta_{d,e}}$. After r is fixed and $d = \ulcorner R \urcorner$, write $\mathcal{F}_{R,e} := \mathcal{F}_{r,d,e}$. By [Proposition 17.3](#), the joint map $(r, d, e) \mapsto \langle \mathcal{F}_{r,d,e} \rangle$ is total computable, while its dependence on (d, e) is primitive recursive after r is fixed.

The testing machine encoded by $\beta_{d,e}$ fails to halt exactly when $R(e, n)$ holds for every n . Combining the certificate transformations in [Theorem 17.5](#) with [Lemma 17.2](#) identifies this condition with the absence of a balanced improvement witness, equality with the fixed baseline value, and the asymptotic optimality of $K_{*,r}$. Together with the primitive-recursive compiler in [Proposition 17.3](#), this gives the following normal form.

Theorem 17.6. *There is one fixed finite template $K_{*,r}$ such that, uniformly and primitive recursively from $(\ulcorner R \urcorner, e)$, one can construct a finite family $\mathcal{F}_{R,e}$ satisfying the following equivalent conditions.*

- (i) $\forall n \ R(e, n)$.
- (ii) $\text{BalSharp}_{\tau_*}(\mathcal{F}_{R,e})$.

(iii) $\pi(\mathcal{F}_{R,e}) = \lambda_r(K_{*,r})$.

(iv) $K_{*,r}$ is asymptotically optimal for $\mathcal{F}_{R,e}$.

Moreover, if the predicate $\forall n R(e, n)$ fails, let n_0 be the least integer such that $R(e, n_0) = 0$. From the finite evaluations certifying that $R(e, i) = 1$ for every $i < n_0$ and that $R(e, n_0) = 0$, one can compute a finite $\mathcal{F}_{R,e}$ -hom-free r -graph H satisfying $\lambda_{\text{bal}}(H) > \tau_*$.

Proof. Take $K_{*,r}$ from Lemma 17.1, put $d := \lceil R \rceil$, and recall that $\mathcal{F}_{R,e} = \mathcal{F}_{\beta_{d,e}}$. Since R is primitive recursive, every test performed by the program with index $\text{Idx}_{\text{test}}(d, e)$ terminates. This program therefore halts on input 0 exactly when $R(e, n) = 0$ for some n . By the defining property of tr_U ,

$$\forall n R(e, n) \iff U(\beta_{d,e}) \uparrow.$$

The final equivalence in Theorem 17.5 identifies the latter condition with $\text{BalSharp}_{\tau_*}(\mathcal{F}_{R,e})$. Moreover, Lemma 17.1 shows that $K_{*,r}$ is $\mathcal{F}_{R,e}$ -hom-free and that $\lambda_r(K_{*,r}) = \tau_*$. Hence Lemma 17.2 makes balanced sharpness equivalent to $\pi(\mathcal{F}_{R,e}) = \lambda_r(K_{*,r})$ and to the asymptotic optimality of $K_{*,r}$. The uniform primitive-recursive construction of $\mathcal{F}_{R,e}$ follows from Proposition 17.3.

Now suppose that n_0 is the least integer with $R(e, n_0) = 0$. The supplied evaluation certificates form a finite bad run in the notation of Appendix A. The forward trace transformation in Lemma A.4 computes from this bad run a time bound t satisfying $\text{Trace}_U(\beta_{d,e}, t)$. Part (i) of Theorem 17.5 then gives $H := \text{Build}_r(\beta_{d,e}, t)$ with $\text{Improve}_{\tau_*}(\mathcal{F}_{R,e}, H)$. By definition, H is $\mathcal{F}_{R,e}$ -hom-free and $\lambda_{\text{bal}}(H) > \tau_*$. This also proves the asserted effective construction from the finite evaluations. ■

18 Internal ZFC bridges

The preceding section proves the certificate equivalence as an ordinary mathematical theorem in the usual ZFC foundations. The purpose of this section is to record the same argument as one arithmetized formula with a ZFC proof uniform in the input codes. Here and below, $T \vdash \varphi$ means that φ has a formal proof from the axioms of T in a fixed classical proof calculus, and $T \not\vdash \varphi$ means that no such proof exists. A theory is *computably axiomatized* when an algorithm enumerates its axioms, and it is *consistent* when it does not prove $0 = 1$.

18.1 The numerical bridge

Fix the primitive recursive syntax and evaluation certificates described in Appendix A. The formula

$$\text{All}_{\text{PR}}(d, e) \iff \forall n \text{ Eval}_{\text{PR}}(d, e, n, 1)$$

says that every test encoded by d succeeds for the parameter e . Let $\text{CodeFam}_r^{\text{num}}(d, e, f)$ say that the fixed- r numerical compiler returns the canonical family code f .

For a canonical family code f , write $\mathcal{F}_f$ for its filtered decode. Define $\text{Improve}_{\tau_*}^{\text{code}}(f, H)$ to assert that H is a legal finite $\mathcal{F}_f$ -hom-free r -graph satisfying $b_* r! |H| > a_* v(H)^r$, and put

$$\text{BalSharp}_{\tau_*}^{\text{code}}(f) \iff \forall H \neg \text{Improve}_{\tau_*}^{\text{code}}(f, H).$$

To express exact density arithmetically, put $m_* := v(K_{*,r})$. For $s \geq 1$, let m_s be the canonical multiple of m_* and let $\text{Ex}_s(f)$ be the corresponding finite extremal number defined in Section A.1. Then

$$\text{TuranEq}_{\tau_*}(f) \iff \text{CanFam}_r(f) \wedge \forall s \geq 1 [b_* r! \text{Ex}_s(f) \leq a_* m_s^r \wedge a_* \binom{m_s}{r} \leq b_* \text{Ex}_s(f)].$$

Finally, $\text{Base}_r(f)$ says that f is canonical and $K_{*,r}$ is $\mathcal{F}_f$ -hom-free. These four formulas are the public interface of the numerical formalization. In words, the next proposition says that ZFC proves uniformly that the compiled family has density τ_* exactly when every test encoded by the primitive recursive predicate succeeds, and that the fixed baseline template is admissible for every compiled family.

Proposition 18.1. *Fix $r \geq r_{\text{num}}$ and the corresponding compiler data. ZFC proves the following statements.*

- (i) We have $\forall d \forall e \exists! f \text{ CodeFam}_r^{\text{num}}(d, e, f)$.

(ii) We have the following single formula:

$$\begin{aligned} \forall d \forall e \forall f \big(& \text{Valid}_{\text{PR}}(d) \wedge \text{CodeFam}_r^{\text{num}}(d, e, f) \big) \longrightarrow \big[\quad \big(\text{All}_{\text{PR}}(d, e) \iff \text{BalSharp}_{\tau_*}^{\text{code}}(f) \big) \\ & \wedge \big(\text{BalSharp}_{\tau_*}^{\text{code}}(f) \iff \text{TuranEq}_{\tau_*}(f) \big) \\ & \wedge \big(\text{All}_{\text{PR}}(d, e) \iff \text{TuranEq}_{\tau_*}(f) \big) \\ & \wedge \text{Base}_r(f) \big]. \end{aligned} \quad (18.1)$$

(iii) We have the fixed finite-dimensional identity $\lambda_r(K_{*,r}) = \tau_*$. Consequently, $\pi(\mathcal{F}_f) \geq \tau_*$ whenever $\text{Base}_r(f)$ holds.

Proof. Part (i) of [Lemma A.7](#) gives the totality and uniqueness of the compiler output. Under the antecedent of (18.1), bad-run completeness and the two typed certificate maps give

$$\neg \text{All}_{\text{PR}}(d, e) \iff \exists n \exists u \text{BadRun}(d, e, n, u) \iff \exists H \text{Improve}_{\tau_*}^{\text{code}}(f, H).$$

Taking negations yields $\text{All}_{\text{PR}}(d, e) \iff \text{BalSharp}_{\tau_*}^{\text{code}}(f)$. The compiler invariant in [Lemma A.5](#) gives $\text{Base}_r(f)$, and the arithmetic interface in [Lemma A.1](#) converts balanced sharpness into $\text{TuranEq}_{\tau_*}(f)$. This proves all four conjuncts in (18.1).

The fixed algebraic certificates in [Lemma A.2](#) prove that the uniform weighting of $K_{*,r}$ attains τ_* and that no weighting has larger value. Its balanced blowups give the asserted lower bound under $\text{Base}_r(f)$. ■

18.2 Numerical Gödel–Rosser incompleteness

Theorem 18.2. *Let r_{num} be as in [Theorem 1.1](#). There is a total algorithm which, on input an integer $r \geq r_{\text{num}}$ and a natural number a , outputs a rational number $\tau_r \in \mathbb{Q} \cap (0, 1)$, a finite r -graph $K_{*,r}$, and a finite family $\mathcal{F}_{r,a}$ of r -graphs. The first two outputs depend only on r , the uniform weighting of $K_{*,r}$ is optimal, and the balanced blowups of $K_{*,r}$ have limiting density τ_r . In particular, $\lambda_r(K_{*,r}) = \tau_r$.*

Suppose that a indexes an algorithm enumerating the axioms of a computably axiomatized theory T_a extending ZFC. If T_a is consistent, then the following statements hold.

- (i) *The theory T_a proves that $K_{*,r}$ is $\mathcal{F}_{r,a}$ -hom-free, that its uniform weighting is optimal with normalized value τ_r , and hence that $\pi(\mathcal{F}_{r,a}) \geq \tau_r$.*
- (ii) *The equality $\pi(\mathcal{F}_{r,a}) = \tau_r$ is true, so $K_{*,r}$ is asymptotically optimal for $\mathcal{F}_{r,a}$.*
- (iii) *The theory T_a proves neither $\pi(\mathcal{F}_{r,a}) = \tau_r$ nor $\pi(\mathcal{F}_{r,a}) > \tau_r$. Consequently, the asymptotic optimality of $K_{*,r}$ is independent of T_a .*

Taking a fixed canonical index for an enumerator of the axioms of ZFC in [Theorem 18.2](#) proves [Theorem 1.4](#). We use Rosser’s theorem [63] under ordinary consistency alone. In particular, we do not assume that every arithmetical theorem of the theory is true in the standard natural numbers.

Proof of Theorem 18.2. On inputs with $r < r_{\text{num}}$, let the algorithm return a fixed default output triple. Now let $r \geq r_{\text{num}}$. Run the parameter-selection algorithm in [Lemma 6.9](#) and the rational calibration in [Lemma 6.8](#). This computes τ_r and all input-independent compiler data. Then use [Lemma 17.1](#) to compute $K_{*,r}$. These two outputs are computed before a is used and therefore depend only on r .

Given the axiom-enumerator index a , the witnessed proof coding supplies the primitive recursive proof predicate required by Rosser’s theorem. The effective syntactic diagonal construction used in that theorem [63] computes from a a code $d_a = \ulcorner R_a \urcorner$ of a primitive recursive predicate; see, for example, Boolos, Burgess, and Jeffrey [8, Sections 15.1 and 17.1]. Let ρ_a be the sentence $\forall n R_a(0, n)$, the resulting Rosser sentence for the theory enumerated by a .

Let $\text{Idx}_{\text{test}}(d_a, 0)$ be the testing-program index from [Theorem 17.6](#), and put $\beta_a := \text{tr}_U(\text{Idx}_{\text{test}}(d_a, 0), 0)$ and $\mathcal{F}_{r,a} := \mathcal{F}_{r,\beta_a}$. The parameter theorem, the universal-machine translator, and the fixed- r compiler are total. By [Proposition 17.3](#), their composition with the total computation of the r -dependent setup is total computable in (r, a) . In particular, the algorithm does not test whether the enumerated theory is consistent or whether it extends ZFC; these are hypotheses only for the correctness conclusion.

Now suppose that a enumerates a consistent theory $T_a \supseteq \text{ZFC}$, and write $T = T_a$ and $\rho_T = \rho_a$. Rosser’s theorem gives $T \not\vdash \rho_T$ and $T \not\vdash \neg \rho_T$. The concrete syntax code d_a is valid, and ZFC verifies this primitive

recursive fact. Since $d_a = \ulcorner R_a \urcorner$, correctness of the evaluation predicate for this fixed primitive recursive code also gives $\text{ZFC} \vdash (\rho_T \iff \text{All}_{\text{PR}}(d_a, 0))$.

Let $f_{r,a} := \langle \mathcal{F}_{r,a} \rangle$ be the compiler's canonical family code. Then ZFC proves $\text{CodeFam}_r^{\text{num}}(d_a, 0, f_{r,a})$. Freeze the already computed value of r and apply the first two equivalences in (18.1). They give $T \vdash (\text{All}_{\text{PR}}(d_a, 0) \iff \text{TuranEq}_{\tau_r}(f_{r,a}))$. Combining the preceding two equivalences yields $T \vdash (\rho_T \iff \text{TuranEq}_{\tau_r}(f_{r,a}))$.

The same instance of (18.1) proves $\text{Base}_r(f_{r,a})$, and hence in particular that $f_{r,a}$ is a legal family code. Part (i) of Lemma A.1 therefore proves in ZFC that the right-hand side is equivalent to the usual set-theoretic sentence denoted by $\pi(\mathcal{F}_{r,a}) = \tau_r$. Hence $T \vdash (\rho_T \iff \pi(\mathcal{F}_{r,a}) = \tau_r)$. A proof in T of the equality or its negation would therefore prove ρ_T or $\neg\rho_T$, respectively, contradicting Rosser independence.

The same fixed- r ZFC verification proves that $K_{*,r}$ is $\mathcal{F}_{r,a}$ -hom-free and that its uniform weighting attains $\lambda_r(K_{*,r}) = \tau_r$. Its balanced blowups are therefore $\mathcal{F}_{r,a}$ -free and give the T -provable lower bound $\pi(\mathcal{F}_{r,a}) \geq \tau_r$. Consequently, T proves that the negation of the equality is equivalent to the strict inequality. Since the hom-freeness and baseline identity are also provable in T , it proves that the equality is equivalent to the asymptotic optimality of $K_{*,r}$.

Finally, under the standing assumption that T is consistent, ρ_T is true for the usual natural numbers. Indeed, if it were false, its negation would be a true Σ_1^0 arithmetic sentence and hence would be provable in Robinson arithmetic [8, Section 16.4]. ZFC proves the von Neumann interpretation of Robinson arithmetic, so T would prove the translated sentence, contradicting $T \not\vdash \neg\rho_T$. Since ρ_T is the sentence $\forall n R_a(0, n)$, its truth implies that the testing program with index $\text{Idx}_{\text{test}}(d_a, 0)$ does not halt on input 0. By the defining property of tr_U and the definition of β_a , it follows that $U(\beta_a) \uparrow$. Since $\mathcal{F}_{r,a} = \mathcal{F}_{r,\beta_a}$, Theorem 1.1 now gives $\pi(\mathcal{F}_{r,a}) = \tau_r$. This final implication is external and uses no soundness assumption on T . ■

18.3 The structural bridge

The structural compiler uses the base uniformity $k = r_{\text{num}}$ and the threshold $\tau_{\text{base}} = \tau_k$, independently of the target structural uniformity s . Appendix B defines, uniformly from a finite family code g , five set-theoretic formulas $\text{Str}_{i,s}(g)$. They express, in order, uniqueness of the extremal limit, connectedness of the extremal limit space, Sep_s -symmetry breaking, a Sep_s -separated two-phase decomposition, and Erdős–Simonovits stability. The two computational branches of the structural compiler produce the following alternatives. The set-theoretic predicates introduced below encode this table.

Simulated computation	unique extremal limit	connected	Sep_s -symmetry breaking	Sep_s -separated two phases	Erdős–Simonovits stability
$U(\beta) \uparrow$	yes	yes	no	no	yes
$U(\beta) \downarrow$	no	no	yes	yes	no

For the Rosser input in the theorem, consistency places the actual family in the nonhalting row, although the corresponding structural sentences cannot be decided in T . Here $\text{TuranEq}_{\tau_{\text{base}},k}$ and $\text{Improve}_{\tau_{\text{base}},k}^{\text{code}}$ denote the base- k specializations of the preceding numerical formulas. Write $\tau_{\text{base}} = a_{\text{base}}/b_{\text{base}}$ in lowest terms and define

$$\text{Zero}_k(f) := \text{TuranEq}_{\tau_{\text{base}},k}(f), \quad \text{Pos}_k(f) := \exists H \text{Improve}_{\tau_{\text{base}},k}^{\text{code}}(f, H).$$

The two structural rows are

$$\begin{aligned} \text{ZeroStr}_s(g) &\iff \text{Str}_{1,s}(g) \wedge \text{Str}_{2,s}(g) \wedge \neg\text{Str}_{3,s}(g) \wedge \neg\text{Str}_{4,s}(g) \wedge \text{Str}_{5,s}(g), \\ \text{PosStr}_s(g) &\iff \neg\text{Str}_{1,s}(g) \wedge \neg\text{Str}_{2,s}(g) \wedge \text{Str}_{3,s}(g) \wedge \text{Str}_{4,s}(g) \wedge \neg\text{Str}_{5,s}(g). \end{aligned}$$

Let Lift_s be the fixed primitive recursive postprocessing map from a numerical family code to the structural output, and write $\text{CodeLift}_s(f, g)$ for its graph. Thus the synchronized structural compiler relation is

$$\text{CodeFam}_s^{\text{str}}(d, e, g) \iff \exists f (\text{CodeFam}_k^{\text{num}}(d, e, f) \wedge \text{CodeLift}_s(f, g)).$$

The exact factorization and all typed structural formulas are given in Section B.4. Here $\text{E}_s(g)$ and Close_s are the coded extremal space and profile-approximation predicate from Section B.1; $\mathfrak{B}_p(V)$ and w_p denote the corresponding profile class and limit vector. The following elementary obstruction is the only local structural argument needed in the main text.

Lemma 18.3. *ZFC proves that if $x, y \in E_s(g)$ and $\text{Sep}_s(x) > 0 > \text{Sep}_s(y)$, then $\neg \text{Str}_{5,s}(g)$.*

Proof. We use the coded removal, edit-density, and profile-limit facts proved in Lemma B.1; the proof of that lemma is deferred to Appendix B. Suppose instead that p witnesses $\text{Str}_{5,s}(g)$. By (B.1), choose asymptotically extremal sequences $\mathbf{G}^+$ and $\mathbf{G}^-$ converging to x and y , respectively. The formula $\text{Close}_s(p, \mathbf{G}^+)$ has the quantifier order

$$\forall \varepsilon > 0 \exists N \forall n \geq N \exists B.$$

For each j , choose an index $n_j^+ \geq j$, larger than the thresholds for $\varepsilon = 1/i$ with $i \leq j$, and then choose the least canonical $B_j^+ \in \mathfrak{B}_p(V(G_{n_j^+}^+))$ satisfying

$$\text{edit}(G_{n_j^+}^+, B_j^+) \leq j^{-1} v(G_{n_j^+}^+)^s.$$

Thus $v(B_j^+) \rightarrow \infty$. Equation (B.3) and convergence of the selected subsequence show that $\mathbf{B}^+$ converges to x , while part (iii) of Lemma B.1 shows that it converges to w_p . Hence $x = w_p$.

Apply the same diagonal choice to $\mathbf{G}^-$ to obtain $y = w_p$. This contradicts $\text{Sep}_s(x) > 0 > \text{Sep}_s(y)$. Choose each subsequence index as the least integer satisfying the displayed finite threshold conditions, and choose each $B_j^\pm$ from the nonempty finite set of canonical graphs on the selected ground set. These least choices are definable in ZFC. $\blacksquare$

Proposition 18.4. *Fix $s \geq r_{\text{str}}$ and put $k = r_{\text{num}}$. ZFC proves the following statements.*

- (i) *We have $\forall f \exists! g \text{ CodeLift}_s(f, g)$, and (B.4) agrees with the structural compiler on every numerical compiler output.*
- (ii) *We have the single coded bridge*

$$\begin{aligned} \forall f \forall g (\text{Base}_k(f) \wedge \text{CodeLift}_s(f, g)) &\longrightarrow [(\text{Zero}_k(f) \iff \text{ZeroStr}_s(g)) \\ &\wedge (\text{Pos}_k(f) \iff \text{PosStr}_s(g))]. \end{aligned} \quad (18.2)$$

Proof. The postprocessing map consists of the bounded basis-completion, witness-catalogue, star-blocker, image, and pair-covering operations from Part II. Section B.4 gives their canonical code, and Section B.5 proves totality in ZFC. Under $\text{Base}_k(f)$, (B.7) shows that the numerical alternatives $\text{Zero}_k(f)$ and $\text{Pos}_k(f)$ are exhaustive.

The uniform substitution ledger in Section B.5 applies to every such lift pair. In the zero case, (B.8), (B.10), and (B.11) give the unique neutral limit $W_{0,s}$ and stability with respect to the fixed profile $\mathbf{P}_{0,s}$. Hence $\text{ZeroStr}_s(g)$ holds. In the positive case, (B.9) and (B.10) give two nonempty compact phases on which Sep_s has opposite signs. Uniqueness and connectedness fail, both sign properties hold, and Lemma 18.3 rules out one-profile stability. Hence $\text{PosStr}_s(g)$ holds. The two structural rows are contradictory, so exhaustiveness of the numerical alternatives gives the reverse implications. All of these implications use the fixed real-algebraic certificates recorded in the same ledger. $\blacksquare$

For a primitive recursive predicate code d and a parameter e , recall $\beta_{d,e} = \text{tr}_U(\text{Idx}_{\text{test}}(d, e), 0)$. Define $\mathbf{P}_{i,s}(d, e)$ to mean that there is a g for which $\text{CodeFam}_s^{\text{str}}(d, e, g)$ and $\text{Str}_{i,s}(g)$ both hold. After numerals are substituted for d, e , this is an ordinary set-theoretic sentence.

The next proposition composes the numerical bridge with the direct coded structural bridge. Here $\bigwedge$ denotes conjunction over the indicated finite index set.

Proposition 18.5. *Fix $s \geq r_{\text{str}}$. The following statements hold.*

- (i) *ZFC proves the following single formula directly on the structural output code:*

$$\begin{aligned} \forall d \forall e \forall g (\text{Valid}_{\text{PR}}(d) \wedge \text{CodeFam}_s^{\text{str}}(d, e, g)) &\longrightarrow \left[\bigwedge_{i \in \{1, 2, 5\}} (\text{Str}_{i,s}(g) \iff \text{All}_{\text{PR}}(d, e)) \right. \\ &\left. \wedge \bigwedge_{j \in \{3, 4\}} (\text{Str}_{j,s}(g) \iff \neg \text{All}_{\text{PR}}(d, e)) \right]. \end{aligned} \quad (18.3)$$

(ii) ZFC proves $\forall d \forall e \exists ! g \text{ CodeFam}_s^{\text{str}}(d, e, g)$.

(iii) Consequently, ZFC proves the following formula uniformly in d, e :

$$\forall d \forall e \left(\text{Valid}_{\text{PR}}(d) \longrightarrow \left[\bigwedge_{i \in \{1, 2, 5\}} (\text{P}_{i,s}(d, e) \iff \text{All}_{\text{PR}}(d, e)) \right. \right. \\ \left. \left. \wedge \bigwedge_{j \in \{3, 4\}} (\text{P}_{j,s}(d, e) \iff \neg \text{All}_{\text{PR}}(d, e)) \right] \right). \quad (18.4)$$

(iv) The defining data for Sep_s , $W_{0,s}$, and $\mathbf{P}_{0,s}$ are fixed before (d, e) is given.

Proof. The numerical compiler has a unique output and [Proposition 18.4](#) gives a unique lift, proving totality and uniqueness of the structural output. For the unique intermediate numerical code f , [Proposition 18.1](#) gives

$$\text{All}_{\text{PR}}(d, e) \iff \text{Zero}_k(f), \quad \neg \text{All}_{\text{PR}}(d, e) \iff \text{Pos}_k(f).$$

Applying the coded structural bridge and expanding the two row definitions gives (18.3). Uniqueness of g then gives (18.4). The objects Sep_s , $W_{0,s}$, and $\mathbf{P}_{0,s}$ are fixed by [Lemma B.2](#) before d, e are quantified. The detailed typed composition is recorded in [Section B.6](#). ■

For each fixed $s \geq r_{\text{str}}$, the formula in (18.4) has a ZFC proof uniform in d and e ; we do not identify an explicit finite fragment of ZFC sufficient for the structural argument.

18.4 Structural Gödel–Rosser incompleteness

Theorem 18.6. Fix $s \geq r_{\text{str}}$. For every consistent computably axiomatized theory $T \supseteq \text{ZFC}$, one can effectively construct, from an index enumerating the axioms of T , a finite family $\mathcal{G}_{s,T}$ of s -graphs such that each of the following five set-theoretic sentences is independent of T , with the terminology fixed in [Section 8](#). Here Sep_s is fixed before T is given.

- (i) $\mathcal{G}_{s,T}$ has a unique extremal limit;
- (ii) its extremal limit space is connected;
- (iii) it exhibits Sep_s -symmetry breaking;
- (iv) it has a Sep_s -separated two-phase decomposition;
- (v) it has Erdős–Simonovits stability.

Properties (i), (ii), and (v) hold, whereas properties (iii) and (iv) fail.

Proof of Theorem 18.6. Fix the supplied axiom-enumerator index a for T , and recall the primitive recursive Rosser predicate R_a computed from a in the proof of [Theorem 18.2](#). Let ρ_T be the corresponding Rosser sentence $\forall n R_a(0, n)$, and set $\beta_T := \beta_{\neg R_a \neg, 0}$. Thus the simulated program for the fixed universal machine successively tests $R_a(0, 0), R_a(0, 1), \dots$ and halts at the first failure. The parameter theorem and the universal simulation are primitive recursive, so ZFC proves that ρ_T holds if and only if $\text{U}(\beta_T)$ does not halt. Set $\mathcal{G}_{s,T} := \mathcal{G}_{s,\beta_T}$.

For $i \in [5]$, let P_i denote the corresponding sentence in the statement for $\mathcal{G}_{s,T}$. Apply [Proposition 18.5](#) with $d = \ulcorner R_a \urcorner$ and $e = 0$. This is a valid primitive recursive syntax code, and ZFC proves its validity. With this substitution, $\beta_{d,e} = \beta_T$ and $\text{P}_{i,s}(d, e) = P_i$ by definition. Moreover, ZFC proves that ρ_T is equivalent to $\text{All}_{\text{PR}}(\ulcorner R_a \urcorner, 0)$. Since T extends ZFC, we obtain

$$T \vdash P_i \iff \rho_T \quad \text{for } i \in \{1, 2, 5\}, \quad \text{and} \quad T \vdash P_j \iff \neg \rho_T \quad \text{for } j \in \{3, 4\}.$$

A proof or refutation in T of any one of these five sentences would therefore prove either ρ_T or $\neg \rho_T$, contradicting Rosser independence.

Finally, consistency of T implies that ρ_T is true in the standard natural numbers, as shown in the proof of [Theorem 18.2](#). Hence $\text{U}(\beta_T)$ does not halt. The first alternative of [Theorem 1.2](#) gives the asserted truth values of the five properties. ■

Part IV

Further properties of Turán densities

This part records several further properties of Turán densities. We first discuss uniform approximation and exact comparison for finite forbidden families, followed by quantitative bounds on improvement witnesses, density gaps, and compiler output. We then turn to recursive and c.e. forbidden families, characterize their density spectrum, and determine the complexity of exact comparison, including the oracle-relative classification in [Theorem 23.2](#). In [Section 24](#), we study two finite semialgebraic value schemes and show that both produce only algebraic values; known transcendental finite-family densities therefore lie outside both schemes.

The opening approximation result applies to every $r \geq 2$. Whenever the discussion fixes $r \geq r_{\text{num}}$, we continue to use $\mathcal{F}_\beta$ and τ_* for $\mathcal{F}_{r,\beta}$ and τ_r , respectively, as in [Theorem 1.1](#).

19 Computability and exact comparison for finite forbidden families

We first use the explicit estimate in [Proposition 2.6](#) to obtain a uniform approximation algorithm for every finite forbidden family. We then determine the exact arithmetical complexity of comparing two such densities and show that the corresponding hardness persists for the stated relations when one family or the rational threshold is fixed. Exact equality remains undecidable because a positive gap need not have a computable lower bound. By contrast, if an instance is supplied with a rational $\eta > 0$ and the promise $|\pi(\mathcal{F}) - \pi(\mathcal{G})| \geq \eta$, then approximating both densities to error less than $\eta/3$ decides their order. The section on quantitative density gaps later shows that the tableau reduction has no uniform computable positive gap.

Corollary 19.1. *There is a total algorithm which, on input an integer $r \geq 2$, a finite family $\mathcal{F}$ of r -graphs, possibly empty, and $j \in \mathbb{N}$, outputs a rational interval $[L, U]$ with $\pi(\mathcal{F}) \in [L, U]$ and $U - L < 2^{-j}$. In particular, every finite-family Turán density is a computable real, uniformly in the uniformity and the forbidden family.*

Proof. Given $(r, \mathcal{F}, j)$, the algorithm has the following four steps.

- (i) For every $F \in \mathcal{F}$, enumerate the partitions of $V(F)$ that are injective on every edge. For each partition, form the corresponding image r -graph, compare all vertex relabelings, retain its lexicographically least canonical code, and remove repeated codes. This computes $\mathcal{Q}(\mathcal{F})$.
- (ii) Let m be the least integer satisfying $m > 2^{j-1}r(r-1)$.
- (iii) Enumerate all r -graphs on $[m]$, retain those containing no member of $\mathcal{Q}(\mathcal{F})$, and take the largest edge count among the retained r -graphs. This computes $\pi^{\text{hom}}(m, \mathcal{F})$ exactly.
- (iv) Output $L := \theta_{r,m}\pi^{\text{hom}}(m, \mathcal{F})$ and $U := \pi^{\text{hom}}(m, \mathcal{F})$.

The conclusion follows from [Proposition 2.6](#). In the definition of a computable real, one may output the midpoint $(L + U)/2$, whose distance from $\pi(\mathcal{F})$ is at most $(U - L)/2 < 2^{-j}$. ■

Uniform approximation therefore does not by itself decide exact equality. We now prove the comparison theorem stated in the Introduction. We use the total filtered decoding convention fixed in [Section 2](#), so every natural number denotes a finite family and the comparison relations below are genuine subsets of $\mathbb{N}$ or $\mathbb{N}^2$, rather than promise problems about well-formed codes.

Let $K_{\text{diag}} := \{e: \psi_e(e) \downarrow\}$ be the standard diagonal halting set, which is Σ_1^0 -complete; see Rogers [62]. The normalization in the direct tableau construction fixed the total computable translator tr_U , which preserves halting. Put $\beta_e := \text{tr}_U(e, e)$. Consequently, $e \in K_{\text{diag}}$ exactly when $U(\beta_e)$ halts.

Fix $r \geq r_{\text{num}}$ and the corresponding rational threshold τ_* from [Theorem 1.1](#). Choose one fixed binary word β_∞ for which $U(\beta_\infty) \uparrow$. Then $\pi(\mathcal{F}_{\beta_\infty}) = \tau_*$.

Proof of Theorem 1.5. For the upper bounds, use Corollary 19.1 to compute rational interval approximations at every stage. Intersecting the approximations through stage s gives nested intervals $I_s(\mathcal{F}) = [L_s(\mathcal{F}), U_s(\mathcal{F})]$ and $I_s(\mathcal{G}) = [L_s(\mathcal{G}), U_s(\mathcal{G})]$ of length less than 2^{-s} containing the respective densities.

The relation $\pi(\mathcal{F}) > \pi(\mathcal{G})$ holds exactly when $L_s(\mathcal{F}) > U_s(\mathcal{G})$ at some stage s . Indeed, the inequality at one stage certifies the strict comparison, while a positive difference between the limiting values eventually exceeds the combined interval lengths. Thus $>$ is c.e., and swapping the two families gives the same conclusion for $<$. Their union is disequality, so $\neq$ is c.e. Equivalently, equality holds exactly when $I_s(\mathcal{F}) \cap I_s(\mathcal{G}) \neq \emptyset$ at every stage. The three complementary relations are therefore co-c.e. The same interval argument applies to comparison with any fixed computable real, in particular with τ_* .

For hardness, given e , output the pair $(\mathcal{F}_{\beta_e}, \mathcal{F}_{\beta_\infty})$. By the defining property of tr_U , Theorem 1.1, and the choice of β_∞ , we have $e \in K_{\text{diag}}$ if and only if $\pi(\mathcal{F}_{\beta_e}) > \pi(\mathcal{F}_{\beta_\infty})$, while $e \notin K_{\text{diag}}$ if and only if $\pi(\mathcal{F}_{\beta_e}) = \pi(\mathcal{F}_{\beta_\infty})$. Thus $>$ and $\neq$ are Σ_1^0 -hard, and equality is Π_1^0 -hard. Swapping the two families gives hardness for $<$, with the first family fixed to be $\mathcal{F}_{\beta_\infty}$. On all produced instances, we have $\pi(\mathcal{F}_{\beta_e}) \geq \pi(\mathcal{F}_{\beta_\infty})$, so $\pi(\mathcal{F}_{\beta_e}) \leq \pi(\mathcal{F}_{\beta_\infty})$ if and only if $\pi(\mathcal{F}_{\beta_e}) = \pi(\mathcal{F}_{\beta_\infty})$. This gives Π_1^0 -hardness for $\leq$; swapping gives hardness for $\geq$, again with the first family fixed to be $\mathcal{F}_{\beta_\infty}$.

For the fixed rational threshold, use the same reduction $e \mapsto \mathcal{F}_{\beta_e}$. Its output always has density at least τ_* , so equality and non-strict comparison coincide on the range of the reduction. ■

We finish the finite-family comparison analysis by isolating the corresponding fixed-template problem. This formulation records when the baseline template $K_{*,r}$ is optimal and makes the finite obstruction to optimality explicit.

Define

$$\text{Sharp}_{K_{*,r}} := \{f \in \mathbb{N} : K_{*,r} \text{ is } \mathcal{F}_f\text{-hom-free and } \pi(\mathcal{F}_f) = \lambda_r(K_{*,r})\}.$$

Proposition 19.2. *The following statements hold.*

- (i) *The set $\text{Sharp}_{K_{*,r}}$ is Π_1^0 -complete.*
- (ii) *Its complement inside the decidable promise class $\{f \in \mathbb{N} : K_{*,r} \text{ is } \mathcal{F}_f\text{-hom-free}\}$ is Σ_1^0 -complete.*

Within this promise class, failure of sharpness for a code f has an explicit finite certificate: a finite $\mathcal{F}_f$ -hom-free template G with $\lambda_r(G) > \lambda_r(K_{,r})$, or equivalently, by Lemma 17.2, a balanced witness H satisfying (17.1).*

Proof. Fix a code f . The promise is decidable. If it fails, then f is immediately outside $\text{Sharp}_{K_{*,r}}$. Under the promise, membership is exactly the Π_1^0 condition $\text{BalSharp}_{\tau_*}(\mathcal{F}_f)$ by Lemma 17.2. More explicitly, enumerate all finite r -graphs H with $v(H) > 0$, first by vertex count and then by the fixed edge-list coding. For each H , test the finitely many maps from every $F \in \mathcal{F}_f$ to H . If H is $\mathcal{F}_f$ -hom-free, test the integer inequality $b_* r! |H| > a_* v(H)^r$ from (17.1). By Lemma 17.2, this search halts exactly when the promised baseline template is not optimal. Together with the decidable promise test, this semidecides the complement of $\text{Sharp}_{K_{*,r}}$.

For hardness, map e to the canonical code of $\mathcal{F}_{\beta_e}$. The promise always holds, and this code belongs to $\text{Sharp}_{K_{*,r}}$ exactly when $e \notin K_{\text{diag}}$. Thus sharpness is Π_1^0 -hard, and its complement inside the promise class is Σ_1^0 -hard. ■

20 Improvement witnesses and Busy-Beaver growth

Fix $r \geq r_{\text{num}}$ throughout this section, and retain the abbreviations $\mathcal{F}_\beta$ and τ_* from the beginning of this part.

The certificate correspondence in Theorem 17.5 has a quantitative consequence. The Introduction defined the halting time and the order of the smallest improving template attached to an input. We now introduce the minimum area of a periodic reset tableau and compare all three quantities. The reset-period estimate supplies the upper construction, while the certificate-recovery argument supplies the converse. Maximizing over inputs will then prove Theorem 1.6.

For an input word β , define the minimum periodic-tableau area by

$$\text{Area}_{\text{per}}(\beta) := \min \{p(W+2) : W, p \in \mathbb{N}, \quad W \geq |\beta| + 2, \quad p \geq 1, \quad \text{and} \quad \Phi_W^p(c_{\beta,W}) = c_{\beta,W}\},$$

with value ∞ if no such pair (W, p) exists. Here Φ_W and $c_{\beta,W}$ are the reset cellular automaton and canonical initial row from [Section 3](#). The reset lemma shows that $\text{Area}_{\text{per}}(\beta) < \infty$ exactly when $\mathcal{U}(\beta)$ halts. Recall that $T(\beta)$ is the running time of $\mathcal{U}$ on β , with value ∞ in the nonhalting case.

In the abbreviations of this part, the definition of ω_r in the Introduction is equivalently

$$\omega_r(\beta) = \min \{v(G) : G \text{ is } \mathcal{F}_\beta\text{-hom-free and } \lambda_r(G) > \lambda_r(K_{*,r}) = \tau_*\},$$

with value ∞ if the set is empty.

The next theorem compares the least order of an improving template with the halting time and the minimum periodic-tableau area.

Theorem 20.1. *There is a constant $C_r > 0$ for which the following statements hold.*

- (i) $\omega_r(\beta) < \infty$ exactly when $\mathcal{U}(\beta)$ halts;
- (ii) If $\mathcal{U}$ halts on β , then $T(\beta) \leq \omega_r(\beta)$;
- (iii) If $\mathcal{U}$ halts on β , then $\sqrt{\text{Area}_{\text{per}}(\beta)} \leq \omega_r(\beta)$;
- (iv) If $\mathcal{U}$ halts on β , then $\omega_r(\beta) \leq C_r \text{Area}_{\text{per}}(\beta) + C_r \leq C_r(T(\beta) + |\beta| + 1)^2 + C_r$.

We first supply the upper construction in [Theorem 20.1](#): the reset dynamics produces a periodic tableau whose area is quadratically bounded in the running time.

Lemma 20.2. *There are constants C_0, C_1 , depending only on the fixed universal machine, such that, whenever $T(\beta) < \infty$, there are integers W, p in the definition of $\text{Area}_{\text{per}}(\beta)$ for which $W \leq T(\beta) + |\beta| + C_0$ and $p \leq T(\beta) + C_1 W + C_0$. Consequently, for a fixed constant C , whenever $T(\beta) < \infty$, we have*

$$\text{Area}_{\text{per}}(\beta) \leq C(T(\beta) + |\beta| + 1)^2 \tag{20.1}$$

Proof. In $T(\beta)$ steps the head, which starts in the first work cell, visits no cell beyond $T(\beta) + 1$. If s is the largest visited cell, choose $W := \max\{s + 1, |\beta| + 2\}$. Then $W \leq T(\beta) + |\beta| + C_0$ for an absolute constant C_0 , and the canonical orbit does not reach the artificial right wall. The boot sequence in [\(3.2\)](#) has at most $2W + C_0$ transitions. The simulation contributes $T(\beta)$ transitions. In [\(3.3\)](#), the left, right, and blanking sweeps each have length at most $W + C_0$. Thus the return to $c_{\beta,W}$ in [\(3.3\)](#) has period $p \leq T(\beta) + C_1 W + C_0$ for a machine-dependent constant C_1 . Multiplying this bound by $W + 2$ and using the bound on W proves [\(20.1\)](#). ■

We now combine the reset-period estimate with the certificate-recovery argument to prove the quantitative equivalence stated at the beginning of the section.

Proof of Theorem 20.1. Part (i) follows from [Lemma 17.1](#). We prove the quantitative bounds in two directions. First, a periodic reset tableau produces a better finite template of comparable order. Conversely, any better template contains a closed relation component whose width and period are controlled by the order of the template.

For the forward construction, take a periodic reset tableau of area $A = p(W + 2) = \text{Area}_{\text{per}}(\beta)$. Every physical type in the five-rail realization constructed from this tableau has A vertices, and the number of physical types is fixed. Passing to tagged records adds five-set records but no new data vertices; the rooted construction adds only a fixed number of root vertices. Thus the halting tableau produces an r -graph $G \in \mathcal{C}_\beta$ of order at most $C_r A + C_r$ with $\lambda_r(G) > \tau_*$. The pair-covering lower bound makes every blowup of G $\mathcal{F}_\beta$ -free, so G is $\mathcal{F}_\beta$ -hom-free. This proves the first inequality in part (iv); the second is [\(20.1\)](#).

For the converse, let G be a witness in the definition of $\omega_r(\beta)$ of order n . By [Lemma 17.4](#), there are integers W, p such that $\Phi_W^p(c_{\beta,W}) = c_{\beta,W}$, $W + 2 \leq n$, and $p \leq n$. Therefore, $\text{Area}_{\text{per}}(\beta) \leq p(W + 2) \leq n^2$, which proves part (iii) with constant one. The final assertion of the same lemma gives $T(\beta) \leq n$, proving part (ii), again with constant one. ■

Taking maxima in parts (ii) and (iv) of [Theorem 20.1](#) gives the two inequalities in [Theorem 1.6](#).

For the first noncomputability consequence recorded after [Theorem 1.6](#), suppose that a total computable function $B(m)$ bounded $\text{BB}_r^{\text{Tur}}(m)$. Given β , enumerate all r -graphs of order at most $B(|\beta|)$, decide $\mathcal{F}_\beta$ -hom-freeness, and decide the strict inequality $\lambda_r(G) > \tau_*$ by quantifier elimination. By [Theorem 20.1](#), such an r -graph exists exactly when $\text{U}(\beta)$ halts. This would decide the halting problem, a contradiction.

For the second consequence, suppose that a total computable function g satisfied $\omega_r(\beta) \leq g(|\langle \mathcal{F}_\beta \rangle|)$ for every input β on which U halts. Given β , compute the finite family $\mathcal{F}_\beta$ and the bound $g(|\langle \mathcal{F}_\beta \rangle|)$, then carry out the same finite exhaustive search through templates up to that order. Again, a better $\mathcal{F}_\beta$ -hom-free template exists exactly when $\text{U}(\beta)$ halts, so this procedure would decide the halting problem.

21 Quantitative density gaps and effective output size

Fix $r \geq r_{\text{num}}$ throughout this section, with $\mathcal{F}_\beta$ and τ_* carrying their usual r -dependence.

This section records three quantitative features of the explicit tableau construction. An individual halting certificate gives an inverse-polynomial lower bound for the corresponding density gap, but no positive lower bound is computable uniformly from the input. By contrast, for fixed uniformity, the forbidden r -graphs output by the compiler have polynomially bounded order.

We first express the positive density gap in terms of the area of a periodic tableau.

Use the notation of [Sections 3](#) and [4](#). Recall the compiler-interface constants $r_{\text{rec}} = r - 5$, N_{rt} , and $s_{\text{phys}} = 16$ fixed at the beginning of [Part III](#); the calibrated total root weight is $s_0 = N_{\text{rt}}/(N_{\text{rt}} + 1)$.

Theorem 21.1. *The following statements hold.*

- (i) *If $W \geq |\beta| + 2$, $p \geq 1$, and $\Phi_W^p(c_{\beta,W}) = c_{\beta,W}$, then*

$$\pi(\mathcal{F}_\beta) - \tau_* \geq \frac{r!N_{\text{rt}}}{r_{\text{rec}}(N_{\text{rt}} + 1)^r(5s_{\text{phys}})^5} (p(W + 2))^{-5}.$$

- (ii) *Consequently, there is a constant $c_r > 0$, depending only on the fixed construction, such that whenever U halts on β , we have $\pi(\mathcal{F}_\beta) - \tau_* \geq c_r(T(\beta) + |\beta| + 1)^{-10}$.*

- (iii) *There is no total computable function $\eta: \{0, 1\}^* \rightarrow \mathbb{Q}_{>0}$ such that, whenever U halts on β , we have $\pi(\mathcal{F}_\beta) - \tau_* \geq \eta(\beta)$.*

Recall from [\(6.6\)](#) that, in the notation of this part, the one-variable function has the form

$$\Upsilon_z(s) = a_{\text{bg}}s^r + \frac{C_{\text{typ}}}{N_{\text{rt}}^{r-1}}(1-s)s^{r-1} + \frac{z}{r_{\text{rec}}N_{\text{rt}}^{r_{\text{rec}}-1}}(1-s)^5s^{r_{\text{rec}}},$$

with $\Upsilon_z^* = \max_{0 \leq s \leq 1} \Upsilon_z(s)$. By [Lemma 6.8](#), at the baseline z_0 , the point s_0 is the unique maximizer and $\tau_* = r!\Upsilon_{z_0}^*$. For $z > z_0$, evaluating at the fixed point s_0 gives the explicit slope estimate

$$\Upsilon_z^* - \Upsilon_{z_0}^* \geq \Upsilon_z(s_0) - \Upsilon_{z_0}(s_0) = \frac{N_{\text{rt}}}{r_{\text{rec}}(N_{\text{rt}} + 1)^r} (z - z_0).$$

For the proof, put $A := p(W + 2)$, the area of the periodic reset tableau. This tableau produces five isomorphic rails. Give each rail total weight $1/5$ and distribute it uniformly over its $s_{\text{phys}}A$ vertices; denote this weighting by $\mathbf{x}$. Every selected anchor has weight $1/(5s_{\text{phys}}A)$, so $\lambda_{\text{rel}}(\mathbf{M}; \mathbf{x}) = 1/(5s_{\text{phys}}A)^5$. All matching defects vanish. Thus the semantic value exceeds z_0 by at least $(5s_{\text{phys}}A)^{-5}$.

Proof of [Theorem 21.1](#). The periodic construction has no matching defect, so the exact weighted identity gives $\Phi_\beta^* - z_0 \geq \lambda_{\text{rel}}(\mathbf{M}; \mathbf{x}) = 1/(5s_{\text{phys}}A)^5$. The pair-covering transfer and the numerical construction give the normalized chain

$$\pi(\mathcal{F}_\beta) - \tau_* = r!(\Upsilon_{\Phi_\beta^*}^* - \Upsilon_{z_0}^*) \geq r! \frac{N_{\text{rt}}}{r_{\text{rec}}(N_{\text{rt}} + 1)^r} (\Phi_\beta^* - z_0) \geq \frac{r!N_{\text{rt}}}{r_{\text{rec}}(N_{\text{rt}} + 1)^r(5s_{\text{phys}})^5} A^{-5},$$

which is the first bound in the theorem. If $\text{U}(\beta)$ halts, choose the period from [Lemma 20.2](#); then $A \leq C(T(\beta) + |\beta| + 1)^2$, and the second bound follows after changing the construction-dependent constant.

It remains to prove the third assertion. Suppose such a function η existed. By [Corollary 19.1](#), compute a rational approximation to $\pi(\mathcal{F}_\beta)$ with error less than $\eta(\beta)/3$ and compare it with $\tau_* + \eta(\beta)/2$. In the nonhalting case the approximation lies below this threshold, while in the halting case it lies above it. This would decide halting, a contradiction. $\blacksquare$

The exponent 10 comes from combining the degree-five marked reward, which gives a gap of order A^{-5} , with the quadratic bound $A = O((T(\beta) + |\beta| + 1)^2)$. We do not claim that either estimate is optimal.

Thus no positive lower bound can be computed uniformly from β alone. Such a bound becomes available once a finite halting certificate, such as the periodic tableau specified by W and p , is given.

Finally, although the smallest better template has Busy-Beaver growth, the r -graphs in the resulting forbidden family have polynomially bounded order for fixed r . We track the order through the root-elimination step and then through the pair-covering extension.

Theorem 21.2. *For each fixed output uniformity r , there are effectively computable constants $A_r, B_r > 0$ and an effectively computable integer $d_r > 0$ for which the following statements hold for every input word β .*

- (i) *Every r -graph $F \in \mathcal{F}_\beta$ satisfies $v(F) \leq A_r(|\beta| + 1)^{d_r}$.*
- (ii) *The number of members of $\mathcal{F}_\beta$ is at most $2^{B_r(|\beta|+1)^{r d_r}}$.*

We now give explicit choices for the exponent and constants by tracking the root-elimination and pair-covering steps. In the estimates below, r is fixed, so N_{rt} and every constant carrying a subscript r are independent of β . Recall that b_β is the explicit raw-certificate bound in [\(5.1\)](#). The tests (Q1)–(Q4) and all elementary profile and matching-conflict obstructions have fixed size. The input-dependent tests in (Q5) use anchored paths of length at most $|\beta| + 3$. Since every logical relation is expanded along a relation path of fixed length, the quantities v_β and a_β in the proof of [Lemma 5.4](#) are both at most a fixed multiple of $|\beta| + 1$. Choose an effectively computable constant $C_r^{\text{raw}} \geq 2$ so that $b_\beta \leq C_r^{\text{raw}}(|\beta| + 1)$.

Recall that the persistent-witness search uses $w_\beta = r \max\{2, b_\beta\}$ and outputs unrooted obstructions of order at most $B_\beta = w_\beta \sum_{j=0}^{N_{\text{rt}}} (N_{\text{rt}} w_\beta)^j$. Set $C_r^{\text{bas}} := (N_{\text{rt}} + 1) N_{\text{rt}}^{N_{\text{rt}}} (r C_r^{\text{raw}})^{N_{\text{rt}} + 1}$. Since $w_\beta \leq r C_r^{\text{raw}}(|\beta| + 1)$ and $N_{\text{rt}} w_\beta \geq 1$, this formula for B_β gives $B_\beta \leq C_r^{\text{bas}}(|\beta| + 1)^{N_{\text{rt}} + 1}$.

Every pair-covering extension of an obstruction J has order at most $v(J) + (r - 2) \binom{v(J)}{2}$. Accordingly, set $M_\beta := B_\beta + (r - 2) \binom{B_\beta}{2}$. Since $B_\beta \leq C_r^{\text{bas}}(|\beta| + 1)^{N_{\text{rt}} + 1}$ and $M_\beta \leq r B_\beta^2$, we have $M_\beta \leq r (C_r^{\text{bas}})^2 (|\beta| + 1)^{2N_{\text{rt}} + 2}$. Thus set $d_r := 2N_{\text{rt}} + 2$ and $A_r := r (C_r^{\text{bas}})^2$. For the family-size estimate, also set $B_r := 2(C_r^{\text{bas}})^r + 2A_r$. With these choices, the more precise bounds are $v(F) \leq M_\beta \leq A_r(|\beta| + 1)^{d_r}$ for every $F \in \mathcal{F}_\beta$ and $|\mathcal{F}_\beta| \leq 2^{B_r(|\beta|+1)^{r d_r}}$.

The four stages of the count are summarized in [Table 7](#). The cardinality of the raw local-obstruction list affects the running time of the construction but not the final count, because the finite obstruction family is obtained by enumerating all candidate r -graphs through order B_β .

Stage	Maximum order or support	Number relevant to the output enumeration
Raw local obstructions	$b_\beta \leq C_r^{\text{raw}}(\beta + 1)$	A finite effective list; its cardinality is not used in the final family-size bound.
Persistent-witness obstruction family	$B_\beta \leq C_r^{\text{bas}}(\beta + 1)^{N_{\text{rt}} + 1}$	Search depth N_{rt} and branching at most $N_{\text{rt}} w_\beta$; at most $(B_\beta + 1) 2^{B_\beta}$ candidate isomorphism types through this order.
Pair-covering extensions of one J	$M_\beta \leq A_r(\beta + 1)^{d_r}$	At most $(M_\beta + 1) 2^{M_\beta}$ candidate isomorphism types through this order.
Final family $\mathcal{F}_\beta$	Every member has order at most M_β .	At most the product of the preceding two candidate counts.

Table 7: Stages in the output-size estimate.

Proof of Theorem 21.2. Applying [Proposition 2.8](#) with $w = w_\beta$ and $|\mathcal{M}| = N_{\text{rt}}$ gives $v(J) \leq B_\beta$ for every $J \in \mathcal{A}_\beta$. By the definition of $\text{Ext}(J)$ immediately before [Lemma 2.9](#), every $F \in \text{Ext}(J)$ satisfies $v(F) \leq v(J) + (r - 2) \binom{v(J)}{2} \leq M_\beta$. Since $B_\beta \leq C_r^{\text{bas}}(|\beta| + 1)^{N_{\text{rt}} + 1}$ and $M_\beta \leq r B_\beta^2$, the definitions of

d_r and A_r give $v(F) \leq A_r(|\beta| + 1)^{d_r}$, proving the first part. For the count, the enumeration used to construct the finite obstruction family $\mathcal{A}_\beta$ gives $|\mathcal{A}_\beta| \leq (B_\beta + 1)2^{B_\beta^r}$. For each $J \in \mathcal{A}_\beta$, every member of $\text{Ext}(J)$ has order at most M_β , and the number of its isomorphism classes is at most the number of labeled r -graphs through that order. Hence $|\text{Ext}(J)| \leq (M_\beta + 1)2^{M_\beta^r}$. Since $B_\beta, M_\beta \geq 1$ and $r \geq 2$, the two polynomial prefactors are bounded by $2^{B_\beta^r}$ and $2^{M_\beta^r}$, respectively. Therefore we have

$$|\mathcal{F}_\beta| \leq 2^{2B_\beta^r + 2M_\beta^r} \leq 2^{B_r(|\beta| + 1)^{r d_r}},$$

which is the asserted family-size bound. ■

22 The density spectrum

Every Turán density produced by a finite forbidden family is computable. By contrast, allowing forbidden families that may be infinite and are specified by a membership algorithm or an enumerator changes the spectrum. This section gives a complete effective characterization of the resulting Turán densities.

With the decoder fixed in Section 2, the legal universe for an effectively presented forbidden family consists of the canonical r -graph codes accepted by dec_r . Such a family is *recursive* if a total algorithm rejects every illegal code and decides membership on the legal codes, and it is *computably enumerable*, or *c.e.*, if its legal canonical codes can be enumerated. Enumeration eventually certifies membership, but the continued absence of a code does not certify nonmembership. A program index e denotes the c.e. family obtained by applying dec_r to every number printed by the program and discarding the outputs decoded as $\perp$. Thus syntactically invalid outputs and outputs encoding edgeless r -graphs are ignored.

Both labels describe the presentation; the family itself is not required to be infinite, so both classes include all finite families. Following Weihrauch [73, Chapter 4], a real number α is *right-c.e.* if there is a computable nonincreasing rational sequence converging to it. Equivalently, its strict upper cut $\{q \in \mathbb{Q}: \alpha < q\}$ is computably enumerable. Such an approximation gives a certified upper bound at every stage, but in general no computable indication of the distance from that bound to the limit. The term *left-c.e.* is defined analogously using a computable nondecreasing rational approximation. We write $\mathbb{R}_{\text{right-c.e.}}$ for the class of right-c.e. reals.

Given any program index d , let $\text{Enum}(d)$ interleave the computations of d on all legal canonical r -graph codes, so that each computation is eventually advanced, and print precisely those codes on which d halts with answer yes. If d is a total membership decider, then $\text{Enum}(d)$ enumerates the same family. The standard *s-m-n* theorem, which allows fixed parameters to be inserted effectively into program codes, makes $d \mapsto \text{Enum}(d)$ a total computable transformation of program indices; the same holds for oracle programs without the transformation itself querying the oracle [62].

Let $\Pi_\infty^{(r)} := \{\pi(\mathcal{F}): \mathcal{F} \text{ is an arbitrary family of } r\text{-graphs}\}$. Let $\Pi_{\text{rec}}^{(r)}$ and $\Pi_{\text{ce}}^{(r)}$ denote the corresponding sets when $\mathcal{F}$ is recursive or c.e., respectively. These spectrum symbols are unrelated to the logical classes Π_n^0 .

Finite initial segments of a c.e. enumeration give decreasing upper approximations to its Turán density. The next theorem shows that this necessary condition is also sufficient, even when the realizing family is required to be a recursive antichain.

Theorem 22.1. *For every $r \geq 2$, the following statements hold.*

- (i) *The three spectra satisfy $\Pi_{\text{rec}}^{(r)} = \Pi_{\text{ce}}^{(r)} = \Pi_\infty^{(r)} \cap \mathbb{R}_{\text{right-c.e.}}$.*
- (ii) *If $\alpha \in \Pi_\infty^{(r)}$ is given by a computable decreasing rational sequence converging to α , then an index for a recursive antichain of r -graphs with Turán density α can be computed uniformly from an index for that sequence.*

Consequently, for every $r \geq 3$, there exists a recursive antichain of r -graphs with noncomputable Turán density.

Proof. The proof has two directions. We first extract a right-c.e. upper approximation from a c.e. presentation. For the converse, let $\alpha \in [0, 1]$ and let $q_1 \geq q_2 \geq \dots \downarrow \alpha$ be a nonincreasing rational sequence, and define

$$\mathcal{R}_q := \{G: v(G) \geq r \text{ and } \lambda_r(G) > q_{v(G)}\}.$$

Let $\mathcal{M}_{\mathbf{q}}$ consist of those $G \in \mathcal{R}_{\mathbf{q}}$ for which no proper subgraph of G , whether obtained by deleting edges, vertices, or both, belongs to $\mathcal{R}_{\mathbf{q}}$.

Claim 22.2. *If $\mathcal{F}$ is a c.e. family of r -graphs, then $\pi(\mathcal{F})$ is right-c.e., uniformly from an index enumerating $\mathcal{F}$.*

Proof. At stage s , run the enumerator for s steps and let $\mathcal{F}_s$ be the finite set of legal r -graphs printed by that time. Then $\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \dots$ and $\bigcup_s \mathcal{F}_s = \mathcal{F}$. By [Theorem 2.3](#), we have $\pi(\mathcal{F}_s) \downarrow \pi(\mathcal{F})$. By [Corollary 19.1](#), compute the upper endpoint u_s of an approximation interval of length less than 2^{-s} for $\pi(\mathcal{F}_s)$, and define $q_s := \min_{1 \leq t \leq s} u_t$. Then (q_s) is computable and nonincreasing. Since $\pi(\mathcal{F}_s) \downarrow \pi(\mathcal{F})$ and the approximation errors tend to zero, we have $q_s \downarrow \pi(\mathcal{F})$. ■

Claim 22.3. *If $\alpha \in \Pi_{\infty}^{(r)}$ and (q_n) is a computable nonincreasing rational sequence converging to α , then $\mathcal{R}_{\mathbf{q}}$ and $\mathcal{M}_{\mathbf{q}}$ are recursive uniformly from an index for (q_n) , the family $\mathcal{M}_{\mathbf{q}}$ is an antichain, and $\pi(\mathcal{M}_{\mathbf{q}}) = \pi(\mathcal{R}_{\mathbf{q}}) = \alpha$.*

Proof. We first verify the effective assertion. On a raw input code a , compute $\text{dec}_r(a)$ and reject if the result is $\perp$. If the result is an n -vertex r -graph G , reject when $n < r$; otherwise the condition $\lambda_r(G) > q_n$ is equivalent to the first-order sentence over the real closed field

$$\exists (x_v)_{v \in V(G)} \left[x_v \geq 0 \text{ for every } v \in V(G), \quad \sum_v x_v = 1, \quad \text{and} \quad r! \sum_{e \in G} \prod_{v \in e} x_v > q_n \right].$$

Basu, Pollack, and Roy [6] give a quantifier-elimination procedure over real closed fields that decides this sentence. Hence $\mathcal{R}_{\mathbf{q}}$ is recursive uniformly from the given index for (q_n) .

For the density upper bound, let $n \geq r$ and let H be an n -vertex $\mathcal{R}_{\mathbf{q}}$ -free r -graph. Since $H \notin \mathcal{R}_{\mathbf{q}}$, we have $\lambda_r(H) \leq q_n$. The uniform weighting gives $\lambda_r(H) \geq r!|H|/n^r = \theta_{r,n}|H|/\binom{n}{r}$. Therefore, $|H|/\binom{n}{r} \leq q_n/\theta_{r,n}$. Taking the maximum over H and letting $n \rightarrow \infty$ yields $\pi(\mathcal{R}_{\mathbf{q}}) \leq \alpha$.

For the reverse inequality, choose a family $\mathcal{A}$ with $\pi(\mathcal{A}) = \alpha$ and replace it by its homomorphic closure, continuing to denote the resulting family by $\mathcal{A}$; the density is unchanged by [Theorem 2.3](#). For each $j \geq r$, let G_j be a j -vertex extremal r -graph for this homomorphism-closed family. Every blowup of G_j is $\mathcal{A}$ -free. Hence $\lambda_r(G_j) \leq \alpha$; otherwise a blowup with vertex-class proportions close to a maximizing weighting would have density greater than α . On the other hand, the uniform weighting gives

$$\lambda_r(G_j) \geq \frac{r!|G_j|}{j^r} = \theta_{r,j} \frac{|G_j|}{\binom{j}{r}},$$

where $\theta_{r,j} = r!\binom{j}{r}/j^r$, as in (2.1). The final factor tends to α and $\theta_{r,j} \rightarrow 1$, so $\lambda_r(G_j) \rightarrow \alpha$.

If $F \in \mathcal{R}_{\mathbf{q}}$, then $\lambda_r(F) > q_{v(F)} \geq \alpha \geq \lambda_r(G_j)$. By [Lemma 2.2](#), there is no homomorphism $F \rightarrow G_j$. Hence every blowup of G_j is $\mathcal{R}_{\mathbf{q}}$ -free, and $\pi(\mathcal{R}_{\mathbf{q}}) \geq \lambda_r(G_j)$. Letting $j \rightarrow \infty$ gives $\pi(\mathcal{R}_{\mathbf{q}}) \geq \alpha$, and hence $\pi(\mathcal{R}_{\mathbf{q}}) = \alpha$.

It remains to pass to the minimal forbidden r -graphs. If $G \in \mathcal{R}_{\mathbf{q}}$ is a subgraph of H , extend a weighting of G by zero on the additional vertices of H . Then $\lambda_r(H) \geq \lambda_r(G) > q_{v(G)} \geq q_{v(H)}$, so $H \in \mathcal{R}_{\mathbf{q}}$. Thus $\mathcal{R}_{\mathbf{q}}$ is upward closed under taking supergraphs.

Membership in $\mathcal{M}_{\mathbf{q}}$ can therefore be decided by first testing whether $G \in \mathcal{R}_{\mathbf{q}}$ and then testing every proper subgraph of G , including non-spanning subgraphs. There are only finitely many such r -graphs, so this gives a decider for $\mathcal{M}_{\mathbf{q}}$ uniformly from the decider for $\mathcal{R}_{\mathbf{q}}$. Distinct subgraph-minimal members cannot contain one another, so $\mathcal{M}_{\mathbf{q}}$ is an antichain. Finally, every member of $\mathcal{R}_{\mathbf{q}}$ contains a minimal member by finite descent. Hence $\text{Forb}(\mathcal{M}_{\mathbf{q}}) = \text{Forb}(\mathcal{R}_{\mathbf{q}})$ and $\pi(\mathcal{M}_{\mathbf{q}}) = \pi(\mathcal{R}_{\mathbf{q}}) = \alpha$. ■

By [Claim 22.2](#), we have $\Pi_{\text{rec}}^{(r)} \subseteq \Pi_{\text{ce}}^{(r)} \subseteq \Pi_{\infty}^{(r)} \cap \mathbb{R}_{\text{right-c.e.}}$. Conversely, [Claim 22.3](#) produces a recursive antichain of density α uniformly from every computable decreasing rational sequence converging to a right-c.e. real $\alpha \in \Pi_{\infty}^{(r)}$. This proves both assertions.

For the consequence, fix $r \geq 3$. Liu and Pikhurko [46, Theorem 1.1] give $\delta_r > 0$ such that $[1 - \delta_r, 1] \subseteq \Pi_{\infty}^{(r)}$. Fix once and for all a positive rational $\varepsilon_r < \delta_r$. This choice is nonuniform in r ; no algorithm computing ε_r from r is asserted.

Recall that $K_{\text{diag}} = \{e: \psi_e(e) \downarrow\}$ is the diagonal halting set, and put $\xi_{\text{diag}} := \sum_{e \in K_{\text{diag}}} 4^{-(e+1)}$. This real is left-c.e. and noncomputable. For every e , the tail satisfies $\sum_{j>e} 4^{-(j+1)} = 4^{-(e+1)}/3$. After the membership of $0, \dots, e-1$ has been determined and the corresponding prefix has been subtracted, the remaining value is at least $4^{-(e+1)}$ if $e \in K_{\text{diag}}$ and at most $4^{-(e+1)}/3$ if $e \notin K_{\text{diag}}$. Consequently, a computable value of ξ_{diag} would distinguish the two cases and recover membership in K_{diag} inductively. This is the classical Specker-sequence phenomenon [67], here written as an explicit base-4 coding of the halting set.

Since $0 \leq \xi_{\text{diag}} \leq 1/3$ and $0 < \varepsilon_r < \delta_r$, the real $1 - \varepsilon_r \xi_{\text{diag}}$ is noncomputable and right-c.e., and it belongs to $[1 - \delta_r, 1]$.

Choose a computable decreasing rational sequence converging to this value. Applying Claim 22.3 produces a recursive antichain with noncomputable Turán density and proves the consequence. More generally, the spectrum identity shows that passing from finite forbidden families to recursive or c.e. presentations enlarges the possible densities from a subset of the computable reals to exactly the right-c.e. points of $\Pi_{\infty}^{(r)}$. ■

23 Exact comparison in the arithmetical hierarchy

At the second level of the arithmetical hierarchy, a set $A \subseteq \mathbb{N}$ is Σ_2^0 if there exists a decidable relation $R \subseteq \mathbb{N}^3$ such that, for every $e \in \mathbb{N}$, $e \in A$ if and only if $\exists u \forall v R(e, u, v)$. It is Π_2^0 if there exists a decidable relation $S \subseteq \mathbb{N}^3$ such that, for every $e \in \mathbb{N}$, $e \in A$ if and only if $\forall u \exists v S(e, u, v)$. More generally, after all number quantifiers are moved to the front, a Σ_n^0 definition has n alternating quantifier blocks beginning with an existential block and a decidable matrix, whereas a Π_n^0 definition begins with a universal block.

Each reduction in this section computes from a source index one recursive-family decider or one c.e.-family enumerator and makes no query to the target decision problem. When a construction first produces a recursive-family decider d but the target problem is presented by c.e.-family indices, it outputs $\text{Enum}(d)$. The total decoders fixed in Section 2 ensure that these transformations remain total even on syntactically invalid inputs.

Fix $r \geq 3$. Fix once and for all rational numbers $q \in (0, 1)$ and $c > 0$ such that

$$[q - c, q + c] \subseteq \Pi_{\infty}^{(r)}; \quad (23.1)$$

this is possible by the terminal-interval theorem of Liu and Pikhurko [46, Theorem 1.1]. This choice is nonuniform in r ; no algorithm computing q and c from r is asserted. We study indices of c.e. families of r -graphs. All hardness constructions below actually produce recursive families via Theorem 22.1.

Since $q \in \Pi_{\infty}^{(r)}$ by (23.1), applying Theorem 22.1 to the constant sequence $q_n = q$ gives one fixed recursive family $\mathcal{H}_q$ of density q .

Theorem 23.1. *For every fixed $r \geq 3$, there are a rational threshold $q \in (0, 1)$ and a recursive family $\mathcal{H}_q$ of r -graphs with $\pi(\mathcal{H}_q) = q$ such that the following statements hold.*

- (i) *For c.e. forbidden families $\mathcal{F}$ of r -graphs, comparison with q has the classification*

$$\begin{array}{l|l} \pi(\mathcal{F}) < q & \Sigma_1^0\text{-complete} \\ \pi(\mathcal{F}) \geq q & \Pi_1^0\text{-complete} \\ \pi(\mathcal{F}) > q & \Sigma_2^0\text{-complete} \\ \pi(\mathcal{F}) \leq q & \Pi_2^0\text{-complete} \\ \pi(\mathcal{F}) = q & \Pi_2^0\text{-complete} \\ \pi(\mathcal{F}) \neq q & \Sigma_2^0\text{-complete.} \end{array}$$

- (ii) *For pairs $(\mathcal{F}, \mathcal{G})$ of c.e. forbidden families of r -graphs, the relations $>$, $<$, and $\neq$ between $\pi(\mathcal{F})$ and $\pi(\mathcal{G})$ are Σ_2^0 -complete, whereas $\leq$, $\geq$, and $=$ are Π_2^0 -complete. For $>$, $\neq$, $=$, and $\leq$, hardness remains true when $\mathcal{G}$ is fixed to be $\mathcal{H}_q$; for $<$ and $\geq$, it remains true when $\mathcal{F}$ is fixed to be $\mathcal{H}_q$.*

In both parts, all variable families in the hardness constructions may be chosen recursive.

Its upper bounds follow from uniform right-c.e. approximations. Hardness at the first level comes from a single halting computation, whereas hardness at the second level comes from totality. We then pass to comparison between two c.e. families and finally relativize the construction.

Proof. We first establish the upper bounds. A sequence $(x_e)_{e \in \mathbb{N}}$ of real numbers is *uniformly right-c.e.* if there is a computable rational array $(u_{e,s})$ such that, for every e , the sequence $(u_{e,s})_s$ is nonincreasing and converges to x_e . For such a sequence and every rational q , we have

$$\begin{aligned} x_e < q &\iff \exists s \, u_{e,s} < q, \\ x_e > q &\iff \exists k \, \forall s \, u_{e,s} \geq q + 2^{-k}, \\ x_e > x_f &\iff \exists k \, \exists N \, \forall s \geq N \, u_{e,s} \geq u_{f,s} + 2^{-k}. \end{aligned}$$

For the first equivalence, an upper approximation falls below q exactly when its limit does. For the second, if $x_e > q$, choose k with $2^{-k} < x_e - q$ and use $u_{e,s} \geq x_e$ for every s ; the converse follows by taking limits. If $x_e > x_f$, choose k with $2^{-k} < x_e - x_f$ and then choose N so that $u_{f,s} < x_e - 2^{-k}$ for every $s \geq N$. The reverse implication again follows by taking limits.

By [Claim 22.2](#), from an index e enumerating a family $\mathcal{F}_e$ one obtains a computable nonincreasing rational sequence $u_{e,1} \geq u_{e,2} \geq \dots \downarrow \pi(\mathcal{F}_e)$. The first equivalence places $\pi(\mathcal{F}) < q$ in Σ_1^0 , and the second places $\pi(\mathcal{F}) > q$ in Σ_2^0 . Taking complements gives the two non-strict relations, while conjunction and complementation give the claimed upper bounds for equality and disequality.

First-level hardness. For each Turing-machine index e , put

$$\alpha_e^< := \begin{cases} q - c, & \psi_e(e) \downarrow, \\ q, & \psi_e(e) \uparrow. \end{cases}$$

The sequence $(\alpha_e^<)_{e \in \mathbb{N}}$ is uniformly right-c.e.: a computable decreasing approximation begins at q and drops permanently to $q - c$ if the computation halts. By [Theorem 22.1](#), one can uniformly output an index of a recursive family of density $\alpha_e^<$. Thus the output family has density less than q exactly when $e \in K_{\text{diag}}$.

Second-level hardness. Let $\text{TOT} := \{e : \forall x \, \psi_e(x) \downarrow\}$ be the standard Π_2^0 -complete totality set [\[62\]](#).

For a machine index e , define

$$\delta_e := \sum_{\psi_e(x) \uparrow} 2^{-x-1}. \quad (23.2)$$

The sequence $(\delta_e)_{e \in \mathbb{N}}$ is uniformly right-c.e. A convenient decreasing approximation is

$$\delta_{e,s} := \sum_{\substack{x < s \\ \psi_e(x) \text{ has not halted within } s \text{ steps}}} 2^{-x-1} + 2^{-s}. \quad (23.3)$$

To verify this, observe that at stage $s + 1$ each old summand either remains or disappears, while the possible new summand for $x = s$ and the new final term have total weight $2^{-s-1} + 2^{-(s+1)} = 2^{-s}$. Hence $\delta_{e,s+1} \leq \delta_{e,s}$. For each fixed x , the summand 2^{-x-1} remains in the limit exactly when $\psi_e(x)$ never halts; the summable geometric weights therefore give $\delta_{e,s} \downarrow \delta_e$. Consequently, $\delta_e = 0$ exactly when $e \in \text{TOT}$.

Put $\alpha_e^> := q + c\delta_e$. Then $\alpha_e^> \in [q, q + c]$, and the sequence $(\alpha_e^>)_{e \in \mathbb{N}}$ is uniformly right-c.e. Moreover, $e \in \text{TOT}$ exactly when $\alpha_e^> = q$, whereas $e \notin \text{TOT}$ exactly when $\alpha_e^> > q$. By [Theorem 22.1](#), the values $\alpha_e^>$ are realized uniformly by recursive families. Thus equality with q codes totality, whereas strict excess over q codes non-totally.

The first construction gives Σ_1^0 -hardness for $< q$ and Π_1^0 -hardness for $\geq q$. Since $\alpha_e^> \geq q$, the second construction gives Σ_2^0 -hardness for $> q$ and $\neq q$, and Π_2^0 -hardness for $\leq q$ and $= q$. Together with the upper bounds, this proves part (i).

Two-family comparison. The sequence $(\pi(\mathcal{F}_e))_e$ is uniformly right-c.e. when $(\mathcal{F}_e)_e$ is presented by a uniform sequence of c.e. family indices. The third equivalence above makes each strict comparison Σ_2^0 . The non-strict relations are their complements, equality is the conjunction of the two non-strict relations, and disequality is the disjunction of the two strict relations. For $>$, $\neq$, $=$, and $\leq$, hardness follows by comparing the variable family from part (i) with the fixed family $\mathcal{H}_q$. For $<$ and $\geq$, reverse the order of the comparison. The variable families supplied by the two hardness constructions are recursive, and so is $\mathcal{H}_q$. ■

Together with [Theorem 1.5](#), this result shows that allowing c.e. presentations raises exact two-family comparison by one level of the arithmetical hierarchy. The hardness in the second row already holds when the variable families are recursive. The comparison is summarized in the following table.

forbidden-family presentation	$>, <, \neq$	$\leq, \geq, =$
finite	Σ_1^0 -complete	Π_1^0 -complete
c.e.	Σ_2^0 -complete	Π_2^0 -complete

Finally, we record the oracle-relative form. It shows how the same classification shifts uniformly through the arithmetical hierarchy.

The spectrum theorem relativizes because the computation occurs in the decreasing approximation and the family presentation. An *oracle* $A \subseteq \mathbb{N}$ is an auxiliary set whose membership a machine may query in one step; relativizing a computability notion to A means allowing the relevant algorithms to make these queries. Using a fixed enumeration of oracle programs, write ψ_e^A for the unary partial function computed by the e th program with oracle A . An A -right-c.e. real is the limit of an A -computable decreasing rational sequence, and a family is A -recursive or A -c.e. according to its presentation. We write $\mathbb{R}_{\text{right-c.e.}}^A$ for the class of A -right-c.e. reals. For $i \geq 1$, a Σ_i^A definition has i alternating number-quantifier blocks beginning with an existential block and an A -computable matrix, whereas a Π_i^A definition begins with a universal block. Write δ_e^A and $\delta_{e,s}^A$ for the quantities in (23.2) and (23.3), respectively, with ψ_e replaced by ψ_e^A .

For $d \geq 0$, put $\mathbf{0}^{(0)} := \emptyset$ and $\mathbf{0}^{(d+1)} := \{e \in \mathbb{N} : \psi_e^{\mathbf{0}^{(d)}}(e) \downarrow\}$. Thus $\mathbf{0}^{(d)}$ is the d th iterated halting oracle, and $\mathbf{0}^{(d)}$ -c.e. means enumerable by a machine allowed to query this oracle. When we say below that a transformation does not query A , we mean that the code-producing transformation is ordinarily computable; the oracle program whose index it outputs may still query A .

Theorem 23.2. *The following statements hold.*

- (i) *For every oracle A and every fixed $r \geq 2$, we have*

$$\{\pi(\mathcal{F}) : \mathcal{F} \text{ is } A\text{-recursive}\} = \{\pi(\mathcal{F}) : \mathcal{F} \text{ is } A\text{-c.e.}\} = \Pi_\infty^{(r)} \cap \mathbb{R}_{\text{right-c.e.}}^A.$$

If $\alpha \in \Pi_\infty^{(r)}$ is given by an index for an A -computable decreasing rational sequence converging to α , then an index for an A -recursive forbidden family of density α is obtained uniformly by a computable transformation of oracle-program indices that does not query A .

- (ii) *For every $d \geq 0$, every fixed $r \geq 3$, and all rationals $q \in (0, 1)$ and $c > 0$ satisfying $[q - c, q + c] \subseteq \Pi_\infty^{(r)}$, the comparison problems for $\pi(\mathcal{F})$, where $\mathcal{F}$ ranges over the standard indices of $\mathbf{0}^{(d)}$ -c.e. families of r -graphs, have the following classification under ordinary many-one reductions:*

$$\begin{array}{l|l} \pi(\mathcal{F}) < q & \Sigma_{d+1}^0\text{-complete} \\ \pi(\mathcal{F}) \geq q & \Pi_{d+1}^0\text{-complete} \\ \pi(\mathcal{F}) > q, \pi(\mathcal{F}) \neq q & \Sigma_{d+2}^0\text{-complete} \\ \pi(\mathcal{F}) \leq q, \pi(\mathcal{F}) = q & \Pi_{d+2}^0\text{-complete.} \end{array} \quad (23.4)$$

In every hardness construction, the variable family may be chosen $\mathbf{0}^{(d)}$ -recursive.

Proof. For part (i), relativize the proof of [Theorem 22.1](#). Quantifier elimination remains computable without the oracle, while the sequence (q_n) and the family presentation use A . The standard parameterization theorem for oracle programs produces the required A -decider uniformly without querying A ; applying the transformation $d \mapsto \text{Enum}(d)$ defined in [Section 22](#) gives an A -c.e.-family index when needed.

For part (ii), put $A := \mathbf{0}^{(d)}$. The upper bounds follow by relativizing the right-c.e. normal-form argument in the proof of [Theorem 23.1](#). Post's theorem [56] identifies $\Sigma_i^{\mathbf{0}^{(d)}}$ and $\Pi_i^{\mathbf{0}^{(d)}}$ with Σ_{d+i}^0 and Π_{d+i}^0 , respectively, for $i \geq 1$, and hence gives the levels in (23.4); Soare [66] gives a modern account. We use the standard uniform normal forms in which the reductions to the relative index sets below are ordinary computable maps: the source parameter is compiled into an oracle-program index without querying A , and only the compiled program may query A .

For first-level hardness, the set $\{e : \psi_e^A(e) \downarrow\}$ is Σ_1^A -complete under these ordinary index reductions. The A -computable decreasing approximation that begins at q and drops to $q - c$ if $\psi_e^A(e)$ halts gives hardness for $\pi(\mathcal{F}) < q$ and, by complementation, for $\pi(\mathcal{F}) \geq q$.

For second-level hardness, put $\text{TOT}^A := \{e: \forall x \psi_e^A(x) \downarrow\}$, which is Π_2^A -complete under the same ordinary index convention [66]. The approximation $\delta_{e,s}^A$ decreases to δ_e^A , which is zero exactly when $e \in \text{TOT}^A$. Hence the corresponding real $q + c\delta_e^A$ equals q exactly on TOT^A and is larger otherwise. This gives hardness for the four second-level relations.

Part (i) realizes all these values by A -recursive forbidden families. The oracle s - m - n theorem makes the remaining approximation and family-index transformations ordinary computable maps without querying A . Composing them with the preceding normal-form reductions therefore gives the claimed ordinary many-one reductions. ■

24 Limits of two semialgebraic value schemes

The preceding sections concern effective approximation, exact comparison, and quantitative witnesses. We end with a different question about finite descriptions of exact Turán densities. Here “recursive” refers to repeatedly applying finite construction rules, not to the computability-theoretic notion of a recursive forbidden family from Section 22.

Pikhurko [54, Theorem 3] showed that the recursive construction associated with a single minimal pattern can be realized by a finite forbidden family. Liu and Pikhurko [45, Theorem 1.2] extended this result to arbitrary finite collections of minimal patterns that may be mixed recursively. These results motivate the question considered here. Can the density of every finite forbidden family be represented by either of two particular finite recursive value schemes?

A set $S \subseteq \mathbb{R}^m$ is *semialgebraic over $\mathbb{Q}$* if it can be described by a finite Boolean combination of polynomial equalities and inequalities with rational coefficients.

The first scheme takes finite maxima of values obtained by recursively repeating hypergraph patterns. The second allows finitely many states and semialgebraically parameterized production rules. The two schemes are defined precisely below. Their common limitation is the main result of this section.

Proposition 24.1. *For every sufficiently large r , there is a finite family of r -graphs whose ordinary Turán problem has neither a finite mixed-pattern value description nor a proper finite-state semialgebraic recursive value description over $\mathbb{Q}$.*

This proposition is an obstruction to these two schemes, not to every possible finite description formalism. Its proof is independent of the tableau and structural constructions.

The proof has two independent algebraicity steps, one for each value scheme defined below. The final deduction then applies the transcendental-density theorem of Li, Liu, and Liu.

We begin with recursive patterns. Such a pattern divides the vertex set into finitely many parts, inserts the prescribed top-level edges, and then repeats the construction inside selected parts. The resulting density is therefore governed by a one-variable fixed-point inequality.

For this section, a *Bellman r -pattern descriptor* is a triple $P = (m, E, R)$, where $m \geq 1$, E is a finite family of r -multisets on $[m]$, and $R \subseteq [m]$ is the set of recursive parts. The word “descriptor” distinguishes this broader syntax from the standard minimality-based notion of a pattern. Its profile polynomial is

$$\lambda_E(\mathbf{x}) := r! \sum_{D \in E} \prod_{i=1}^m \frac{x_i^{D(i)}}{D(i)!}.$$

Here the factor $r!$ places λ_E in the ordinary edge-density normalization. Put

$$S_P := \left\{ z \in [0, 1] : \lambda_E(\mathbf{x}) + z \sum_{j \in R} x_j^r \leq z \text{ for every } \mathbf{x} \in \Delta_m \right\}.$$

For the present purpose, the descriptor is *Bellman-admissible* when S_P is nonempty, and we define its *Bellman value* by $\lambda_P := \min S_P$. The minimum exists because S_P is a closed subset of $[0, 1]$, by continuity of its defining polynomials. Thus λ_P is the least supersolution of the defining fixed-point inequality. Bellman admissibility automatically rules out $i \in R$ together with the pure multiset $\{i, \dots, i\} \in E$, since the inequality would fail at the i th vertex of the simplex.

A finite family $\mathcal{F}$ of r -graphs has a *finite mixed-pattern value description* if there are Bellman-admissible descriptors $P_1, \dots, P_t$ such that $\pi(\mathcal{F}) = \max_{1 \leq i \leq t} \lambda_{P_i}$.

Lemma 24.2. *The value of every Bellman-admissible finite descriptor is a real algebraic number. Consequently, every Turán density admitting a finite mixed-pattern value description is algebraic.*

Proof. By Tarski’s quantifier-elimination theorem [69], in the algorithmic form developed by Basu, Pollack, and Roy [6], the set S_P is a semialgebraic subset of $\mathbb{R}$ over $\mathbb{Q}$. Bellman admissibility says that $\lambda_P = \min S_P$. Every finite endpoint of a semialgebraic subset of $\mathbb{R}$ defined over $\mathbb{Q}$ is algebraic. After quantifier elimination, an endpoint is either rational or a zero of one of the finitely many rational polynomials appearing in the sign decomposition. Hence λ_P is algebraic. This argument also covers the boundary values 0 and 1. The maximum of finitely many such values is therefore algebraic. Thus every Turán density admitting a finite mixed-pattern value description is algebraic. Liu and Pikhurko [45, Lemma 3.4] use precisely this finite-maximum operation in their mixing construction. ■

We next verify that this descriptor language contains the usual proper recursive patterns. The descriptor notion is deliberately broader than the minimality-based notion of a proper pattern in the sense of Pikhurko. For such a proper pattern, the usual recursive value is precisely the least supersolution of the defining fixed-point inequality for S_P , so the pattern is Bellman-admissible. Pikhurko’s recursive-pattern construction [54, Section 2 and Lemma 14] implicitly uses this fixed-point formulation.

For completeness, this identification follows directly from the recursive construction. Define the monotone continuous map

$$\mathcal{B}_P(z) := \max_{\mathbf{x} \in \Delta_m} \left(\lambda_E(\mathbf{x}) + z \sum_{j \in R} x_j^r \right) \quad \text{for } 0 \leq z \leq 1.$$

Here recursion depth zero means that no edges are inserted. Inductively, a construction of depth at most $d + 1$ applies the top-level pattern and, inside each recursive part, independently uses a construction of depth at most d . If V_d denotes the largest possible asymptotic density among constructions of depth at most d , then $V_0 = 0$. The construction rule immediately gives $V_{d+1} \leq \mathcal{B}_P(V_d)$.

Conversely, fix $\varepsilon > 0$, choose $\mathbf{x} \in \Delta_m$ maximizing $\mathcal{B}_P(V_d)$, approximate $\mathbf{x}$ by rational part proportions, and in every recursive part of positive weight use independently a depth-at-most- d construction whose asymptotic density is at least $V_d - \varepsilon$. Taking large integer realizations of the rational proportions and then letting $\varepsilon \downarrow 0$ gives the reverse inequality. Hence $V_{d+1} = \mathcal{B}_P(V_d)$.

It follows that $V_d = \mathcal{B}_P^d(0)$. Allowing one additional recursion level cannot decrease the optimum, and every density lies in $[0, 1]$, so $0 \leq V_d \leq V_{d+1} \leq 1$. Hence V_d increases to the usual recursive value $\lambda_P^{\text{rec}} = \sup_d V_d = \sup_d \mathcal{B}_P^d(0)$, and continuity gives $\mathcal{B}_P(\lambda_P^{\text{rec}}) = \lambda_P^{\text{rec}}$.

Finally, if $z \in S_P$, then $\mathcal{B}_P(z) \leq z$; monotonicity and induction give $\mathcal{B}_P^d(0) \leq z$ for every d . Hence $\lambda_P^{\text{rec}} \leq z$ for every $z \in S_P$, while the fixed-point identity shows that $\lambda_P^{\text{rec}} \in S_P$. Therefore $\lambda_P^{\text{rec}} = \min S_P$, as claimed.

Thus finite mixtures of proper patterns are included in the Bellman-descriptor class. The mixing construction of Liu and Pikhurko [45, Theorem 1.2] provides one such example.

The second scheme replaces a single recursive pattern by finitely many states. At each state, one chooses among finitely many semialgebraically parameterized productions, and the Bellman operator records the value of one recursive step.

A finite-state semialgebraic recursive grammar has states $[s]$ and the following three layers of data.

- (i) For every state i , the production set A_i is finite and nonempty.
- (ii) For every $i \in [s]$ and every $a \in A_i$, there are an integer $d_{i,a} \geq 0$ and a nonempty compact semialgebraic parameter set $K_{i,a} \subseteq \mathbb{R}^{d_{i,a}}$ defined over $\mathbb{Q}$.
- (iii) For every $i \in [s]$ and every $a \in A_i$, there are polynomials $p_{i,a}, q_{i,a,j} \in \mathbb{Q}[\mathbf{x}]$ in $d_{i,a}$ variables for $j \in [s]$.

The coefficient functions $q_{i,a,j}$ may take either sign on $K_{i,a}$. This is a deliberate relaxation: the algebraicity argument below uses only the semialgebraicity of the fixed-point set and its uniqueness, not monotonicity of the Bellman operator. Its Bellman operator is

$$(\mathcal{B}\mathbf{z})_i := \max_{a \in A_i} \max_{\mathbf{x} \in K_{i,a}} \left(p_{i,a}(\mathbf{x}) + \sum_{j=1}^s q_{i,a,j}(\mathbf{x}) z_j \right).$$

We require $\mathcal{B}([0, 1]^s) \subseteq [0, 1]^s$. The grammar is *proper* if $\mathbf{z} = \mathcal{B}\mathbf{z}$ has a unique solution in $[0, 1]^s$; mutual recursion among the states is fully allowed through the coefficients $q_{i,a,j}$.

A finite family of r -graphs has a *proper finite-state semialgebraic recursive value description* over $\mathbb{Q}$ if its density is a designated coordinate of this unique fixed point.

Both notions concern exact value descriptions. They impose no restriction on finite approximations or on descriptions in broader recursive formalisms.

Lemma 24.3. *Every coordinate of the unique fixed point of a proper finite-state semialgebraic recursive grammar is algebraic.*

Proof. The fixed-point condition is first-order over the real closed field. For every state i it requires

$$\bigwedge_{a \in A_i} \left(\forall \mathbf{x} \in K_{i,a} \quad p_{i,a}(\mathbf{x}) + \sum_{j=1}^s q_{i,a,j}(\mathbf{x}) z_j \leq z_i \right),$$

$$\text{and} \quad \bigvee_{a \in A_i} \left(\exists \mathbf{x} \in K_{i,a} \quad p_{i,a}(\mathbf{x}) + \sum_{j=1}^s q_{i,a,j}(\mathbf{x}) z_j = z_i \right).$$

The conjunction and disjunction are finite because A_i is finite. Thus the set of fixed points in $[0, 1]^s$ is semialgebraic over $\mathbb{Q}$. Properness makes it a singleton, and every coordinate of a semialgebraic singleton over $\mathbb{Q}$ is algebraic. ■

Proof of Proposition 24.1. Li, Liu, and Liu [38, Theorem 1.2] construct a finite family with transcendental Turán density for every sufficiently large uniformity. A finite mixed-pattern value description would force this density to be algebraic by Lemma 24.2, and a proper finite-state semialgebraic recursive value description would do the same by Lemma 24.3. Hence the family admits neither description. ■

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Declaration on the use of AI

The authors used generative AI tools to discuss proof organization, test local calculations, check proofs, and improve exposition.

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Appendix

A Numerical arithmetization and certificate ledgers

This appendix supplies the finite codes and proof ledgers used by the numerical bridge. We first define the arithmetic Turán interface, then record the fixed algebraic certificates, the typed certificate maps, the numerical compiler invariants, and the two formal certificate translations. Throughout, fix $r \geq r_{\text{num}}$, write $\mathcal{F}_\beta = \mathcal{F}_{r,\beta}$ and $\tau_* = \tau_r$, and retain $K_{*,r}$ and the compiler data selected at the beginning of Part III. Every map is total on natural-number codes, with an explicit default outside its intended domain.

A.1 Finite graph and family codes

Arithmetization assigns natural-number codes to finite formulas and derivations so that syntax and proof checking become arithmetic predicates. We use one fixed effective proof calculus and the standard arithmetization of syntax and proofs; see Boolos, Burgess, and Jeffrey [8, Chapters 15–17]. When the axioms are supplied by an enumerator, a proof code records, beside each nonlogical axiom line, a stage by which that axiom has appeared in the enumeration. Consequently, the witnessed relation saying that p is a proof of the formula with code q from the theory enumerated by index a is primitive recursive, uniformly in (a, p, q) . Its existential projection over p is the usual theoremhood relation, which is in general c.e., rather than primitive recursive.

We represent the natural numbers inside set theory by the finite von Neumann ordinals. Throughout this section, every variable and quantifier described as numerical is understood to range over ω . With this convention, ZFC formalizes primitive recursive functions, finite hypergraph codes, and Turing computations in the usual way. The provability notation in this section refers to the witnessed proof coding just described.

We now give the finite codes and certificate transformations that allow ZFC to verify the normal-form equivalence in [Theorem 17.6](#).

We use the following finite coding conventions.

- (i) A canonical code for an r -graph consists of its vertex number m and the characteristic bit string of its edge set in the lexicographic ordering of $\binom{[m]}{r}$.
- (ii) A finite-family code is a finite sequence of canonical r -graph codes.
- (iii) In a forbidden-family presentation, the filtered decoder dec_r sends syntactically invalid entries or entries encoding edgeless r -graphs to $\perp$ and discards them.
- (iv) A witness variable H is read by the unfiltered canonical r -graph decoder, which accepts every well-formed finite r -graph code, including an edgeless hypergraph.

For fixed r , legality, the relation $F \rightarrow H$, and hom-freeness are primitive recursive because all relevant vertex maps form a finite search space bounded by the input code.

For every $f \in \mathbb{N}$, write $\mathcal{F}_f := \text{DecFam}_r(f)$ for its total filtered decode, and write $\text{CanFam}_r(f)$ when f is a canonical finite-family code. Let $\text{ImgCode}_r(f)$ be the canonical code of $\mathcal{Q}(\mathcal{F}_f^\circ)$ when $\text{CanFam}_r(f)$ holds, and let it be the canonical empty-family code otherwise. On canonical inputs this code lists, up to isomorphism, the edgewise-injective homomorphic images of F° for $F \in \mathcal{F}_f$, with repeated image edges merged. Both the decoding relation and $f \mapsto \text{ImgCode}_r(f)$ are primitive recursive. For canonical f and every r -graph G with $v(G) \geq r$, the r -graph G is $\mathcal{F}_{\text{ImgCode}_r(f)}$ -free exactly when it is $\mathcal{F}_f$ -hom-free, which is also exactly when it is $\mathcal{Q}(\mathcal{F}_f)$ -free.

Write $\tau_* = a_*/b_*$ in lowest terms. Let $\text{Improve}_{\tau_*}^{\text{code}}(f, H)$ be the conjunction of $\text{CanFam}_r(f)$, legality of the unfiltered witness code H , hom-freeness of H for $\mathcal{F}_f$, and the integer inequality $b_* r! |H| > a_* v(H)^r$. Define $\text{BalSharp}_{\tau_*}^{\text{code}}(f)$ to mean $\forall H \neg \text{Improve}_{\tau_*}^{\text{code}}(f, H)$. Thus $\text{BalSharp}_{\tau_*}^{\text{code}}(f)$ says that no finite balanced construction certifies density larger than τ_* .

To express exact equality arithmetically, put $m_* := v(K_{*,r})$. For $s \geq 1$, let m_s be the least positive multiple of m_* satisfying $m_s > 2^{s-1}r(r-1)$, and put $\text{Ex}_s(f) := \text{ex}(m_s, \mathcal{F}_{\text{ImgCode}_r(f)})$. Set $m_0 = m_*$ and $\text{Ex}_0(f) = 0$ as the off-domain defaults. For fixed r , the functions $s \mapsto m_s$ and $(f, s) \mapsto \text{Ex}_s(f)$ are

primitive recursive; on inputs $s \geq 1$, the value $\text{Ex}_s(f)$ is computed by bounded enumeration of all r -graphs on $[m_s]$.

Define $\text{TuranEq}_{\tau_*}(f)$ by

$$\text{TuranEq}_{\tau_*}(f) \iff \text{CanFam}_r(f) \wedge \forall s \geq 1 \left[b_* r! \text{Ex}_s(f) \leq a_* m_s^r \wedge a_* \binom{m_s}{r} \leq b_* \text{Ex}_s(f) \right]. \quad (\text{A.1})$$

The two inequalities say that the lower and upper endpoints supplied by [Proposition 2.6](#) lie on the appropriate sides of τ_* . Finally, define $\text{Base}_r(f)$ to mean that $\text{CanFam}_r(f)$ holds and $K_{*,r}$ is $\mathcal{F}_f$ -hom-free.

The next lemma is the formal interface between these finite arithmetic predicates and the usual Turán-density statement.

Lemma A.1. *For every family code f , ZFC proves the following statements.*

- (i) *If $\text{CanFam}_r(f)$ holds, then $\text{TuranEq}_{\tau_*}(f)$ holds if and only if $\pi(\mathcal{F}_f) = \tau_*$.*
- (ii) *If $\text{Base}_r(f)$ holds, then $\text{BalSharp}_{\tau_*}^{\text{code}}(f)$ holds if and only if $\text{TuranEq}_{\tau_*}(f)$ holds.*

Proof. For part (i), [Proposition 2.6](#) places $\pi(\mathcal{F}_f)$ in the interval whose endpoints are encoded by the two inequalities in (A.1). The choice of m_s makes the interval length less than 2^{-s} , so all these inequalities hold exactly when the common limiting value is τ_* . This argument uses only bounded finite searches and the averaging and blowup proofs of [Proposition 2.6](#), and is therefore formalizable in ZFC.

For part (ii), suppose first that $\text{BalSharp}_{\tau_*}^{\text{code}}(f)$ holds. An m_s -vertex $\mathcal{Q}(\mathcal{F}_f^\circ)$ -free r -graph with $\text{Ex}_s(f)$ edges is $\mathcal{F}_f$ -hom-free, so its balanced value is at most τ_* . This gives the first integer inequality in (A.1). Since $m_* \mid m_s$, the balanced blowup of $K_{*,r}$ on exactly m_s vertices exists. Under $\text{Base}_r(f)$ it is $\mathcal{F}_f$ -hom-free and hence $\mathcal{Q}(\mathcal{F}_f^\circ)$ -free. The fixed edge-list identity, certified by bounded arithmetic in [Lemma A.2](#), gives its edge number $\tau_* m_s^r / r!$, so its normalized edge density is $\tau_* / \theta_{r,m_s} \geq \tau_*$. This gives the second integer inequality.

Conversely, suppose that $\text{Improve}_{\tau_*}^{\text{code}}(f, H)$ holds. Put $h := v(H)$ and $\Delta(H) := b_* r! |H| - a_* h^r > 0$. Choose s such that $m_s > 2r b_* h^{r+1} / \Delta(H)$. Since $0 < \Delta(H) \leq b_* h^r$, we also have $m_s > 2rh \geq h$. Replace every vertex of H by either $\lfloor m_s/h \rfloor$ or $\lceil m_s/h \rceil$ vertices, with total order m_s . The resulting r -graph remains $\mathcal{F}_f$ -hom-free and hence is $\mathcal{Q}(\mathcal{F}_f^\circ)$ -free. Its balanced density is at least $\frac{r! |H|}{h^r} \left(1 - \frac{h}{m_s}\right)^r > \tau_*$. Indeed, the strict inequality follows from $(1-x)^r \geq 1-rx$, the bound $r! |H| / h^r \leq 1$, and $r! |H| / h^r - \tau_* = \Delta(H) / (b_* h^r)$. Consequently $b_* r! \text{Ex}_s(f) > a_* m_s^r$, so (A.1) fails. This proves part (ii) using only finite r -graph codes and integer inequalities. $\blacksquare$

A.2 Fixed algebraic and semantic certificates

The direct numerical compiler uses several assertions about fixed rational polynomials. To keep their formal role separate from the combinatorial coding, fix a complete proof calculus for real closed fields whose formulas have integer coefficients. Let $\text{ClosedRCF}(q)$ mean that q codes a closed sentence in this language, let $\text{RCFPrf}(c, q)$ mean that c is a derivation of that sentence, and let $\text{Sat}_{\mathbb{R}}(q)$ be the usual recursively defined satisfaction predicate for such closed codes in the set-theoretic real field. Proof checking is primitive recursive. Induction on the length of a derivation gives the single uniform soundness formula

$$\text{ZFC} \vdash \forall c \forall q \left(\text{ClosedRCF}(q) \wedge \text{RCFPrf}(c, q) \longrightarrow \text{Sat}_{\mathbb{R}}(q) \right).$$

Quantifier elimination decides whether a closed sentence is true [\[69, 6\]](#). When it is true, completeness of the fixed calculus and enumeration of its derivations return the first proof code. In every displayed member of a certificate list, vector notation, finite sums, rational coefficients, and constrained quantifiers abbreviate the fully expanded scalar ordered-field formula after positive denominators have been cleared.

For the fixed r -dependent setup, let $\mathcal{S}_r^{\text{alg}}$ be the following four named closed sentences after all compiler parameters have been replaced by their selected rational values:

- (i) Cal_r , the universal closure of the exact calibration test in [\(6.14\)](#);
- (ii) Root_r , the universal closure of the three implications involving A_N in the parameter-selection test following that equation;
- (iii) Five_r , the five-variable assertion that $\rho e_4(\rho) - 5e_5(\rho) \leq 4\rho^5/625$ whenever $\rho_1, \dots, \rho_5 \geq 0$ and $\rho = \sum_i \rho_i$;

(iv) **BaseLag_r**, the finite-dimensional upper bound for the fixed baseline template in [Lemma 17.1](#).

The rational parameter identities not contained in these formulas are checked by bounded integer arithmetic. The boundary-moment estimate, Hölder’s inequality over arbitrary finite vertex sets, and the matching, component, and tableau assertions remain uniform set-theoretic arguments rather than members of this finite real-closed-field list.

Write $V(K_{*,r}) = [m_*]$. The last sentence in the list is

$$\forall \mathbf{x} \in \mathbb{R}^{m_*} \left[\left(x_i \geq 0 \text{ for every } i \in [m_*] \wedge \sum_{i \in [m_*]} x_i = 1 \right) \longrightarrow b_* r! \sum_{E \in K_{*,r}} \prod_{i \in E} x_i \leq a_* \right]. \quad (\text{A.2})$$

Let $q_{r,i}^{\text{alg}}$ be the Gödel code of the i th sentence in $\mathcal{S}_r^{\text{alg}}$, for $i \in [4]$.

Lemma A.2. *The fixed r -dependent setup can be augmented by a finite sequence $\mathbf{c}_r^{\text{alg}}$ containing one real-closed-field proof code for each sentence in $\mathcal{S}_r^{\text{alg}}$. ZFC proves the exact checker statement*

$$\bigwedge_{i \in [4]} (\text{ClosedRCF}(q_{r,i}^{\text{alg}}) \wedge \text{RCFPrf}((\mathbf{c}_r^{\text{alg}})_i, q_{r,i}^{\text{alg}})).$$

In particular, ZFC proves (A.2) and the integer identity

$$b_* r! |K_{*,r}| = a_* m_*^r.$$

Consequently, ZFC proves $\lambda_r(K_{*,r}) = \tau_*$ and that the uniform weighting of $K_{*,r}$ is optimal.

Proof. The parameter-selection procedure already checks **Cal_r** and **Root_r**, and the proofs of [Theorem 4.1](#) and [Lemma 17.1](#) establish **Five_r** and **BaseLag_r** externally. After the rational parameters have been selected, use quantifier elimination to decide each of these four closed sentences, then enumerate derivations in the complete calculus until the first derivation of that sentence is found. Retain the four resulting proof numerals. This changes neither a parameter nor a forbidden family. The codes are independent of the later variables d , e , and f .

For the final sentence, $K_{*,r}$ and τ_* are fixed finite rational data. The proof of [Lemma 17.1](#) shows externally that (A.2) is true: the preliminary template is evaluated at the fixed nonhalting word $\beta_\infty = 0$, and projection sends every weighting of $K_{*,r}$ to a weighting of that template. Completeness of the real-closed-field calculus therefore gives a finite derivation of the closed sentence (A.2); retain the first such derivation in the fixed proof-code ordering. ZFC checks the resulting numeral and applies the soundness implication above. The integer identity is verified by bounded arithmetic on the canonical edge list of $K_{*,r}$. It shows that the uniform weighting attains $a_*/b_* = \tau_*$, while (A.2) gives the matching upper bound for every probability weighting. $\blacksquare$

Lemma A.3. *ZFC proves, uniformly in the finite codes β , $\mathfrak{R}$ and the finite real vector $\mathbf{x} = (x_v)_{v \in V(\mathfrak{R})}$, that if $\mathfrak{R}$ is a legal five-rail realization for β , $\mathbf{x}$ is nonnegative with total weight ρ , and every coordinate of every $\mathbf{M}$ -tuple lies in a grounded relation component, then*

$$\Phi_\beta(\mathfrak{R}, \mathbf{x}) \leq z_0 \rho^5.$$

Proof. The relation components and their boundaries are obtained by bounded search in the finite relation tables. Expanding the square in the definition of the matching defect proves, term by term,

$$\lambda_{\text{rel}}(D; \mathbf{x}) + e_2(X) + e_2(Y) = \frac{1}{2}(e_1(X)^2 + e_1(Y)^2) - \frac{1}{2} \text{Def}(D).$$

Finite summation over the fixed regular rail signature gives $\mathcal{E}_i = d\rho_i^2/2 - \text{Def}_i$. These are identities in finite sums, so ZFC proves them by induction on the coded edge lists.

For each grounded component, bounded search chooses a shortest path from a maximum-weight vertex to a vertex v missing a prescribed relation edge e . Append the private zero-weight leaf $\partial_{v,e}$ from the definition of Def_C . The running-minimum sequence along this path is finite. The discrete calculation in

[Lemma 4.2](#), equivalently its piecewise-linear interpolation coded by finitely many intervals, is an instance of Cauchy–Schwarz and proves

$$\sum_{v \in C} x_v^5 \leq \frac{9}{4} \mu_C^3 \text{Def}_C.$$

Summing over the coded component list, applying the finite form of Hölder’s inequality, and then the five-term arithmetic–geometric mean inequality gives

$$\lambda_{\text{rel}}(\mathbf{M}; \mathbf{x}) \leq \frac{9}{10} \sum_{i \neq j} P_{ij} \text{Def}_i.$$

ZFC proves the finite forms of Cauchy–Schwarz, Hölder, and arithmetic–geometric mean by induction on the number of summands; no compactness or unbounded satisfaction predicate is used here.

Substitution in the semantic objective leaves

$$\Phi_\beta(\mathfrak{R}, \mathbf{x}) \leq \frac{d}{2} (\rho e_4(\rho) - 5e_5(\rho)).$$

The remaining five-variable inequality is one of the fixed verified scalar sentences and gives the upper bound $2d\rho^5/625 = z_0\rho^5$. Every construction in this proof is uniform in the lengths of the finite vertex and relation lists, which proves the displayed formula in ZFC. $\blacksquare$

A.3 Fixed baseline and primitive-recursive certificate maps

We first give the effective details suppressed in the proof of the fixed baseline lemma.

Details for [Lemma 17.1](#). We first construct one template that is admissible for every input word, then compute its exact blowup value, and finally obtain a better template in the halting case.

Given $r \geq r_{\text{num}}$, run the parameter-selection algorithm in [Lemma 6.9](#) and use the first accepted choices, as in the proof of [Theorem 1.1](#). Take the class-size parameter in part (i) of [Lemma 3.5](#) to be one. Thus every physical vertex class has one vertex, the horizontal and temporal permutations on the cell class are the identity, every directed bipartite graph D_e is the unique perfect matching between its endpoint classes, every cell has the fixed background color, and $\mathbf{M} = \emptyset$. The five rails already present in the fixed signature are isomorphic. Give the vertices within each rail equal weight and give every rail total weight $1/5$. Every β -dependent local obstruction arises from (Q5) and requires an $\mathbf{M}$ -tuple. Hence none can occur when $\mathbf{M} = \emptyset$, so this same baseline five-rail realization is legal for every β .

Apply part (ii) of [Lemma 5.5](#), always using the first available choice in the fixed finite order, to obtain a fully typed record-admissible tagged 5-graph $\mathfrak{X}^q$, and denote the probability weighting supplied there by $\mathbf{x}^q$. The equality case in [Theorem 4.1](#) and the exact polynomial identity in [Lemma 5.2](#) show that the semantic value of the original five-rail realization and the record value of $\mathfrak{X}^q$ are both z_0 .

Now follow the reverse construction in the proof of [Theorem 6.3](#). Adjoin the total root frame and all completion edges prescribed by the selected root data, then forget the root labels while retaining the underlying root injection as a witness to admissibility. Denote the resulting ordinary r -graph by K^0 . Since the same record system $\mathfrak{X}^q$ is admissible for every β , this rooting witnesses $K^0 \in \mathcal{C}_\beta$ for every β .

Retain the probability weighting $\mathbf{x}^q$ on the data vertices of $\mathfrak{X}^q$. Define a probability weighting $\mathbf{w}^q$ on K^0 by giving each of the N_{rt} roots weight s_0/N_{rt} , where $s_0 = N_{\text{rt}}/(N_{\text{rt}} + 1)$, and each data vertex v weight $(1 - s_0)x_v^q$. Every coordinate of $\mathbf{w}^q$ is positive. The calculation in the reverse inequality of [Theorem 6.3](#) is exact for this admissible rooted construction of K^0 , so $\lambda(K^0; \mathbf{w}^q) = \Upsilon_{z_0}(s_0) = \Upsilon_{z_0}^* = \tau_*/r!$. Thus $\lambda(K^0) \geq \tau_*/r!$. For the fixed known nonhalting input $\beta_\infty = 0$, we also have $K^0 \in \mathcal{C}_{\beta_\infty}$, and therefore $\lambda(K^0) \leq \Lambda(\mathcal{C}_{\beta_\infty}) = \Upsilon_{z_0}^* = \tau_*/r!$. Hence equality holds and $\lambda_r(K^0) = \tau_*$.

The normalization of $\mathbf{U}$ sends β_∞ into a permanent loop. This is a finite transition-table calculation, and ZFC proves by induction on the running time that no configuration in this loop is halting. Thus the conclusion $\lambda_r(K^0) = \tau_*$, which uses $\mathbf{U}(\beta_\infty)\uparrow$, is available to the ZFC formalization, not merely as an external semantic fact.

Since $K^0 \in \mathcal{C}_\beta$ for every β , the lower-bound direction of the pair-covering transfer shows that every blowup of K^0 is $\mathcal{F}_\beta$ -free. Equivalently, K^0 is $\mathcal{F}_\beta$ -hom-free by [Lemma 2.1](#).

Let D be a common positive denominator of the coordinates of $\mathbf{w}^q$, and form an integer blowup $K_{*,r}$ of K^0 by replacing each vertex v with a class of size Dw_v^q . All these class sizes are positive integers. The projection of this blowup gives a homomorphism $K_{*,r} \rightarrow K^0$, while choosing one vertex from every class gives a homomorphism $K^0 \rightarrow K_{*,r}$. By Lemma 2.2, these two maps give $\lambda_r(K_{*,r}) = \lambda_r(K^0) = \tau_*$. Moreover, $v(K_{*,r}) = D$ and $|K_{*,r}| = D^r \lambda(K^0; \mathbf{w}^q)$, so the uniform weighting of $K_{*,r}$ attains τ_* and its balanced blowups have limiting density τ_* . Finally, if some $F \in \mathcal{F}_\beta$ admitted a homomorphism to $K_{*,r}$, composition with the projection $K_{*,r} \rightarrow K^0$ would give a homomorphism $F \rightarrow K^0$, contrary to the preceding paragraph. Hence $K_{*,r}$ is $\mathcal{F}_\beta$ -hom-free for every β .

All steps, including clearing the fixed rational denominators and writing the final blowup in canonical edge-list form, are finite and effective from the selected r -dependent data. Since the parameter-selection search terminates for every $r \geq r_{\text{num}}$, this defines the asserted total algorithm.

If $U(\beta) \downarrow$, then the class Lagrangian is strictly larger than $\tau_*/r!$. By the definition of a supremum, some finite r -graph $G \in \mathcal{C}_\beta$ satisfies $\lambda_r(G) > \tau_*$. Again the pair-covering lower bound and Lemma 2.1 show that G is $\mathcal{F}_\beta$ -hom-free. ■

We next record the explicit bounds behind the primitive-recursive compiler statement.

Details for Proposition 17.3. On input $r < r_{\text{num}}$, or when the second coordinate is not a valid binary-word code, define the first map in part (i) to return the fixed default family code. On input $r \geq r_{\text{num}}$ and a valid word β , run the terminating parameter searches in Lemma 6.9. They compute the first accepted value of every r -dependent parameter, the required finite Baranyai decompositions, and all input-independent compiler data. Given these data and β , the proof of Theorem 1.1 performs finite local-obstruction expansion, bounded-root elimination, and enumeration of the pair-covering extensions. Hence it computes $\langle \mathcal{F}_{r,\beta} \rangle$ and terminates for every pair (r, β) . Composing this procedure with the total decoder for primitive recursive syntax, the parameter theorem, and the translator tr_U gives the asserted total computable map on (r, d, e) .

Now fix r and its input-independent data. Only the anchored paths in the local test list depend on β , and their length is $O(|\beta|)$. The raw-certificate bound is linear in $|\beta|$. The explicit search-tree bound in the proof of Lemma 7.2 and the bound $B_{\text{pc}}(J)$ in Lemma 2.9 then give primitive recursive bounds on the order of every r -graph enumerated by the compiler. The quantitative polynomial simplification of these bounds is recorded in Theorem 21.2.

The construction therefore consists of bounded enumerations of local obstructions, admissible type-preserving homomorphic images, frozen-core identifications, partial rootings, r -graphs, vertex maps, and pair-covering extensions. Every loop has a primitive recursive bound. Canonicalization and deletion of duplicate isomorphism types are likewise finite bounded searches. The specialization procedure in the fixed programming system is primitive recursive, so the predicate-code version is primitive recursive once r is fixed. The proposition makes no primitive-recursive claim about the outer parameter-selection search as r varies. ■

Finally, the following proof records the canonical construction of Build_r and the exact off-domain conventions used by both certificate maps.

Details for Theorem 17.5. Suppose first that $\text{Trace}_U(\beta, t)$ holds. From the finite computation transcript, the fixed reset construction computes, primitive recursively in (β, t) , integers W, p with $\Phi_W^p(c_{\beta, W}) = c_{\beta, W}$. More explicitly, bounded simulation finds the first halting time and the rightmost visited cell, after which the fixed boot and reset phases determine W and p by bounded recursion. Put $A := p(W + 2)$. Use the periodic five-rail structure from part (ii) of Lemma 3.5, complete its record system, and apply the rooted compiler. Denote the resulting finite uncolored r -graph by G .

Recall that one rail has s_{phys} physical types. Give each root weight $1/(N_{\text{rt}} + 1)$ and each of the $5s_{\text{phys}}A$ data vertices weight $1/((N_{\text{rt}} + 1)5s_{\text{phys}}A)$. Denote this rational probability weighting of G by $\mathbf{x}$. All matching defects vanish and the marked five-tuple has positive weight. The calculation of Φ_β in the proof of Theorem 4.1 therefore gives a normalized record value larger than z_0 . For $s_0 = N_{\text{rt}}/(N_{\text{rt}} + 1)$, the map $z \mapsto \Upsilon_z(s_0)$ is strictly increasing, and $r!\Upsilon_{z_0}(s_0) = \tau_*$. Hence the completed rooted r -graph satisfies $r!\lambda(G; \mathbf{x}) > \tau_*$. Clear denominators by replacing every vertex by a vertex class of size equal to its numerator over the common denominator $D = (N_{\text{rt}} + 1)5s_{\text{phys}}A$. The resulting ordinary r -graph H has $v(H) = D$ and $\lambda_{\text{bal}}(H) = r!\lambda(G; \mathbf{x}) > \tau_*$.

The construction uses the fixed canonical vertex ordering, so the canonical code of H is primitive recursive in (β, t) . On inputs satisfying $\text{Trace}_U(\beta, t)$, define $\text{Build}_r(\beta, t)$ to be this code. The r -graph G belongs to $\mathcal{C}_\beta$, so the lower-bound direction of the pair-covering transfer makes every blowup of G $\mathcal{F}_\beta$ -free. By [Lemma 2.1](#), both G and its integer blowup H are therefore $\mathcal{F}_\beta$ -hom-free. This proves the required improvement assertion.

On the remaining inputs, define $\text{Build}_r(\beta, t)$ to be the code of the fixed one-edge r -graph $K_r^{(r)}$. Since Trace_U is primitive recursive, this makes Build_r a total primitive recursive function.

Conversely, suppose that H is an improvement witness. Since $\lambda_r(H) \geq \lambda_{\text{bal}}(H) > \tau_*$, [Lemma 17.4](#) shows that $U(\beta)$ halts within $v(H)$ steps. Define $\text{Bound}_r(\beta, H)$ to be $v(H)$ when H is a legal r -graph code and 0 otherwise. This is a total primitive recursive function; under the premise it is a valid halting-time bound, and the corresponding transcript is obtained by bounded simulation. Taking negations gives the final equivalence in the theorem. $\blacksquare$

A.4 Typed trace maps

We next make the two finite certificate transformations explicit. Fix a depth-first stack-machine interpreter for the chosen primitive recursive syntax. An evaluation certificate records the complete finite tree of recursive calls made in a computation. Let $\text{EvalCert}_{\text{PR}}(d, e, n, b, w)$ say that w is a correct evaluation tree showing that the primitive recursive syntax code d , on input (e, n) , has Boolean value b . The tree lists recursive calls in the execution order of this interpreter, so its correctness can be checked by a bounded computation. Define the predicates Eval_{PR} and All_{PR} by $\text{Eval}_{\text{PR}}(d, e, n, b) \iff \exists w \text{ EvalCert}_{\text{PR}}(d, e, n, b, w)$ and $\text{All}_{\text{PR}}(d, e) \iff \forall n \text{ Eval}_{\text{PR}}(d, e, n, 1)$, respectively. ZFC proves by induction on valid primitive recursive syntax and on the recursion argument that every valid code has exactly one Boolean value on every input. For each fixed valid code d , this identifies $\text{All}_{\text{PR}}(d, e)$ with a literal Π_1^0 predicate in e whose primitive recursive matrix is the Boolean function represented by d . This is a code-by-code assertion and does not assert the existence of a primitive recursive universal evaluator uniform in d .

For a finite-sequence code u , let $\text{len}(u)$ be its length and let $(u)_i$ be its i th entry. Define $\text{BadRun}(d, e, n, u)$ by

$$\begin{aligned} \text{BadRun}(d, e, n, u) \iff & \text{len}(u) = n + 1 \\ & \wedge \forall i < n \text{ EvalCert}_{\text{PR}}(d, e, i, 1, (u)_i) \\ & \wedge \text{EvalCert}_{\text{PR}}(d, e, n, 0, (u)_n). \end{aligned}$$

Thus a bad run records successful evaluations on $0, \dots, n-1$ followed by the first failed test at n .

Recall that $\beta_{d,e} = \text{tr}_U(\text{Idx}_{\text{test}}(d, e), 0)$. We choose the source program with index $\text{Idx}_{\text{test}}(d, e)$ to use this stack-machine interpreter. On a valid code d , it evaluates $(d, e, 0), (d, e, 1), \dots$ in order and halts immediately after the first completed call with Boolean value 0.

Fix a primitive recursive pairing function $\langle \cdot, \cdot \rangle$ and primitive recursive projections fst and snd . The interfaces used below are summarized in [Table 8](#). Every coded map is total; the final column records its behavior when a certificate or syntactic code is malformed.

Define the total primitive recursive map $T^+ : \mathbb{N}^4 \rightarrow \mathbb{N}$ as follows. On input (d, e, n, u) , first check $\text{BadRun}(d, e, n, u)$. If the check fails, return 0. If it succeeds, replay the evaluation trees in u in their recorded execution order, continue through the fixed source-machine halt instruction, and apply the forward simulation used by tr_U . Return the time of the terminal configuration in the resulting U -trace. All loops are bounded by the finite certificates in u , so this is primitive recursive.

Define $T^- : \mathbb{N}^3 \rightarrow \mathbb{N}$ by bounded reconstruction. On input (d, e, b) , simulate U on $\beta_{d,e}$ for at most b steps. If no halting configuration occurs, return $\langle 0, 0 \rangle$. Otherwise use the fixed simulation relation for tr_U to reconstruct the finite execution of the source testing program. Scan this execution until its first zero-valued completed call, record the preceding one-valued evaluation trees and the final zero-valued tree, and return their bad-run code $\langle n, u \rangle$. The simulation, reconstruction, and scan have bounds primitive recursively computable from (d, e, b) .

Lemma A.4. *ZFC proves that T^+ and T^- are total primitive recursive maps with the displayed types and that*

$$\text{Valid}_{\text{PR}}(d) \wedge \text{BadRun}(d, e, n, u) \longrightarrow \text{Trace}_U(\beta_{d,e}, T^+(d, e, n, u)),$$

Map	Formal sort	Intended output	Behavior off the intended domain
$(d, e) \mapsto \beta_{d,e}$	$\mathbb{N}^2 \rightarrow \mathbb{N}$	binary-word code	invalid d denotes the constant true predicate
$\text{Basis}_r(d, e)$	$\mathbb{N}^2 \rightarrow \mathbb{N}$	canonical code of $\mathcal{A}_{\beta_{d,e}}$	compiles the total decode of d
$\text{PairExt}_r(a)$	$\mathbb{N} \rightarrow \mathbb{N}$	code of all bounded pair extensions	canonical empty-family code
$\text{Comp}_r^{\text{num}}(d, e)$	$\mathbb{N}^2 \rightarrow \mathbb{N}$	canonical final-family code	compiles the total decode of d
$\text{T}^+(d, e, n, u)$	$\mathbb{N}^4 \rightarrow \mathbb{N}$	halting-time bound	0
$\text{Build}_r(\beta, t)$	$\mathbb{N}^2 \rightarrow \mathbb{N}$	canonical r -graph code	the fixed one-edge r -graph
$\text{Bound}_r(\beta, H)$	$\mathbb{N}^2 \rightarrow \mathbb{N}$	halting-time bound	0
$\text{T}^-(d, e, b)$	$\mathbb{N}^3 \rightarrow \mathbb{N}$	pair code $\langle n, u \rangle$	$\langle 0, 0 \rangle$
$\text{ImgCode}_r(f)$	$\mathbb{N} \rightarrow \mathbb{N}$	canonical finite-family code	canonical empty-family code
$(f, s) \mapsto \text{Ex}_s(f)$	$\mathbb{N}^2 \rightarrow \mathbb{N}$	nonnegative integer	uses $\text{ImgCode}_r(f)$

Table 8: Typed interfaces in the fixed- r numerical formalization.

$$\text{Valid}_{\text{PR}}(d) \wedge \text{Trace}_{\text{U}}(\beta_{d,e}, b) \longrightarrow \text{BadRun}\left(d, e, \text{fst}(\text{T}^-(d, e, b)), \text{snd}(\text{T}^-(d, e, b))\right).$$

Proof. The first implication follows by induction through the evaluation trees listed in u . Their ordering is the execution ordering of the chosen interpreter, so the replay reaches the first zero-valued call and then the source halt instruction. Correctness of the forward simulation gives the asserted U-trace and its terminal time.

For the second implication, bounded simulation finds the least halting time not exceeding b . Correctness of the translator identifies the corresponding finite source execution. A valid testing program has no halt instruction other than the one following a completed zero-valued call. Totality and uniqueness of primitive recursive evaluation therefore show that the reverse scan recovers one-valued certificates for $0, \dots, n-1$ and a zero-valued certificate for n . These are exactly the three clauses in the definition of BadRun . All recursions and searches are bounded by the supplied finite trace, which also proves primitive recursiveness and totality in ZFC. ■

A.5 The numerical compiler ledger

Fix r and its compiler data. Let $\text{Basis}_r(d, e)$ be the canonical code of the finite obstruction basis $\mathcal{A}_{\beta_{d,e}}$ obtained after root elimination, and let $\text{PairExt}_r(a)$ enumerate and canonicalize the bounded pair-covering extensions of the family decoded from a . Both maps are total primitive recursive after the fixed- r bounds have been retained; malformed family codes are sent to the canonical empty-family code. Choose the numerical compiler so that the identity

$$\text{Comp}_r^{\text{num}}(d, e) = \text{PairExt}_r(\text{Basis}_r(d, e)) \tag{A.3}$$

holds by definition. Define $\text{CodeFam}_r^{\text{num}}(d, e, f)$ to mean $f = \text{Comp}_r^{\text{num}}(d, e)$. The total decoder fixed before [Theorem 17.6](#) interprets syntactically invalid predicate codes as the constant true predicate.

For use in the coded proofs, let $\text{AdmClass}_r(d, e, G)$ say that G has an admissible partial rooting for the direct compiler at $\beta_{d,e}$. This is one set-theoretic formula obtained by expanding the finite rooted incidence and local-obstruction predicates. Let K_r^0 denote the fixed preliminary baseline template constructed before the integer blowup defining $K_{*,r}$.

Lemma A.5. *For fixed r , ZFC proves uniformly in d, e that, with $a = \text{Basis}_r(d, e)$ and $f = \text{Comp}_r^{\text{num}}(d, e)$,*

- (i) *for every finite r -graph code G ,*

$$\text{AdmClass}_r(d, e, G) \iff G \in \text{Forb}(\text{DecFam}_r(a));$$

every member of $\text{DecFam}_r(a)$ has an edge. Moreover, K_r^0 is a fixed edge-bearing graph and

$$\text{AdmClass}_r(d, e, K_r^0)$$

holds;

- (ii) $f = \text{PairExt}_r(a)$, and the family decoded from f is the union of the bounded pair-covering extensions of the members of the family decoded from a ;
- (iii) $\text{Base}_r(f)$ holds.

Proof. The root-elimination program first computes the order bound supplied by Lemma 7.2. It then enumerates every canonical r -graph through that order and retains exactly the codes having no admissible partial rooting, finally removing duplicate isomorphism types. The order bound is a fixed primitive recursive function of (d, e) after r has been fixed, and admissibility is a bounded predicate on the finite incidence and rooting lists. The persistent-witness argument in that lemma proves that an arbitrary finite graph fails $\text{AdmClass}_r(d, e, \cdot)$ exactly when it contains one of these retained basis codes. The edgeless graph is admissible, so every retained code has an edge. The fixed completed baseline rooting has an edge and is admissible for every input. These bounded checks discharge the two nondegeneracy hypotheses of Lemma 2.9. This proves part (i) and the totality of Basis_r .

For each decoded basis member J , the bound $B_{\text{pc}}(J)$ makes the search over extensions, injections of the distinguished J -core, isolated vertices, and pair coverage finite. Hence ZFC verifies the output of PairExt_r by bounded enumeration. Equation (A.3) then proves part (ii).

For part (iii), let K_r^0 be the fixed preliminary baseline template and let $\kappa_r: K_{*,r} \rightarrow K_r^0$ be the fixed blowup projection from the proof of Lemma 17.1. The fixed completed rooting whose record system has $\mathbf{M} = \emptyset$ is admissible for every input, and its finite incidence table proves $\text{AdmClass}_r(d, e, K_r^0)$. By part (i), no decoded basis member J embeds in K_r^0 . Suppose that some $Q \in \text{Ext}(J)$ admitted a homomorphism to K_r^0 . Every pair in the distinguished J -core lies in an edge of Q , so a homomorphism to a simple r -graph cannot identify two vertices of that core. Its restriction would therefore embed J in K_r^0 , a contradiction. Thus K_r^0 is hom-free for the family decoded from f . Composition with κ_r shows that $K_{*,r}$ is also hom-free for that family. The output f is canonical by construction, so this is exactly $\text{Base}_r(f)$. ■

Finally, define

$$\begin{aligned} \text{BuildTr}_r(d, e, n, u) &:= \text{Build}_r(\beta_{d,e}, \mathbf{T}^+(d, e, n, u)), \\ \text{RunTr}_r(d, e, H) &:= \mathbf{T}^-(d, e, \text{Bound}_r(\beta_{d,e}, H)). \end{aligned}$$

Thus $\text{BuildTr}_r: \mathbb{N}^4 \rightarrow \mathbb{N}$ returns a canonical r -graph code, whereas $\text{RunTr}_r: \mathbb{N}^3 \rightarrow \mathbb{N}$ returns a pair code $\langle n, u \rangle$.

We next expose the reverse certificate argument at the level of its intermediate objects. All finite graphs, rootings, record systems, relation systems, tuples, and finite maps below use the canonical sequence coding fixed above. A finite real vector is represented by a finite sequence of set-theoretic real numbers. Thus each relation in the next definition is one formula of set theory. No assertion of primitive-recursive decidability is made for relations containing a maximizing real weighting.

For a totally rooted finite graph (C, μ) and a weighting $\mathbf{y}$, let $\text{DataWt}(C, \mu, \mathbf{y})$ be the restriction of $\mathbf{y}$ to $V(C) \setminus \text{im } \mu$, in the inherited vertex order, and let $\text{DataMass}(C, \mu, \mathbf{y})$ be the sum of this vector. These are set-theoretic terms defined by finite replacement and finite summation.

Define $\text{SuppCore}_r(d, e, f, H, D, \iota_D, \mu_0, \mathbf{x}_H, \mathbf{x}_D)$ to mean that the following clauses hold.

- (i) $\text{ValidPR}(d)$, $\text{CodeFam}_r^{\text{num}}(d, e, f)$, and $\text{Improve}_{\tau_*}^{\text{code}}(f, H)$ hold.
- (ii) The vector $\mathbf{x}_H$ is a probability weighting of H which maximizes $\lambda(H; \mathbf{x}_H)$, and among all maximizing weightings its support has minimum cardinality.
- (iii) The code D is the canonical relabeling of $H[\text{supp}(\mathbf{x}_H)]$, and ι_D is the resulting edge-preserving injection from $V(D)$ into $V(H)$. The positive probability weighting $\mathbf{x}_D$ of D is defined by $x_{D,v} = x_{H, \iota_D(v)}$.
- (iv) The graph D covers pairs, μ_0 is an admissible partial rooting witnessing $\text{AdmClass}_r(d, e, D)$, and

$$r! \lambda(D; \mathbf{x}_D) = r! \lambda(H; \mathbf{x}_H) = \lambda_r(H) > \tau_*.$$

In particular, $v(D) \leq v(H)$. The last admissibility clause is the finite rooted incidence predicate constructed by the compiler, not an appeal to an unencoded class.

Define $\text{CompletionStep}_r(D, \mu_0, C_0, \mu)$ to mean that μ is the canonical total extension of μ_0 obtained by adding one new vertex for every missing root label and that C_0 is obtained from D by adding exactly the following edges: every permitted root-only edge whose support belongs to none of the fixed hole matchings; every generic one-data type edge whose support belongs to none of the fixed type matchings; every selected type-support edge prescribed by an existing type declaration; and every root-support edge prescribed by an existing raw-record declaration. No other edge is added, and the declared record system is unchanged. The canonical coding keeps the label of every old vertex of D and gives every added root a new label. All subsequent repair and projection maps preserve these labels; consequently the eventual record and relation vertex sets are literally subsets of $V(D)$. Define $\text{AverageRootWeights}_r(C_0, \mu, \mathbf{y}_0, \mathbf{y}_1, s)$ to mean that $\mathbf{y}_0$ and $\mathbf{y}_1$ are probability weightings of C_0 , s is the total root weight of $\mathbf{y}_0$, $\mathbf{y}_1$ agrees with $\mathbf{y}_0$ on every data vertex, and $y_{1, \mu(i)} = s/N_{\text{rt}}$ for every root label i .

Define $\text{TypingRepairSeq}_r(d, e, C_0, \mu, \mathbf{y}_1, \sigma, C)$ to mean that σ is a finite sequence of rooted record systems starting with the system declared by (C_0, μ) and ending with the fully typed system whose terminal graph code is literally C . At every successor stage, σ records the least previously untyped data vertex, all raw-record bundles containing it, the deletion of precisely those bundles, and the least type for which the admissibility check and the nondecreasing-value inequality from Lemma 6.7 hold, followed by the addition of exactly the corresponding type bundles. The old data and root vertex sets and the vector $\mathbf{y}_1$ are unchanged, and the value at every successor stage is at least its value at the preceding stage. Each successor clause is a bounded incidence and polynomial check, and the length of σ is at most the number of data vertices of C_0 .

Define $\text{ClosedCoord}(\beta, \mathfrak{R}, \mathbf{a}, j)$ to mean that $\mathbf{a} = (a_1, \dots, a_5)$ is an M -tuple of the legal relation system $\mathfrak{R}$, $j \in [5]$, and a_j belongs to a closed relation component of rail j .

Define

$$\text{CompletedRec}_r(d, e, f, H, D, \iota_D, \mu_0, \mathbf{x}_H, \mathbf{x}_D, C_0, \mu, C, \mathfrak{X}, \mathbf{y}, s, \rho)$$

to mean that $\text{SuppCore}_r(d, e, f, H, D, \iota_D, \mu_0, \mathbf{x}_H, \mathbf{x}_D)$ holds and there are $\mathbf{y}_0, \mathbf{y}_1, \sigma$ satisfying the following clauses.

- (i) The following three relations hold:

$$\begin{aligned} &\text{CompletionStep}_r(D, \mu_0, C_0, \mu), \quad \text{AverageRootWeights}_r(C_0, \mu, \mathbf{y}_0, \mathbf{y}_1, s), \\ &\text{TypingRepairSeq}_r(d, e, C_0, \mu, \mathbf{y}_1, \sigma, C). \end{aligned}$$

The vector $\mathbf{y}_0$ is the zero extension of $\mathbf{x}_D$ to C_0 before root equalization. No one of these operations introduces or identifies a data vertex, although a typing repair may delete raw-record bundle edges.

- (ii) The vector $\mathbf{y} = \mathbf{y}_1$ is the resulting probability weighting of C , and $\rho = \text{DataMass}(C, \mu, \mathbf{y}) = 1 - s$.
- (iii) The system $\mathfrak{X} = \mathfrak{X}_{C, \mu}$ is fully typed and record-admissible for $\beta_{d, e}$, and

$$\lambda_{\text{rec}}(\mathfrak{X}; \text{DataWt}(C, \mu, \mathbf{y})) > z_0 \rho^5, \quad \lambda(C; \mathbf{y}) > \Upsilon_{z_0}^*.$$

Define

$$\text{ClosedMark}_r(d, e, f, H, D, \iota_D, C, \mu, \mathfrak{X}, \mathbf{y}, \mathfrak{X}^+, \mathfrak{R}, \mathbf{a}, j)$$

to mean that there are $\mu_0, \mathbf{x}_H, \mathbf{x}_D, C_0, s, \rho$ for which

$$\begin{aligned} &\text{CompletedRec}_r(d, e, f, H, D, \iota_D, \mu_0, \mathbf{x}_H, \mathbf{x}_D, \\ &C_0, \mu, C, \mathfrak{X}, \mathbf{y}, s, \rho) \end{aligned}$$

holds and the following clauses hold.

- (i) The system $\mathfrak{X}^+$ is the canonical filler saturation of $\mathfrak{X}$, and $\mathfrak{R}$ is its projected five-rail matching system. They have the same data-vertex set as $\mathfrak{X}$.
- (ii) The system $\mathfrak{R}$ is legal for $\beta_{d, e}$ and

$$\Phi_{\beta_{d, e}}(\mathfrak{R}, \text{DataWt}(C, \mu, \mathbf{y})) = \lambda_{\text{rec}}(\mathfrak{X}^+; \text{DataWt}(C, \mu, \mathbf{y})) \geq \lambda_{\text{rec}}(\mathfrak{X}; \text{DataWt}(C, \mu, \mathbf{y})) > z_0 \rho^5.$$

(iii) The relation $\text{ClosedCoord}(\beta_{d,e}, \mathfrak{R}, \mathbf{a}, j)$ holds.

Finally, define $\text{TableauRec}_r(d, e, H, D, \iota_D, \mathfrak{R}, \mathbf{a}, j, W, p)$ to mean that $\text{ClosedCoord}(\beta_{d,e}, \mathfrak{R}, \mathbf{a}, j)$ holds, the logical relation component of a_j contains h cell vertices for some h , these vertices belong to $V(\mathfrak{R}) \subseteq V(D)$ and map injectively into $V(H)$ through ι_D , and

$$W \geq |\beta_{d,e}| + 2, \quad \Phi_W^p(c_{\beta_{d,e}, W}) = c_{\beta_{d,e}, W}, \quad W + 2 \leq h \leq v(D) \leq v(H), \quad 1 \leq p \leq h.$$

The value h is definable by bounded counting in the finite component. These four relations retain the original improvement code H and the support embedding ι_D precisely so that the last two bounds do not lose their common source.

In the statements below, every displayed implication is its universal closure over all free variables, and every phrase “extends to a witness” abbreviates the corresponding existential quantifiers over exactly the additional variables displayed in its conclusion.

Lemma A.6. *For the fixed r -dependent setup, ZFC proves, uniformly in the displayed variables, the following implications.*

(i) *If $\text{ValidPR}(d)$, $\text{CodeFam}_r^{\text{num}}(d, e, f)$, and $\text{Improve}_{\tau_*}^{\text{code}}(f, H)$ hold, then there are $D, \iota_D, \mu_0, \mathbf{x}_H, \mathbf{x}_D$ such that*

$$\text{SuppCore}_r(d, e, f, H, D, \iota_D, \mu_0, \mathbf{x}_H, \mathbf{x}_D).$$

(ii) *Every witness to $\text{SuppCore}_r(d, e, f, H, D, \iota_D, \mu_0, \mathbf{x}_H, \mathbf{x}_D)$ extends to a witness to*

$$\text{CompletedRec}_r(d, e, f, H, D, \iota_D, \mu_0, \mathbf{x}_H, \mathbf{x}_D, C_0, \mu, C, \mathfrak{X}, \mathbf{y}, s, \rho).$$

(iii) *Every witness to CompletedRec_r extends to a witness to*

$$\text{ClosedMark}_r(d, e, f, H, D, \iota_D, C, \mu, \mathfrak{X}, \mathbf{y}, \mathfrak{X}^+, \mathfrak{R}, \mathbf{a}, j).$$

(iv) *Every witness to ClosedMark_r extends to integers W, p satisfying*

$$\text{TableauRec}_r(d, e, H, D, \iota_D, \mathfrak{R}, \mathbf{a}, j, W, p) \quad \text{and} \quad \text{Trace}_U(\beta_{d,e}, v(H)).$$

Proof. For part (i), the finite simplex of probability weightings of H is compact and the edge polynomial is continuous, so a maximizing weighting exists. Choose one whose support has least cardinality and call it $\mathbf{x}_H$. The standard first-variation argument used in Lemma 2.9 shows that its support covers pairs: if two support vertices were not contained together in an edge, transferring the smaller of their two weights to the other vertex in the favorable direction would preserve maximality and remove one support vertex. Canonically relabel the support, record the inverse relabeling as ι_D , and pull the positive support weights back to $\mathbf{x}_D$. The support core D has the same Lagrangian as H .

Let $a = \text{Basis}_r(d, e)$. Since H is $\mathcal{F}_f$ -hom-free, it is $\mathcal{F}_f$ -free. Parts (i)–(ii) of Lemma A.5 identify f with the union of the extension lists of the basis decoded from a . Suppose that some decoded basis member J embedded in the pair-covering support D . For each pair of vertices in its displayed copy, choose the least edge of D containing that pair and take the union of these edges with the copy of J . This union has at most

$$v(J) + (r - 2) \binom{v(J)}{2}$$

vertices and canonically yields a member $Q \in \text{Ext}(J)$ embedded in D . Every auxiliary vertex lies in a selected edge, while every core vertex lies in a selected pair-covering edge; here $v(J) \geq r \geq 2$ because J has an edge. Thus Q has no isolated vertex. Composing with ι_D embeds Q in H , contrary to $\mathcal{F}_f$ -freeness. Hence D avoids every decoded basis member. Part (i) of the compiler invariant gives $\text{AdmClass}_r(d, e, D)$, and unpacking that formula supplies the admissible partial rooting μ_0 . Every choice in this paragraph is a bounded least-code search. Finally,

$$r! \lambda(D; \mathbf{x}_D) = r! \lambda(H; \mathbf{x}_H) = \lambda_r(H) \geq \lambda_{\text{bal}}(H) > \tau_*.$$

Every assertion in this paragraph is a theorem about a fixed finite simplex, finite vertex maps, and finite edge lists. Compactness of a closed bounded subset of a finite power of $\mathbb{R}$, continuity of a polynomial,

and the first-variation calculation are theorems of ZFC. Hence the choices just made give the asserted set-theoretic witness, including $v(D) \leq v(H)$.

For part (ii), apply the canonical algorithm in [Lemma 6.2](#) to (D, μ_0) and record its output by the relation

$$\text{CompletionStep}_r(D, \mu_0, C_0, \mu).$$

Extend $\mathbf{x}_D$ by zero on the added roots, obtaining $\mathbf{y}_0$. Completion does not decrease the value, so

$$\lambda(C_0; \mathbf{y}_0) \geq \lambda(D; \mathbf{x}_D) > \frac{\tau_*}{r!} = \Upsilon_{z_0}^*.$$

Let $\mathbf{u}$ be the root-weight vector of $\mathbf{y}_0$, let $s = \sum_a u_a$, and put $\rho = 1 - s$. In the partially typed record system declared by (C_0, μ) , let η_i be the total weight of type i and let q_α be the α th tagged record polynomial evaluated on $\text{DataWt}(C_0, \mu, \mathbf{y}_0)$. Disjointness of the type classes gives $\eta_i \geq 0$ and $\sum_i \eta_i \leq \rho$. Since each tagged record class is a simple 5-graph,

$$0 \leq q_\alpha \leq e_5(\text{DataWt}(C_0, \mu, \mathbf{y}_0)) \leq \frac{\rho^5}{120}.$$

The exact master-polynomial identity [\(6.3\)](#) gives

$$\lambda(C_0; \mathbf{y}_0) = \Psi_{N_{\text{rt}}}(\mathbf{u}, \rho, \boldsymbol{\eta}, \mathbf{q}) > \frac{J_0}{(N_{\text{rt}} + 1)^r}.$$

Thus every conjunct of $\mathbf{A}_{N_{\text{rt}}}$ holds. The verified Root_r certificate gives $\rho < 2/(N_{\text{rt}} + 1)$ and $s > 1/2$, and says that replacing $\mathbf{u}$ by its equal vector cannot decrease the value. Make this direct replacement and call the resulting weighting $\mathbf{y}_1$. Hence $\lambda(C_0; \mathbf{y}_1) \geq \lambda(C_0; \mathbf{y}_0)$; no finite termination assertion about pair averaging is used. The two displayed bounds and the fixed choice of ℓ_{typ} are exactly the numerical hypotheses of [Lemma 6.7](#). Apply that lemma to one untyped data vertex at each stage. Record the chosen vertex, deleted record bundles, selected type, and added type bundles in σ . There are at most $v(C_0)$ stages, every stage preserves admissibility and does not decrease the value, and the final system $\mathfrak{X}$ is fully typed. It yields C , $\mathbf{y} = \mathbf{y}_1$, and

$$\lambda(C; \mathbf{y}) > \Upsilon_{z_0}^*.$$

At the equal root vector, the exact completed-polynomial identity, with the baseline record contribution subtracted, is

$$\lambda(C; \mathbf{y}) - \Upsilon_{z_0}(s) = \frac{s^{r_{\text{rec}}}}{r_{\text{rec}} N_{\text{rt}}^{r_{\text{rec}}-1}} (\lambda_{\text{rec}}(\mathfrak{X}; \text{DataWt}(C, \mu, \mathbf{y})) - z_0 \rho^5). \quad (\text{A.4})$$

Here N_{rt} is the root-set size denoted by N in the direct compiler. Since $\Upsilon_{z_0}(s) \leq \Upsilon_{z_0}^*$, the left side is positive. The localization clause gives $s > 1/2$, so the coefficient on the right is positive. The bracket is therefore positive; in particular, $\rho > 0$. The finite sequence σ verifies TypingRepairSeq_r , and all the clauses of CompletedRec_r follow.

For part (iii), perform the bounded filler-saturation operation in [Lemma 5.3](#). It changes no vertex or weight and does not decrease the record value. Part (i) of [Lemma 5.5](#) verifies legality of the projected relation system, and [Lemma 5.2](#) gives the displayed equality between its semantic score and the saturated record value. Suppose that no required pair $(\mathbf{a}, j)$ existed. The finite component dichotomy for a legal partial-matching system would then put every coordinate of every $\mathbf{M}$ -tuple in a grounded component. The uniform finite-sum result [Lemma A.3](#) would give

$$\Phi_{\beta_{d,e}}(\mathfrak{R}, \text{DataWt}(C, \mu, \mathbf{y})) \leq z_0 \rho^5,$$

contrary to the strict inequality obtained from [\(A.4\)](#). Bounded search through the finite tuple and component lists therefore supplies $\mathbf{a}$ and j .

For part (iv), apply part (iii) of [Lemma 3.5](#) to the closed component containing a_j . The recovery walks through finite functional relation tables, so it is formalized by bounded recursion and produces W, p with the asserted periodicity and with $W + 2, p \leq h$. Completion adds only roots, while root equalization, typing repair, filler saturation, and projection neither add nor identify data vertices. Hence the h cell vertices inject into $V(D)$ and then, through ι_D , into $V(H)$. Thus $h \leq v(D) \leq v(H)$. By the converse implication in [Lemma 3.2](#), the periodic return contains a genuine halting configuration before the reset

begins. Each temporal update before that configuration simulates at most one step of U , so U halts on $\beta_{d,e}$ within $p \leq h \leq v(H)$ steps. This is exactly $\text{Trace}_U(\beta_{d,e}, v(H))$.

All invocations above are uniform formulas after r and its finite data have been fixed. The combinatorial operations are bounded recursions over finite codes, the arbitrary-length energy estimate is supplied by Lemma A.3, and every remaining fixed real-algebraic inequality has its retained proof code in $\mathcal{C}_r^{\text{alg}}$. Thus ZFC proves the four implications uniformly, rather than only each external instance. ■

A.6 Formal certificate translation

Lemma A.7. *For the fixed r -dependent setup, ZFC proves the following statements.*

- (i) *We have $\forall d \forall e \exists! f \text{ CodeFam}_r^{\text{num}}(d, e, f)$.*
- (ii) *If $\text{Valid}_{\text{PR}}(d)$, then $\neg \text{All}_{\text{PR}}(d, e)$ holds if and only if $\text{BadRun}(d, e, n, u)$ holds for some n, u .*
- (iii) *The two certificate transformations satisfy*

$$\begin{aligned} & \text{Valid}_{\text{PR}}(d) \wedge \text{BadRun}(d, e, n, u) \wedge \text{CodeFam}_r^{\text{num}}(d, e, f) \\ & \implies \text{Improve}_{\tau_*}^{\text{code}}(f, \text{BuildTr}_r(d, e, n, u)), \\ & \text{Valid}_{\text{PR}}(d) \wedge \text{CodeFam}_r^{\text{num}}(d, e, f) \wedge \text{Improve}_{\tau_*}^{\text{code}}(f, H) \\ & \implies \text{BadRun}(d, e, \text{fst}(\text{RunTr}_r(d, e, H)), \text{snd}(\text{RunTr}_r(d, e, H))). \end{aligned}$$

Proof. The input-dependent bounds for the local-obstruction lists, root-elimination search, pair-covering extensions, and canonicalization are primitive recursive functions of (d, e) . ZFC proves these functions total by induction on their defining recursions, so it proves that every bounded search in the compiler terminates with a unique output. This proves part (i).

The relation BadRun is primitive recursive. Assume $\text{Valid}_{\text{PR}}(d)$. If $\neg \text{All}_{\text{PR}}(d, e)$, let n be least such that no one-valued evaluation certificate exists. Totality and uniqueness give a zero-valued certificate at n , while minimality gives one-valued certificates at every $i < n$; finite sequence coding packages these certificates into a code u satisfying $\text{BadRun}(d, e, n, u)$. Conversely, if $\text{BadRun}(d, e, n, u)$ holds, its final zero-valued certificate and uniqueness exclude every one-valued certificate at n , so the n th conjunct in $\text{All}_{\text{PR}}(d, e)$ fails. This proves part (ii).

For the first implication in part (iii), Lemma A.4 gives

$$\text{Trace}_U(\beta_{d,e}, \mathbb{T}^+(d, e, n, u)).$$

Bounded simulation reconstructs its least halting prefix. The reset rules then compute W, p and the periodic tableau by bounded recursion. The five rail edge lists, the completed record system, and the rooted r -graph are finite functions of this tableau. Give their roots and data vertices the rational weights specified in the proof of part (i) of Theorem 17.5. ZFC checks the perfect-matching incidences, the one positive marked monomial, the calibration identity, and the strict value comparison using the retained algebraic proof codes. Clearing the displayed common denominator is an integer blowup; bounded multiplication verifies

$$b_* r! |\text{BuildTr}_r(d, e, n, u)| > a_* v(\text{BuildTr}_r(d, e, n, u))^T.$$

The rooted graph G constructed before the integer blowup satisfies $\text{AdmClass}_r(d, e, G)$. Put

$$B := \text{BuildTr}_r(d, e, n, u),$$

and let $\pi_B: B \rightarrow G$ be its blowup projection. Put $a = \text{Basis}_r(d, e)$. Parts (i)–(ii) of Lemma A.5 show that G avoids the family decoded from a and that every member of the family decoded from f is isomorphic to some $Q \in \text{Ext}(J)$ with $J \in \text{DecFam}_r(a)$.

Suppose that $\varphi: Q \rightarrow B$ were a homomorphism. Then $\pi_B \circ \varphi: Q \rightarrow G$ would be a homomorphism. This composite is injective on the distinguished J -core. Indeed, any two distinct core vertices lie together in an edge of Q , and identifying them would send that edge to a set of fewer than r vertices, which cannot be an edge of the simple r -graph G . The restriction of $\pi_B \circ \varphi$ to the distinguished core would therefore

embed J in G , a contradiction. Hence B is hom-free for the compiled family. Together with the strict integer inequality, this is precisely

$$\text{Improve}_{\tau_*}^{\text{code}}(f, \text{BuildTr}_r(d, e, n, u)).$$

All maps, projections, and core-injectivity checks in this argument are bounded operations on finite codes. For the reverse implication, [Lemma A.6](#) gives

$$\text{Trace}_U(\beta_{d,e}, v(H)).$$

By definition, $\text{Bound}_r(\beta_{d,e}, H) = v(H)$ on this legal graph code. The second implication in [Lemma A.4](#), with $b = v(H)$, now gives the displayed bad run with the projections fst and snd. This proves part (iii) with all intermediate sorts fixed. $\blacksquare$

Finite proof fragment. For the fixed r -dependent setup, the uniform derivation used in [Proposition 18.1](#) is finite. Consequently, it uses only a finite collection of ZFC axioms and axiom-scheme instances, independent of d , e , and f .

B Structural arithmetization and the coded phase bridge

This appendix defines the five structural formulas used in the main text and verifies the structural compiler inside ZFC. It separates dense-limit coding and stability from the fixed scalar certificates, the compiler factorization, and the coded weighted-to-ordinary transfer. Throughout, fix $s \geq r_{\text{str}}$, put $k = r_{\text{num}}$ and $\tau_{\text{base}} = \tau_k$, and retain the corresponding structural compiler data from Part II.

B.1 Dense-limit and extremal-space formulas

Throughout this subsection, every variable and quantifier described as numerical is understood to range over ω , represented by the finite von Neumann ordinals. Let $\text{FinGraph}_s(G)$ denote the set-theoretic predicate saying that G is a finite s -graph.

Dense-limit codes. Fix an effective enumeration $(H_\ell)_{\ell \in \mathbb{N}}$ of the finite s -graphs. We represent a dense s -graph limit by its homomorphism-density vector $(t(H_\ell, W))_{\ell \in \mathbb{N}}$ in the closure of the finite s -graph vectors in $[0, 1]^\mathbb{N}$. This is a definable compact metric space in ZFC.

For every $g \in \mathbb{N}$, write $\mathcal{F}_g := \text{DecFam}_s(g)$ for its total filtered decode, and let $\ell(F)$ be the least index of F in the fixed enumeration. Define

$$\mathbf{X}_s := \left\{ x \in [0, 1]^\mathbb{N} : \forall m \forall q \in \mathbb{Q}_{>0} \exists G \left(\text{FinGraph}_s(G) \wedge v(G) > 0 \wedge \max_{\ell \leq m} |x_\ell - t(H_\ell, G)| < q \right) \right\}.$$

If ℓ_0 indexes the one-edge s -graph, put

$$\begin{aligned} \mathbf{X}_s(g) &:= \{x \in \mathbf{X}_s : x_{\ell(F)} = 0 \text{ for every } F \in \mathcal{F}_g\}, \\ \pi_s(g) &:= \max_{x \in \mathbf{X}_s(g)} x_{\ell_0}, \\ \mathbf{E}_s(g) &:= \{x \in \mathbf{X}_s(g) : x_{\ell_0} = \pi_s(g)\}. \end{aligned}$$

Note that these definitions are formulas of set theory. The maximum defining $\pi_s(g)$ exists because $\mathbf{X}_s(g)$ is a nonempty compact set.

Profiles and edit approximation. A finite profile is represented set-theoretically by its finite real weight tuple and finite edge-type set. More precisely, $\text{Profile}_s(p)$ asserts that, for some $q \geq 1$, one has $p = ((a_i)_{i \in [q]}, \mathcal{M})$, where $a_i \geq 0$ for every $i \in [q]$, $\sum_{i \in [q]} a_i = 1$, and $\mathcal{M}$ is a collection of multisets of size s on $[q]$. We fix a set-theoretic formula for its integer realizations as follows. Given a ground set of order n , the formula defines the class sizes by the largest-fractional-parts rule in [Section 8](#), including its fixed tie-breaking by type index, and then quantifies over the corresponding ordered vertex partitions and

prescribed edge types. We use $B \in \mathfrak{B}_p(V)$ as an abbreviation for this fixed formula. For $p = ((a_i)_{i \in [q]}, \mathcal{M})$ satisfying $\text{Profile}_s(p)$, define its limit code $w_p \in [0, 1]^\mathbb{N}$ coordinatewise by

$$(w_p)_\ell := \sum_{\phi: V(H_\ell) \rightarrow [q]} \left(\prod_{v \in V(H_\ell)} a_{\phi(v)} \right) \left(\prod_{e \in H_\ell} \mathbf{1}_{\{\{\phi(v): v \in e\} \in \mathcal{M}\}} \right).$$

A sequence of finite s -graphs is coded as a function from $\mathbb{N}$ to the canonical codes of finite s -graphs. We define $\text{AExt}_s(g, \mathbf{G})$ as the conjunction of the following three set-theoretic conditions.

- (i) Every G_n is $\mathcal{F}_g$ -free.
- (ii) For every integer M , all sufficiently late terms have order at least M .
- (iii) For every positive rational ε , all sufficiently late edge densities differ from $\pi_s(g)$ by less than ε .

Define convergence to a limit code by

$$\text{Conv}_s(\mathbf{G}, x) \iff \forall m \in \mathbb{N} \forall q \in \mathbb{Q}_{>0} \exists N \forall n \geq N \max_{\ell \leq m} |t(H_\ell, G_n) - x_\ell| < q.$$

Let $\text{Close}_s(p, \mathbf{G})$ denote

$$\forall \varepsilon \in \mathbb{Q}_{>0} \exists N \forall n \geq N \exists B \in \mathfrak{B}_p(V(G_n)) \quad \text{edit}(G_n, B) \leq \varepsilon v(G_n)^s.$$

We also use the finitary stability formula

$$\begin{aligned} \text{FStab}_s(g, p) \iff \forall \varepsilon \in \mathbb{Q}_{>0} \exists \delta \in \mathbb{Q}_{>0} \exists N \forall G \left[G \text{ is a canonical } \mathcal{F}_g\text{-free } s\text{-graph, } v(G) \geq N, \right. \\ \left. |G| \geq \text{ex}(v(G), \mathcal{F}_g) - \delta v(G)^s \right. \\ \left. \implies \exists B \in \mathfrak{B}_p(V(G)) \quad \text{edit}(G, B) \leq \varepsilon v(G)^s \right]. \end{aligned}$$

The five structural predicates. The fixed statistic Sep_s is evaluated on a limit code by its finite rational linear combination of coordinates. Define $\text{Str}_{1,s}(g)$ by $|\mathbf{E}_s(g)| = 1$. For $\text{Str}_{2,s}(g)$, we use the fixed formula asserting that there do not exist two disjoint nonempty subsets whose union is $\mathbf{E}_s(g)$ and which are relatively open there; relative openness is expressed using rational basic cylinder sets in $[0, 1]^\mathbb{N}$. Thus connectedness here is a set-theoretic formula rather than an external topological abbreviation. Define $\text{Str}_{3,s}(g)$ by

$$\text{Str}_{3,s}(g) \iff \exists x, y \in \mathbf{E}_s(g) \quad \text{Sep}_s(x) > 0 > \text{Sep}_s(y),$$

and define $\text{Str}_{4,s}(g)$ by

$$\begin{aligned} \text{Str}_{4,s}(g) \iff \exists q \in \mathbb{Q}_{>0} \left[\forall x \in \mathbf{E}_s(g) \left(\text{Sep}_s(x) \geq q \vee \text{Sep}_s(x) \leq -q \right) \right. \\ \wedge \exists x \in \mathbf{E}_s(g) \text{Sep}_s(x) \geq q \\ \left. \wedge \exists y \in \mathbf{E}_s(g) \text{Sep}_s(y) \leq -q \right]. \end{aligned}$$

This expands both coverage and nonemptiness of the two sign parts. Finally, define $\text{Str}_{5,s}(g)$ by

$$\text{Str}_{5,s}(g) \iff \exists p \left[\text{Profile}_s(p) \wedge w_p \in \mathbf{E}_s(g) \wedge \forall \mathbf{G} \left(\text{AExt}_s(g, \mathbf{G}) \longrightarrow \text{Close}_s(p, \mathbf{G}) \right) \right].$$

B.2 Stability formulas and removal

The following lemma verifies that the preceding formulas represent the five properties used in the structural theorem.

Lemma B.1. *For fixed s , ZFC proves the following statements uniformly from g .*

- (i) *The predicates $\text{Str}_{i,s}(g)$ are formulas of set theory and are equivalent to properties (i)–(v) in Section 8 for the decoded family $\mathcal{F}_g$.*
- (ii) *For every limit code x ,*

$$x \in \mathbf{E}_s(g) \iff \exists \mathbf{G} \left(\text{AExt}_s(g, \mathbf{G}) \wedge \text{Conv}_s(\mathbf{G}, x) \right). \tag{B.1}$$

- (iii) *If $\text{Profile}_s(p)$, $v(B_n) \rightarrow \infty$, and $B_n \in \mathfrak{B}_p(V(B_n))$ for every n , then $\text{Conv}_s(\mathbf{B}, w_p)$.*

(iv) If $\text{Profile}_s(p)$, then

$$\text{FStab}_s(g, p) \iff \forall \mathbf{G} (\text{AExt}_s(g, \mathbf{G}) \longrightarrow \text{Close}_s(p, \mathbf{G})). \quad (\text{B.2})$$

Proof. By definition, $\mathbf{X}_s$ is the closure of the finite homomorphism-density vectors in the countable product $[0, 1]^\mathbb{N}$, hence is the usual compact dense-limit space in ZFC. The filtered family $\mathcal{F}_g$ is finite. The simultaneous hypergraph removal lemma, in the forms proved by Gowers [25] and Rödl and Schacht [61, Theorem 3], therefore applies to its finitely many members inside ZFC.

We first prove part (ii). Suppose that $\text{AExt}_s(g, \mathbf{G})$ and $\text{Conv}_s(\mathbf{G}, x)$ hold. If $F \in \mathcal{F}_g$, every homomorphism from F to the F -free graph G_n identifies two vertices. Hence

$$t(F, G_n) \leq \frac{\binom{v(F)}{2}}{v(G_n)} \longrightarrow 0,$$

so $x_{\ell(F)} = 0$. If H_{ℓ_0} is the one-edge s -graph, then

$$t(H_{\ell_0}, G_n) = \frac{s!|G_n|}{v(G_n)^s} = \theta_{s, v(G_n)} \frac{|G_n|}{\binom{v(G_n)}{s}}.$$

The order condition makes $\theta_{s, v(G_n)} \rightarrow 1$, while asymptotic extremality makes the last ordinary density tend to $\pi_s(g)$. Thus $x \in \mathbf{X}_s(g)$ and $x_{\ell_0} = \pi_s(g)$, proving $x \in \mathbf{E}_s(g)$.

Conversely, let $x \in \mathbf{E}_s(g)$. For $j \geq 1$, put

$$m_j := \max(\{j, \ell_0\} \cup \{\ell(F) : F \in \mathcal{F}_g\})$$

and choose a positive rational η_j satisfying

$$\eta_j < \frac{1}{3j s! (1 + \max_{\ell \leq m_j} |H_\ell|)}.$$

Let $\delta_j > 0$ be the simultaneous-removal threshold for the finite family $\mathcal{F}_g$ and edit tolerance η_j . The definition of $\mathbf{X}_s$, together with $x_{\ell(F)} = 0$ for every $F \in \mathcal{F}_g$, gives a finite graph whose first m_j coordinates are within $\min\{1/(3j), \delta_j/2\}$ of those of x . Take a sufficiently large balanced blowup. It still has the same homomorphism-density vector, has order at least j , and has $t(F, \cdot) < \delta_j$ for every $F \in \mathcal{F}_g$. Simultaneous removal deletes at most $\eta_j v(G)^s$ edges and produces a genuinely $\mathcal{F}_g$ -free graph. A union bound over the edges of H_ℓ gives

$$|t(H_\ell, G) - t(H_\ell, G')| \leq |H_\ell| \frac{s! \text{edit}(G, G')}{v(G)^s}. \quad (\text{B.3})$$

For $\ell \leq j$, the choice of η_j makes the right side less than $1/(3j)$. Among the graphs obtained this way, take the least canonical code and call it G_j . Then $v(G_j) \geq j$, the first j coordinates converge to those of x , and the one-edge coordinate tends to $x_{\ell_0} = \pi_s(g)$. Hence $\text{AExt}_s(g, \mathbf{G})$ and $\text{Conv}_s(\mathbf{G}, x)$ hold. This proves (B.1) with an explicit set-theoretic choice.

For part (iii), fix H_ℓ . Expanding a homomorphism into B_n according to its profile types gives the finite sum defining $(w_p)_\ell$. The largest-fractional-parts rule changes every type proportion from its prescribed weight by $O_p(1/v(B_n))$. Assignments in which two adjacent vertices use the same ordinary vertex, and assignments using a class whose prescribed weight is zero, contribute $O_{H_\ell, p}(1/v(B_n))$. Therefore

$$|t(H_\ell, B_n) - (w_p)_\ell| = O_{H_\ell, p}(1/v(B_n)).$$

For every fixed initial coordinate segment the maximum error tends to zero, which is $\text{Conv}_s(\mathbf{B}, w_p)$.

For part (iv), suppose first that $\text{FStab}_s(g, p)$ holds and that $\mathbf{G}$ is asymptotically extremal. Since both $|G_n|/\binom{v(G_n)}{s}$ and $\text{ex}(v(G_n), \mathcal{F}_g)/\binom{v(G_n)}{s}$ tend to $\pi_s(g)$, their difference is $o(v(G_n)^s)$. Given $\varepsilon > 0$, apply $\text{FStab}_s(g, p)$ with that ε and then use this deficit bound; all sufficiently late G_n have a profile realization at edit distance at most $\varepsilon v(G_n)^s$. Thus $\text{Close}_s(p, \mathbf{G})$ holds.

Conversely, suppose that $\text{FStab}_s(g, p)$ fails. There is a rational $\varepsilon_0 > 0$ such that, for every $j \geq 1$, a canonical $\mathcal{F}_g$ -free graph G_j of order at least j satisfies

$$|G_j| \geq \text{ex}(v(G_j), \mathcal{F}_g) - \frac{1}{j} v(G_j)^s$$

but has edit distance larger than $\varepsilon_0 v(G_j)^s$ from every member of $\mathfrak{B}_p(V(G_j))$. Choose the least such canonical code. The resulting sequence is asymptotically extremal, but $\text{Close}_s(p, \mathbf{G})$ fails. This proves the reverse implication in (B.2).

It remains to verify part (i). Equation (B.1) identifies $\pi_s(g)$ and $\mathbf{E}_s(g)$ with the ordinary Turán density and extremal limit space. Connectedness is the nonexistence of a separation into two nonempty relatively open sets, and the two expanded sign formulas are exactly the corresponding definitions. Parts (iii)–(iv) identify the profile formula with one-construction Erdős–Simonovits stability. Thus the five predicates have the claimed meanings, and every definition and implication above is uniform in the finite family code g . $\blacksquare$

B.3 Fixed structural algebraic certificates

For fixed $s \geq r_{\text{str}}$, retain the structural notation m , $I_s = [1/(2(m+1)), 2/(m+1)]$, and $\varrho_s(u) = (1-u)/m$. Expand every named algebraic constant by its rational defining polynomial and rational isolating interval. After the rational parameters have been substituted, write

$$\Phi_{s,z}(u, a, b) = \Phi_{\text{base},s}(u) + \alpha_s(u)(z - \tau_{\text{base}})(a^k + b^k) - \gamma_s(u)(a + b)^{k+1}.$$

Define the following scalar real-closed-field formulas:

$$\begin{aligned} \Omega_s(u, a, b) &\iff u \in I_s \wedge a \geq 0 \wedge b \geq 0 \wedge a + b \leq u, \\ \text{Max}_s(z, u, a, b) &\iff \Omega_s(u, a, b) \wedge \forall u' \forall a' \forall b' (\Omega_s(u', a', b') \longrightarrow \Phi_{s,z}(u', a', b') \leq \Phi_{s,z}(u, a, b)), \\ \mathbf{S}_s(z, u, a, b) &:= \varrho_s(u)^m z(a^k - b^k). \end{aligned}$$

Let $\text{PosRoof}_s(z, u, t)$ abbreviate the conjunction

$$\begin{aligned} &\tau_{\text{base}} < z \wedge z \leq 1, \quad u \in \text{int } I_s, \quad t > 0, \\ &(k+1)\gamma_s(u)t = k\alpha_s(u)(z - \tau_{\text{base}}), \\ &\forall u' \forall a \forall b [\text{Max}_s(z, u', a, b) \iff (u' = u \wedge ((a = t \wedge b = 0) \vee (a = 0 \wedge b = t)))] . \end{aligned}$$

Let Θ_s be the finite list of named real algebraic constants occurring in the scalar formulas and the four certificate sentences below, including u_0 . For each $\vartheta \in \Theta_s$, fix an integer polynomial P_ϑ and rationals $\ell_\vartheta < u_\vartheta$ such that ϑ is the unique real zero of P_ϑ in $(\ell_\vartheta, u_\vartheta)$, and write

$$\text{Iso}_\vartheta(x) \iff P_\vartheta(x) = 0 \wedge \ell_\vartheta < x \wedge x < u_\vartheta \wedge \forall y (P_\vartheta(y) = 0 \wedge \ell_\vartheta < y \wedge y < u_\vartheta \longrightarrow y = x).$$

We use the following closure convention. If a displayed ordered-field sentence is written as $\varphi(\vartheta_1, \dots, \vartheta_h)$, its formal code is the code of

$$\exists x_1 \cdots \exists x_h \left[\bigwedge_{j \in [h]} \text{Iso}_{\vartheta_j}(x_j) \wedge \varphi(x_1, \dots, x_h) \right].$$

Repeated constants use one variable, rational coefficients are cleared by positive denominators, and interval membership and finite conjunctions are expanded into scalar ordered-field formulas. Thus no named algebraic constant is treated as an additional symbol of the real-closed-field language.

Let $\mathcal{S}_s^{\text{str}}$ consist of the following four closed real-closed-field sentences under this convention.

- (i) Par_s is the conjunction of $\Phi_{\text{loc}}(s) < 1$, $\Phi_{\text{eq}}(s) < 1$, the four rational inequalities in (10.19), and

$$\begin{aligned} &\forall u \forall z [u \in I_s \wedge \tau_{\text{base}} \leq z \wedge z \leq 1 \\ &\quad \longrightarrow (\gamma_s(u) > 0 \wedge 0 \leq k\alpha_s(u)(z - \tau_{\text{base}}) \wedge \\ &\quad \quad k\alpha_s(u)(z - \tau_{\text{base}}) \leq \tfrac{1}{2}(k+1)\gamma_s(u)u)]. \end{aligned}$$

- (ii) Roof_s is the conjunction

$$\forall u \forall a \forall b [\text{Max}_s(\tau_{\text{base}}, u, a, b) \iff (u = u_0 \wedge a = 0 \wedge b = 0)]$$

and

$$\forall z [\tau_{\text{base}} < z \wedge z \leq 1 \longrightarrow \exists u \exists t \text{PosRoof}_s(z, u, t)].$$

(iii) Sign_s is the conjunction of $S_s(\tau_{\text{base}}, u_0, 0, 0) = 0$ and

$$\forall z \forall u \forall t \text{ PosRoof}_s(z, u, t) \longrightarrow \left[(S_s(z, u, t, 0) = \varrho_s(u)^m z t^k) \wedge \varrho_s(u)^m z t^k > 0 \right. \\ \left. \wedge (S_s(z, u, 0, t) = -\varrho_s(u)^m z t^k) \wedge -\varrho_s(u)^m z t^k < 0 \right].$$

(iv) Weight_s is the finite conjunction

$$u_0 \in \text{int } I_s, \quad \Phi'_{\text{base},s}(u_0) = 0, \quad \forall u (u \in I_s \longrightarrow \Phi_{\text{base},s}(u) \leq \Phi_{\text{base},s}(u_0)), \\ u_0 \geq 0, \quad \bigwedge_{i \in M} (\varrho_s(u_0) \geq 0), \quad u_0 + \sum_{i \in M} \varrho_s(u_0) = 1.$$

The associated zero-profile terms are, by definition, $a_* = u_0$ and $a_i = \varrho_s(u_0)$ for $i \in M$; the last line is exactly their nonnegativity and normalization, with no free weight variables.

The finite design tables, the Maclaurin estimate, pair averaging and root equalization, completion, reduced realization, and the quantum-graph type-assignment expansions are uniform coded proofs in ZFC; they are not additional members of this fixed-dimensional real-closed-field list. In particular, the identity (11.2) is verified by bounded expansion over the fixed type-assignment table and is not folded into Sign_s . Let $q_{s,i}^{\text{str}}$ be the Gödel code of the i th sentence in $\mathcal{S}_s^{\text{str}}$, for $i \in [4]$.

Lemma B.2. *The fixed structural setup can be augmented by a sequence $\mathbf{c}_s^{\text{str}}$ containing one real-closed-field proof code for every sentence in $\mathcal{S}_s^{\text{str}}$. ZFC proves*

$$\bigwedge_{i \in [4]} (\text{ClosedRCF}(q_{s,i}^{\text{str}}) \wedge \text{RCFPrf}((\mathbf{c}_s^{\text{str}})_i, q_{s,i}^{\text{str}}))$$

and also proves

$$\text{Profile}_s(\mathbf{P}_{0,s}), \quad W_{0,s} = w_{\mathbf{P}_{0,s}}, \quad \text{Sep}_s(W_{0,s}) = 0.$$

The maps $s \mapsto q_{s,i}^{\text{str}}$ and $s \mapsto \mathbf{c}_s^{\text{str}}$ are total computable for $s \geq r_{\text{str}}$, and the three displayed objects depend only on s .

Proof. The parameter search and the rational estimates in Proposition 10.9 prove Par_s . The scalar maximization in Lemma 9.1 proves Roof_s , direct substitution in the displayed definition of S_s proves Sign_s , and the isolating data for u_0 prove Weight_s . Quantifier elimination decides these four closed sentences. Enumerate derivations in the fixed complete real-closed-field calculus and retain the first proof of each one. The uniform soundness formula preceding Lemma A.2 lets ZFC verify the retained numerals.

The rational isolating intervals name unique real roots, so the zero-profile weights and $W_{0,s}$ are sets definable without parameters. Bounded arithmetic checks the finite edge-type table $\mathcal{M}_{0,s}$, and the finite type-assignment expansion in part (iii) of Lemma B.1 identifies its limit code with $W_{0,s}$. Applying Weight_s gives $\text{Profile}_s(\mathbf{P}_{0,s})$. The bounded quantum-graph identity (11.2), followed by the zero case of Sign_s , gives $\text{Sep}_s(W_{0,s}) = 0$. ■

B.4 Structural compiler factorization

Write $\tau_{\text{base}} = a_{\text{base}}/b_{\text{base}}$ in lowest terms and put $\mathcal{F}_f^{(k)} := \text{DecFam}_k(f)$. For every family code f , define

$$\mathcal{C}_f^{\text{in}} := \text{Forb}(\text{DecFam}_k(\text{ImgCode}_k(f))), \quad z_f := \Lambda_k(\mathcal{C}_f^{\text{in}}).$$

We now separate the base-family input from the structural postprocessing. Let $\text{Lift}_s: \mathbb{N} \rightarrow \mathbb{N}$ be the following fixed program. On input f , it computes $\text{ImgCode}_k(f)$, uses this code for the two sign projections in the rooted structural construction, and then performs the bounded basis-completion, witness-catalogue, star-blocker, edgewise-injective-image, and pair-covering-extension operations of Part II. It returns the final canonical s -family code. Define

$$\text{CodeLift}_s(f, g) \iff g = \text{Lift}_s(f),$$

and define the structural compiler relation by the synchronized formula

$$\text{CodeFam}_s^{\text{str}}(d, e, g) \iff \exists f (\text{CodeFam}_k^{\text{num}}(d, e, f) \wedge \text{CodeLift}_s(f, g)). \quad (\text{B.4})$$

For a lift pair (f, g) , let $\text{SAdm}_s(f, D, \mu)$ be the finite set-theoretic formula obtained by substituting $\text{ImgCode}_k(f)$ into the rooted structural incidence tables: it asserts that D is a finite s -graph and that μ is an admissible partial rooting of D . Put

$$\mathcal{D}_s(f) := \{D : \exists \mu \text{ SAdm}_s(f, D, \mu)\}.$$

Whenever ordinary data vertices and root labels occur in one rooted graph below, $U \dot{\cup} M$ denotes their fixed tagged disjoint union. We suppress the two canonical injections: thus id_M denotes the injection of M into the root summand, and intersections with U or M mean inverse images under the corresponding injection. We retain the finite presentation of an ordinary canonical completed model. Let $\text{SComp}_s(f, \hat{D})$ be the bounded formula asserting that $\hat{D}$ is a totally M -rooted s -graph, that $\text{SAdm}_s(f, \hat{D}, \text{id}_M)$ holds, and that applying the fixed completion operation of [Lemma 10.3](#) adds no edge. Thus $\hat{D}$ is a completed member of $\mathcal{D}_s(f)$.

For an ordered partition $\mathcal{V} = (U, (V_i)_{i \in M})$ and a completed rooted graph $\hat{D}$ on $U \dot{\cup} M$, let $\text{Exp}_{\mathcal{V}}(\hat{D})$ be the ordinary root-class expansion whose edge set is

$$\bigcup_{e \in E(\hat{D})} \left\{ (e \cap U) \cup \{v_i : i \in e \cap M\} : (v_i)_{i \in e \cap M} \in \prod_{i \in e \cap M} V_i \right\}.$$

If one of the root classes indexed by $e \cap M$ is empty, the corresponding product is empty and contributes no ordinary edge.

Let $\text{CompWit}_s(f, N, c)$ be the bounded finite-code formula asserting that c is the normalized code of a pair $(\mathcal{V}, \hat{D})$ such that $\mathcal{V} = (U, (V_i)_{i \in M})$ is an ordered partition of $V(N)$, the data vertex set of $\hat{D}$ is exactly U , the formula $\text{SComp}_s(f, \hat{D})$ holds, and $N = \text{Exp}_{\mathcal{V}}(\hat{D})$. Fix a primitive recursive bound $B_s(f, N)$ exceeding every normalized code of a partition and rooted edge table carried by $V(N) \dot{\cup} M$, and define

$$\text{CanComp}_s(f, N) \iff \exists c < B_s(f, N) \text{ CompWit}_s(f, N, c).$$

Indeed, every component of c is carried by $V(N) \dot{\cup} M$, and the edge table of $\hat{D}$ is a subset of the finitely many s -sets on this ground set, so the displayed bound is obtained by primitive recursion once s is fixed. Thus both CompWit_s and CanComp_s are bounded formulas. They record a presentation rather than asserting that a canonical partition is intrinsically unique.

Whenever $\text{CompWit}_s(f, N, c)$ holds, decode c and write its partition as

$$\mathcal{V}_{f, N, c} = (U_{f, N, c}, (V_{i, f, N, c})_{i \in M}).$$

Write $D_{f, N, c}$ for the completed rooted graph encoded by c . For $v(N) > 0$, give each data vertex of $D_{f, N, c}$ weight $1/v(N)$ and each root i weight $|V_{i, f, N, c}|/v(N)$; denote this weighting by $\mathbf{x}_{f, N, c}$. Also put

$$u_f(N, c) := \frac{|U_{f, N, c}|}{v(N)}, \quad y_{i, f}(N, c) := \frac{|V_{i, f, N, c}|}{v(N)}, \quad a_{\sigma, f}(N, c) := \frac{|U_{\sigma, f, N, c}|}{v(N)},$$

where $U_{\sigma, f, N, c}$ is the covered vertex set of the σ -projection of $D_{f, N, c}$. For $v(N) = 0$, assign fixed empty default values to these objects and zero to the scalar parameters.

Unpacking the finite expansion, ZFC proves uniformly that $\text{CompWit}_s(f, N, c)$ with $v(N) > 0$ implies

$$D_{f, N, c} \in \mathcal{D}_s(f), \quad t(H, N) = t(H, D_{f, N, c}, \mathbf{x}_{f, N, c}) \quad \text{for every finite } s\text{-graph } H,$$

and

$$\lambda(D_{f, N, c}; \mathbf{x}_{f, N, c}) = \frac{|N|}{v(N)^s}.$$

An edge of $D_{f, N, c}$ meeting an empty root class contributes neither an ordinary edge to N nor a positive monomial to either weighted identity. Thus the identities retain the complete rooted presentation without requiring it to be recoverable from the underlying ordinary graph. The localization, root-equalization, and finite-roof estimates are pointwise statements about every presentation witness and therefore apply to $(D_{f, N, c}, \mathbf{x}_{f, N, c})$. Thus $\mathcal{D}_s(f)$, $\text{CanComp}_s(f, N)$, and all presentation data used below are formulas of set theory uniform in f .

Let $\mathfrak{W}_s(f)$ be the closure in X_s of the homomorphism-density vectors of all weighted members of $\mathcal{D}_s(f)$, and put

$$\mathfrak{L}_s(f) := \{x \in \mathfrak{W}_s(f) : x_{\ell_0} = \Lambda_s(\mathcal{D}_s(f))\}.$$

Here $\Lambda_s(\mathcal{D}_s(f)) = s! \Lambda(\mathcal{D}_s(f))$. Under $\text{Base}_k(f)$, the scalar calculation in [Section B.5](#) is in the unnormalized convention and gives $\Lambda(\mathcal{D}_s(f)) = \Phi_{s,z_f}^*$, where Φ_{s,z_f}^* is the value singled out by Max_s ; in contrast, $\mathfrak{L}_s(f)$ is defined using normalized one-edge density.

B.5 The coded structural bridge

We now give the full substitution ledger behind [Proposition 18.4](#).

Details for Proposition 18.4. For part (i), the bounds in [Proposition 2.8](#) and [Lemmas 2.9, 10.3](#) and [12.6](#) bound every enumeration in Lift_s by a primitive recursive function of f after s has been fixed. Canonicalization is a bounded search through vertex permutations. ZFC proves totality and uniqueness by induction on these recursions. If $f = \text{Comp}_k^{\text{num}}(d, e)$, the compiler invariant identifies its decode with $\mathcal{F}_{k,\beta_{d,e}}$, and $\text{ImgCode}_k(f)$ is the canonical code of $\mathcal{Q}(\mathcal{F}_{k,\beta_{d,e}}^\circ)$. This is exactly the input-dependent family used in the structural construction; its members have no isolated vertices. Every later choice is the same lexicographically first bounded choice fixed in [Part II](#). Hence the composite in [\(B.4\)](#) returns the canonical code of $\mathcal{G}_{s,\beta_{d,e}}$.

Fix f, g under the antecedent of [\(18.2\)](#) and recall $\mathcal{C}_f^{\text{in}}$ and z_f from the definitions above. The map $f \mapsto \text{ImgCode}_k(f)$ is primitive recursive, since it is computed by bounded enumeration of finite vertex maps and edgewise-injective images. The Brown–Simonovits blowup and approximation argument used in [Lemma 2.4](#), formalized in ZFC, gives

$$z_f = \pi(\mathcal{F}_f^{(k)}). \quad (\text{B.5})$$

Moreover, $\mathcal{C}_f^{\text{in}}$ is blowup closed. Under $\text{Base}_k(f)$, the code f is canonical, and the filtered decoder retains only k -graphs having an edge. Deleting isolated vertices and then taking edgewise-injective homomorphic images therefore gives a family whose members have an edge and no isolated vertices. Composition of the defining maps shows that this family is closed under further edgewise-injective homomorphic images. These three assertions are bounded statements about the finite decoder and vertex maps, so ZFC proves them uniformly from $\text{Base}_k(f)$.

Under the same predicate, the fixed template $K_{*,k}$ belongs to $\mathcal{C}_f^{\text{in}}$ and has normalized value τ_{base} ; hence

$$\tau_{\text{base}} \leq z_f \leq 1.$$

Thus the inner optimizer space is nonempty and compact, and the fixed structural certificates apply at $z = z_f$.

We record the uniform substitution ledger used below. Under $\text{Base}_k(f) \wedge \text{CodeLift}_s(f, g)$, ZFC proves the following four assertions.

(U1) The basis-completion relation defines $\mathcal{D}_s(f)$, and the family decoded from g satisfies both conclusions of [Lemma 13.2](#); in particular, its free class is blowup invariant, and every graph satisfying $\text{CanComp}_s(f, \cdot)$ is $\mathcal{F}_g$ -free.

(U2) The exact-density identity is

$$\pi_s(g) = \Lambda_s(\mathcal{D}_s(f)). \quad (\text{B.6})$$

(U3) The optimizer, converse-realization, and determination implications in [Lemma 10.10](#) hold with $\mathcal{C}_f^{\text{in}}$, z_f , $\mathcal{D}_s(f)$, and $\mathfrak{L}_s(f)$ in place of the corresponding β -indexed objects.

(U4) For all positive rationals ε, η , there are a positive rational δ and an integer n_0 such that every canonical $\mathcal{F}_g$ -free s -graph G satisfying

$$v(G) \geq n_0, \quad |G| \geq \text{ex}(v(G), \mathcal{F}_g) - \delta v(G)^s$$

has codes N, c satisfying

$$\begin{aligned} \text{CompWit}_s(f, N, c), \quad V(N) &= V(G), \\ \text{edit}(G, N) &\leq \varepsilon v(G)^s, \quad |N| \geq \text{ex}(v(G), \mathcal{F}_g) - \eta v(G)^s. \end{aligned}$$

For part (U1), the proof of Lemma 10.3 is a bounded enumeration with the finite inner-family code as a parameter. The (F5) case of Lemma 12.6 and the proof of Lemma 12.7 use, respectively, the absence of isolated vertices and closure under further edgewise-injective images, both verified above uniformly from $\text{Base}_k(f)$. The remaining arguments in Lemma 13.2 use the code only through the resulting admissibility predicate. Part (U2) is the same parameter substitution in the two inequalities of Proposition 13.1. Part (U3) follows by substituting the same code into the three displayed implications of Lemma 10.10; its hypotheses are precisely blowup closure and $z_f \in [\tau_{\text{base}}, 1]$, verified above.

For part (U4), blowup invariance is part (U1). The fixed reference model D_{bal} in Lemma 10.7 has positive density, so (B.6) gives nondegeneracy. By Lemma 12.2 and part (ii) of Lemma 13.2, the coded canonical class is closed under isomorphisms and arbitrary subgraphs and consists of $\mathcal{F}_g$ -free graphs. The substitutions in Propositions 14.2 and 14.3 give strong symmetrized stability and vertex extendability. Hence Theorem 15.1 gives degree stability, and the finitary degree-stability-to-edit argument in the proof of part (ii) of Theorem 15.2 gives the graph N together with a witnessing partition and the final completed rooted graph whose root-class expansion is N . These two finite objects form a code c satisfying $\text{CompWit}_s(f, N, c)$ and prove exactly the quantified implication in part (U4). This proves the four ledger entries as formulas of ZFC; no halting predicate occurs in them.

Part (i) of Lemma A.1, specialized to k , shows that $\text{Zero}_k(f)$ implies $z_f = \tau_{\text{base}}$. If $\text{Pos}_k(f)$ holds, the balanced witness and its blowups give $z_f = \pi(\mathcal{F}_f^{(k)}) > \tau_{\text{base}}$. Conversely, under $\text{Base}_k(f)$, part (ii) of that lemma gives

$$\text{Zero}_k(f) \iff \neg \text{Pos}_k(f). \quad (\text{B.7})$$

Coded weighted phases. For $c > 0$, define the two closed level sets

$$\mathfrak{L}_{s,\pm}(f, c) := \{x \in \mathfrak{L}_s(f) : \text{Sep}_s(x) = \pm c\}.$$

We claim first that the weighted optimizer space has the following two descriptions:

$$z_f = \tau_{\text{base}} \implies \mathfrak{L}_s(f) = \{W_{0,s}\}, \quad (\text{B.8})$$

$$z_f > \tau_{\text{base}} \implies \exists c_f > 0 \left[\mathfrak{L}_s(f) = \mathfrak{L}_{s,+}(f, c_f) \dot{\cup} \mathfrak{L}_{s,-}(f, c_f), \quad \mathfrak{L}_{s,+}(f, c_f) \neq \emptyset \neq \mathfrak{L}_{s,-}(f, c_f) \right]. \quad (\text{B.9})$$

The two sets in (B.9) are compact because $\mathfrak{L}_s(f)$ is compact and Sep_s is continuous.

To verify the claim, take a coded optimizing sequence of completed weighted models. Root localization and equalization are the uniform coded arguments in Lemmas 10.7 and 10.8; the fixed inequalities needed there are certified by Par_s . Part (U3) shows that every subsequential scalar limit (u, a, b) maximizes the fixed polynomial Φ_{s,z_f} and that its nonzero sign projections converge to optimizer limits of $\mathcal{C}_f^{\text{in}}$. When $z_f = \tau_{\text{base}}$, Roof_s makes $(u_0, 0, 0)$ the unique scalar limit. The determination implication in part (U3) then identifies the entire weighted limit with $W_{0,s}$, while its converse implication constructs that limit. This proves (B.8).

When $z_f > \tau_{\text{base}}$, let u_f and $t_f > 0$ be the positive-branch scalar parameters certified by Roof_s at $z = z_f$. That certificate leaves exactly the two pure-sign triples $(u_f, t_f, 0)$ and $(u_f, 0, t_f)$. Compactness of the inner optimizer space and the optimizer clause of Lemma 10.10 place every weighted optimizer in one of the two asserted compact sets. The converse clause realizes every inner optimizer with either sign, so both sets are nonempty. The determination clause shows that their union is the entire optimizer space. Finally, the finite quantum-graph expansion in Lemma 11.2, followed by Sign_s , gives

$$c_f = \varrho_s(u_f)^m z_f t_f^k > 0$$

on the positive phase and its negative on the other phase. This proves (B.9). Each scalar implication used here is verified by $\mathbf{c}_s^{\text{str}}$. The remaining steps are the compactness, cloning and completion, and finite homomorphism-density expansions in Lemma 10.10, all expressed by the displayed set-theoretic formulas.

Coded weighted-to-ordinary transfer. We next prove the exact set-theoretic identity

$$\mathbf{E}_s(g) = \mathfrak{L}_s(f). \quad (\text{B.10})$$

Let $x \in \mathbf{E}_s(g)$. Part (ii) of Lemma B.1 gives a coded asymptotically extremal free sequence $\mathbf{G}$ converging to x . For $j \geq 1$, apply part (U4) with $\varepsilon = \eta = 1/j$ and replace the resulting δ_j by $\min\{\delta_j, 1/j\}$. Since $\mathbf{G}$

is asymptotically extremal and its orders tend to infinity, there are increasing indices M_j such that every $n \geq M_j$ satisfies the order and deficit hypotheses for level j . Put

$$j(n) := \max(\{1\} \cup \{j \leq n : M_j \leq n\}).$$

Then $j(n) \rightarrow \infty$. For each sufficiently large n , choose the least coded pair (N_n, c_n) supplied at level $j(n)$. Part (U4) gives

$$\text{edit}(G_n, N_n) = o(v(G_n)^s), \quad |N_n| = \text{ex}(v(G_n), \mathcal{F}_g) - o(v(G_n)^s).$$

Equation (B.3) shows that N_n has the same limit x . The presentation c_n and the identities following the definition of CompWit_s give

$$s!\lambda(D_{f, N_n, c_n}; \mathbf{x}_{f, N_n, c_n}) = \frac{s!|N_n|}{v(N_n)^s} \rightarrow \Lambda_s(\mathcal{D}_s(f)).$$

Every homomorphism density is preserved by this compression. The sequence $(D_{f, N_n, c_n}, \mathbf{x}_{f, N_n, c_n})$ is therefore a weighted optimizing sequence converging to x , so $x \in \mathfrak{L}_s(f)$.

Conversely, take $x \in \mathfrak{L}_s(f)$ and put

$$\varepsilon_j = \frac{1}{j(1 + s! \max_{i \leq j} |H_i|)}.$$

By the definition of $\mathfrak{L}_s(f)$ and continuity of the first j homomorphism coordinates and the normalized one-edge density, there is a weighted member $(D_j, \mathbf{x}^{(j)})$ of $\mathcal{D}_s(f)$, together with an admissible partial rooting, such that

$$\begin{aligned} \max_{i \leq j} |t(H_i, D_j, \mathbf{x}^{(j)}) - x_i| &< \varepsilon_j, \\ s!\lambda(D_j; \mathbf{x}^{(j)}) &> \Lambda_s(\mathcal{D}_s(f)) - \varepsilon_j. \end{aligned}$$

Countable choice in ZFC selects one such pair and rooting for every j . Complete this partial rooting, obtaining $\overline{D}_j$, and extend $\mathbf{x}^{(j)}$ by zero on every root added by the completion, retaining the same notation for the extended vector. Since no weighted member of $\mathcal{D}_s(f)$ has normalized value above $\Lambda_s(\mathcal{D}_s(f))$, the normalized gain is less than ε_j . The finite edge-union estimate therefore gives, for $i \leq j$,

$$0 \leq t(H_i, \overline{D}_j, \mathbf{x}^{(j)}) - t(H_i, D_j, \mathbf{x}^{(j)}) \leq |H_i| \varepsilon_j.$$

On the fixed completed graph, choose a rational probability vector $\mathbf{x}_{\text{rat}}^{(j)}$ that is positive on every vertex, including every root, and changes the normalized value and each of the first j coordinates by less than ε_j . This is possible because the relative interior of the finite probability simplex is dense and the finitely many relevant polynomials are continuous.

Take a common denominator for $\mathbf{x}_{\text{rat}}^{(j)}$ and multiply it and all numerators so that the denominator n_j is at least j . Clone each data vertex according to the numerator of its weight, complete the cloned rooted graph, and replace each root by a class whose size is the numerator of its root weight. Data cloning and the final root blowup preserve every weighted homomorphism density exactly. Before the second completion the normalized value exceeds $\Lambda_s(\mathcal{D}_s(f)) - 2\varepsilon_j$; hence that completion gains less than $2\varepsilon_j$ and changes the i th coordinate by at most $2|H_i|\varepsilon_j$. These operations also produce a normalized presentation code c_j satisfying $\text{CompWit}_s(f, N_j, c_j)$. Hence part (U1) makes the resulting ordinary canonical completed model N_j $\mathcal{F}_g$ -free, and it satisfies

$$|t(H_i, N_j) - x_i| \leq (2 + 3|H_i|)\varepsilon_j \quad (i \leq j),$$

and

$$\Lambda_s(\mathcal{D}_s(f)) - 2\varepsilon_j < \frac{s!|N_j|}{n_j^s} \leq \Lambda_s(\mathcal{D}_s(f)).$$

Thus $s!|N_j|/n_j^s \rightarrow \Lambda_s(\mathcal{D}_s(f)) = \pi_s(g)$. Since $n_j \rightarrow \infty$ and $s! \binom{n_j}{s} / n_j^s \rightarrow 1$, the ordinary edge densities of N_j also tend to $\pi_s(g)$. Hence $\text{AExt}_s(g, \mathbf{N})$ and $\text{Conv}_s(\mathbf{N}, x)$ hold. The reverse implication in (B.1) gives $x \in \mathbf{E}_s(g)$. This proves (B.10) without invoking an external truth predicate for the analytic statement.

The zero branch. Suppose $\text{Zero}_k(f)$. Equations (B.5), (B.8), and (B.10) give

$$\mathbf{E}_s(g) = \{W_{0,s}\}, \quad W_{0,s} = w_{\mathbf{P}_{0,s}}, \quad \text{Sep}_s(W_{0,s}) = 0.$$

The last two assertions and $\text{Profile}_s(\mathbf{P}_{0,s})$ are verified in Lemma B.2.

It remains to verify the profile quantifier in $\text{Str}_{5,s}$. We prove the stronger finitary formula

$$\begin{aligned} \text{ZFC} \vdash \quad & \forall f \forall g \left(\text{Base}_k(f) \wedge \text{CodeLift}_s(f, g) \wedge z_f = \tau_{\text{base}} \right) \\ & \longrightarrow \text{FStab}_s(g, \mathbf{P}_{0,s}). \end{aligned} \quad (\text{B.11})$$

Fix the antecedent and a positive rational ε . Choose a positive rational $\kappa < \varepsilon/12$. First, the coded substitution in Lemma 10.7, applied presentationwise, supplies a deficit threshold $\eta_{\text{loc}} > 0$ and an order threshold N_{loc} beyond which $u_f(N, c) \in I_s$ and the selected root vector lies in the fixed coordinate box whenever $\text{CompWit}_s(f, N, c)$ holds. Within that box, the finite-roof estimate in Lemma 15.4, the zero clause of Roof_s , the positive root-equalization gap in (15.4), and compactness give a positive rational $\eta_0 \leq \eta_{\text{loc}}$ and an integer $N_1 \geq N_{\text{loc}}$ with the following property: if $\text{CompWit}_s(f, N, c)$ holds, $v(N) \geq N_1$, and

$$0 \leq \Lambda_s(\mathcal{D}_s(f)) - \frac{s!|N|}{v(N)^s} \leq \eta_0,$$

then

$$a_{+,f}(N, c) + a_{-,f}(N, c) < \kappa, \quad |u_f(N, c) - u_0| + \sum_{i \in M} |y_{i,f}(N, c) - \varrho_s(u_0)| < \kappa. \quad (\text{B.12})$$

This is a coded implication: after a proposed κ is fixed, failure for every rational η_0 and every N_1 produces presentation codes (N, c) and hence a maximizing sequence in the compact scalar box; Roof_s forces its scalar limit to be $(u_0, 0, 0)$, while (15.4) forces the selected root vector to be equalized, contradicting the negation of (B.12). Taking the least code of an admissible rational-integer pair in the fixed enumeration makes η_0 and N_1 definable in ZFC.

Apply part (U4) with edit tolerance $\varepsilon/3$ and edge-count deficit tolerance $\eta_0/(4s!)$. Increase its order threshold so that

$$\left| \pi_s(g) - \frac{s! \text{ex}(n, \mathcal{F}_g)}{n^s} \right| < \frac{\eta_0}{2}$$

and every fixed rounding error is smaller than $\varepsilon n^s/6$. For every graph G satisfying the resulting finitary antecedent, that ledger entry supplies codes N, c with $\text{CompWit}_s(f, N, c)$ and $V(N) = V(G)$ such that the normalized deficit of N is less than $3\eta_0/4$ and

$$\text{edit}(G, N) \leq \frac{\varepsilon}{3} v(G)^s \quad \text{and} \quad (N, c) \text{ satisfies (B.12)}.$$

Let $S := U_{+,f,N,c} \cup U_{-,f,N,c}$ be the active data vertices in the presentation encoded by c . The first inequality in (B.12) gives $|S| < \kappa v(G)$. Delete every raw signed-record edge from $D_{f,N,c}$, and complete the resulting rooted model through the same partition $\mathcal{V}_{f,N,c}$; denote its ordinary expansion by N' . Every raw σ -record edge of $D_{f,N,c}$ has all its data vertices in $U_{\sigma,f,N,c} \subseteq S$. Hence the old and new sign projections, and their covered vertex sets, agree outside S . By Lemma 15.5, the graphs N and N' can differ only on edges meeting S .

The second inequality in (B.12) permits the root and neutral classes of N' to be changed to the largest-fractional-parts realization of the fixed profile by moving a set T of fewer than $\kappa v(G) + O_s(1)$ vertices. Let B be the completion for this new partition. The two partitions agree outside T , and both N' and B have empty sign projections. A direct check of the fixed rooted edge types shows that every s -set disjoint from T has the same edge status in the two completions: its root/data role profile is unchanged, the frame and dispersion rules are unchanged, every threshold and penalty rule sees the same empty covered sets, and no raw record edge is present. Thus N' and B can differ only on edges meeting T , so N and B can differ only on edges meeting $S \cup T$. The graph B belongs to $\mathfrak{B}_{\mathbf{P}_{0,s}}(V(G))$, and, after the preceding threshold enlargement,

$$\text{edit}(N, B) \leq |S \cup T| \binom{v(G) - 1}{s - 1} < \frac{2\varepsilon}{3} v(G)^s.$$

The triangle inequality gives $\text{edit}(G, B) < \varepsilon v(G)^s$. This proves (B.11) with exactly the quantifier order in the definition of FStab_s . Equation (B.2) then yields the profile quantifier in $\text{Str}_{5,s}$. The singleton is

connected, neither sign predicate can hold at a zero value, and the displayed profile witnesses $\text{Str}_{5,s}(g)$. Hence $\text{ZeroStr}_s(g)$ holds.

The positive branch. Suppose $\text{Pos}_k(f)$. Fix $c_f > 0$ supplied by (B.9). Equations (B.5), (B.9), and (B.10) give two nonempty compact ordinary phase sets with $\text{Sep}_s = \pm c_f$. Choose a rational q with $0 < q < c_f$. Then every member of $E_s(g)$ has $\text{Sep}_s \geq q$ or $\text{Sep}_s \leq -q$, and both alternatives occur. These relatively closed sets are also relatively open because their complement is the other one. Thus $\text{Str}_{1,s}(g)$ and $\text{Str}_{2,s}(g)$ fail, whereas $\text{Str}_{3,s}(g)$ and $\text{Str}_{4,s}(g)$ hold. The two selected points with opposite signs and Lemma 18.3 give $\neg \text{Str}_{5,s}(g)$. Hence $\text{PosStr}_s(g)$ holds.

We have proved the implications $\text{Zero}_k(f) \rightarrow \text{ZeroStr}_s(g)$ and $\text{Pos}_k(f) \rightarrow \text{PosStr}_s(g)$. The two structural rows are contradictory already in their first conjunct, while (B.7) says that exactly one numerical row holds. If, for example, $\text{ZeroStr}_s(g)$ held while $\text{Zero}_k(f)$ failed, then $\text{Pos}_k(f)$ and hence $\text{PosStr}_s(g)$ would hold, a contradiction. The other reverse implication is identical. This proves (18.2). ■

B.6 The uniform structural bridge

The remaining composition keeps the numerical and structural compiler codes synchronized.

Details for Proposition 18.5. The numerical compiler is a total primitive recursive function, and part (i) of Proposition 18.4 gives a unique lift. Equation (B.4) therefore proves part (ii).

Fix d, e, g under the antecedent of (18.3). By (B.4), there is an f satisfying

$$\text{CodeFam}_k^{\text{num}}(d, e, f) \wedge \text{CodeLift}_s(f, g).$$

The defining program for $\text{CodeFam}_k^{\text{num}}$ is functional, so this f is unique. Apply Proposition 18.1 at the base uniformity k . It gives $\text{Base}_k(f)$ and

$$\text{All}_{\text{PR}}(d, e) \iff \text{Zero}_k(f).$$

Bad-run completeness and the certificate transformations give

$$\neg \text{All}_{\text{PR}}(d, e) \iff \text{Pos}_k(f).$$

Part (ii) of Proposition 18.4 converts these two formulas into $\text{ZeroStr}_s(g)$ and $\text{PosStr}_s(g)$, respectively. Expanding the two row definitions gives all five code-level equivalences in (18.3), proving part (i). Uniqueness of g makes the existential definition of each $P_{i,s}(d, e)$ equivalent to the corresponding predicate on this one code, and hence proves part (iii).

Finally, Lemma B.2 fixes the three named objects and all of their defining codes before d, e are quantified. This proves part (iv) and completes the proposition. ■